THE PSYCHOLOGY OF SOCIABILITY

Understanding Human Attachment

THE SYDNEY SYMPOSIUM OF SOCIAL PSYCHOLOGY

EDITED BY JOSEPH P. FORGAS,
WILLIAM D. CRANO, AND KLAUS FIEDLER

THE PSYCHOLOGY OF SOCIABILITY

This edited volume brings together the latest research in understanding the nature, origins, and evolution of human sociability, one of the most intriguing aspects of human psychology.

Sociability—our sophisticated ability to interact with others, imagine, plan, and execute interdependent behaviors—lies at the heart of our evolutionary success, and is the most important prerequisite for the development of increasingly elaborate civilizations. With contributions from internationally renowned researchers in areas of social psychology as well as anthropology and evolutionary psychology, this book demonstrates the role of social psychology in explaining how human sociability evolved, how it shapes our mental and emotional lives, and how it influences both large-scale civilizational practices and intimate interpersonal relations. Chapters cover the core psychological characteristics that shape human sociability, including such phenomena as the role of information exchange, affective processes, social norms, power relations, personal relationships, attachment patterns, personality characteristics, and evolutionary pressures.

Featuring a wide variety of empirical and theoretical backgrounds, the book will be of interest to students and researchers in all areas of the social sciences, as well as practitioners and applied professionals who deal with issues related to sociability in their daily lives.

Joseph P. Forgas is Scientia Professor at the University of New South Wales. His research focuses on affective influences on social cognition and behavior. For his work, he received the Order of Australia and the Distinguished Scientific Contribution Award, and he has been elected Fellow of the Australian and Hungarian Academies of Science.

William Crano is Oskamp Professor of Psychology at Claremont Graduate University. He was Liaison Scientist for the US Office of Naval Research, NATO Senior Scientist, and Fulbright Senior Scholar. His research focuses on attitude development and attitude change and their applications.

Klaus Fiedler is Professor of Psychology at the University of Heidelberg and Fellow of the German Academies of Science, the Association for Psychological Sciences, and Society for Personality and Social Psychology. His research focuses on social cognition, language, judgments, and decision-making. He received several awards, including the Leibniz Award, and he is on the editorial boards of leading journals.

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Dorottya Lantos and Oliver Keenan Goldsmiths, Joachim I. Krueger, David J. Grüning, Jan-Willem van Prooijen, Arie W. Kruglanski, Erica Molinario, Gilda Sensales, Klaus Fiedler, Michael A. Hogg, Oluf Gøtzsche-Astrup, Joseph P. Forgas, Dorottya Lantos, Péter Krekó, Eotvos Lorand, Leonie Huddy, Alessandro Del Ponte, Michele J. Gelfand, Rebecca Lorente, Amber M. Gaffney, Joel Cooper and Joseph Avery, Robin R. Vallacher and Eli Fennell, Stanley Feldman, William D. Crano and Klaus Fidler.

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Understanding Human Attachment

Edited by Joseph P. Forgas, William Crano, and Klaus Fiedler

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CONTRIBUTORS

Baumeister, Roy F. University of Queensland, Australia

Crano, William D. Claremont Graduate University

Dunn, Elizabeth W. University of British Columbia

Ellenberg, Molly, University of Maryland

Fiedler, Klaus, University of Heidelberg

Fiske, Alan Page, University of California, Los Angeles

Forgas, Joseph P. University of New South Wales

Gaffney, Amber M. Humboldt State University

Grüning, David J. University of Heidelberg

Hogg, Michael A. Claremont Graduate University

Hütter, Mandy, University of Tübingen

Itzchakov, Guy, University of Haifa

Krueger, Joachim I. Brown University

Kruglanski, Arie W. University of Maryland

Lee, Karisa Y. University of Rochester

Lok, Iris, University of British Columbia

Lyubomirsky, Sonja, University of California, Riverside

Maranges, Heather M. Concordia University, Canada

Mikulincer, Mario, Interdisciplinary Center (IDC) Herzliya

Prislin, Radmila, San Diego State University

Regan, Annie, University of California, Riverside

Reis, Harry T. University of Rochester

Ruan, Yan, Facebook, Inc.

Shaver, Phillip R. University of California, Davis

Smith, Nicholas M. A. University of Queensland

Sudnar, Tanushri, Brown University

Von Hippel, William, University of Queensland

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1

THE PSYCHOLOGY OF HUMAN SOCIABILITY

From Individuals to Community

Joseph P. Forgas, William D. Crano, and Klaus Fiedler

Abstract: In this chapter we review different alternative approaches to understanding sociability, including nativist (evolutionary), environmentalist (learning), and interactional models. Next, different views about human sociability by classic psychologists, and the contributions of empirical social psychology to understanding the social nature of humans are reviewed. We then turn to discussing the importance of sociability for human flourishing and well-being, and the role of sociability in generating, disseminating, and managing collective knowledge and the transmission of cultural information. In the final section of the chapter, contributions to the book are summarized, organized into four topic areas such as (1) basic approaches to sociability, (2) sociability and well-being, (3) sociability and knowledge, and (4) sociability and human relationships.

Introduction

Sociability is a defining feature of *homo sapi*ens—our survival and stunning evolutionary success as the dominant creatures on this planet are due to our unparalleled ability to cooperate and organize ourselves into ever-more efficient and sophisticated social groups (Dunbar, 1998; Harari, 2014; see also von Hippel & Smith, this volume). There are other species whose survival strategy is based on intense sociability and group cooperation, but these patterns of sociability are rigid, reflexive, and inflexible. In contrast, humans are unique in having developed highly sophisticated patterns of symbolic sociability that can be readily

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shaped and adapted to social, environmental and cultural challenges (see also Prislin & Crano, this volume).

Human sociability has given rise to ever-changing cultural and civilizational adaptations, turning individuals into sophisticated and flexible members of ever-larger social groups and communities (Durkheim, 1997; Tönnies, 2001). The way we interact with each other can be modified, revised, and transmitted, and we are unique in having the symbolic cognitive capability to represent fictional realities (Harari, 2014; see also Baumeister & Maranges; Fiske, this volume). The processes of gathering, transmitting, exchanging, and modifying information among individuals are the core functions of human sociability (see also Fiedler & Hütter, this volume). In this way, a shared corpus of taken-for-granted knowledge accumulates that is permanently refined and transmitted to subsequent generation, making cultural evolution possible (Pinker, 2018).

Sociability: Individualism versus Collectivism

A fascinating feature of human sociability is that it provides the mechanism that helps to reconcile the conflict between our selfish, individual, and altruistic, collective motivations. Sociability is the glue that holds human groups together and turns individuals into communities, and that allows groups to thrive and succeed, thus protecting individuals and enhancing their chances of survival (Buss, 2019; von Hippel, 2018; von Hippel & Smith, this volume). Yet the very same mechanisms of sociability that generate group integration can also be employed to produce selfish benefits for the individual. In a fundamental sense, sociability is the name of a perpetual power game, where groups and individuals vie with each other for improved opportunities (see also Krueger et al., this volume). Individuals in their daily social interactions are continuously exposed to dilemmas where selfish benefits to themselves must be balanced against expectations and benefits to their groups, decisions that must be made rapidly and instinctively (see also Baumeister & Maranges, this volume). Navigating this maze of sociability that pits selfishness against cooperation is one of the most cognitively demanding tasks humans face (Dunbar, 1998; Haidt, 2012).

In extreme cases of often spontaneous sociability, individuals become submerged in a total collective experience, as is the case in some religious rituals, sports events, rock concerts, or demonstrations. This feeling of oneness with a group has an intense emotional quality that has long-fascinated researchers such as Durkheim, Le Bon, Tarde, and Tajfel. Alan Page Fiske (this volume) offers a careful analysis of the features and functions of the "kama muta" feeling of intense bonding and group attachment. It seems that humans are endowed with a deep sensitivity for powerful imitative social experiences that is unique to our species.

The objective of this book is to survey the latest research on what we now know about the psychology of human sociability. Chapters are organized into

four sections, dealing with (1) the nature and basic approaches to sociability, (2) the importance of sociability for human well-being and flourishing, (3) the role of sociability in generating and disseminating knowledge, and (4) how sociability shapes our relationships. This chapter lays the foundations for much of what follows.

Approaches to Sociability

We may start by asking a fundamental question: How can we explain the intense and all-encompassing sociability so characteristic of humans? Several approaches have been proposed. Perhaps the most common explanation is based on nativist, evolutionary approaches.

Nativist Approaches

Humans are sociable because in our ancestral environment, sociability was adaptive and essential for survival (Buss, 2019; Harari, 2014). Accordingly behavioral tendencies for sociability have been naturally selected and are now part of our shared evolutionary heritage, defining our species. One issue with this theory is that humans are also quite capable for antisocial behavior as well as dishonesty and dissimulation, so *faking* sociability for selfish ends often works just as well as genuine sociability (Buss, 2019). Indeed, fake sociability may offer special survival benefits to successful but selfish practitioners.

The nativist approach was traditionally adopted by French sociologists like Le Bon (1895/2009) and Gabriel Tarde (1898) among others, who thought that humans are born with a powerful "instinct" to follow and imitate each other. It is this innate tendency to coordinate and imitate that explains the remarkable human ability to spontaneously coordinate complex social behaviors even in very large groups and crowds. According to Le Bon, crowds can be seen as new emerging psychological entities, shaped by the spontaneous but unconscious imitative reactions of its members (see also Fiske, this volume). Some of Le Bon's and Tarde's ideas found new relevance in the psychological analysis of modern-day Internet influencers who are also in the business of producing powerful patterns of spontaneous imitation and modeling.

Another major voice in the development of ideas about sociability was *Wilhelm Wundt*, who also thought that sociability is a fundamental part of being human (e.g., Grotius, 1625/1901; Wundt, 1900). For Wundt, human sociability explains our tendency to form interactive, supportive social groups, the subject matter of *social* psychology. Wundt thought that social psychology could not be an experimental discipline owing to the nature of its subject matter, and should be concerned with the study of tribes, cultures, nations, etc., as later indeed undertaken by social anthropologists. Despite Wundt's recommendations, the study of sociability has eventually become part of experimental social

psychology, focusing on the study of the social nature of individuals, as illustrated by contributions to this book as well.

William James also saw sociability as an inherited instinct, a potent but malleable tendency that responds to environmental triggers. Evolutionary psychologists today owe a large debt to James' observations (often with little data to support them) who considered humans gregarious animals who were excited by both the presence *and* the absence of conspecifics. Consistent with the Hobbesian view James considered human sociability as driven by self-serving desires that could only be satisfied when engaging with others (see also Kruglanski & Ellenberg; and Krueger et al., this volume). This idea was also echoed in the work of *William McDougall* (1908), who proposed a "gregarious instinct" as the basis of all human sociability that could be satisfied only by interacting with others, foreshadowing later work of Festinger (1954), Sherif (1936), and Tajfel and Turner (1979).

Environmentalist Approaches

A complementary theoretical view of sociability emphasizes the acquired, learned aspects of being social. The rather simple-minded *learning/reinforcement* views advocated by Watson (1924) and Skinner (1974) posit that sociability is acquired from childhood as infants learn to associate positive reinforcements (food, care, support) with other people, thus becoming sociable. This unnecessarily doctrinaire explanation denies the influence of inborn predispositions, as well as the very real possibility that others are not always the source of rewards but can be highly aversive and punitive as well.

Subsequently, Floyd Allport attempted a synthesis by recognizing on the one hand the inherited bases of sociability, but on the other hand seeing these enduring tendencies as modified by learning. He viewed gregariousness as an instinctual cause of sociability, driven by the desire to satisfy basic needs (Allport, 1924). Influential contemporary theories of sociability such as attachment theory suggest a similar view: early childhood experiences may have a disproportionate influence on establishing enduring patterns of sociability, but these can be modified as a result of life-long influences we receive in our developing personal relationships (see also Mikulincer & Shaver, this volume). Whatever early experiences shape our patterns of attachment, these are not fixed but to some extent malleable.

Social exchange theory offers a more mercenary example of a flexible model that sees human sociability as driven by the mutually rewarding nature of many social interactions, where both participants can derive real benefits by engaging with each other (Homans, 1961). Exchange theory is based on an economic market metaphor, relying on a cost-benefit analysis suggesting that by interacting with others, humans can exchange resources that are more valuable to the recipient than their cost for the donor, thus producing mutual benefits or "profit."

Mutually beneficial exchanges occur in romantic relationships, friendships, professional relationships, and even exchanging words with a customer at the cash register (see also Regan & Lyubomirsky; Dunn & Lok, this volume).

Interactionist Approaches

A popular philosophical view of sociability considered it to be the outcome of a "social contract," in which individualism and collectivism, and selfishness and cooperation are in a dynamic balance and complement each other. On the one hand, sociability is essential to control unconstrained self-serving desires (e.g., Hobbes, 1651/2020). On the other hand, individual creativity and achievement can only flourish when freed from the repressive influence of society (Rousseau, 1775/1950). Kant (1784/2008) joined these views in his *Universal History*, where he coined the term unsocial sociability to describe a dynamic in which the need for sociability is threatened by selfish assertiveness, risking social disorganization or destruction. In a sense, the fragile evolution of liberal democracy, the most successful civilization in human history, can also be seen as the result of a dynamic but fragile evolutionary equilibrium path between too much individualism and potential chaos on the one hand, and too much collectivism and repressive state control on the other (Acemoglu & Robinson, 2019; Pinker, 2018).

Symbolic interactionism. Perhaps the most sophisticated interactive theory of human sociability was developed by *George Herbert Mead* in his "symbolic interactionism" (Mead, 1934). Mead tried to understand the fundamental links that bind unique individuals to the larger systems of sociability that shape them. He argued that both individual personalities and larger social systems can only emerge over the course of repeated symbolic social interactions between individuals. It is the unique human ability to mentally represent and symbolize our interactive experiences that makes this possible. For Mead, the actual, face-to-face interaction between people is the crucible in which both unique individual personalities and larger social systems are formed.

As humans are endowed with the capacity to represent their experiences symbolically, their interactions necessarily affirm, or sometimes modify the way they see themselves (their personalities), as well as the social norms and expectations that they bring to the situation. We become the people that others see and accept us as being—and this knowledge can only come from interactions with others (see Kruglanski & Ellenberg, this volume). This is also the idea behind Charles Cooley's "looking glass self"—those of our interactive performances that are successful and accepted by others become our view of ourselves, our personality. Those that fail are discarded. Although rarely recognized, Festinger's social comparison theory owes much to Mead's pioneering ideas. For Mead, there can be no meaningful personality without regular social interactions and sociability to create and refine it, just as there can be no larger social system

or culture without social interaction to sustain and modify it (see also Fiske, this volume).

Perhaps unfortunately, Mead's (1934) symbolic interactionism, identifying sociability as the source of both the individual person and the social collective had relatively little influence on empirical social psychology. However, symbolic interactionism gave rise to a thriving micro-sociological tradition in the works of Erving Goffman and Rom Harre, and the ethnomethodological work of Harold Garfinkel, who sought to discover the subtle patterns of taken-for-granted expectations that regulate everyday social life, as a mixture of both predictable rule-following and creative innovations. It is indeed unfortunate that these endeavors have not yet been properly integrated into the mainstream of social psychological inquiry.

Theory of mind. As symbolic interactionists discovered, the most dramatic feature of human sociability is its infinitely flexible and adaptive nature that makes social interactions enormously variable, and makes cultural evolution possible (Acemoglu & Robinson, 2019; Pinker, 2018). In order to develop this level of sophistication and complexity, humans uniquely among species must possess a *theory of mind*—the capacity to understand that other people live in different subjective universes, experience different mental states, beliefs, and intentions, and that these states are only imperfectly accessible to outsiders. Having a theory of mind is absolutely essential for success in everyday human social interactions.

For example, internalized expectations about partner responsiveness to our needs play an essential role in managing successful personal relationships (see Reis et al., this volume). In practical terms, theory of mind is the sum total of the cognitive capacities we use to analyze, infer, and predict others' social behaviors. Theory of mind appears to be an innate potential ability in humans that requires intensive social and other experience over many years for its full development. Different people may develop more, or less, effective theories of mind, and this will impact on their sociability.

Sociability is often problematic precisely because our theory of mind is so limited. Our ability to mentally model others is woefully inadequate. The subjective world of others is barely accessible to outsiders, and so misunderstanding, mis-communication, and misjudgment are often the norm rather than the exception (Kahneman, 2013). To make things even more problematic, attempts at dissimulation, dishonesty, and deception have always been part and parcel of human social life, and are an essential feature of our evolutionary heritage (Buss, 2019; von Hippel & Smith, this volume). That we manage to get on as well as we do and have been able to progress is truly remarkable. Several chapters in this volume focus on the inherent cognitive limitations that impact on human sociability (see Baumeister & Maranges; Fiedler & Hütter; Krueger et al., this volume).

Group identification. Individual sociality is also profoundly shaped by our attachment to various meaningful groups and collectives that define our sense

of identity and connectedness (Fukuyama, 2018). The social identity perspective offers clear evidence for a close interaction between individual characteristics and group relations (Tajfel & Forgas, 2000; see also Gaffney & Hogg, this volume). The powerful desire to belong to and identify with an ingroup has been a major factor in our evolutionary success, and people employ sophisticated strategies of sociability in order to obtain the kind of warmth, belonging, identity, and epistemic certainty they can only get from their group memberships (see Kruglanski & Ellenberg, this volume). So, in a sense, human sociability is strongly shaped by a clear preference for rewarding ingroup associations that in turn can often generate hostility to outgroups (Tajfel & Forgas, 2000). It is the unique human symbolic ability to internalize, represent, and feel at one with one's identity group that makes patterns of sociability so effective in promoting group cohesion, and ultimately, survival (Gaffney & Hogg, this volume).

Most likely we can view these alternative approaches to human sociability as complementary to each other. There is no reason why an inborn tendency for sociability could not be shaped by additional learning experiences and maintained as a result of various rewarding exchange relationships. Rather than seeing them as conflicting, theories of sociability offer genuine insights into the complementary facets of our very complex social nature.

Sociability in Experimental Social Psychology

From these theoretical approaches, and despite Wundt's pessimistic views about the prospects of an empirical social psychology, in fact a thriving scientific discipline devoted to the study of human sociability has emerged since the beginning of the 20th century. Empirical social psychology by now has much of importance to say about the nature of human social behavior. The classic works of Sherif (1936), Asch (1951), Milgram (1974), and Tajfel (Tajfel & Turner, 1979; Tajfel & Forgas, 2000) shed new empirical light on the evolutionary fundamentals of human sociability, producing dramatic results that demonstrate the basic human need for social acceptance and interaction.

Sherif's (1936) norm-formation studies were among the first to demonstrate the elementary human tendency to construct and maintain a consensual reality and eliminate ambiguity when facing uncertainty. When confronted with an intrinsically ambiguous perceptual experience (the autokinetic effect), instead of reporting their authentic perceptions, judges spontaneously accept estimates by others and automatically establish a consensual norm. Sherif's work is essentially a classic demonstration of spontaneous norm creation. It almost seems that social ambiguity and the absence of shared norms are intrinsically aversive for humans, one of the evolutionary fundamentals of human sociability. We are predisposed instinctively to rely on others for information when faced with ambiguous situations, a process Deutsch and Gerard (1955) identified as *informational conformity*.

Subsequent experiments by Sherif (1936) also showed that once a shared arbitrary judgmental norm is established, later generations of naïve judges will faithfully follow the norm, even if they had no part in shaping it, and have no idea about its origins or validity. Doing as others do, thinking as others think seems a universal human imperative, as also recognized by French sociologists like Gustave Le Bon and Gabriel Tarde, who considered imitation and modeling as fundamental human evolutionary tendencies.

Subsequent conformity studies by Asch (1951) that considered judgments of unambiguous stimuli demonstrate another basic aspect of sociability: the need to "fit in," or *normative* dependence on others (Deutsch & Gerard, 1955). Asch's findings were often interpreted, contrary to his own views, as indicative of human frailty in the face of implicit social pressure. Their message could be better understood as illustrating the fundamental evolutionary principle, that for humans, normative agreement with others and maintaining group consensus is often more important than truth or reality (Buss, 2019; Deutsch & Gerard, 1955; see also Gaffney & Hogg, and von Hippel & Smith, this volume).

Following Asch's work, Milgram's (1974) dramatic findings showing the power of obedience to direct commands illustrate yet another striking evolutionary facet of sociability: obedience, the powerful universal tendency to accept and follow commands, irrespective of their consequences. The underlying principle is the same—fitting in and not deviating from the demands of leaders is more important for human survival than seeking truth and discovering reality (Buss, 2019). These basic tendencies of sociability promoting social cohesion appear essential parts of human nature.

The role of group identification in shaping sociability was demonstrated in the classic experiments of Henri Tajfel, himself a Holocaust survivor, who wanted to understand the minimal requirements for people to identify with a group, often producing spontaneous discrimination against other groups (Tajfel & Forgas, 2000; see also Gaffney & Hogg, this volume). In his "minimal group" experiments, people are assigned to groups based on transparently meaningless criteria, such as flipping a coin. Next, participants have to allocate resources (money, etc.) between two unknown individuals, identified only by their randomly assigned group membership. Numerous experiments found that even this minimal and meaningless group assignment produced a strong tendency for ingroup favoritism and outgroup discrimination. People seem motivated to discriminate against others simply because they belong to a different group, even if that group has no meaning, no past, no future—the label alone is enough. Tajfel's studies offer a dramatic illustration of yet another evolutionary principle of human sociability: our powerful need to be identified with an ingroup.

The classic works of empirical researchers such as Asch, Sherif, Milgram, and Tajfel emphasize the surprising power of evolved patterns of sociability in affecting our judgments of behaviors. These experiments suggest that following

others, reinforcing group cohesion and cooperation and avoiding dissension must have been of important evolutionary benefit in our ancestral environment (Buss, 2019). To demonstrate these effects, in these experimental situations, there is some degree of uncertainty that allows the underlying effects to arise. When situations are more clear-cut, concrete situational pressures often dominate our reactions. Sherif (1936), for example, found that those with prior experience in the autokinetic judgment task were highly resistant to information provided by others. Asch also found that a single dissenting experimental subject was sufficient to reduce the normative conformity effect, and obedience in the Milgram situation dramatically declined when participants witnessed a previous participant refusing to obey.

Research on persuasion, another common strategy of sociability, also shows subtle effects. For example, messages provided by outsiders are not usually effective but may sometimes produce "indirect change" of attitudes to enhance cognitive consistency (Crano & Alvaro, 2013). Outsiders can also make a difference when providing new knowledge the group does not have. These effects are subtle, as what is considered truth is often in the eye of the beholder, with maintaining group consensus a priority. Dissenting ingroup sources are often relegated to outgroup status, and not considered (see also Gaffney & Hogg; and Prislin & Crano, this volume).

Experimental research using economic games such as the ultimatum game and the dictator game has also produced important insights into fundamental aspects of human sociability. Trade-offs between benefiting the self and benefiting others are shaped by these conflicting demands (see also Krueger et al., this volume). For example, seeking status and significance as a visible resource is a selfish objective, that paradoxically can only be achieved within the confines of a cooperative group that validates status and power displays (see also Baumeister & Maranges; Gaffney & Hogg; Kruglanski & Ellenberg, this volume).

Other experimental work showed that affective states can also have a subtle effect on strategies of sociability, for example, how selfish versus fair decisions are made by people in the dictator game or the ultimatum game (Forgas, this volume). This research found that negative mood recruiting more accommodative processing tends to promote the fairer treatment of others. Positive mood in turn tends to produce a more internally focused and selfish mental frame, and accordingly, happy players were more selfish in these games.

Overall, experimental social psychology has yielded important insights into the nature of human sociability. These classic experiments can be regarded as ingenious manipulations designed to allow fundamental features of human sociability to emerge. As a result of these investigations we now know far more about human nature than at any time in the past. In the next sections we will consider the importance of sociability for human well-being, and the generation and transmission of knowledge and information.

Sociability and Well-Being

Sociability is intrinsically rewarding. Even superficial encounters with others produce an improvement in well-being (Regan & Lyubomirsky, this volume). As was argued, humans are unique in having a "theory of mind"—an understanding that others live in a slightly different subjective universe from ourselves. Perhaps part of the hedonistic benefit that accrues from sociability is the experience of pleasure we get from understanding other minds when interacting with them. The human capacity for sociability is an essential prerequisite to maintain and manage a reliable network of rewarding long-term relationships that prevail because they continue to fulfill the cognitive and emotional expectations of participants and provide a sense of safety, security, and belonging (see also Dunn & Lok; Mikulincer & Shaver; Reis et al., this volume).

Judgments and evaluations. The study of sociability also offers fascinating insights into how positive and rewarding social judgments and evaluations are formed. Social comparison processes play a key role in many forms of judgments where people evaluate themselves against social expectations and the performance of others (Festinger, 1954). Most judgments of others (as well as of ourselves) are formed along the "big two" dimensions that underlie the majority of our social relationships: (a) social evaluation (warmth, liking) and (b) task evaluation (competence, achievement: Fiske, Cuddy, & Glick, 2007; Reeder & Brewer, 1979). It is the ever-present process of social comparison that helps to establish shared communal standards, and can also trigger social competition and stimulate individual achievement. Interacting with others not only is a source of pleasure and satisfaction (see also Lok & Dunn; Regan & Lyubomirsky; Reis et al., this volume), but also can stimulate achievement, epistemic vigilance, and problem-solving (Zajonc, 1965; Sperber et al., 2010; Misyak, Melkonyan, Zeitoun, & Chater, 2014).

As George Herbert Mead (1934) suggested, human sociability is the crucible that both produces communal cohesion and shapes individual identities and agentive behavior (see also Fiske, this volume). Communal support can enhance individual performance in sports, arts, and intellectual domains. Sociability—interaction with others—not only improves subjective well-being but also makes people more competitive, strong, and resilient (Regan & Lyubomirsky, this volume). Thus, sociability and comparisons with others function like catalysts promoting both individual progress and communal bonding (Lewis, 2016; Stroebe & Stroebe, 1983).

Trust. Another major function of sociability promoting well-being is to generate trust between people. The unique human capacity for symbolic thought—our ability to imagine and mentally represent fictional, future realities—plays a critical role in the evolution of social trust and communal integration. Humans, alone among animals, are capable of mentally representing future interactions and events that allow the evolution of complex reciprocal behaviors

over long time periods (Buss, 2019). This gives rise to the emergence of much more complex and mutually rewarding trusting relationships than would be possible without the ability for mental abstraction. Interacting with others creates mental expectations about future behaviors, generates norms and standards that make future interactions more predictable, and produces mutual expectations that create trust and bonding.

It has been argued, for example, that free market-based exchange transactions are the major engine for creating interpersonal trust, and ultimately, social cohesion. In the process of trading with each other, people come to trust that fair exchange is taking place, and that future transactions will be honored (Acemoglu & Robinson, 2019). In a more fundamental way, as Durkheim (1997) in his classic work on the division of labor in society argued, growing specialization and the increasing reliance on personally unknown others to produce the goods and services we use generates a new form of social integration and organic solidarity that no longer relies on personal, face-to-face experience (mechanical solidarity).

Cohesion. A very similar point was made by the classic German sociologist Tönnies (2001), who distinguished between two fundamentally different forms of social integration: community (Gemeinschaft) and association (Gesellschaft). Community relies on direct face-to-face interaction between members of a social group to generate a shared identity and effective group integration, corresponding to Durkheim's concept of "mechanical solidarity." When it comes to integrating larger groups, direct personal interaction is supplanted by abstract, symbolic interaction where common expectations about shared rules and norms come to regulate relationships (organic solidarity). Both Toennis and Durkheim recognized that human sociability is the fundamental mechanism that produces social bonds and trust that promotes well-being and is the basis of group integration. The impersonal, abstract interdependence that the division of labor produces is only possible because humans have the ability to mentally represent the norms of their common shared social world and act upon such shared symbolic representations.

Sociability and Knowledge Sharing

What is the role of sociability in the acquisition of knowledge, and the evolution of human cognitive abilities? Getting information from others—a key feature of sociability and social comparison—is very important in acquiring a better understanding of reality (see Fiedler & Hütter, this volume). Humans are intensely social creatures, and our very survival and evolutionary success owes much to our rational ability to understand reality, and to form ever-more sophisticated and well-integrated cooperative groups. These two processes are intricately related. Sociability is the mechanism that promotes knowledge sharing that leads to more accurate discoveries and cultural evolution, consistent

with the notion of the wisdom of crowds (Herzog & Hertwig, 2009; Surowiecki, 2004; see also Fiedler & Hütter, this volume). Yet social hypothesis testing is not exclusively motivated by the epistemic goal to produce the best decisions (Snyder, 1984). An equally important goal is to get along with other people.

The evolution of the prodigious intellectual capacities of our species probably owes much to the cognitive demands of sociability integrating ever-larger and ever-more elaborate groups and cultures (Dawkins, 2009; Dunbar, 1998). There is now good evidence that in primates, there is a direct correlation between group size and brain size, consistent with the idea that the primary function of the human brain is to manage sociability. This claim however is in apparent contrast with the other fundamental requirement for survival: to know and to manage our physical environment, a task that requires a very different set of cognitive skills, involving independence, deduction, rationality, prediction, and causality.

Sometimes, the pressures of sociability—the need to maintain shared beliefs—can also be the source for irrationality. The tolerant and open-minded examination of reality and the search for truth is not exactly a universal hallmark of human cognitive endeavor (see Gaffney & Hogg; and von Hippel & Smith, this volume). Cognitive researchers like Kahneman (2013) demonstrated the powerful role that heuristics, shortcuts, gut feelings, and automatic responses play in our thinking, and historians such as Harari (2014) suggested that it is this capacity for abstract symbolic thought and shared fictional realities that underlies the human propensity for collective delusions.

As it turns out, the shared cultural knowledge produced by sociability is not all about better understanding reality and seeking truth. As several chapters here will show, in human affairs, discovering truth often comes a distant second to the primary evolutionary imperative of keeping a group together (Baumeister & Maranges; Gaffney & Hogg; Kruglanski & Ellenberg, this volume). Our intense cultural sociability is as likely to produce and maintain bizarre consensual delusions that hold groups together as it is to produce genuine insights about reality (Albright, 2018). Sociability is often driven by the demands of obtaining, transmitting, and using information, irrespective of the truth value of that information. A nice piece of gossip can fulfill a useful role in cementing relationships and affirming shared norms, irrespective of its veracity.

Throughout human history, a colorful variety of weird, shared belief systems have been invented and provided cohesive norms for sociability defining successful cultures. Most such fictional belief systems feature not real knowledge, but comprise strange creation stories, religions, and unsubstantiated myths and rituals (Forgas & Baumeister, 2019; Harari, 2014). The history of human cultural evolution is a history of shared misconceptions, fake news, delusions, and superstitions, sometimes integrating entire civilizations for centuries. Belief in human sacrifice was central to successful meso-American civilizations, and millions of

European citizens were slaughtered in religious wars because of arcane disagreements about virgin birth, trans-substantiation, or papal infallibility. Efficiently propagated sectarian beliefs about divine revelations, and religious practices prescribing social rituals, prayer, dietary and sexual restrictions, pilgrimages, and self-flagellation worked remarkably well in regulating day-to-day sociability and integrating cultural groups, despite having no discernible basis in reality.

The human quest for shared beliefs, however unsubstantiated, is still with us today. Influential but unfalsifiable modern theories claiming to be "scientific" such as Marxism or psychoanalysis continue to exert enormous influence on many people's social lives, despite having no scientific status as Karl Popper (1947) convincingly showed. Even today in our elite universities, tenured professors may lose their jobs because they dare question absolutist neo-Marxist tribal movements advocating woke ideology, cancel culture, and political correctness (Murray, 2019). It seems then that many of the norms guiding our intense sociability are as much about preserving and promoting group integration and consensual delusions as they are about understanding reality and discovering truth.

Consensually shared beliefs can have important causal consequences as do objective facts. For example, naïve theories about how to detect lies can be more important in what people pay attention to than the actual objective diagnostic value of different cues (Hartwig & Bond, 2011). In many spheres of social life, doing what is expected triumphs over what actually works, or what is factually established. Sociability can thus be a source of gullibility and can create alternative realities that are as influential as objective realities (Forgas & Baumeister, 2019). In many domains, objective criteria of accuracy, validity, and achievement may be superseded by consensual but incorrect beliefs (Kahneman, 2013).

Sociability thus plays a dual role in the acquisition of knowledge. On the one hand, sharing information with others can be a powerful force for improving accuracy and producing better judgments and predictions, as Fiedler and Hütter in their chapter show. But often, in our pursuit of effective group cooperation, the search for truth may come off second best (see also Gaffney & Hogg, this volume). Even within science, consensus-seeking and loyalty to established paradigms sometimes triumphs over new, but unexpected ideas, as Thomas Kuhn (1962) argued. Humans tend to think as others do, and as a result can easily assimilate false incorrect or misleading information.

Examples of striking gullibility, self-deception, hubris, and wishful thinking maintained by intense ties of sociability often characterize human affairs to this day, including where one would least expect it, in the halls of academia (Forgas & Baumeister, 2019). In a justly famous hoax the physicist Alan Sokal submitted an intentionally meaningless text to a post-modernist theoretical journal to investigate whether adherents of post-modern cultural studies would publish an article full of nonsense if (a) it used the right (albeit meaningless) jargon and (b) it confirmed the group's preconceptions (Sokal, 1996). The article was duly accepted and published (Sokal, 1994), confirming that many academic

departments in the humanities and social sciences now committed to an unfalsifiable ideology of post-modernism will accept meaningless verbiage as valuable intellectual product as long as it confirms their cherished world view.

Our innate need for sociability can thus become a source of bias and distortion (see also Baumeister & Maranges; Fiedler & Hütter; Krueger et al.; and Kruglanski & Ellenberg, this volume). In the economic sphere, shared if irrational beliefs can produce recurring investment "bubbles" at least since the famous "tulip bulb" craze in the 18th century. Advertising and marketing explicitly seek to exploit the kind of irrationality that the demands of sociability can often produce, and populist political communication does the same (Forgas, 2021; Forgas, Fiedler, & Crano, 2015). These examples, and countless others, suggest that the human need for sociability can be a powerful force for creating and maintaining irrational beliefs.

The progress of civilization partly depends on managing this inherent conflict between sociability-producing improved real knowledge and sociability-reinforcing consensual delusions. The ideas of the enlightenment produced a rational resolution to this conflict. John Stuart Mill, in his famous essay "On Liberty," offered perhaps the most important advice to advance genuine progress (Mill, 1859/1982): we must have strict rules to promote free speech and open exchange in order to avoid being swayed by group think, listening to and incorporating what is valid from contrary positions. In order to avoid the tyranny of unquestioned and often misleading group knowledge, we must emphasize the importance of individual freedom, the right to think otherwise, and to be heard.

Overview of the Volume

After this chapter, our book is organized into four sections. Part 1 considers some general ***approaches to sociability***, covering evolutionary (von Hippel & Smith), psychological (Prislin & Crano), and cultural (Fiske) perspectives analyzing the nature and functions of sociability. Part 2 looks at the importance of sociability in ***human well-being***, showing how social contact improves happiness (Dunn & Lok), can improve health (Regan & Lyubomirsly), and how people derive a sense of significance (Kruglanski & Ellenberg) and achieve a sense of social identity and belonging from their social experiences (Gaffney & Hogg).

Part 3 discusses the links between sociability and ***knowledge***, looking at how humans function as information agents (Baumeister & Maranges), how shared information produces superior knowledge (Fiedler & Hütter), and how reputational concerns are managed in social decisions (Krueger, Gruening & Sundar). Finally, part 4 turns to the role of sociability in **personal relationships**, discussing attachment theory (Mikulincer & Shaver), the importance of partner responsiveness in relationships (Reis, Itzchakov, Lee & Ruan), and the influence of affective states on managing relationship behaviors (Forgas). A brief summary of the chapters follows.

After this chapter, Part 1 of the book deals with basic approaches to sociability.

Chapter 2 **by von Hippel and Smith** reviews evolutionary explanations of sociability, and the adaptive challenge humans face in understanding and managing others is analyzed. It is by understanding the minds of others that mutually beneficial outcomes can be achieved, and social perception plays a key role in this process. The chapter argues that sociability is one of our most important characteristics that drove the evolution of our cognitive abilities, and is also the key to our evolutionary success as a species.

In Chapter 3, **Prislin and Crano** survey classic work by psychology's founders on the nature of human sociability, attempting to link the individual and the communal domains. Sociability is a multivariate trait shaping people's desire for meaningful connection with others, and early theorizing played a crucial role in placing the study of sociability into the focus of our discipline.

In Chapter 4, **Fiske** takes a social anthropological perspective and suggests that human sociability is organized around culturally evolved practices that are inherently motivating and emotionally engaging. He discusses the "kama muta" emotion, a feeling of oneness, connectedness, love, and belonging promoted by cultural practices designed to intensify communal sharing, commitment, and devotion.

Part 2 features contributions that discuss the role of sociability in promoting well-being and human flourishing.

Chapter 5 **by Regan and Lyubomirsky** suggests that sociability can improve well-being by deliberately engaging in connection-boosting activities, indicating that sociability is an important tool in positive emotional interventions. They review research showing that asking participants to act more extraverted, engaged, kind, and interactive is an efficacious method to induce sociability and well-being.

Chapter 6 **by Dunn and Lok** suggests that even trivial social interactions can contribute to happiness and well-being. The chapter identifies causes that lead to interaction avoidance and suggests methods for overcoming these biases. They describe a Stranger Engagement Model offering a road map for facilitating connection in a world where loneliness is increasingly viewed as an epidemic.

Chapter 7 **by Kruglanski and Ellenberg** argues that it is through effective strategies of sociability that people can satisfy the deeply felt human need for recognition and significance. They develop a novel significance quest theory (SQT) as a major contribution to understanding the functions of human sociability. The chapter demonstrates how the quest for significance impacts on several domains of sociability, and how achieving significance is a major requirement for achieving well-being and life satisfaction.

Chapter 8 **by Gaffney & Hogg** explores the importance of group identification in producing rewarding patterns of sociability and the development of meaningful group relationships. They argue that group identification creates warm feelings toward group members and promotes sociability within groups establishing closer and more rewarding links between ingroup members. The

chapter reviews evidence for the positive influence of group identification for sociability, and describes how group identification can also shape a collective understanding of reality.

In Part 3, chapters discuss the important role that sociability plays in the generation and dissemination of knowledge.

In Chapter 9 **Baumeister and Maranges** discuss the role of sociability in how humans function as information agents, creating cultural stability and change. The functions of human information agents are analyzed including curiosity, communication, criticizing, and maintaining shared reality. Strategies of withholding information and disseminating false information are also discussed, processes that shape collectively shared understanding.

Chapter 10 **by Fiedler and Hütter** explores how sociability can promote superior collective knowledge. They review evidence showing that collective estimates are often more accurate than individual estimates, and collective decisions are more often correct than individual decisions. Research on advice taking supports the wisdom of crowds, and the dangers of misleading and biased advice and metacognitive shortcomings are also considered.

Chapter 11 **by Krueger, Gruening, and Sundar** review informational biases in social perception as related to the management of conflicting needs for power and sociability. Research using the ultimatum game suggests that observers are sensitive to both motives when judging others. The chapter describes how individuals might incorporate reputational concerns into their social decision-making and how the power-sociability dilemma can be transcended even if it cannot be fully solved.

Part 4 contains chapters that analyze the intricate patterns of sociability that influence human relationships.

In Chapter 12**, Mikulincer and Shaver** discuss attachment theory, conceptualizing human sociability and social relationships in terms of two major dimensions, attachment anxiety and avoidance. Evidence for these dimensions is reviewed, showing that they are mediated by cognitive-motivational predispositions, mental scripts, and patterns of information-processing biases, ultimately determining people's relationship outcomes and strategies of sociability.

In Chapter 13**, Reis, Itzchakov, Lee, and Ruan** show that people's desire for partners who are responsive to their needs and preferences is a key aspect of understanding sociability in general, and close relationships in particular. The chapter reviews research demonstrating how perceived partner responsiveness influences core processes such as self-enhancing social cognitions, attitude structure, and emotion regulation, and argues that a greater focus on perceived partner responsiveness can enhance psychological theories of sociability.

Chapter 14 **by Forgas** looks at the role of mild affective states in regulating important relationships processes, such as the perception and evaluation of others, and the performance of relationship behaviors. The important evolutionary functions of affective states in fine-tuning sociability and relationship processes

are considered, and numerous experiments are described demonstrating how affective states may influence the strategies adopted in interpersonal judgments, decisions, and interpersonal communication strategies. The adaptive benefits of moods for promoting sociability and relationships are considered.

Conclusion

Sociability is one of the most important and at the same time most complex and still incompletely understood characteristics of our species. Our aim with this book is to contribute to a better understanding of the nature and psychological characteristics of human sociability. Sociability lies at the heart of our greatest cultural and civilizational achievements, yet our capacity to affiliate with ingroups and propensity to favor socially shared narratives in favor of reality has also resulted in some disastrous historical developments. At the most extreme, unquestioning attachment to cohesive tribal collectives can produce discrimination, hatred, violence, and bloodshed against outgroups, as our history amply demonstrates.

We hope that these chapters will help to highlight the complex patterns of human sociability that produce such extreme outcomes, a particularly important task in our time when populist politics and tribal allegiances are once again on the rise even in liberal democracies (Forgas, Crano, & Fiedler, 2021). The last few decades of research in social psychology and related disciplines have produced genuine breakthroughs in our understanding of human nature in general, and sociability in particular. In this chapter we tried to argue for the importance of our topic, and survey some of the most important features and functions of sociability in promoting human well-being and attachment, facilitating the transmission of knowledge and shaping human relationships. As editors, we are deeply grateful to all our contributors for accepting our invitation to contribute to this, the 23rd volume of the Sydney Symposium of Social Psychology Series and sharing their valuable ideas with our readers. We sincerely hope that the insights contained in these chapters will contribute to a better understanding of the crucial role that sociability plays in shaping us both as individuals and as communities.

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2

EVOLUTIONARY ORIGINS AND CONSEQUENCES OF HUMAN SOCIABILITY

William von Hippel and Nicholas M. A. Smith

Abstract: In this chapter, we propose that the hyper-sociability exhibited by our species originated in our expulsion to the savannah and our subsequent need to band together for our mutual protection. The value of cooperation was then enhanced once our ancestors began using it for predation as well as protection, particularly once *Homo erectus* gained the capacity for division of labor. Cooperation, in turn, was facilitated when group members shared an understanding of their environment and goals, and hence humans evolved to seek a sense of shared reality within their groups. Sociability further drove the tendency to share reality and the evolution of language greatly increased their capacity for information sharing, which in turn amplified the positive impact of their sociability. Finally, sociability greatly enhanced learning and problem-solving, as language enabled people to leverage the knowledge and experience of others in service of their own understanding of the world. As a result of these processes, sociability played a central role in creating the cognitive niche that we occupy and the immensely positive sum nature of human cooperation.

The most important challenge people face in life is understanding and managing others. People who can understand the goals of the important people around them can put themselves in a position to benefit from their peers' likely actions. People who can leverage that understanding by also managing others effectively are even more likely to succeed. By planting their own goals in the minds of others, effective social managers create a world in which others work toward mutually beneficial outcomes. In contrast, people who cannot understand others are buffeted by the unexpected winds of social change. People who can understand others but cannot manage them effectively have the advantage that they can see the bad news coming, but they have limited capacity to do anything

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about it. Because humans are a highly cooperative species who achieve most of their goals together, social perception and persuasion are the most reliable paths to success now and in our deep past.

In other words, *sociability is one of our most important proclivities* and it underlies our most important cognitive abilities (see also Fiedler & Hütter, this volume). It is no coincidence that sociability is also the key to our success as a species. The goal of this chapter is to discuss how our evolutionary history is entwined with our sociability—how sociability first evolved and then became fundamental to the human success story. In doing so, we describe how our ancestors created and then exploited the *social cognitive niche* (Whiten & Erdal, 2012).

The Savannah Hypothesis

Humans and chimpanzees share a common ancestor from six or seven million years ago (Steiper & Young, 2006). There are numerous possible reasons why our ancestors might have split into different species; such an important event is likely to have multiple causes. Of the various possibilities, the explanation we find the most compelling focuses on the East African Rift Valley (Coppens, 1994). Over the last 40-some million years, tectonic activity along the East African Rift Valley has slowly and sporadically raised vast portions of Ethiopia, Kenya, and Tanzania to an elevated plateau (George, Rogers, & Kelley, 1998). These changes in topography led to localized changes in climate, with the rainforests on the east side of the Rift Valley drying out one by one, to be replaced by open woodlands and savannah. As a consequence, our chimp-like ancestors were gradually forced out of the trees and onto the ground in search of food.

Because chimpanzees are safe from predation when with their group in the forest canopy, but vulnerable to attack by lions, leopards, hyenas, etc., when on the ground (Boesch, 1991), the gradual replacement of rainforest with savannah would have put our chimp-like ancestors at great risk. Furthermore, the diet of chimpanzees comprised predominantly of fruits, berries, leaf buds, etc., which are widely available in the rainforest but nearly absent in the savannah. Thus, the move to the savannah would have put enormous evolutionary pressure on our ancestors as they tried to find new ways to survive and thrive.

A population of chimpanzees in Fongoli, Senegal, spends much of its time in the savannah and hence provides some clues about how our ancestors might have initially reacted to this changed environment. Compared to their forest-dwelling cousins, savannah chimpanzees tend to travel in larger groups (Pruetz & Bertolani, 2009), use caves as shelters (Pruetz, 2007), occasionally use sharpened sticks to spear prey in tree hollows (Pruetz & Bertolani, 2007), and share with each other more readily (Pruetz & Lindshield, 2012). Similar responses to life on the savannah may well have played a role in the survival of our chimp-like ancestors as they were forced out of the trees. Indeed, in many ways these

behaviors appear to be precursors to important differences between chimpanzees and humans.

Whether our earliest hominin[1] ancestors adopted these strategies is unknown, but within a few million years of their split from chimpanzees we do know that our ancestors' bodies had adapted to life on the ground via the evolution of bipedalism. Fossil evidence of *Australopithecines* reveals numerous signs of an upright, bipedal gait within a few million years of our separation from chimps (Leakey & Leakey, 2020). Additionally, computer modeling of *Australopithecine* footprints preserved in ash from 3.5 million years ago also suggests bipedal locomotion (Crompton et al., 2012). As a by-product of bipedalism, *Australopithecines* evolved a longer and more flexible waist, musculature that was aligned laterally rather than vertically, and arm, hand, and shoulder structures that were more flexible once they were no longer required to bear their weight on the branches (Leakey & Leakey, 2020; Marzke, 1983).

An emergent consequence of these biomechanical changes that evolved in response to life on the ground is that by three million years ago, our ancestors had developed a body type that was much more suitable for the motions involved in effective stone throwing (Roach, Venkadesan, Rainbow, & Lieberman, 2013). Stone throwing might not seem like much of an advance, but the most important invention in military history is the capacity to kill from a distance, as it allows a larger group of weaker individuals to defeat a smaller force of stronger individuals from a position of relative safety. Thus, the most effective solution to the increased risk of predation would have been the capacity to throw stones (Calvin, 1982; Isaac, 1987). Although evidence for projectile use only appears much later in the archaeological record, support for the possible early use of such a strategy can be seen in changes to the hands of *Australopithecus afarensis*, which enable better throwing and clubbing than the hands of chimpanzees (Marzke, 1983; Napier, 1993; Young, 2003).

Consistent with this possibility, the historical record also indicates that throwing can be remarkably effective. There are numerous descriptions of encounters between European and indigenous populations in which conflict ensued and the indigenous population was armed only with stones. The Europeans typically relied on guns and armor, but on some occasions they lost these skirmishes. By way of example, consider this historical account reported in Isaac (1987):

> Many a time, before the character of the natives was known, has an armed soldier been killed by a totally unarmed Australian. The man has fired at the native, who, by dodging about has prevented the enemy from taking correct aim, and then has been simply cut to pieces by a shower of stones, picked up and hurled with a force and precision that must be seen to be believed . . . the Australian will hurl one after the other with such rapidity that they seem to be poured from some machine; and as he throws them

> he leaps from side to side so as to make the missiles converge from different directions upon the unfortunate object of his aim.
>
> *John Wood (1870)*

Historical accounts such as this one highlight the potential deadliness of stone throwing, but they gloss over a crucial fact; even if *Australopithecines* had evolved into highly effective stone throwers, a lone individual throwing stones at lions, leopards, or even hyenas would have stood little chance of survival. The only way for such a strategy to be useful would be if other individuals cooperated in a joint effort to drive off predators.

Roots of Human Sociability in the Need for Collective Action

Chimpanzees are more likely to compete with one another than they are to cooperate (Boesch, 1994; Gilby, 2006; Hare & Tomasello, 2004), and thus it would have been difficult for our distant ancestors to act collectively to protect themselves from large predators. Once our ancestors evolved the physical capacity to defend themselves by throwing, there would have been much greater pressure than in the past for them to evolve the psychological proclivity to cooperate. Cooperation in mutual defense would have been much less likely to evolve prior to stone throwing, as the only way for earlier hominins to protect themselves would have been in direct physical contact, which would put all cooperators at great risk. Furthermore, the presence of dangerous predators would have provided a strong and mutually understood orienting signal regarding the value of stone throwing, as all members of the group would have stood a greater chance of survival if they banded together in their mutual defense. Indeed, the tendency to orient toward threatening predators initiates collective action in other group-living species such as elephants (e.g., McComb et al., 2011).

This account of our evolutionary history raises the possibility that the beginnings of the hyper-sociability we see among humans lie in the collective action that emerged millions of years ago when our ancestors faced deadly predators on the savannah. Importantly, once this move toward collective action was initiated, it would have set our ancestors on a new evolutionary trajectory, as new social abilities created new needs and opportunities, which in turn led to the evolution of novel abilities. Thus, evolution would have favored any subsequent genetic and psychological changes that supported the quality of the individual's relationship with the group and the group's collective response.

For example, once our ancestors learned to fend off predators by throwing stones, they would have soon discovered that they could also hunt via collective stone throwing. Collective stone throwing requires little advance planning or coordination, and thus would have been possible even with the comparatively

limited cognitive abilities of our *Australopithecine* ancestors. Stone throwing not only enhanced the benefits of cooperation, but it also created new means to enforce it (Bingham, 2000). The greatest challenge to cooperation is free riding, which would have been much less likely in the face of hominins who were capable of collective action and killing at a distance.

Finally, once our ancestors moved to the savannah, expulsion from the group would have been akin to a death sentence, as lone hominins on the grasslands are at much greater risk than lone hominins in the rainforest. Thus, ostracism would have become another highly effective tool for enforcing cooperation (Boehm, 2009). Those hominins who evolved a strong and negative emotional reaction to ostracism would have done their utmost to avoid it, which would have enhanced their survival. Such an evolutionary pathway might explain why people today are so sensitive to rejection and ostracism (Baumeister & Leary, 1995; Williams, 2007; see also Baumeister & Maranger; and Gaffney & Hogg, this volume). As a result of these changes, hominins who were predisposed to sociability would have had a great advantage over hominins who adopted a competitive or asocial orientation. Although we do not know when our ancestors began to behave in such an interdependent fashion, the result of these processes was dramatically increased sociability that is evident in many aspects of our psychology (von Hippel, 2018).

Sociability and the Cognitive Explosion

Many animals live in large groups but do little more than coordinate their movements with each other. In contrast, once our ancestors began to engage in collective action, their interdependent style of group living would have created evolutionary pressure to develop more sophisticated cognitive abilities. Although most animals would benefit if they were smarter, greater intelligence can only evolve if it brings additional opportunities to source calorie-dense foods. Because brain tissue has greater ongoing (and unavoidable) metabolic costs than muscle, the trade-off between brain and brawn is typically resolved in favor of brawn, which is more likely to bring in enough calories to sustain itself.

In the case of our hominin ancestors, the scales appear to have been tipped toward brain over brawn just prior to the advent of the various *Homo* species, a few million years ago. Our first three or four million years on the savannah resulted in only limited cranial expansion, from the 380-gram brain of a chimpanzee to a 450-gram brain of an *Australopithecus afarensis*. In the next 1.5 million years, hominin brains more than doubled in size, to the 960-gram brain of *Homo erectus* (which also has a larger body, so the increase in brain size is not as dramatic when compared to body size). Over the 1.5 million years since then, we have added almost an entire chimpanzee brain to our *Homo erectus* ancestors, with the current average around 1,350 grams. This dramatic increase

in brain expansion coincides well with the social opportunities our ancestors would have gained if they began engaging in collective action after the advent of bipedalism.

There are numerous opportunities to enhance group effectiveness with increased intelligence, but by way of example consider two abilities that appear to have emerged for the first time in *Homo erectus* (who appeared two million years ago and looks much more like a modern human than like a chimpanzee; Herries et al., 2020). *Homo erectus* is the first animal on this planet which has left behind evidence of the ability to envision unfelt needs.[2] The clearest evidence for this capacity lies in the fact that the tools made by *Homo erectus* can often be found at great distances from the sites at which they were made, suggesting that *Homo erectus* knew that the tools they made would be useful in the future and hence worth bringing along when they traveled (Shipton, 2013). Prior to *Homo erectus*, there is no evidence that our ancestors' tools were ever carried any great distance from the location at which they were quarried and made.

The capacity to envision unfelt needs, which are not currently experienced but likely to emerge in the future, is a critical ability underlying future planning. An animal that can only envision a world that reflects its current needs will be severely limited in its preparations for future opportunities. Thus, once *Homo erectus* was capable of envisioning potential future needs, it gained the advantages of increased preparation and long-term planning (Suddendorf & Corballis, 1997). One of those advantages was the capacity to divide labor in such a way to make groups more than just the sum of their parts, and indeed *Homo erectus* appears to have invented division of labor. There are numerous hints of this capacity in the fossil record, but some of the clearest evidence can be found at a 1.2-million-year-old site in India. At this particular location, *Homo erectus* manufactured stone tools in clusters that showed clear spatial separation of different aspects of production (Shipton, 2013). The evidence from this site indicates that flakes were knocked loose in one location and then retouched into their finished forms in another location. If one hominin were making each tool from start to finish, there would be no reason to separate the production of different stages into different locations (Shipton & Nielsen, 2015). Thus, the systematic use of the site suggests that several individuals were dividing the tasks involved in the manufacture of their rather complex *Acheulean* stone tools.

Sociability and Shared Reality

The data from the aforementioned *Homo erectus* sites suggest that the opportunities inherent in sociability created a selection pressure on our ancestors to become more intelligent and also created opportunities to garner greater caloric benefits from that intelligence. Division of labor and planning for the future gave *Homo erectus* enormous advantages over the other predators on the planet, and these

abilities then enhanced the value of other skills as well. Perhaps the next social cognitive skill that emerged from our increasing sociability was the desire and ability to share reality. For a group to work together effectively, it is important that its members agree about potential threats, opportunities, and the strategies that will be most effective in avoiding the former and pursuing the latter. In other words, it becomes important for its members to share the same reality.

The first step toward achieving a shared reality is shared attention. Numerous birds and mammals have evolved the capacity for shared attention, which is often achieved via alarm calls. In primates this ability is often quite sophisticated, as the various calls can reflect the spatial orientation of the threat (Seyfarth, Cheney, & Marler, 1980). The greater cognitive abilities of gorillas and chimpanzees have allowed them to ratchet up their ability for shared attention one step further, through their capacity to infer what another animal can perceive from its vantage point (Okamoto-Barth, Call, & Tomasello, 2007). This capacity opens enormous opportunities for the effective use of shared attention, but the great apes obfuscate the direction of their attention via the brown sclera of their eyes. The fact that they hide their gaze suggests that for the most part, the other great apes benefit by not letting members of their group know what has captured their attention. That state of affairs, in turn, suggests that they are more likely to compete rather than cooperate with their fellow group members, and a great deal of evidence is consistent with this interpretation (Tomasello, 2016).

In contrast, humans have evolved white sclera to their eyes, which visibly advertise the direction of our attention. The fact that we advertise the direction of our gaze in such a manner provides clear evidence that our ancestors typically gained more from other members of their group knowing what had captured their attention than from keeping it a secret (Tomasello, Hare, Lehmann, & Call, 2007). In other words, we evolved to cooperate with each other; otherwise, our sclera would never have evolved away from those of the other apes. Although in principle it is possible that such adaptations can emerge because they benefit the group, and hence indirectly benefit the individual (as a member of the group), the group benefit would need to be very large and the individual cost very small for such a system to evolve (Williams, 1992). When group and individual goals collide, individual goals are almost always evolutionarily paramount. Individual success determines what genes get passed to the next generation, even if individual success comes at a cost to the group (Williams, 1992).

Shared attention was the first step toward a shared reality, but it is only an initial building block. Shared interpretation comes next, and as with shared attention, there are both simple and complex forms. Zebras, wildebeest, and almost every other group-living animal are also capable of shared interpretation, but their version of this ability is a rather simple product of observing the behavior of others and following suit. If zebras see their neighbors take off at high speed when they thought the world was momentarily safe from predators, they infer from their fleeing conspecifics that they are in imminent danger. This

capacity to harvest the knowledge of the crowd is highly adaptive, but not very computationally challenging, and is only of utility with regard to immediate threats (see also Fiedler & Hütter, this volume).

In contrast, by virtue of their capacity to see into the future and envision unfelt needs, our ancestors gained the capacity to respond to threats and opportunities that were spatially and temporally distant. An awareness of other times and places is much more useful if it can be shared, but of course it is also much more difficult to share. By simply running away, a zebra can communicate everything his neighbors want to know through his direction and velocity, but such actions are meaningless in the face of more distant threats and opportunities. Hence, from *Homo erectus* forward there was also substantial selection pressure on our ancestors to create more sophisticated forms of communication. Numerous language theorists have suggested that our communicative abilities likely began with increasingly complex forms of gesture, eventually resulting in the grammar and spoken language that enable humans to communicate anything they are thinking. For example, social transmission of how to fashion our ancestors' early stone tools could be readily accomplished with gestures, providing an initial impetus for the development of both teaching and language (Morgan et al., 2015). On the basis of their capacity to envision the future, it is possible that the evolutionary roots of language took hold somewhat later with our *Homo erectus* ancestors and their need to share information that was increasingly difficult to communicate.

But sharing information is only half of the battle; a truly shared reality also depends on sharing emotional responses to that information (see Fiske, this volume). If I tell you that members of another group are coming toward us, but if I fail to communicate whether their impending arrival is a dreadful or joyous occasion, we will not be prepared for a joint response. As was noted, group members need to agree on whether various events are good or bad, safe or dangerous, etc., for the group to act effectively as a unit. As a result of these evolutionary pressures, humans appear to be the only species on the planet that continually seeks opportunities to share the contents of their minds with each other, even when there is no immediate benefit from doing so (Suddendorf, 2013).

It might appear from this line of reasoning that the evolution of the proclivity to share reality relies on group selection—greater information sharing leads to more effective group action—but individual selection also pushes people to share their own perspective and be curious about the perspectives of others. If I can implant my ideas in the minds of others, I increase the probability that the group will share my motives and hence act in a way that is favorable to me. Similarly, if I can learn of the ideas of my fellow group members, I increase the probability that I can predict what they will do next, and hence benefit from their actions (see Gaffney & Hogg, this volume). Both of these individual goals push me to information sharing, and importantly, both of these individual

goals align perfectly with group goals as well. It is this frequent alignment between individual and group goals that has made groups of humans such a potent force, and the lack of such alignment that has prevented chimpanzees from making similar advances.

This desire to share the contents of our minds may have played the single most important role in elevating us to the top of the food chain. We are the fiercest predator on the planet by virtue of the power of our minds, but even human minds are not that special on their own—they gain their power in groups (see also Krueger et al., this volume). By way of example, imagine your reaction if you were told that your flight over the jungle was going to crash land and you had to survive alone in the wilderness for the next six months. Setting aside your likely loneliness, the most notable thought going through your mind would probably be the concern that you would end up as a tasty snack. In contrast, imagine your reaction if you were told that you and a 100 others were similarly to be stranded in the wilderness. Even if you were returning from an academic conference and your fellow group members were no more prepared than yourself for six months in the forest, your reaction to this latter scenario would be far less fear and far more focused on tasks the group needs to accomplish.

Unshared Reality: Information Hoarding, Deception, and Exaggeration

The desire to share knowledge and emotions made our groups effective, but there are costs associated with our proclivity to share information as well. First and foremost, individual goals may be broadly aligned with group goals, but they will often come into conflict. The most notable and common source of individual-group conflict is status concerns. Because status is relative, and hence one of the few domains that are inevitably zero-sum (my gain in status is only achievable at someone else's loss), individual goals and group goals often fall out of alignment when status is at play (see also Kruglanski & Ellenberg, this volume). The most common source of status concerns is sexual selection. Because humans are a pair-bonding species, males and females compete with other members of their sex to attract the interest of members of the opposite sex. People who are deemed the best catch—better than other members of their group and hence higher in status—are more likely to win this competition than people who are deemed the worst catch. Because sexual reproduction is the currency of evolution, status concerns are deeply ingrained in our psychology and a constant threat to group harmony.

Despite the fact that individuals and their groups benefit in many ways from complete information sharing, individuals often benefit in *relative* terms from incomplete information sharing. Humans evolved to exploit the cognitive niche they created, and hence information is one of our most important commodities.

As a result, if I want to achieve higher status than other group members, it will sometimes behoove me to hoard information or to pass along inaccurate information. Both types of curtailed information sharing are a threat to my group, but both types can potentially allow me to rise in status compared to other group members by enabling me to exploit opportunities of which others are unaware.

The tendencies to hoard or distort information are threats that arise when people want to know things that others do not. But distortion to group knowledge can also emerge when people are overly keen to ensure that others share their emotional reaction. As was noted, the desire to share our understanding and experience goes beyond mere knowledge, as sharing emotional experience with others is also critical. Consequently, there are few experiences more frustrating than describing an emotional event to someone who reacts with indifference or an emotion opposite to one's own. The failure of a group member to share your emotional reaction is a direct threat to unity and effectiveness, and perhaps more importantly, a clear sign that your goals are not likely to be advanced by the behaviors of those around you. As a consequence, we are highly motivated to convey our emotional reactions and convince others of their appropriateness (see also Forgas, this volume).

This need to share emotional experience is a root cause of exaggeration. One of the easiest ways to ensure that people share your emotional reaction is to exaggerate your story along the critical emotional dimension. If you are worried that your friend might not be sufficiently outraged by your date's rude behavior, then your date becomes ruder in the telling. This need to share emotions lies at the heart of urban legends, which are a modern incarnation of gross exaggerations. The more disgusting and outrageous a story becomes, the more likely people are to say they would pass it on to others. For example, Heath, Bell, and Sternberg (2001) presented people with different versions of an urban legend in which a family returns from vacation to find photos in their camera that the bellboy took of himself fouling their toothbrushes. In one version, the photos show him cleaning his nails with their toothbrushes, in another he has their toothbrushes stuck in his armpits, and in the coup de grace, they see the bellboy with the family toothbrushes "up his bootie." Despite the fact that each version of the story becomes increasingly implausible, the more disgusting it was the more likely people were to say they would share it.

Sociability, Story-Telling, and Cumulative Culture

As was noted, once *Homo erectus* gained the capacity to envision the future, and hence plan for it, there was increased selection pressure on our communicative abilities to enable us to share this complex information with each other. Hence, it seems likely that language evolved in part to facilitate planning and division of labor among *Homo erectus* group members. The evolution of language in

humans is a major transition in evolution (Maynard-Smith & Szathmary, 1995), and it comes as no surprise that language has many benefits beyond planning.

Perhaps the most notable benefit of language is that it lies at the heart of our capacity for cumulative culture. Because language allows humans to communicate any and all ideas, it also allows us to incorporate the learning and accomplishments of others into our own understanding of the world. In contrast to every other animal on this planet, who learn almost entirely through direct experience or direct observation, language extends the power of sociability to people who are spatially or temporally distant from the events in question. The end result of this process is that animal societies do not accumulate knowledge or move forward in any but the most rudimentary ways, whereas human societies continually ratchet themselves forward on the experiences and wisdom of their prior members (Dean, Vale, Laland, Flynn, & Kendal, 2014).

The oldest form of this type of learning is our rich oral tradition. Storytelling can be found in all human cultures, and probably began when our hunter-gatherer ancestors gathered around the fire at the end of the day for warmth, safety, and cooking. Given that the control of fire is at least a million years old (Berna et al., 2012), it is thus possible that storytelling traditions began to a limited degree among our *Homo erectus* ancestors. Indeed, storytelling around the fire remains an important activity in hunter-gatherer communities, as people set aside their immediate economic and social concerns and focus on the broader patterns, rules, and lessons for achieving a productive life in their community (Wiessner, 2014).

As a consequence of their enormous benefit to their group, good storytellers would have gained status for the quality of information they provided, and hence people would have evolved a desire to tell stories to each other. People would also have evolved a proclivity to enjoy listening to stories, as hearing about the trials and tribulations of others provides a risk-free way to learn the difficult and sometimes expensive lessons garnered from the experiences of others. Finally, storytelling connected community members to each other through shared emotional experiences, a sense of shared reality, and a common knowledge of how to approach the world.

Conclusions

In this chapter, we have argued that human sociability has its origins in the exigencies of our expulsion to the savannah some six or seven million years ago. Once our ancestors evolved bipedalism and thereby gained the capacity to kill at a distance, it became much more advantageous for them to cooperate in their mutual defense, which at some point led to a dramatic increase in their sociability. This sociability, in turn, created novel selection pressures and opportunities that led to many of our uniquely human proclivities and abilities. Although we can

only speculate about the timeline of many of these events, data from numerous fields of research converge on the centrality of sociability in human evolution.

Notes

1. *Hominin* is a term for modern humans and our extinct ancestors back to our common ancestor with chimpanzees. *Hominid* is a term for all existing and extinct great apes, including modern gorillas, orangutans, chimpanzees, and humans.
2. Keeping in mind that absence of evidence is not evidence of absence.

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3

SOCIABILITY

A Foundational Construct in Social Psychology

Radmila Prislin and William D. Crano

Abstract: This chapter differentiates lay conceptualizations of sociability from psychological theories and research on the construct, highlighting the influence of research on sociability across various fields of social psychology. Classic authors such as Wilhelm Wundt and William James considered sociability as either part of human nature in the Aristotelian tradition, or a product of a social contract required to counter unconstrained human nature. Later researchers applied the construct in service of their goal to build a science of human behavior. Theory and research on conformity and normative behavior by Asch and Sherif, and Moscovici's work on minority influence deserve special consideration. The chapter discusses Crano's recent work on the importance of the relation between a message source and the target's social group in attitude change as factors illustrating the role of sociability in human behavior. We also review Prislin's work on the role of minority groups in competitions for group power, documenting how minority and majority identities could be shaped in dynamic social interactions. The links between this work and classic formulations by Wundt and James are considered, and the importance of understanding the role of social influence processes in sociability is discussed.

Sociability has occupied the attention of social psychologists from the earliest days of the discipline. In this chapter, we survey some of the earliest work of psychology's founders in developing social psychology as an entity related to, but distinct from general experimental psychology as envisioned at its birth as a scientific discipline. Intense intellectual efforts were necessary to foster development of a psychology of social (human) beings. Sociability means more than its usual denotations of amiability, conviviality, or gregariousness, a multivariate trait reflecting people's characteristic desire for meaningful connection with

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others (Boswell et al., 2020). It is that, and more. A widely accepted understanding of the construct in social psychology is foundational, a necessary element of our discipline. It represents a shared understanding of the essential nature of people individually and in groups, an indispensable building block of the discipline. No doubt sociability can involve amiability or gregariousness, and higher levels of the trait generally are seen as reflecting a positive orientation to others—*if not all others*, at least those in one's ingroup. Levels of sociability vary from person to person, context to context, and society to society. Social psychology is built on these considerations. We are concerned with how interpersonal, environmental, and contextual factors affect this inherent (or learned) human trait to produce the enormous variations seen in social behavior. To do justice to sociability as a foundational element of social life, it is useful to study the trains of thought that carried sociability from the early days of the discipline to social psychology's adolescence to today's theory and research.

Historical Antecedents

We begin our consideration with some of the earliest writings on psychology (James, 1890; Wundt, 1907) and social psychology (Allport, 1924; McDougall, 1908; Wundt, 1902, Wundt & Schaub, 1916). These early authors drew from the conceptualizations of sociability that philosophers had discussed from antiquity, especially those who emerged during the early modern period that advanced two distinct conceptualizations: one considered sociability as part of human nature (Grotius, 1625/1901/2005), the other considered sociability a product of the *social contract*, an explicit or tacit agreement among individuals to form communities, to counter or control unconstrained human nature (Hobbes, 1651/2020; Pufendorf, 1678/1990). Wundt subscribed to the former, arguing that natural sociability, reflected in the development of mental communities and mental products, was the principal domain of social psychological inquiry. His idea that people's natural sociability was foundational for a sense of individuality, individual rights, and normative behavior would be echoed widely in subsequent social psychological theorizing and research (see also Gaffney & Hogg, this volume).

Antecedents: Sociability Considered by the Founders of the Science

Wilhelm Maximilian Wundt

Unsurprising for a polymath, who tackled topics ranging from the sodium chloride component of urine (Kim, 2016) to ethics (Wundt, 1886), a founder of scientific psychology, Wilhelm Wundt (1832–1920) wrote on human sociability too. Though he never used the word, some of his capital writings elaborate

ideas that refer to sociability. In his treatise "On principles of psychology," Wundt postulated the existence of "mental products of a general character," which "presuppose as their condition the existence a *mental community* composed of many individuals, though, of course, their deepest sources are the physical attributes of the individual" (Wundt, 1907, p. 27). Wundt argued that there was nothing supra-individual in the concept of a mental community, as some later argued (e.g., McDougall, 1908).

These mental communities are presumed relatively enduring and existing independently of the observer. Wundt (1907, p. 27) assigned the study of these products to social psychology, "Because of this dependence on the community, in particular on the social community, the . . . investigation here involved is designated as social psychology." By delegating investigation of mental communities to social psychology, Wundt declared the field an essential component of psychology as a scientific discipline, critical for understanding its object of study. Expressing this position in the language of his time, Wundt (1886, p. 389) wrote, "We know man only as a social being."

Wundt argued that a psychological analysis of the development of mental communities and their intellectual accomplishments (e.g., language, myths, customs) can identify their motivational underpinnings and law-like regularities (see also Fiske, this volume). Commenting on the instrumentality of mental communities in satisfying physical needs and "mental ends," he noted great variability in the latter. They evolve, just as specific forms of mental communities do, "according to the special conditions of life." Lest this is misunderstood as a utilitarian approach to mental communities as an expression of human sociability, Wundt made it clear that he had little use for those philosophical approaches that

> treat all mental product of communities. . . . as voluntary inventions designated from the first for certain utilitarian ends. This view . . . is the doctrine of social contract, according to which a mental community is not something original and natural, but is derived from voluntary union of a number of individuals. This position is psychologically untenable, and completely helpless in the process of the problems of social psychology (Wundt, 1902, p. 35).

By conceptualizing mental communities as "something original and natural," Wundt appeared to argue for sociability as a constitutive element of human nature rather than an intentionally developed survival mechanism to guard against human innate unsociability. Less represented in philosophical thought, his idea of "natural sociability" did have its proponents. Tracing its origins to Aristotle, it was elaborated in one of the first early modern-era theories of sociability by Dutch humanist Hugo Grotius, who postulated the existence of a need-like "appetite for the social life" (*appetitus societatis*). Writing that "among the things

peculiar to man, is his desire of society, that is, a certain inclination to live with those of his own kind, not in any manner whatever, but peaceful . . ." He further proposed that sociability serves as the foundation of individual rights (Grotius, 1625/1901/2005, p. 253). Without explicitly acknowledging Grotius's influence, common for his times, Wundt echoed his ideas speculating that a mental community of humanity, grounded in "natural sociability," is characterized by a universal norm acknowledging human worth (see Dunn & Lot; and also Regan & Lyubomirsky, this volume).

This position ran counter to a doctrine of the contractual nature of sociability. The origins of the idea of intentional sociability grounded in self-interest are traceable ultimately to the Epicurean philosophers of the 4th century B.C. Their intellectual heritage was embraced and developed in the 17th and 18th centuries' Enlightenment era. True to their maxim about the power of reason, Enlightenment thinkers argued that sociability could overcome human inherently ("natural") unsociable nature to protect humans against themselves. Likely the best-known proponent of this position, Thomas Hobbes (1651/2020) elaborated this idea in his masterwork, *Leviathan*. He detailed how various expressions of contractual sociability, including political systems and entire societies put brakes on otherwise antisocial human nature, which, if left unchecked, would lead to "*bellum omnium contra omnes*" (a war of all against all). Building on the Hobbesian notion, another thinker of the era, Samuel Pufedorf (1632–1694) offered a more nuanced conception. Although subscribing to the self-regarding basis of human motivation, Pufendorf focused on the process by which sociability developed, through interactions among unsociable individuals (Pufendorf, 1678/1990). He emphasized the constructive role of social interactions, particularly spontaneous ones outside organized and regulated political arenas, which bear traces of later social psychological theories on habits and norm formation.

The 17th-century discussions of sociability carried into the 18th century with marked vigor. Joining the Anglo-German philosophers' chorus, Genevan Jean-Jacques Rousseau made his contribution with a fine French flare. While acknowledging sociability as contractual in the title of his masterpiece, *The Social Contract*, Rousseau (1775/1950) posited the social contract served a function different from that of his predecessors. The contract was not a defense against human "natural unsociability" but rather a control over society's negative influence on otherwise noble human nature. To prevent destruction from socially imposed constant competition, humans enter a social contract, subjecting themselves to the authority of the general will (people as a whole), trading their natural rights for social rights. Rousseau thought of the social contract as a mechanism to ensure the greater good for the collective rather than individual (self-)interests as earlier social contract proponents argued.

Closing the 18th-century cross-European debate on sociability, Immanuel Kant (1724–1804) addressed the issue in his *Universal history from a cosmopolitan viewpoint* (Kant, 1784/2008). Faced with the seemingly antagonistic viewpoints

on human nature (antisocial versus social), Kant acknowledged both. He maintained the "capacity for social existence is clearly embedded in human nature" but so is "a great capacity to isolate himself, for he finds in himself the unsociable characteristic of wanting everything to go according to his own desires." Conjoining the viewpoints in a dialectical marriage, Kant coined a wondrous oxymoron, "unsocial sociability" defined as the human "propensity to enter into society, bound together with a mutual opposition which constantly threatens to break up society" (pp. 31–32). This complexity makes for intricate social interactions and leads to development of roles and rules that become "another nature." Humans, then, are as much producers as products of their development. This idea is closer to Wundt's notion of sociability, which he recognized when he claimed Kant as his intellectual ancestor.

Wundt acknowledged the Kantian complexity of social outcomes stemming from sociability in a ten-volume opus, *Völkerpsychologie* (Wundt, 1900; Wundt & Schaub, 1916). Credited as a foundational document in social psychology, the work was published over a two-decade period from 1900 to 1920. It focused on the history of *mental* development and may be understood better as a "psychological-developmental theory of mind" than folk psychology (Fahrenberg, 2019). Wundt's developmental theory of mind placed critical emphasis on its social dimension, implying that "natural sociability" underlies the development of specific forms of mental communities throughout human history. His analysis of mental communities based on an immense volume of sources ranging from anthropological and ethnographic to economic and geographic identified three universal "mental products": Language, which contains the general forms of representations (ideas) and the rules of their connections; myths (based on language), to give form to humans' capacity for imagination; and customs. Their relatively slow changes over time are regular and lawful because the individual psychological processes whose interaction produces them are themselves regular and lawful. This continuity, grounded in universal dispositions indicative of sociability, is evident throughout developmental phases, from the earliest "primitive man" to his aspirational "humanity." Sociability, the heart of this psychogenesis, does invariably lead neither to positive social interactions nor to uninterrupted social progress, as Wundt was careful to warn. Yet, he allowed that the long arch of human history appeared to bend toward the development of a (still aspirational) mental community of humanity where "appreciation of human worth shall have become a universal norm" and "a recognition of universal human rights to which each of the members of the human race may lay claim, and of duties which he, in turn, owes to human society" (p. 473; see also Fiske, this volume). The idea of sociability as a basis for both social and individual identity (or in Wundt's terms, cosmopolitanism and individuality) anticipated later social psychological theories concerned with the seemingly opposite motivations for inclusion and differentiation that can be reconciled only in social contexts (see also Gaffney & Hogg, this volume).

Wundt's and Kant's views on the dual nature of sociability are echoed in today's social psychology. In a rare disagreement with Campbell's views, for example, Brewer (2012, p. 82) remarked she

> could not accept the idea that the extent of sociality and sustained group living that characterizes human beings could have been maintained solely to external constraints embodied in social institutions, traditions, and practices selected at the group level and operation in opposition to biological selection. . . . My point was that human beings are neither inherently purely selfish nor purely altruistic but instead characterized by a kind of functional antagonism between self-interested and group-interested behavior. Because of the resultant variability in motives underlying human social behavior . . . the distribution of human sociality might best be depicted in terms of a "golden standard deviation" rather than a "golden mean" (see also von Hippel & Smith, this volume).

Wundt subscribed to the idea that sociability is inherent to human nature rather an intentional, contractual mechanism developed to guard against the chaos otherwise resulting from inherently unsociable human nature or the invariably corrupting force of society. Human sociability is reflected in the development of mental communities, whose forms vary across history, ultimately leading to a universal community of humanity. No less aspirational today than it was at the time of Wundt's writing, this idealized mental community provides a necessary context of the development of both normative equality and individuality.

William James

On the other side of the Atlantic, Wundt's younger contemporary, William James (1842–1910), also wrote on a range of topics, allowing psychologists, philosophers, and historians to claim him as their own. Like Wundt, James came to psychology via medicine, reflected in both men's emphasis on the physiological bases of psychological phenomena. Both considered sociability an essential aspect of mental life, albeit in different ways. James accorded sociability the status of instinct—one of a dozen along with the physiological reflexes, he considered in his major work, *The principles of psychology* (James, 1890, Volumes I and II). An enthusiastic subscriber to Darwin, James made instinct an essential ingredient in the study of the human mind. Instincts were flexible "springs to action," responsive to environmental variations triggered by contextual stimuli. More than the residue of animal ancestry, he viewed instincts malleable and transitory.

Characteristically, James explained sociability as a non-reflexive instinct by describing rather than defining it: "As a gregarious animal, man is excited both by the absence and by the presence of his kind" (p. 430). Social deprivation was thought so painful that "Solitary confinement is by many regarded as a

mode of torture too cruel and unnatural for civilized countries to adopt" (p. 430). Ostracism, being excluded and ignored by individuals and groups, was a variant of blocked sociability considered more painful than physical torture. A century later, Williams (2001) documented many expressions of the pain of ostracism, which Eisenberger and Lieberman (2005) grounded in the same neural mechanism as physical pain.

James elaborated the role of sociability in his theory of the self. Differentiating the "me" self-component—a separate object an individual refers describing experiences—from the "I" as a thinking individual reflecting on the self as the subject of experience. He defined social self as "recognition which he gets from his mates," stemming from "an innate propensity to get ourselves noticed . . . favorably, by our kind" (p. 294). He postulated that self-knowledge arose from social relations. Sensitivity to others is crucial in developing self-knowledge and a sense of self-worth. Sociability informs (who we are) and affords (a sense of worth).

Given the critical role of sociability, it is surprising that James never considered its implications for social interactions. Granted, social self-seeking was motivated by "amativeness and friendliness," but in all expressions of sociability, the fundamentally individualistic motive is invariably selfish, evident in favor-seeking friendliness and in any and every act of sympathy, including those seemingly altruistic: "Our unselfishness is . . . apt to be a virtue of necessity," and "altruistic passions and interests are contradictory to the nature of things . . . if they appear anywhere to exist" (p. 314). For James, prosocial tendencies existed only as instrumental servants to the self in a Hobbesian world of perpetual war of all against all. "Hence the gory cradle, the bellum omnium contra omnes, in which our race was reared . . . we . . . still carry about with us, ready at any moment to burst into flame . . . harming others, but themselves unharmed" (p. 410).

For all the richness of observations of modern psychology's predecessors, they never served as a springboard to systematic theorizing about social interaction *processes*, much less group dynamics. James's concept of sociability, like his psychology in general, remained firmly individualistic. This position was early recognized and criticized as a weakness (e.g., Spoerl, 1942, p. 17). It is worth noting that James's work on sociability focused on its conceptualization, limiting its verification to anecdotal evidence some would find of only literary value, but in doing so, he provided a bridge between his roots in philosophy and the empirical psychology his work foresaw.

Social Psychology's Pioneers: Early Approaches to Sociability

William McDougall

One of the earliest texts in the field, McDougall's (1908) *An Introduction to Social Psychology* saw 23 editions in his lifetime and remains in print in spite of his

pariah status in his adopted USA (Heidbreder, 1939). McDougall agreed with Wundt that psychology is the basis of all social sciences:

> The fundamental problem of social psychology is the moralization (socialization) of the individual by the society into which he is born as a creature in which the non-moral and purely egotistic tendencies are so much stronger than any altruistic tendencies (McDougall, 1908, p. 16).

His view of sociability reflects his overarching instinct theory of motivation. Crediting James, McDougall posited the *gregarious instinct* as the foundation of sociability. This foundation does not necessarily imply amiability or pacific temperament. Sociability is more complex, more highly developed. Nonetheless, gregarious instinct was accorded special status in the development of social life and societies, in that it motivated different behavior toward those perceived similar or dissimilar. The instinct could be satisfied "only in the presence of animals similar to itself, and the closer the similarity the greatest is the satisfaction." One of the most developed cognitive processes that activated the gregarious instinct was "consciousness of kind" (p. 258). This consciousness, and a postulate about differential reactions to (dis)similar others, contains the seeds of Tajfel and Turner's (1979) social identity theory (see also Gaffney & Hogg, this volume).

Floyd Allport

Judging by the number of books whose titles referenced social psychology, the roaring 1920s were roaring for social psychology, too. Among the more than 20 books that appeared in that decade, Allport's (1924) *Social psychology* was one of the most significant. It championed behaviorism as the fundamental theoretical framework for social behavior and the experiment as the crucial method of scientific inquiry.

Allport recognized that behavior may have some previously recognized inherited bases (instincts), but emphasized habits as learned, purposeful integrations of these "innate neural coordinations" (p. 43). He reinterpreted gregariousness, which earlier authors (e.g., James, McDougall) considered the instinctual root cause of sociability as an effect, not a cause, and an overrated one at that. Allport acknowledged that "man is a society-forming animal," emphasizing the instrumentality of any such formation (e.g., "There is always definite aim or interest, other than merely 'getting together,' in every congregation of people," p. 77), which, in no way diminished its importance. On the contrary, its importance in the development of "permanent habits of reposes which are in accord with the necessities of group life" was evident in the process of socialization (p. 123). Expressions of these permanent habits ranged from unselfishness completely benefiting a group to a passive selfishness reflecting "unmodified self-seeking of human nature" (p. 123). This most clearly reflects Allport's adoption of the

notion that human nature is selfish but modifiable through social interactions that reinforce sociability. Pufendorf's ideas at last receive their behaviorist translation!

Emerging from this process is "personality," which Allport defined as "the individual's characteristic reactions to social stimuli, and the quality of his adaptation to the social features of his environment" (p. 101; see also Schmitt, this volume). He discussed the social aspect of the self as a reflection of others' behavior toward an individual, which once adopted are projected as the individual's "own personal ideals and aims" (p. 326). By postulating social as a condition sine qua non for individuality, Allport built on the foundations established by Wundt and James—without ever acknowledging them as intellectual predecessors.

Although considered a manifesto for experimental social psychology. Allport's book, understandably, had little experimental work to present. It was limited to research on recognition of facial expressions of emotions and Allport's own work on social facilitation, both reflective of human sociability. Allport's work often but by no means always guaranteed facilitative effects of co-workers on an individual's performance on "mental abilities" and judgment tasks foretold the now canonical work by Sherif and Gordon Allport.

Sociability: Creating and Maintaining Social Reality

Muzafer Sherif

From the start of his career, Sherif focused on the importance of the key ingredients of sociability—attitudes, norms, roles, status, (Sherif, 1936; Sherif & Cantril, 1945, 1946), and ego-involvement, or self-interest (e.g., Sherif & Cantril, 1947; Sherif & Hovland, 1961). He subscribed to his mentor's definition of his chosen field:

> Social psychology is the science which studies the behavior of the individual insofar as his behavior stimulates other individuals, or is itself a reaction to their behavior . . . it is a consciousness of social objects and social reactions (Allport, 1924, p. 12).

All his pursuits reflected his belief in the human and societal need for sociability (inherited or otherwise), for normative effects assume axiomatically that actors care about and are influenced by the expectations of others in defining social reality (Sherif, 1936, 1961). Roles and status depended on norms, which are determined by socially acknowledged superior and subordinate roles (Sherif, 1935a). Attitudes, too, presume effects of social groups on individual beliefs and behavior, and ego involvement plays a role in attitude-behavior-consistency, depending on the strength of involvement and presumed social group expectancies (Sherif & Hovland, 1961; Sherif & Sargent, 1947).

Sherif and his wife, Carolyn Wood Sherif, powerfully influenced much of today's social psychology, having absorbed what came in the field's formative days to inform their own theories and research. One of their most widely discussed contributions is the "Robber's Cave Experiment" (Sherif, Harvey, White, Hood, & Sherif, 1988), whose focus on norm formation, especially under competitive stress, leads to considerations of ingroup solidarity or leniency and outgroup benign neglect, rejection, or outright hostility (Alvaro & Crano, 1997; Crano & Alvaro, 2013).

More individualistically oriented research predated Robber's Cave, and represents well the Sherifs' longer term research focus, which might be described as people's search for ways to describe and understand their social realities. One of the most ingenious of these investigations of the social basis of consensually defined reality is found in Sherif's (1935b) study of norm development. In an elegant series, Sherif studied people's perceptions of movement of a pinpoint of (stationary) light in an otherwise completely dark room. Surprisingly, the light will appear to move after briefly fixating on it. The *autokinetic phenomenon* is a profound illusion—perception of movement is undeniable even if one knows it's an illusion. Sherif found over a series of judgment trials that each individual's judgments followed a similar pattern. At first, judgments from one to the next were highly variable, but as trials progressed, judgments converged within a relatively narrow channel: Variability was attenuated, and the subject seemed to apprehend the light's "normal" movement. Although each subject's judgment of normal differed, all rapidly fell within narrow self-defined ranges, thereby lending consistency in their perceptions of a complete illusion—and this is what norms do. The response norm lent consistency to otherwise highly complex perceptual information. By extension, in social contexts, we (and Sherif) proposed that norms foster differentiation of normal or appropriate and lend order to social interactions. But what does the judgment of light movement have to do with social norms? Not much, were it not for Sherif's second and third studies, in which *pairs* of naïve subjects made their judgments in tandem across trials.

As in Study 1, each participant's initial judgments were highly variable, with low correlation between judgments within dyads. However, after a few trials, paired subjects' estimates began to converge within a narrow range. Imagine, when judging a complete illusion, participants in each dyad began to report seeing what their partner saw! These results emboldened Sherif and appeared to confirm his ideas of the role of social/perceptual norms, and hence, sociability, which is assumed if norms make any difference whatsoever in human social behavior.

As reasonable as his interpretation seems, Sherif must have worried about the alternative, that judgmental convergence resulted from respondents agreeing with their partners to mitigate potential conflict. To alleviate this rival hypothesis, he conducted Study 3, which paired non-naïve respondents from previous

judgment dyads—but the new dyads were drawn from those that earlier had formed quite different perceptual norms. Rapid convergence of reported judgments within pairs would suggest simple conflict avoidance as the most parsimonious interpretation of Study 2 and deal the norm formation interpretation a fatal blow. But within-dyad judgments did not converge rapidly, thus supporting the proposition that socially developed norms can resist counterpressure and wield strong influence.

Solomon Asch

At its most basic, Asch's (1952, 1956, 1961) research demonstrated the power of numeric majorities to influence responses of a minority-of-one on simple judgments. More careful analysis suggests Asch's work served to contextualize Sherif's findings (Prislin & Crano, 2012). In Asch's "line studies," participants indicated which of three lines matched a comparison line, a judgment made so easy that preliminary subjects almost never erred. However, facing a unanimous majority of experimental confederates who made *incorrect unanimous* judgments on some trials, many participants succumbed and agreed with the majority's incorrect option. The errant subjects probably did not think the majority's estimates were valid, and for many readers, the study was taken as evidence of majority power over an individual's capacity to resist, and people's weakness when facing strong social pressure. Asch (1956, p. 3) disagreed: "A theory of social influence should not be a theory of submission to social pressure." Rather than judge his findings the result human frailty, he argued they were the inevitable result of human sociability, comporting with Sherif's views. His post-study debriefing supported his interpretation. Campbell (1990, p. 39), too, disagreed with the "character defect" interpretation, arguing

> Asch has pointed out our ubiquitous dependence upon the reports of others for our knowledge of the world. Most conformity studies should be reinterpreted as illustrating this process (rather than showing the character defect of suggestibility)
>
> (see also Baumeister; and Krueger et al., this volume).

This departure from common interpretations gained Asch's (1990) hearty agreement. Earlier research supported the Asch/Campbell interpretation (e.g., see also Moore, 1921; Prislin & Crano, 2012 for an extended discussion of this issue).

Social Influence, Compliance, and Conformity

Twenty years before Campbell's (1990) interpretation, but not 20 years before, he had voiced it in his social psychology tutorial, where Crano (1970) showed that judgments of the weight given direct perceptions versus socially supplied

information could be influenced by extraneous social/contextual factors. Judgmental difficulty affects people's dependence on socially supplied information. When Asch's line judgment task is made more difficult, greater dependence on the majority is evident. Crano showed that compliance also could be enhanced when the expertise of the source of supplied information was made evident. This experimental expertise manipulation affected naïve subjects' later judgments as well, even when the influence source could not possibly know the responses of the naïve participants (see also Hood & Sherif, 1962, an underappreciated classic study). Crano's (1970) research not only assumed the importance of others of one's own kind, as James had postulated, but also suggested an important qualification to sociability: *perceived information quality* matters immensely – but not any social conspecific will suffice (also see von Hippel & Smith, and Krueger et al., this volume).

Importance of *perceived* information quality is evident in Crano and Alvaro's (2021), recent study, in which expected (or unexpected) information was presented by an identified source. In this study, Republican subjects read a strong message contrary to Republican political philosophy, allegedly authored either by a Republican or a Democrat. Subjects frequently misidentified the ingroup source's political affiliation if his message was contrary to expectations. When the message was attributed to an outgroup source, misidentification was minimal (Table 3.1).

Table 3.1's results provide a dramatic example of misidentifying the group membership of an ingroup deviant who had been described clearly as ingroup only minutes before. The error rate of the outgroup (Democrat) speaker by Republican subjects was only 5%, rendering implausible the rival hypothesis of memory failure.

Lest we throw stones, the same pattern of misidentification was found with Democrat participants, who fared no better than their Republican peers (Table 3.2). Misidentification rates (36%) of Democrats and Republicans were equal. The consistency and predictability of results across both studies do not support either a measurement error or a misremembering hypothesis. Errors were biased, protecting established beliefs from ingroup dissenters, whose dissent threated group cohesion, graphically demonstrating the importance of fidelity

TABLE 3.1 Biased Source Identification by Republican Subjects:
Pass indicates correct identification, Fail indicates misidentification

Source	*Pass*	*Fail*	χ^2
INGROUP	47	27	
			27.42***
OUTGROUP	93	5	

***$p < .0001$

TABLE 3.2 Biased Source Identification by Democrat Subjects: Pass indicates correct identification, Fail indicates misidentification

Source	*Pass*	*Fail*	χ^2
INGROUP	108	62	
			36.29***
OUTGROUP	167	18	

***$p < .0001$

to the ingroup and leading to more nuanced interpretation sociability and the role of perceived information quality.

It is no coincidence that social influence research established social psychology as a scientific discipline (Prislin & Crano, 2012). As the history of research on social influence shows, sociability is at the heart of this process. Social influence creates a socially shared reality, a prerequisite for coordinated action (see also Fiske; Fiedler & Hütter; and Kruglanski & Ellenberg, this volume). The importance of these social creations, reminiscent of Wundt's mental communities, motivates attempts at their formation, maintenance, and change. Because of an asymmetrical distribution of benefits that such communities grant to their majorities and minorities, the latter are the proverbial agents of change (Prislin & Christensen, 2005a; Seyranian, Atuel, & Crano, 2008).

Majority and Minority Group Influence

Modern study of minority group influence was initiated by Moscovici and his colleagues (Moscovici, 1976; Moscovici & Lage, 1976; Moscovici, Lage, & Naffrechoux, 1969; Moscovici, Mugny, & Van Avermaet, 1985; Moscovici & Nemeth, 1974). His work recognized earlier research growing out of considerations of human sociability, for surely sociability plays a role in response to ingroup or outgroup dissidents.

By virtue of what we know of our need for sociability, neglect of features affecting intra- and intergroup interactions unquestionably would retard progress. It is possible for members of an ingroup minority to maintain membership, if tenuous, in the predominant group. Indeed, Moscovici viewed the minority as a necessary check on our interpretation of reality, and the main driver of social innovation. While this might not be so when opinion minorities also hold obvious physical, religious, or ethnic differences with the majority—see Maass, Clark, and Haberkorn (1982)—it is likely that group members may find themselves at odds with a strongly held belief of the majority, but which they nevertheless believe must be considered by the larger group. Often, such (ingroup) dissidents are seen by majority members as somewhat deviant group members, but members nonetheless. Alvaro and Crano (1997, 2017) found ingroup opinion deviants rarely induced immediate change on the attitudes targeted in their

persuasive appeal (the focal attitude); however, the ingroup minority often was met with apparent leniency, insofar as attitudes *associated* with the focal issue changed in the direction of the minority's appeal. They termed this process *indirect attitude change*. Further, if the indirect attitude was strongly associated with the focal attitude, and was affected strongly by the minority's appeal, the focal attitude was likely to change as well, in the direction advocated in the initial appeal (Crano & Chen, 1998). The *leniency contract* theory of minority influence (Alvaro & Crano, 1997; Crano & Seyranian, 2007, 2009) has guided research in laboratory and field settings since its inception, and has garnered strong support (De Dominicis et al., 2014; De Dominicis, Ganucci Cancellieri, Crano, Stancu, & Bonaiuto, 2021; Donaldson, Siegel, & Crano, 2020; Johnson, Siegel, & Crano, 2014; Siegel, Donaldson, & Crano, 2019).

Applications: Dynamic Sociability of Minorities and Majorities

Prislin and colleagues (Prislin & Christensen, 2005a, 2005b, 2009; Prislin, Sawicki, & Williams, 2011) have expanded views on minority group operations when competing for dominance in defining social reality. Focusing on group dynamics in the context of (un)successful minority influence, they have documented how expressions of sociability evolve with changes in minority and majority positions within groups. In a typical research scenario, opinion-based minority and majority factions are created by having a participant exchange opinions on socially relevant issue(s) with 3–7 confederates over a period of time. Confederates' initial patterns of (dis)agreement place the participant in the initial minority (majority) position, which remains stable or is reversed (minority ↔ majority) halfway through the interaction when several confederates switch their alliances. Importantly, participants actively experience their social positions within interacting groups, in contrast to the prevailing methodological approaches in social influence research that reduce such positions to information to be processed (Prislin, Crowder, & Donnelly, 2017).

Another signature feature of this research program is its emphasis on the dynamics of groups in which social influence preserves *or changes* the status quo, fostering study of changing expressions of sociability in the process of the creation, maintenance, and dissolution/re-creation of mental communities. In a nod to Wundt's construct and in recognition of the inherently complex nature of human sociability, this program aims to capture its dynamic, multi-faceted expressions.

A premise underlying much of the theorizing and research on social influence is that others' support is valued and actively sought. To test this assumption, Christensen, Prislin, and Jacobs (2009) had participants rate importance of others' support in a context where such support did not have any tangible correlates, or alternatively, was instrumental for securing power within the group. In the

former condition, which reflected a value of others' support in and of itself, participants who remained in the minority position throughout group interaction rated other's support as significantly less important than all their counterparts: minority whose initial position was changed to majority, stable majority, and majority whose initial position was changed to minority. The latter three did not significantly differ among themselves. However, when others' support was instrumental for securing power, all factions found such support equally highly important.

This pattern of findings suggests noteworthy implications for the exercise of social influence: Factions will continue vying for others' support, no matter how transient, unless their efforts are met with targets' persistent unyielding. Yet, even persistent unyielding will not discourage influence seeking when others' support affords power. The prospect of a desired tangible outcome elevates the value of social support among all group factions. Given that others' support is multifunctional and typically associated with tangible benefits (Prislin, 2010), it likely will fuel social influence in perpetuity, turning it into a psychological equivalent of a perpetuum mobile.

Although considered (almost) uniformly important, others' support acquires different meaning depending on the context in which it was obtained. Prislin and Christensen (2002) demonstrated that rare supporters who sided with a stable minority were valued more than those who jumped on the bandwagon of an increasingly successful minority that managed to turn itself to majority. In contrast, supporters who sided with a majority, stable or changed, were valued equally. Importantly, supporter valuation partially mediated sources' identification with the overall group. Though increased in size, the suspect quality of support secured by the new majority did not translate into increased identification with the group. At the same time, a decrease in size of the valued support retained by a new minority promptly translated into a decreased identification with the group (Prislin & Christensen, 2002, conversion condition).

The asymmetrical effect of gain and loss of social support on the overall identification with a group has been replicated in numerous studies (see Prislin, 2010). This asymmetry, which significantly reduces overall identification with a group in the aftermath of social change, leaves a group vulnerable. A group is likely to lose its members (Prislin & Christensen, 2005b), who, even if they remain in the group, diminish their loyalty (Shaffer & Prislin, 2011). This vulnerability can be gradually reduced with stabilization of the newly established positions over time if the group survives strong centrifugal forces unleashed by social change (Prislin & Christensen, 2005b, Study 2). In a more optimistic scenario, vulnerability could be avoided altogether through the quality (authentic) support for all factions in a changed group—a possibility with a limited likelihood (Prislin et al., 2006).

Social change affects not only reactions to the group as a whole but also interactions between minorities and majorities. Long before this proposition was

empirically tested, it was anticipated by John Stuart Mill. This insightful observer of the changing fortunes of historical minorities and majorities commented that "the majority, or those who succeed at making themselves accepted as the majority, the people, consequently may desire to oppress a part of their number" (Mill, 1859/1956, p. 6), a common feature of populist politics (Forgas, Crano, & Fiedler, 2021). In a study that examined conditions under which this capacity for oppression or maltreatment of a minority may materialize, Prislin et al. (2011) focused on the quality of social support for factions in a group. Reasoning that the quality (authentic) support affords new majority a sense of control over maintaining their position, Prislin et al. hypothesized that a high sense of control would function as a buffer against oppression. By contrast, low control and the resultant uncertainty over their position should unleash the new majority's potential to oppress. In a study that manipulated faction's position within a group (stable minority versus minority-turned majority) and control (high versus low), results revealed that a new majority claimed significantly more material resources and decision-making authority when lacking (versus having) control (Study 1). This oppression in the form of ingroup favoritism escalated into an open outgroup hostility in the context that condoned open contempt for the minority. With a social license to oppress, a new majority ceased the opportunity (Study 2). Obviously, having experienced what it means to be in a minority does little to prevent power abuse by the new majority uncertain about its position.

Although undeniably potent and often practiced, abuse is not the only reaction to minorities or, in general, different others within a group. A more constructive way of negotiating differences includes their tolerance or acknowledgment of their legitimacy even when evaluating them negatively. A still "warmer grade of tolerance:" (Allport, 1924, p. 425) includes their appreciation or recognition that differences are valuable in and of themselves. They are social capital that strengthens and advances a group. Appreciation could be thought of as a "beyond-tribal" manifestation of sociability, likely necessary for the development of Wundt's "universal community of humanity." In a study that examined how tolerance and appreciation were shaped by one's position within a group, Prislin, Davenport, Xu, Moreno, and Honeycutt (2018) employed their standard procedure through which participants actively experienced stable or reversed minority and majority positions in interacting groups. Subsequently, they expressed their tolerance (e.g., *"Live and let live" is the best way of approaching differences*) and appreciation (e.g., *Differences enrich the group*) for differences within their group. Results revealed a significantly higher tolerance in the change compared to stable conditions. Appreciation, however, was more complexly determined. It was significantly stronger among participants in a minority than a majority but only when their positions remained stable. Change diminished appreciation among those initially in a minority who subsequently gained a majority position.

Significant reduction in appreciation for differences among successful minorities suggests that the ideals of inclusiveness, typically advocated by minorities

seeking social change, may not survive once they win the coveted majority position. Their reaction, coupled with a relatively low appreciation maintained by those who lost their majority position, suggests that social change may not be conducive to turning differences into a social capital. Change holds greater promise for increasing tolerance. Although successful minorities appear unlikely to celebrate those with whom they traded places within a group, they seem willing to tolerate them. Idealism gives way to pragmatism, likely functional in consolidating the newly established "social order." This consolidation is especially likely if the unsuccessful majority's (new minority's) increased tolerance is taken as resignation to the new order. This scenario may prevent or least temper new majorities' potential to oppress. Yet, any potential equilibrium is destined to be temporary because, as discussed throughout, social influence is a ubiquitous expression of human sociability.

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4

AN EMOTION NICHE FOR THE CULTURAL EVOLUTION OF SOCIAL PRACTICES

Alan Page Fiske[1]

Abstract: Most of human sociality is organized in culturally evolved practices that have been selected as a function of how readily people invent, notice, remember, report, reproduce, and recruit others to them. Social practices are especially prone to persist and diffuse to the extent that they are inherently. In all of these ways, the kama muta emotion is a psychosocial niche selecting for cultural practices that evoke it. Hence in all sorts of cultures throughout history, we find myriad culturally central and salient kama muta-evoking practices in religion, politics, marketing, media, the arts, and everyday sociality. These practices vary greatly and have culturally and institutionally specific unique features. What they all have in common is the promotion and presentation of sudden intensification of communal sharing. That is, cultural practices are selected when they make immediately and dramatically salient some kind of love, fellowship, solidarity, support, belonging, unity, compassion, or kindness. This elicits kama muta that sustains and spreads the practice. Kama muta generates commitment and devotion to communal sharing. So when the communal sharing relationships that it reinforces enhance fitness at the biological level, then in turn, there is natural selection for the psychosocial disposition to experience kama muta.

Human life takes place in the stream of cultural practices—music, worship, weddings, YouTube, novels, pet keeping, social movement marches, and so on. To understand why any particular cultural practice exists, we need to know

why people invent the practice;
why others notice and remember the practice;

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why people seek to participate in and re-enact the practice;
why people want to tell others about the practice;
why people want to recruit others to join in the practice; and
how the practice cumulatively changes to become more effective
at reproducing itself in each of these ways.

These processes of reproduction of variant cultural practices comprise cultural evolution, as some practices reproduce themselves more than others. Music, for example, culturally evolves: it is a set of practices that replicate to varying degrees in a given human population, with some forms spreading and supplanting others that reproduce less successfully. The cultural evolution of musical practices consists of different rates of composing, performing, listening to, attracting adherents, and copying different kinds of music.

Why do you listen to some kinds of music more than others, purchase certain pieces and recommend them to others, or attend certain concerts and invite others to come with you, while ignoring other music? Perhaps the most important factor, I suggest, is the emotion that the music evokes in you. Music that gives you a warm feeling in the heart, wonderful goosebumps, joyful tears, and a lump in the throat is music that you want to listen to over and over, tell others about, and play for them. If you're a musician, you want to compose and perform music that evokes that emotion. The greater the number of people in whom the music evokes that emotion and the stronger their emotional response to the music, the more the music is perpetuated, spreads to other communities, is valued and copied, and is perfected so as to evoke the emotion more strongly in more people. No person, community, or network can compose, perform, listen to, and copy all the music that exists: most pieces of music are ignored and forgotten, while only a few pieces and genres are taken up. So the prevalent practices of one time and place are supplanted by musical variants that are taken up at higher rates. That's the cultural evolution of music, a typical practice whose survival and reproduction depend on an emotion.

This chapter first defines culture and cultural evolution; then offers a preliminary sketch of the ecological concept of niche and how it can be applied to shed light on cultural evolution and a key aspect of human sociability. The chapter then describes the kama muta emotion that my colleagues and I have characterized (Fiske, Seibt, & Schubert, 2017; Schubert, Zickfeld, Seibt, & Fiske, 2017; Zickfeld, Schubert, Seibt, & Fiske, 2018; Fiske, 2020a). Kama muta is a Sanskrit phrase referring to a brief warm feeling of oneness, love, belonging, of being "moved" or "touched" by a thought or experience. Its evident evolutionary function is to motivate devotion and commitment to the communal sharing relationships that are crucial to human sociality.

Then the chapter more fully presents the ecological concept of niche and shows how kama muta can be conceptualized as the psychological niche in which a great many social practices culturally evolve. After that, we consider just how music,

for example, evokes kama and hence thrives in the niche that the emotion provides. The chapter goes on to theorize how, reciprocally, the psychological proclivity to experience kama muta likely evolved biologically as a genetically encoded adaptation to the cultural practices that it evokes (see also von Hippel & Smith, this volume). That is, we see how human fitness depends in part on the proclivity to feel kama muta in response to a suite of cultural practices. The chapter concludes with consideration of where we can go with this concept of an emotional niche for cultural practices, and what we can learn from it about human sociality.

Some Cultural Practices That Evolved in the Kama Muta Niche

As I will define it, culture consists of those aspects of life that people have, want, do, or use by virtue of participating in a specific community or network. Culture thus includes beliefs, desires, implicit knowledge, capacities and technologies, habits, activities, institutions, modified landscapes and infrastructure, buildings, and artifacts that we take up from our predecessors and peers. For simplicity of exposition, I call all of these things together "practices." People take up variants of these cultural practices at different rates, depending on the features of the variants. Thus there is temporal change in the proportions of the variant practices in the cultural mix, so eventually some replace others. The surviving variants generate further variations, which people take up at differing rates, leading to the extinction of most practices and their replacement by variant practices more frequently taken up. That is cultural evolution (for key approaches to the construct of cultural evolution, see Sperber,1985; Henrich & McElreath, 2003; Mesoudi, Whiten, & Laland, 2006; Boyer, 2007; Henrich, Boyd, & Richerson, 2008; Richerson & Christiansen, 2013; Mesoudi, 2016).

There are many selective forces that make some practices more likely to be taken up than others, but one of the most important kinds of selective forces is human psychology (see also Baumeister & Maranges, this volume; and Forgas, this volume). Human psychology substantially affects what variants people most frequently notice, remember, tell others about, find appealing, seek to use or participate in, recreate, copy and tinker with. This chapter considers one remarkable psychological selective force in cultural evolution, a specific emotion, kama muta. More than just *a* selective force in cultural evolution, kama muta is an entire niche for an ensemble of cultural practices. A niche in cultural evolution is an assemblage of selective forces that a practice or an ensemble of practices depends on. That is, the niche is the assemblage that enables, and shapes, the survival and reproduction of a cultural practice in a particular manner, together with the constraints imposed by that assemblage. This chapter describes that niche. To my knowledge, no one has previously applied the ecological concept of niche developed by Grinnell (1917) to cultural evolution, but it is illuminating to do so.

We can also make use of further developments in the ecological theory of the niche. A niche is not only the assemblage of resources and constraints actually utilized in a particular manner; it also comprises *the effects of this utilization* on the environment. In biological ecology, this concept of utilization of an assemblage of resources and constraints in a particular manner, affecting the environment in particular respects, is due to Elton (1927). This niche construct recognizes the fact that a species changes the environment that it occupies: earthworms digest dead vegetation to produce leaf mould (Darwin, 1881), large browsing mammals help to create and maintain grasslands (Axelrod, 1985). Likewise, cultural practices change the environment they occupy. For example, most musical practices foster communal sharing bonds among performers and among listeners, while some musical practices spread romanticism.

Organisms are not merely passive utilizers of resources, passive subjects of constraints, and incidental producers of by-products. Organisms often actively shape their environment, making modifications to the environment that facilitate their survival and reproduction. Their descendants commonly inherit and gradually adapt to this advantageous niche. Beavers build dams to create ponds for themselves, termites build mounds to protect themselves and ventilate their nests, and pine trees drop needles that make the soil too acidic for most competing species of trees. This active shaping of the environment to suit an organism is called "niche construction" (Odling-Smee, Erwin, Palkovacs, Feldman, & Laland, 2013). Similarly, cultural practices often fashion a niche that favors them. In particular, cultural practices may create new forms of sociability and new technologies that make people take up the practices more readily and transmit them with greater fidelity. For example, music has constructed for itself a niche comprising musical instruments, musical notation, technologies of recording and disseminating, concerts, fan clubs, music-sharing sites, and award ceremonies that attract attention to music. There is an extensive literature on gene-culture co-evolution (the canonical works are Boyd & Richerson, 1988, 2005; Richerson & Boyd, 2008; Durham, 1991; Lumsden & Wilson, 2005; see also von Hippel & Smith, this volume). But so far as I know, the concepts of niche, and niche construction in particular, have not previously been utilized in the conceptualization of gene—culture co-evolution.

Beyond music, let's consider some more cultural practices and identify the psychological niche they occupy. Imagine that you're gathered with family and friends at your older sister's wedding in a church, the couple have just said their vows, placed rings on each other's fingers, kissed, and now you watch them walking back down the aisle together, hand in hand. Would you feel something? Would you likely tell others about that feeling and what evoked it? Would that feeling, and the later memory of it, affect your plans for your own wedding? Or imagine that you are worshipping in a Pentecostal or other charismatic church, or in a Mormon temple, or perceiving an apparition of the Virgin Mary, or worshipping with a Sufi saint, or approaching the Kaaba in Mecca

(e.g., Hermkens, Jansen, & Notermans, 2009; Ibrahim, 2011; Luhrmann, 2012; Oaks, 1997; all of these are described at length, with many additional sources, in Fiske, 2020a). If you've experienced any of these practices, did they make you want to come back and participate again? Did they make you want to bring others along to join you the next time?

Have you ever had an emotional experience in worship that made you *believe* in the church's doctrine? I've read the absorbing *Jataka* tales of the limitless self-sacrificial loving kindness of the Buddha in his past lives (Dhammarama & Bareau, 1963). Though I'm not Buddhist, I find that these stories evoke an emotion in me that makes me just *have* to retell the stories to anyone who will listen. On top of this, when I feel the emotion that the stories evoke, Buddhism becomes more appealing to me, and somehow feel more valid.

If you were engaged in any of these practices you would likely feel kama muta, an emotion sometimes labeled in English *baptism in the Holy Spirit, burning in the bosom*, or, in other contexts, *being moved* or *touched*, *nostalgia*, or a *heart-warming* experience (Fiske, 2020a; Fiske, Seibt et al., 2017; Fiske, Schubert, & Seibt, 2017b; Seibt, Schubert, Zickfeld, Zhu et al., 2017). You might, in fact, feel literal warmth in your heat, have moist eyes or tears, have a lump in your throat, or get goose bumps (Schubert, Zickfeld et al., 2017; Zickfeld, Schubert, Seibt, Blomster et al., 2018). This wonderful feeling would be memorable, and make you want to come back to participate again. The emotion would likely stimulate you to tell others about the practice and recruit them to join you in it, in part because experiencing kama muta *with* others amplifies it, and in part because you would want to give this wonderful feeling to others. Experiencing kama muta would tend to reinforce your commitment to the congregation and your identification with it. It would tend to convince you of the reality of the divinity you connected with, assure you of the divinity's love for you, and help guarantee your devotion to that divinity. Then too, people observing you or hearing you tell about your experience would likely recognize the kama muta you experienced; this would tend to evoke kama muta in them, in turn, but in any case would attract the observers to your worship service, for example, so they could experience it themselves. Likewise, reinforced by their shared kama muta experiences, the congregation would develop solidarity, loyalty, mutual support, and care that would further attract converts for whom belonging to such a group is appealing. This would make the congregations that engage in such practices more stable, more likely to endure, and hence tend to sustain these worship practices—they would be constructing the niche for their kama muta-evoking practices.

In the aggregate and over the long run, all of these psychological and social processes make kama muta-evoking worship practices perdure, and diffuse to other communities. In ecological evolutionary terms, the kama muta emotion is a psychosocial niche to which worship practices culturally adapt; the more they evoke kama muta, the more prevalent they become and the longer they last. Indeed, this appears to be a major cause of the survival, spread, and

prevalence of most of the world religions, supplanting religions that evoke little or no kama muta.

Yet none of this is specific to music and worship—cultural evolution in the kama muta niche also occurs in politics and social movements, war, sports, marketing, social media, theater, television and cinema, dance, music, poetry, novels, sculpture, painting, photography, in family rituals, in interactions with pets, and in engagement with nature (Fiske, 2020a; Fiske, Seibt et al., 2017; Fiske, Schubert, & Seibt, 2017a; Janicke-Bowles, 2021; Ji et al., 2019; Petersen & Martin, 2021; Petersen et al., 2019; Pierre, 2019; Seibt, Schubert, Zickfeld, & Fiske, 2019; Seibt, Schubert, Zickfeld, Zhu et al., 2017; Steinnes, Blomster, Seibt, Zickfeld, & Fiske, 2019). The modern lore of romantic love is centered on kama muta, as is the mythology of heroes. Myriad cultural practices grow in the kama muta niche and continue to perfect their adaptation to it. The cultural practices that depend on this niche are also supported by disseminative technologies such as writing, printing, audio recording, film, electronic media and communications, and the Internet. But the reverse is equally true: to a significant degree these technologies themselves developed, utilized, and diffused because they facilitated kama muta experiences. That is, these practices built technological additions to the kama muta niche, constructing a niche that greatly favored the practices.

So to understand a myriad of salient and pervasive cultural practices, we need to understand kama muta. In each culture there are unique intriguing features of the practices that occupy the kama muta niche. Yet what is striking is the cross-cultural pattern of convergent evolution to fit this niche. Beyond what this reveals about kama muta as such, these processes elucidate the nature of emotions, culture, cultural evolution, and how culture and psychology complement each other to generate integrated practices and institutions (Fiske, 2000). The processes of adaptation to this emotional niche and the construction of this niche to make it better-support the practices also shed light on a hitherto unrecognized mechanism of group formation, growth, and perpetuation, while pointing to one process by which new groups may bud off from existing groups.

The idea of cultural evolution has been well developed in comparison and contrast to organic evolution (Sperber,1985; Henrich & McElreath, 2003; Mesoudi, Whiten, & Laland, 2006; Boyer, 2007; Henrich et al., 2008; Richerson & Christiansen, 2013; Mesoudi, 2016). Moreover, researchers have theorized how cultural and organic evolution shape each other (Boyd & Richerson, 1988, 2005; Richerson & Boyd, 2008; Durham, 1991; Lumsden & Wilson, 2005). For example, it is clear that the human language capacity is a set of organic adaptations, without which human language could not have culturally evolved. Conversely, without culturally evolved languages, organic adaptations for language would have no benefits for biological fitness, and so could not have evolved. The two co-evolved. The niche theory of kama muta-inducing practices is a theory of co-evolution, but to my knowledge it is the first such theory about an *emotion* niche to be well elaborated in conjunction with an extensive

ethnological account comparing and contrasting the practices of sociability that flourish in the niche.

What Kama Muta Is

Kama muta is an emotion that is

- evoked by a sudden intensification of a communal sharing relationship;
- momentary (brief);
- subjectively positive in seven respects:
  - people report that it is a good, pleasant, positive experience,
  - people seek to re-experience it, and may work hard to do so,
  - people like to remember the experience, and that memory often re-evokes it,
  - people like to recount the experienced to others,
  - people like to give it to ("share" with) others, especially those to whom they are attached,
  - people like to experience it *with* others, especially those to whom they are attached, and so recruit others to experience it with themselves,
  - people who expertly evoke it are praised and admired.
- when sufficiently intense, it is often (though not invariably) characterized by some of the following sensations and signs:
  - a warm or other pleasant feeling in the left center of the chest,
  - tears or moist eyes,
  - being choked up (having a lump in the throat or difficulty speaking clearly),
  - goosebumps or chills,
  - feeling buoyant (light),
  - feeling exhilarated, energetic, alert, engaged,
  - making an exclamation such as *awwww*!
  - putting one or both palms on the upper left chest;
- it generates motives to devote and commit to communal sharing, including generalized kindness and compassion;
- if a language has a specific, distinct lexeme for kama muta in a given context, that lexeme may literally mean "passively displaced" (e.g., *moved*), "passively contacted" (e.g., *touched*), "slightly hot" or "passively heated to a moderate degree" (e.g., *warm feeling*), or some combination of the above with regard to the "heart" (e.g., "*heart warming," "burning in the bosom," "touched my heart"*); this tendency is true across language families, but is by no means the basis for all of the lexemes naming the emotion in all languages.

Communal sharing is one of the four fundamental structures of sociality posited by relational models theory (Fiske, 1991, 1992; Fiske & Haslam, 2005). It is a

relationship in which the participants treat each other as socially equivalent with respect to the aspect(s) of the interaction that they are coordinating. Roughly speaking, communal sharing is "love" in the broadest sense, including romantic and familial love, bonds among teammates and members of a platoon, or members of a group who identify with each other, as well as other forms of affection of any intensity. It involves treating each other's needs and suffering as one's own, and taking collective responsibility for each other (see Reis et al., this volume).

The sudden *intensification* of a communal sharing relationship, evoking kama muta, may occur, for example, in a reunion or reconciliation, at a birth, a marriage proposal or wedding, when one feels welcomed or receives—or observes—a notable kindness, when participating in a social movement (Pierre, 2109), in a patriotic ritual, watching a movie or video, reading poetry, or when feeling one with nature (Petersen, Fiske, & Schubert, 2019; Petersen & Martin, 2021). Intensification need be the result of any action; it may consist of a sudden memory; one's heart going out to a cute or needy being (Steinnes et al., 2019); or when, after separation or loss, one feels the love that makes one miss someone who is gone (for many instances of all of these, see Fiske, 2020a).

The communal sharing relationship that suddenly intensifies to evoke kama muta can be between a person and another person, group, or entity such as an animal, deity, deceased person, team, imagined community, nation, or nature (Pierre, 2019; Petersen et al., 2019; Petersen & Martin, 2021; Schubert, Zickfeld et al., 2017). Intriguingly, people also often feel kama muta when they observe, read, or hear about the sudden intensification of communal sharing between two other persons or beings. Thus kama muta is often evoked by narratives and performances (Fiske, Seibt et al., 2017; Janicke-Bowles, Schubert, & Blomster, 2021), social media images and videos (Schubert, Zickfeld et al., 2017; Seibt. Schubert, Zickfeld, & Fiske, 2017; Zickfeld, Schubert, Seibt, Blomster et al., 2018) and imagined pairings such as those of *shipping* (imputing romantic relationships to fictional characters). An unexpected kindness or expression of compassion often evokes kama muta (Blomster Lyshol et al., 2020), as does seeing or cuddling with a cute baby or animal (Steinnes et al., 2019).

Depending on the language, the context, and the speaker, people may have no specific label for kama muta, or they may use a variety of context-specific lexemes, none of which corresponds one-to-one with the emotion. Whatever label people use, their label may extend to other emotions, and always leaves out instances of kama muta evoked in other contexts (Fiske, 2020b). Like any other emotion, kama muta can lead to other emotions (e.g., embarrassment), and it may co-occur simultaneously with many other emotions, moods, and attitudes (e.g., sadness of loss, awe at greatness, sexual arousal). Sequential or simultaneous occurrence inevitably makes subjective vernacular labels ambiguous (Fiske, 2020b).

When kama muta is mild and fleeting, people occasionally fail to notice it, or notice it but soon forget it, especially if they can't label it. Ordinarily,

however, it is subjectively salient and people like bring it to mind and like to tell others about it.

Because kama muta engenders devotion and commitment to communal sharing relationships, it reinforces the dyads and groups in which it occurs (e.g., see Brown, 2013; Pierre, 2019; see also Dunn & Lok; Reis et al., this volume). Presumably the fitness advantages of belonging to such tightly bonded dyads and groups resulted in natural selection for the propensity for this emotion. Indeed, the probable phylogenetic root of the propensity for kama muta is maternal bonding, which further evolved in a few species to pair bonding accompanied by paternal bonding.

We began with an example of how kama muta is the niche for the cultural evolution of some widespread religious practices (for many more, see Fiske, 2020a). Now let's consider how the niche that this emotion provides for political practices and social movements of many kinds (Pierre, 2019. Great oratory often evokes strong kama muta, and is indeed crafted to do so. This can have profound effects not only for good but also for evil. In the 1930s, Adolph Hitler's speeches elicited kama muta that gave the Nazis a strong, enabling identity that overcame the deep anomie of Germany at the time (Fishman, 1964). During World War II, Winston Churchill's radio talks generated highly motivating patriotic kama muta that reinvigorated and sustained the steadfast resolve of Britons (Churchill, 1940). Later the stirring speeches of Martin Luther King, Jr., and then Jesse Jackson, awoke the collective, purposeful pride that powered the civil rights movement, inspiring the patient, persistent courage of millions (Shamir, Arthur, & House, 1994).

These mobilizations probably would not have been so effective, or perhaps not even occurred at all, had they not utilized the psychosocial niche provided by kama muta. Experiencing kama muta—often without being able to put a name to it—motivates the listener or participant to come back for more, to tell others and bring them to join in, to support and care for others in the movement. The collective effervescence gives them a sense of the self as part of something much greater (Durkheim, 1912).

The Five Principal Routes by Which Kama Muta Fosters the Practices That Evoke It

Like other practices in the kama muta niche, the emotion evoked by demonstrations, protests, and marches, in turn supports the practices that produce it by six principal routes.

First, relatively few cultural practices are directly based on—much less generated by—explicitly articulable beliefs or concepts (Bourdieu, 1977). For example, many local religions involve very little theology, and their practitioners, as well as most practitioners of the world religions—may have no significant theory of worship. Much of religious practice is sustained to a significant degree by the kama muta practices it evokes.

Second, beliefs and concepts often do play some role in supporting and shaping some practices in religion and other domains. To the extent that they do, kama muta may reinforce the practices if experiencing the emotion convinces people that the supporting ideas are true. This is especially notable when people have kama muta experiences in loving union with a deity: the intensity of the emotion induces conviction in the reality of the deity and its love for the person experiencing kama muta in loving union with it. People think, for example, "How can Jesus not be real, how can His love not be true, if I felt that amazing feeling of His love!?" Likewise, experiencing kama muta is likely to cement commitment to the ideology of social movements and political parties (Pierre, 2019; Seibt et al., 2019). In general, experiencing kama muta persuades people of the truth of the teachings and stories linked to the kama muta-evoking practices. It makes them deeply meaningful and hence motivating, while the resulting certitude also fosters purpose: people think, "this is the truth, so I must engage in this practice, adhere to these teachings, spread the stories, and testify to their truth." This maintains the engagement of the person in the kama muta-evoking practice and commits them to engage others, preaching, proselytizing, and testifying.

Third, the persistence of a social group created or solidified by the kama muta-evoking practices that it performs, and the copies of itself that the group spawns, maintain the habitat for the kama muta-evoking practices (for political parties, see Seibt et al., 2019). That is, the social groups fostered by shared kama muta experiences provide the habitat for the cultural reproduction of the practices. These groups consist of members' identification with the group, devotion and commitment to the group, and close cooperation mediated by their communal sharing relationship. When the kama muta creates or strengthens a group that performs the practices that evoke the emotion, it has created the niche that in thrives in.

Fourth, people enjoy remembering their kama muta experiences and reporting them to others. Recalling and retelling re-evokes kama muta, which remotivates subsequent engagement in the practices that evoked it, facilitated by the expectation of experiencing it again. This results in the persistence and spread of the practices. The communicative spread may involve oral and written language (including poetry, fiction, theater, cinema, blogs, and podcasts), visual arts, and other media (Ji et al., 2019; Janicke-Bowles et al., 2021).

Fifth, kama muta often operates recursively: observing or hearing of Azami experiencing kama muta evokes kama muta in Binta and Nur, whose exhibition of kama muta or whose recounting the experience evokes kama muta in Sidwaya, Poko, Ju, Dhara, etc., and so on in a potentially exponential spread. This recursion is common to all sorts of kama muta-evoking practices, and notably pervasive in contemporary social media, where much "sharing" consists of passing kama muta on to others. Furthermore, when Azami experiences kama muta, the perception of which evokes kama muta in Binta, then when Binta recounts

this to Exta, Exta not only perceives Azami's kama muta, but also sees Binta's kama muta, getting a double whammy. This recursive exponential spread is greatly facilitated by ever-improving technologies of recording and communication: it is now very easy to show people what you feel and show them what made you feel that way.

In conjunction with these four processes is a sixth, related to the double whammy of the fifth process. This sixth mechanism is the enhancement of kama muta when people experience it together. This is one of the motivations for sharing social media and telling others about one's own kama muta: it is stronger when it is felt together with others. It is also one factor in people's desire to tell others about their own kama muta experiences and share stories and videos of third parties' kama muta. It is a motivation to write and perform music, poetry, novels, theater, cinema, and so on. Moreover, it motivates people to bring others along with them to participate with them in kama muta-evoking practices. By literally bringing people together it also reinforces the social groups and networks that engage in kama muta-evoking practices.

In these and the other ways discussed at the beginning of the chapter, kama muta provides a remarkably productive niche for cultural practices that evoke it—the more a practice evokes kama muta, the more the practice survives and reproduces. This niche supports life-cycle rituals such as weddings, anniversary and birthday celebrations, funerals and commemorative services, as well as the photography and video recording of these and other key moments. In personal communications, print and digital media, many stories are reported because they are *moving*—they evoke kama muta (Fiske, Seibt et al., 2017). Greeting cards and the cut flower industry thrive in this niche, as do memorials of war and disaster, along with spontaneous shrines and vigils for the deaths of public figures and victims of terrorism. So do sports victories where the winners pile on each other or hug in celebration, and award ceremonies in sports, cinema, and music.

Back to Music

Kama muta results from the sudden intensification of a communal sharing relationship. So, to occupy the kama muta niche, a practice must abruptly make people feel "one" with each other, or with some encompassing whole. It must rapidly dissolve the boundary around the individual self. That is, the practice must, at certain moments, all of a sudden "connect" people so that they feel part of something, so they merge into something beyond their individuality. Having this happen gradually does not evoke kama muta; it must be quick.

The emotion does not last long, but a person can feel it again and again; one doesn't feel it any less strongly after previously feeling it a number of times. Because there are innumerably ways that a sense of equivalence can swiftly emerge, there are innumerable practices in the kama muta niche. The most entrenched, enduring, pervasive practices in this niche evoke kama muta in

more than one way—and they create and refine social conditions that favor experiencing it and conveying to others that one feels it. Let's take music again as an example of a practice that is adapted, and continues to adapt, to the kama muta niche in this way. There are quite a number of psychosocial paths by which music evokes kama muta.

Similarity to crying (Panksepp, 1995; Nelson, 2005, pp. 207–209).
Identifying with hardship.
Compassion for the singer.
Being touched by the trust of revelation of vulnerability, pain, or humiliation.
Infatuation with a cute or beautiful singer with an enthralling voice.
Feeling loved by the singer.
Third person kama muta from lyrics about the singer's kama muta (Booker, 2004; Fiske, Sebit et al., 2017; Hogan, 2003).
Nostalgia when reminded of past love or belonging.
Patriotism concerning a nation, a cause or social movement, a military service, a school or team.
Joint production of the music creating communal sharing relationships among the musicians.
Synchronous rhythmic movement making musicians and/or audience feel one.
Perception of sharing kama muta among musicians and/or the audience. Several informants have reported this, and also, conversely, how their own kama muta is undermined by even one other audience member patently displaying that they are not experiencing kama muta.
Merging with the music itself (Frishkopf, 2001; Gabrielsson, 2010; Racy, 2003).

What we can take away from this brief consideration of music is that it evokes kama muta in multiple ways, while affording opportunities for others to see that it is evocative and to easily join it. At the same time, music constructs the kama muta niche by fostering social groups and networks in which kama muta-inducing music thrives, along with facultative institutions and technologies.

Do Other Emotions Offer Niches for Cultural Practices?

Beyond music, religion, politics, and social movements, there are innumerable kama muta practices in war, sports, marketing, social media, theater, television and cinema, dance, poetry, narratives and novels, sculpture, painting, photography, in family rituals, and in nature, as well as the lore of romantic love and the mythology of heroism (Fiske, 2020a).

Do other emotions or mood dispositions provide additional niches for cultural practices? This is a difficult question to answer because, I believe, we do not have a valid taxonomy of emotions—or even know whether "emotion" itself

is a valid category (Fiske, 2020b). However, we can provisionally reflect on some possible emotional niches. An emotion we often denote in English as *mirth* or *amusement* is the niche for the cultural evolution of modern comedy and comics, as well as satire and joke-telling through the ages. Something that one might denominate "moral outrage" is the niche for modern court and penal systems, and myriad premodern vengeance and punitive practices as well as populist politics (Forgas, Crano, & Fiedler, this volume). Something approximated by the English words *shame* and *embarrassment* provides a niche for clothing in warm seasons when it would otherwise serve little purpose, as well as architectural privacy features and practices related to defecation, urination, sex, and perhaps actions to reduce unpleasant body odors. Emotion researchers do not consider sexual arousal to be an "emotion," but it is surely some sort of affect, and it provides a niche in which a great many practices thrive around the world and throughout history. Pain is a niche for analgesics and anesthetics, physiotherapy, and allied practices. Yet none of these affective niches seem to harbor so many or so diverse yet universal cultural practices as kama muta. No other affect appears to have such extraordinarily generative and synergistic features.

The Psychological Capacity for Kama Muta Evolved to Depend on the Cultural Practices That Evoke and Depend upon It

The thesis of this chapter is that a great many cultural practices depend on the kama muta niche—this niche is a primary reason that people engage in these practices, and if this emotion did not exist, these practices likely would not exist, at least in the forms that they actually do take. Over time these practices presumably have culturally evolved to evoke kama muta more reliably and more intensely, replacing less evocative practices that less frequently survived or diffused.

The converse is also partly true, or at least was true on average over the long run: without certain of these practices, the psychosocial disposition to kama muta would not exist. Communal sharing relationships greatly enhance human fitness, providing mutual support, pooling of risks, and the beneficial joint use of resources and knowledge, among other benefits (see also von Hippel & Smith, this volume). Kama muta experiences arising from the intensification of communal sharing reinforce and sustain those relationships. But a person is not born knowing with whom communal sharing relationships of what sort will be advantageous for coordinating what kinds of activities; she must rely on cultural cues that index what communal sharing relationships exist in her community, and which ones she belongs to or may fruitfully form. She must attend to cues to the relative fitness values of the various potential communal sharing relationships she could participate in, and hence how she should prioritize her efforts to form and support each relationship (Fiske, 2000). The genetically based

emotional proclivity to kama muta must have evolved because it enhanced fitness, which could only be the case if, taken all together over many generations, the cultural practices that suddenly intensified sociability and hence evoked kama muta typically enhanced adaptively valuable communal sharing relationships. That is, the kama muta evoked by all cultural practices in all communities in the population, on the whole, must have strengthened fitness-beneficial relationships, resulting in the success of the kama muta proclivity phenotype.

This does not imply that any particular kama muta-evoking practice is beneficial to biological fitness, just that the sum effect of all cultural practices together has been beneficial, over many generations. It does not imply that this sum is beneficial today, or is beneficial at any other particular moment in any particular community. Indeed, many kama muta-evoking practices today surely contribute nothing to fitness: kama muta responses to such things as Pixar movies, cute kitten videos, or Taylor Swift songs surely do not make you more like to survive, and probably do little or nothing to increase your reproductive success. A person *decreases* his fitness if he feels such wonderful kama muta in response to religious practices or texts that he is motivated to devote his life to a deity by leading the life of a celibate priest, monk, or nun.

Cultural practices may interact with the psychological proclivity to kama muta in three ways. In ecological terms, cultural practices that evoke kama muta may be *mutualistic*, that is, beneficial to the spread of the practice and to the fitness of the person (such as kama muta seeing the first ultrasound of your baby). They may be *commensalistic*, that is, beneficial only to the spread of the practice, but not to genetic fitness (such as kitten videos). Or they may be *parasitic*, that is, beneficial to the spread of the practice but detrimental to the biological fitness of the persons with the disposition to experience kama muta (e.g., kama muta that motivates a celibate life). If the sum of the biological fitness effects of all kama muta-evoking cultural practices becomes negative, natural selection will slowly replace the kama muta-susceptibility phenotype with non-susceptible phenotypes. Cultural evolution occurs orders of magnitude faster than genetic evolution, so, as noted above, it is not necessarily the case that the psychological proclivity to experience kama muta in response to the aggregate of contemporary cultural practices is biologically adaptive at this moment in history. But it must have been biologically adaptive in the context of the evocative cultural practices that were pervasive over the last few thousand years.

However it is that kama muta practices affect biological fitness, it is clear that a great many prevalent cultural practices occupy the niche provided by the kama muta emotion. And those kama muta-inducing practices have in turn shaped their cultural-evolutionary niche by fostering the development of technologies and social activities favorable to the spread of the kama muta-inducing practices. To understand music, religion, the arts, narratives, oratory, recording and dissemination practices, marketing, politics and social movements, social

media and kitten videos, we need to see how they depend on the niche constituted by kama muta.

Where Do We Go from Here?

To correctly characterize the kama muta niche, to learn just how cultural practices evolve to adapt to that niche, and to discern how the cultural practices that occupy the niche reconstruct the emotion to their advantage, we need to do historical and long-term prospective studies specifically designed to elucidate the processes: need to observe how practices and emotions change and interact over time. Precisely how does the reproduction of evocative practices depend on their capacity to attract attention, be remembered, be persuasive, elicit the desire to tell others, motivate people to recruit others, evoke needs to re-enact evocative practices and perfect them? What is the nature of competition among practices occupying an emotion niche, and when do practices partition the niche or construct auxiliary niches of their own in which they can avoid competing with practices occupying the original niche?

To further inform us about the adaptation of cultural practices to the kama muta niche, we also need to search for cases of convergent evolution. For example, do religions independently evolve in similar ways to more strongly and reliably evoke kama muta? Do narrative genres independently converge on the same devices to make stories maximally evocative (see Booker, 2004; Fiske, Seibt et al., 2017; Hogan, 2003)? In the contemporary world, ubiquitous channels of intercommunication make it hard to discern whether any two practices are converging on the same means to evoke kama muta independently of each other. But it may be possible to find contemporary cases and to uncover historical ones.

As I indicated earlier, in conjunction with such investigations, it will be interesting to see what other emotions (and other affects) provide fertile niches for cultural practices—provided we can correctly delineate and distinguish among human emotions. When confronted with astounding human performances, magnificent constructions, or majestic natural phenomena, people experience an emotion that English speakers sometimes label *awe*, though *awe* is also used to label other emotions, including kama muta. This sort of "awe" is also positive in various senses, memorable, and inspires people to tell others about it. It also supports authority ranking relations: the gigantic pyramids, temples, palaces, and statues that rulers build evoke "awe" that reinforces the "superiority" of the rulers. So "awe" seems to be a niche for spectacular cultural practices.

To my knowledge, no previous work has mapped the full panoply of cultural practices that depend on one emotion, nor have many previous culture evolution researchers developed the notion of emotion niches, as such. Here I hope that I have indicated the potential merits of treating an emotional proclivity as a niche for the cultural evolution of practices. In particular, illuminating cultural

evolution by borrowing the concepts of niche and niche construction, we enhance our understanding of why humans engage in many aspects of religion, the arts, fiction, social media, iPods, marketing, and so on. This approach sheds light on emotions, culture, cultural evolution, and on intriguing ways in which culture and psychology complement each other to generate practices and institutions. This approach also illuminates a hitherto neglected emotion mechanism of group formation, growth, and perpetuation. I look forward to historical, ethnographic, ethnological, interview, behavior sampling, survey, diary, experimental, and physiological research that digs deeper into this.

Note

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PART 2

Sociability and Well-Being

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5

INDUCING SOCIABILITY

Insights From Well-Being Science

Annie Regan and Sonja Lyubomirsky

Abstract: A growing body of evidence suggests that individuals can increase their well-being by deliberately engaging in connection-boosting positive activities such as expressing gratitude and performing acts of kindness for others. Indeed, social connection—fundamental to both mental and physical health—is a prime candidate for future positive activity interventions. In this chapter, we argue that inducing sociability via interventions has implications for improving individual well-being, while also allowing psychological scientists to carefully examine the mechanisms underlying the relationship between sociable behavior and emotional health. Drawing on three recent studies from our laboratory, we review evidence that suggests that asking participants to act more extraverted, to engage in more social interactions, and to do more acts of kindness are all promising and efficacious methods to experimentally induce sociability. Finally, we summarize open questions and future directions for research on inducing sociability in experimental contexts.

Social connection is central to human experience. Feeling connected to others has been theorized as a basic psychological need, with implications for both mental and physical health (Baumeister & Leary, 1995; Holt-Lunstad, Robles, & Sbarra, 2017; Ryan & Deci, 2000; see also Baumeister & Maranger; Dunn & Lok; Fiske; and von Hippel & Smith, this volume). Accordingly, social dysfunction is a hallmark of mental health disorders such as depression and anxiety, and treatments for these disorders (such as cognitive behavioral therapy) often focus on improving and strengthening one's social ties (Berkman et al., 2003; LaRocca & Scogin, 2015). In this chapter, we argue that it is possible to induce sociability experimentally through self-directed activities, and that doing so

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increases psychological well-being. Furthermore, inducing sociability through this type of intervention allows social psychologists to examine the mechanisms underlying the relationship between sociable behavior and well-being.

It is worth noting that sociability can have two meanings. One is sociable if one tends to seek out and enjoy the company of others. Alternatively, one can be considered sociable if others enjoy their company. For example, an individual might be described as sociable based on their brimming social calendar, or due to the humor and charm they display during their social interactions. Although these components go hand in hand—that is, people are more likely to socialize if they delight in doing so and if others enjoy interacting with them—this chapter focuses on the first meaning of sociability, or instances of sociable behavior.

We begin our analysis by reviewing evidence from well-being science, which demonstrates that individuals can increase their well-being through volitional activities. Then, we present evidence that suggests sociability may be an especially effective strategy for increasing well-being in experimental research and applied contexts. Finally, using three recent studies from our laboratory as a framework, we review literature focused on inducing sociable behavior (i.e., engaging in more social interactions), with an emphasis on how augmenting sociability produces improvements in well-being.

Subjective Well-Being

Subjective well-being stands on two legs, comprising a cognitive component, assessed by one's satisfaction with life, and an affective component, consisting of relatively high positive affect and low negative affect (Diener, 1984). Decades of research have demonstrated that well-being is associated with myriad positive outcomes across life domains (see Diener, Lucas, & Oishi, 2018; Lyubomirsky, King, & Diener, 2005, for reviews). Furthermore, these links are more than correlational. Some longitudinal and experimental evidence suggests that subjective well-being leads people to succeed in the domains of social life (Jacobs Bao & Lyubomirsky, 2013), work life (Walsh, Boehm, & Lyubomirsky, 2018), and physical health (Diener, Pressman, Hunter, & Delgadillo-Chase, 2017).

Happiness is both hedonically and instrumentally rewarding, and as such, many individuals wish to become happier. However, seeking happiness is a notoriously elusive pursuit, and indeed, trying to become happier by pursuing it directly may backfire (Gruber, Mauss, & Tamir, 2011). Fortunately, experimental research has consistently shown that individuals can indirectly increase their well-being by deliberately and effortfully engaging in positive activities such as expressing gratitude or performing acts of kindness for others (Armenta, Fritz, Walsh, & Lyubomirsky, 2020; Fritz, Armenta, Walsh, & Lyubomirsky, 2019; Ko, Margolis, Revord, & Lyubomirsky, 2021; Nelson, Layous, Cole, & Lyubomirsky, 2016). These activities are typically short, self-administered, and non-stigmatizing and

are thus an accessible, low-cost option for happiness seekers. Positive activity interventions have been shown to be effective in increasing well-being and reducing depressive symptoms, with average effect sizes of r = .29 and r =.31, respectively (Sin & Lyubomirsky, 2009; see also Bolier et al., 2013).

Positive activities typically include simple, relatively flexible instructions to maximize adherence and minimize threats to participants' autonomy. An intervention instructing participants to engage in more acts of kindness for others or to act more sociable may provide participants with instances of these behaviors without instructing them to take specific actions. For example, the instructions in Nelson and colleagues' (2016) prosocial behavior intervention included several examples of kind acts—such as doing a chore for a family member, paying for someone's coffee in line behind them, or visiting an elderly family member—while noting that the participants are welcome to diverge from this list. Such instructions are also designed to encourage variety in participants' behavior, which has been shown to impact the efficacy of positive interventions. In one study, for example, participants who engaged in more varied kind acts over the course of 10 weeks experienced greater well-being benefits than those instructed to engage in the same kind acts each week (Sheldon, Boehm, & Lyubomirsky, 2012; see also Dunn & Lok, this volume).

The Positive Activity Model

Positive activities do not impact well-being directly. Rather, they do so through a variety of mechanisms (e.g., by increasing the frequency of positive thoughts and behaviors) and are influenced by moderating variables (e.g., an activity's dosage). The positive activity model (see Figure 5.1) posits specific mediating and moderating variables underlying the success of positive activities (Lyubomirsky & Layous, 2013). For example, expressing gratitude to a loved one may satisfy one's need for social connection, which, in turn, boosts well-being. With respect to moderators, the efficacy of positive activities is influenced by features of the happiness seeker (e.g., their level of effort) and features of the activity itself (e.g., whether a specific practice is social or reflective).

Furthermore, the positive activity model suggests that the efficacy of any particular positive activity hinges on whether the characteristics of the happiness seeker align with the characteristics of the activity. This is called "person-activity fit." For example, if a person engages in a high-fit activity—that is, one they find natural and rewarding—they are likely to muster more effort into the activity and experience greater increases in positive mood or life satisfaction as a result (e.g., Lyubomirsky, Dickerhoof, Boehm, & Sheldon, 2011). Similarly, if an individual engages in an ill-fitting activity, the intervention may be ineffective or even backfire (Fritz & Lyubomirsky, 2018).

Researchers are continuing to identify and test a wide range of positive interventions in order to maximally benefit individuals seeking to purposely

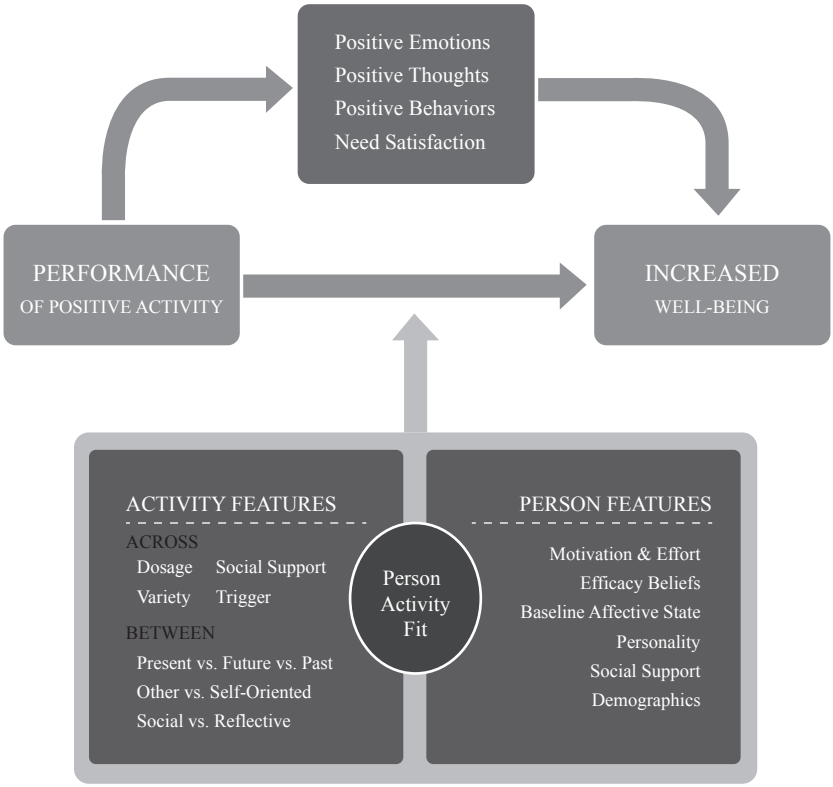

FIGURE 5.1 Model of the psychological mediators and moderators underlying the efficacy of practicing positive activities.

Source: Adapted with permission from "How Do Simple Positive Activities Increase Well-Being" by S. Lyubomirsky & K. Layous, 2013, Current Directions in Psychological Science.

increase their well-being (Carr et al., 2020; Parks & Titova, 2016; Revord, Walsh, & Lyubomirsky, 2018; Shin & Lyubomirsky, 2016). In the following sections, we review evidence supporting social behavior as a promising target for future positive activity interventions.

Increasing Extraverted Behavior

To be extraverted is to be sociable. Indeed, sociability has been repeatedly identified as a critical facet of trait extraversion, along with assertiveness and energy level (Saucier & Ostendorf, 1999; Soto & John, 2017). Although trait extraversion is a multifaceted construct, we propose that one way to induce sociability is by experimentally manipulating extraverted behavior. Before

describing recent experimental evidence from our laboratory, we begin by briefly summarizing the robust relationship between extraversion and well-being.

Extraversion and Well-Being

The relationship between extraversion and well-being is well-established, with dozens of correlational studies demonstrating a positive association between these two variables. A meta-analysis found that extraversion is positively associated with positive affect and negatively associated with negative affect, with average effect sizes of $r = .44$ and $r = -.18$, respectively (Steel, Schmidt, & Shultz, 2008). In addition to greater self-reported positive affect, extraversion is associated with a variety of advantageous social outcomes, including larger social networks and greater satisfaction with friendships (Swickert, Rosentreter, Hittner, & Mushrush, 2002; Wilson, Harris, & Vazire, 2015).

Engaging in extraverted behavior has been shown to increase well-being in laboratory and field settings. Fleeson (2002) applied the principle of state-trait isomorphism in conducting one of the first experiments to induce extraverted behavior. That is, because states (e.g., extraverted behavior) share properties and consequences of traits (e.g., heightened positive affect), increasing the frequency of trait-relevant behavior should likewise increase the consequences of a trait. This hypothesis was supported in a series of studies in which participants instructed to act more extraverted in a 10-minute conversation reported higher levels of positive affect than those instructed to act introverted (Fleeson, Malanos, & Achille, 2002; McNiel & Fleeson, 2006; McNiel, Lowman, & Fleeson, 2010). Subsequent research also suggested that people enjoy acting extraverted more than they believe they will. In one study, participants high in introversion forecasted more negative and less positive affect than they actually experienced when they were asked to behave more extraverted (Zelenski et al., 2013). In another experiment, participants who commuted to work via train or bus were instructed to either talk to a stranger or remain silent. Those who talked to a stranger during their commute experienced greater positive affect than those who remained silent, despite predicting a preference for solitude (Epley & Schroeder, 2014).

Although extraversion is usually considered to be a stable characteristic, a growing literature suggests that people may be able to volitionally change their level of extraversion and other personality traits (see Hudson & Fraley, 2017 for a review). One study specifically sought to test whether individuals could intentionally change their personality traits over 16 weeks (Hudson & Fraley, 2015). The results suggested that participants can do this, but only when they create plans to change their trait-relevant behavior. This work speaks to the importance of trait-relevant behavior (e.g., talking more) in volitional personality change, and indicates that specific change goals (e.g., I will try to talk more at the dinner table) may facilitate this process. Extending this research, a recent between-subjects study assigned participants to an extraversion condition, where they

were asked to act more extraverted (bold, talkative, outgoing, active, and assertive) than usual over the course of a week, or to an active control condition, where they were instructed to act more unassuming, sensitive, calm, modest, and quiet over a week (Jacques-Hamilton, Sun, & Smillie, 2019). These groups were also compared to a neutral comparison condition collected from another study. Participants instructed to act more extraverted reported increases in retrospective positive affect compared to both active and neutral controls. Interestingly, baseline extraversion moderated these effects such that extraverts benefited more from the intervention than introverts.

Inducing Extraverted Behavior Increases Well-Being: Experimental Evidence

Given the mounting correlational and experimental evidence for the link between extraversion and well-being, we designed an experiment to test whether inducing extraverted behavior over the course of a week would impact well-being (Margolis & Lyubomirsky, 2020). To that end, we recruited 150 undergraduate students to participate in an experiment in which they were asked to change their social behavior over 2 weeks. Using a within-subjects design, participants were randomly assigned to act either introverted or extraverted for 1 week, and then to behave in the other manner during the second week.

The behavioral instructions used in this study were designed to minimize effects due to social desirability and maximize the potential for behavior change. Because descriptors such as "outgoing" tend to be more socially desirable in Western cultures than descriptors such as "shy," we used trait adjectives to describe extraversion that had been rated relatively low in social desirability, and we used highly desirable trait adjectives for introversion (cf. Hampson, Goldberg, & John, 1987). Specifically, participants were asked to act more talkative, assertive, and spontaneous in the extraversion instructions, and deliberate, quiet, and reserved in the introversion instructions. To facilitate behavior change, participants were also asked to list five specific ways in which they planned to change their behavior over the next week, using a similar procedure as previous personality change experiments (Hudson & Fraley, 2015; cf. Gollwitzer & Brandstätter, 1997). An example of such a plan was as follows: "When my friends are discussing something important to me, I (will [extraversion instructions]; will not [introversion instructions]) express my opinion."

As hypothesized, our participants' behavior change throughout the study was associated with their self-reported well-being. Participants improved in positive affect during the extraversion week, and they declined in positive affect during the introversion week. Participants prompted to act extraverted experienced other psychological benefits as well—that is, they reported significant boosts in life satisfaction, in psychological need satisfaction (i.e., their sense of connectedness, competence, and autonomy; Sheldon & Hilpert, 2012; cf. Ryan & Deci, 2000), and in flow. Furthermore, our participants recognized that they were

behaving—or *being*—more extraverted during the extraversion week. They reported increases in extraverted behavior and scored higher in trait extraversion (as measured by both the BFI-2 and the NEO) during the extraversion week, and they showed the converse pattern during the introversion week. Interestingly, baseline trait extraversion did not moderate the effects of our experiment. In other words, individuals who scored high or low in trait extraversion when recruited into the intervention were no more or less likely to benefit from it.

Taken together, the results of this study suggest that individuals can volitionally increase their extraverted behavior over the course of a week, and that doing so is associated with well-being benefits, including greater positive affect and life satisfaction. Acting more extraverted than usual also increased participants' sense of autonomy, competence, and connectedness—psychological needs positioned as key mechanisms through which positive activities increase well-being in the positive activity model (Lyubomirsky & Layous, 2013). Furthermore, acting more introverted than usual actually decreased well-being and diminished feelings of connection and autonomy. These results provide evidence for the benefits of acting more extraverted than usual, and position extraversion as a promising target for future well-being intervention studies.

Our study, however, should be considered in the context of several caveats and limitations. First, the well-being benefits could be partially explained by experimenter demand, and, despite our best efforts, by social desirability. Although we avoided describing extraverted behavior in socially desirable terms, extraversion is highly valued in Western culture (Cain, 2013), and participants may have been nudged to respond to the self-report measures accordingly. Furthermore, although our participants reported increases in trait-level extraversion after acting more extraverted for 1 week, the timeline was likely too short to make strong claims about trait-level change, and observer and objective measures were lacking. Additional longitudinal experiments are needed to better understand the processes involved in volitional personality change, and whether increasing the frequency of trait-relevant behavior is sufficient to produce trait-level change.

Finally, although our behavioral instructions were representative of extraversion's three theorized facets, participants may have changed their behavior in accordance with a single facet rather than all three. For example, some participants may have chosen to focus on being more talkative with their friends, while others may have focused on being more assertive at work. Future research is needed to investigate the extent to which changing behavior based on specific facets of extraversion might differentially relate to well-being.

Increasing Prosocial Behavior

Another route to inducing sociability may be via engaging in prosocial behavior. In the following section, we begin by summarizing robust evidence for the relationship between prosocial behavior and well-being. We then turn to two recent experiments from our laboratory comparing the benefits of acting

prosocially compared to engaging in more frequent social interactions and other positive behaviors.

Prosociality and Well-Being

Theorized to be an innate human tendency, prosocial behavior (or acting to benefit another person) is demonstrated throughout the lifespan, starting as young as 14 months of age (Warneken & Tomasello, 2007; see Keltner, Kogan, Piff, & Saturn, 2014 for a review). Prosocial behavior—also referred to as kind or helping behavior—is usually motivated by a desire to improve the well-being or mood of others (although see Zaki, 2020 for examples of counter-hedonic prosocial interactions). Research suggests, however, that doing acts of kindness may be as beneficial to the doer as it is to the recipient. For example, engaging in prosocial behavior has been associated with greater self-reported well-being and physical health in correlational research (see Anderson et al., 2014; Keltner et al., 2014; and Konrath et al., 2014 for reviews).

The benefits of prosocial behavior have also been tested in numerous experimental studies, with a recent meta-analysis identifying a small to medium effect of experimentally induced prosocial behavior on well-being (Curry et al., 2018). For example, participants asked to engage in prosocial spending (versus self-spending) have consistently reported increases in well-being (Aknin et al., 2013; Dunn, Aknin, & Norton, 2008; Lok & Dunn, 2020; Whillans, Aknin, Ross, Chen, & Chen, 2020). Work from our own laboratory has also repeatedly identified well-being and other benefits from engaging in prosocial interventions. One study found that doing acts of kindness (versus a neutral activity) increased peer acceptance among preadolescents (Layous, Nelson, Oberle, Schonert-Reichl, & Lyubomirsky, 2012). In another study, participants were randomly assigned to perform acts of kindness for themselves, other people, or the world, or to keep track of their daily activities for 4 weeks (Nelson et al., 2016). Those who did kind acts for others experienced greater increases in psychological flourishing and positive affect after the study than did those who did kind acts for themselves or neutral activities. Notably, only the two beneficial—prosocial—conditions likely involved interacting with other people.

Comparing the Hedonic Benefits of Acting Socially versus Prosocially

Although prosocial acts can be anonymous or nonsocial (e.g., leaving money in a parking meter or donating to a charity), many prosocial acts involve actual social behavior—that is, a kindness offered toward at least one other person (e.g., helping someone carry their groceries or volunteering at a local food bank). Indeed, prosociality in everyday life frequently includes providing kind

words, advice, or other forms of social support to others. Might it be the case that the hedonic benefits associated with prosociality observed in both correlational and experimental studies are due to its social—as opposed to the prosocial—component? To address this question, we designed an experiment to directly compare prosocial and social behavior against a neutral control activity (Fritz et al., 2022). Furthermore, we sought to test whether technology-mediated interactions differed from interactions that took place in person.

Our participants were 754 employed adults recruited through Amazon Mechanical Turk. To test our research question, participants were randomly assigned to one of five conditions, using a 2X2 design with an additional neutral control condition. Once a week over the course of 4 weeks (e.g., every Monday), they were asked to either engage in three kind acts on the following day (e.g., buying a treat for a coworker or bringing flowers to a romantic partner) or to have three extra social interactions on the following day (e.g., chatting with a stranger on their morning commute or having a non-work conversation with a coworker). Furthermore, participants were assigned to perform their instructed behavior (i.e., prosocial or social) either in person or online. Those in the control condition were asked to keep track of their activities on the following day.

Interestingly, all participants (including those in the control condition) reported increases in both positive affect and life satisfaction across the intervention period. Participants in the experimental groups, however, reported larger increases in feelings of social connectedness than did those in the control condition. In other words, both prosocial behaviors (e.g., helping a colleague with a computer problem) and social behaviors (e.g., chatting with a barista) led to relatively greater perceived social connection. Furthermore, increases in episode-level social connection among participants in the experimental conditions (i.e., feelings of connectedness while helping or chatting) were positively related to feelings of episode-level positive affect and overall feelings of social connectedness.

In sum, this experiment found no differences between social and prosocial behaviors in their effects on well-being or felt social connection. It is possible that our manipulations were not strong enough to detect differences between two types of positive behavior over the course of 4 weeks. For example, a participant may have chosen to offer kind words to a friend experiencing a hardship in response to instructions for *either* the social or prosocial condition, as this act could be construed as both social and prosocial. Alternatively, it could very well be that the well-being benefits previously attributed to prosocial behavior are due in large part to the inherently social nature of kindness. Although prosocial behavior is valuable and important to promote in its own right, sociability may be an equally efficacious target for future positive activity interventions in terms of its potential to boost momentary feelings of social connection and downstream well-being.

How Does Prosocial Behavior Differ from Other Positive Activities?

Thus far, we have largely focused on the relationship between sociability and subjective well-being—specifically, how social interactions make us feel happier and more connected to others. Many questions in this area, however, remain to be addressed. Why might sociable and prosocial behavior result in similar feelings of social connection and life satisfaction? Are acts of kindness unique or are they just another type of social interaction? To explore these questions, we shift our focus to eudaimonic well-being (or eudaimonia).

The concept of eudaimonia dates back to Aristotelian ethics and is roughly defined as the extent to which one is "living well" (Aristotle, 4th c. BCE/2002). Eudaimonia has been operationalized through a variety of constructs and measures (e.g., psychological flourishing, positive social relationships), and as both a cause and a type of well-being, in the psychological literature (Heintzelman, 2018; Sheldon, 2018). The eudaimonic activity model (EAM) offers conceptual clarity by situating eudaimonia as a class of activities that *predict* well-being rather than being a type of well-being itself (Martela & Sheldon, 2019; Sheldon, 2016, 2018). That is, rather than distinguishing between eudaimonic and subjective well-being, the EAM seeks to explain how eudaimonic actions and motives influence subjective well-being by disentangling well-*being* from well-*doing* (Sheldon, 2018).

The EAM posits that eudaimonic activities increase subjective well-being through psychological need-satisfying experiences. For example, volunteering (a eudaimonic activity) may satisfy one's psychological need for connection (a satisfying experience), which in turn promotes subjective well-being. Furthermore, specific eudaimonic motives and activities may evoke distinct proximal feelings (e.g., feeling a sense of meaning while playing with a child), which could have downstream consequences for the well-being benefits derived from these actions.

We tested this hypothesis in a recent experiment designed to compare the momentary eudaimonic feelings that arise from engaging in a variety of positive behaviors (Regan, Margolis, Ozer, Schwitzgebel, & Lyubomirsky, 2022). Specifically, we sought to understand the unique proximal feelings that arise when engaging in acts of kindness for others, given the robust relationship between prosocial behavior and well-being (see Curry et al., 2018, for a meta-analysis). To that end, we randomly assigned participants to one of four positively valenced conditions that varied in their inclusion of potential "active ingredients" of prosocial behavior. Engaging in kind acts for others was compared to engaging in kind acts for oneself (social element removed), extraverted behavior (kindness element removed), and open-minded behavior (both social and kindness elements removed). This research design allowed us to isolate participants' subjective experience of engaging in acts of kindness compared to other positive behaviors.

The participants included 671 Australian adults recruited from PureProfile, an online panel company. The study took place over the course of 15 days. At the end of the first day, participants were given their intervention instructions, and then were asked to list five specific ways in which they planned to incorporate their assigned behavior into their daily life over the next 2 weeks. In a series of twice-weekly assessments, participants were asked to report on their instructed activities completed since the last survey, and to recall how they felt while completing each of their reported activities. To ease participant burden and maximize retention, we relied on one-item measures of eudaimonic feelings in our weekly assessments—specifically, we asked participants to rate their energy level, sense of meaning, self-esteem, and psychological needs.

As hypothesized, participants randomly assigned to the prosocial behavior group reported stronger eudaimonic feelings over the course of the study. Specifically, those who engaged in prosocial behavior reported stronger feelings of competence, self-esteem, and meaning than those in all other conditions. Prosocial behavior—as well as extraverted behavior—also led participants to report feeling more socially connected relative to open-minded behavior and acts of kindness for themselves.

This pattern of results speaks to the unique experience of acting prosocially. Namely, engaging in kind acts toward others is more meaningful and self-bolstering than simply engaging in other types of positively valenced, socially desirable behaviors. Mirroring our previous findings, however, participants in this study felt similarly connected to others when engaging in prosocial behavior as they did when engaging in extraverted (or social) behavior. In other words, asking participants to do acts of kindness for others may be a similarly connecting experience as asking them to have more social interactions. In sum, recent evidence from our laboratory demonstrates that social connection can be fostered by instructing participants to act more extraverted, engage in more social interactions, or to behave prosocially. Furthermore, our results suggest that acting prosocially results in a unique type of connecting moment, which instills one's actions with a sense of meaning, and satisfies important psychological needs.

Future Directions and Further Questions

Can Sociability Be Sustainably Increased?

So far, we have reviewed experimental evidence supporting the relationship between sociable activities (i.e., acting extraverted, engaging in more social interactions, and doing acts of kindness for others) and well-being (see also Dunn & Lok; Prislin & Crano, this volume). Most of the experimental research reviewed in this chapter, however, asked participants to change their behavior

for a relatively short period of time (days to weeks). Future research could explore the longevity of the social and well-being benefits of these interventions by prolonging the interventions or including additional follow-up assessments. In addition to determining how long these effects last, future investigators could test methods to make the effects last longer. For example, our laboratory is in the process of testing whether mental contrasting with implementation intentions (MCII; Oettingen & Gollwitzer, 2010) increases the efficacy and durability of extraversion interventions. MCII is a well-established behavior change strategy that involves first contrasting a desired outcome (e.g., becoming more extraverted) with an obstacle (e.g., feeling too shy to approach others), which helps create an association between one's desired future and current reality, in turn facilitating goal pursuit (e.g., to be a more extraverted person). Numerous studies have shown that MCII is an effective strategy for lasting behavior change in a variety of domains, with effects lasting up to 2 years (Stadler, Oettingen, & Gollwitzer, 2010; Wittleder et al., 2019).

Pharmacological Approaches to Inducing Sociability

This chapter has focused on positive activity interventions as a method of stimulating social behavior in experimental research, but we recognize that this is not the only viable paradigm for inducing sociability. Another exciting approach involves pharmacological intervention—namely, through the use of substances like alcohol, amphetamines, and ±3,4-methylenedioxymethamphetamine (i.e., MDMA or ecstasy). MDMA in particular is a stimulant characterized by its ability to produce intense feelings of social connection, and has been the subject of careful experimental work in this area (Bershad, Miller, Baggott, & de Wit, 2016; Kamilar-Britt & Bedi, 2015). Indeed, a recent meta-analysis identified a moderate-to-large effect (d = 0.86; 95% CI [0.68, 1.04]; r = .39; 95% CI [.32, .46]) of MDMA on sociability-related outcomes in placebo-controlled experiments (Regan, Margolis, de Wit, & Lyubomirsky, 2021). Given its documented ability to stimulate sociability and connectedness, MDMA could allow psychologists to observe and manipulate facets of social interactions (e.g., feeling understood, openness, warmth, trust) that are challenging to induce using behavioral paradigms alone (cf. Lyubomirsky, in press). Furthermore, although the effects of the drug itself are short-lived, MDMA appears to produce lasting, transformative effects in clinical trials for individuals with mental health disorders such as post-traumatic stress disorder (PTSD) and social anxiety when used under the supervision of a trained mental health professional (Danforth et al., 2018; Mithoefer et al., 2019). The results of such trials suggest that MDMA can potentially amplify the effects of sociability interventions to produce durable effects on social behavior and felt social connection.

Tailoring Sociability Interventions

Another important future direction is understanding how sociability interventions might be targeted to be maximally effective, how they might scale, and under what conditions they might backfire. In light of the theorized importance of person-activity fit (Lyubomirsky & Layous, 2013), it follows that some individuals are likely to benefit more than others from sociability-based interventions. One avenue for future research could be to explore whether some individuals are more likely to benefit from the specific mechanisms targeted by particular sociability-based interventions. For example, both acting more generously and acting more extraverted have been shown to increase momentary feelings of connection. However, prosocial behavior may be a better "fit" for some individuals, if they are particularly likely to experience or to benefit from increases in sense of meaning, competence, and self-esteem (Fritz et al., 2022; Regan et al., 2022). Indeed, comparing sociability-based positive activity interventions in well-controlled, replicable experiments may help researchers elucidate key mechanisms underlying the relationship between social interactions and well-being.

In addition to tailoring interventions to specific individuals, future investigators might also consider targeting common barriers to engaging in social interactions. For example, across seven studies, Sandstrom and Boothby (2021) identified several obstacles to engaging in conversations with strangers, including fears that one would not enjoy the conversation, be disliked, or show poor conversational skills. To minimize such fears and encourage more conversations, the authors suggest that future interventions focus on changing participants' beliefs about their conversation partner (e.g., that their partner will take pleasure in talking to them).

Another promising research direction is to examine whether, how, and why inducing sociability might affect introverts and extraverts differently. Although we did not detect differences between introverts and extraverts in our extraversion experiment, a study using a similar extraversion manipulation and timeline found that introverts did not benefit as much as extraverts, and experienced some iatrogenic effects, including increased negative affect and tiredness (Jacques-Hamilton et al., 2019). Future research is needed to investigate the potential costs as well as benefits to increasing sociable behavior, and whether interventions can be specifically tailored to those who might need them and benefit from them most.

Concluding Thoughts

Evidence from well-being science suggests that sociability is a promising target for future interventions. By engaging in more extraverted or prosocial behavior, individuals may experience greater social connection and overall well-being.

Encouraging prosocial behavior may be an especially powerful approach to future interventions, as such behavior provides an opportunity for more meaningful, connecting moments while benefitting another person. To maximize the potential benefits of these interventions, however, more research is needed to understand their optimal population, format, and duration. Accordingly, such work may not only reveal ways to improve mental health and well-being, but also highlight the key mechanisms by which sociability benefits the human experience.

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6

CAN SOCIABILITY BE INCREASED?

Elizabeth W. Dunn and Iris Lok

Abstract: Social interactions are critically important for emotional and physical well-being. Not surprisingly, quality time with loved ones is linked to greater well-being. But even seemingly trivial social interactions—with the barista at Starbucks or fellow corgi owners at the dog park—can contribute to people's feelings of happiness and social connection. And yet, people often forego or even studiously avoid opportunities to engage with people they do not already know. Why? In this chapter, we identify the intrapersonal and interpersonal biases that lead people to miss out on opportunities for interactions with others. We discuss both failed and successful attempts at overcoming these biases. Finally, drawing on Latané and Darley's (1970) classic work on bystander intervention, we propose a broader framework (the Stranger Engagement Model) for understanding the contextual factors that shape whether people engage with strangers in everyday life. Taken together, the research reviewed in this chapter provides a road map for promoting connection in a world where loneliness is increasingly viewed as an epidemic.

Imagine an island filled with delicious, nutritious, and freely available food: plump mangoes and juicy apples growing on every tree, waterfalls of creamy milk, and mountains of freshly baked bread. Now imagine that many of the humans on the island appear to be starving. Despite seeing the abundant food around them, they seem to go out of their way to ignore it. We propose that modern society is not unlike this island—but the nutrient that people achingly need yet willfully ignore is social rather than edible. Human beings have a fundamental need to belong, and failing to satisfy this need can lead to dire consequences for mental and even physical well-being

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(Baumeister & Leary, 1995; see also Baumeister & Maranges, this volume). The most obvious way to satisfy this need is to interact with close others, but the research we review in this chapter suggests that even interacting with strangers may contribute to filling this need. And yet, people often eschew readily available opportunities to connect with the strangers all around them—even as society reckons with what some scholars have termed a "loneliness epidemic" (e.g., Jeste, Lee, & Cacioppo, 2020). In the pages that follow, we discuss why people avoid engaging with strangers and how this knowledge can be used to enhance this abundantly available form of sociability.

The Benefits of Talking to Strangers

Although social psychologists have traditionally focused on close relationships (e.g., Diener & Seligman, 2002; Lyubomirsky, King, & Diener, 2005; Myers, 2000; see also Reis et al.; and Mikulincer & Shaver, this volume), recent research suggests that even seemingly trivial social interactions with strangers and acquaintances can contribute to people's feelings of happiness (e.g., Dunn, Biesanz, Human, & Finn, 2007; Epley & Schroeder, 2014; Gunaydin, Oztekin, Karabulut, & Salman-Engin, 2021; Sandstrom & Dunn, 2014a, 2014b). For example, in one field experiment, Sandstrom and Dunn (2014a) recruited customers on their way into Starbucks and randomly assigned them to have a brief social interaction with the barista or to be efficient by minimizing conversation. By the time they left Starbucks, customers who had simply taken a moment to chat with the barista felt happier and more socially connected. In another study conducted in Turkey, Gunaydin et al. (2021) asked commuters waiting in line for buses to either exchange greetings with the driver or avoid conversation when boarding the bus. Although these greetings took only a few seconds, participants who greeted the driver left the bus in a better mood compared to those who avoided conversation.

Several experiments suggest that even introverts benefit from being more sociable with strangers (McNiel & Fleeson, 2006; McNiel, Lowman, & Fleeson, 2010; Zelenski, Santoro, & Whelan, 2012). In two studies, Zelenski et al. (2012) put strangers into groups of three and asked them to complete a 20-minute discussion task. In each group, one participant was assigned to act extraverted and another participant was assigned to act introverted; the remaining participant did not receive any special instructions. By the end of the discussion, participants who were told to act extraverted were significantly happier—regardless of whether they were an extravert or introvert. Although this study was relatively small and only examined one brief social interaction, more recent studies using larger sample sizes have found that most people benefit from acting more outgoing over a period of one to two weeks, with the possible exception of extreme

introverts (Jacques-Hamilton, Sun, & Smillie, 2019; Margolis & Lyubomirsky, 2020; see also Regan & Lyubomirsky, this volume).

Taken together, a growing body of research suggests that being more sociable with the strangers around us may confer benefits for well-being. However, we routinely ignore the strangers in our midst. As Stanley Milgram and John Sabini noted 40 years ago, there is a strong social norm that strangers thrust into close proximity in public spaces should avoid interacting with each other (Milgram & Sabini, 1978). Consistent with this classic work, the vast majority of Mturk participants say they would avoid talking to a stranger in a waiting room (93%), on a train (76%), or on a plane (68%; Epley & Schroeder, 2014). This paradox poses a puzzle: Why do human beings studiously avoid interactions that would make them feel happier and more socially connected?

Why Do People Avoid Talking to Strangers?

Underestimating Emotional Benefits

People may eschew interactions with strangers because they fail to recognize the benefits such interactions would provide. Offering the first evidence for such an affective forecasting error, Dunn, Biesanz, Human, and Finn (2007) found that students underestimated how happy they would feel after interacting with a stranger in the lab. Since then, similar effects have been observed with more diverse samples in real-world settings (Epley & Schroeder, 2014; Sandstrom & Dunn, 2014a). For example, in a series of field experiments, Epley and Schroeder (2014) instructed Chicago commuters to interact with a stranger sitting next to them on public transit or to keep to themselves. Participants felt significantly happier when they interacted with a stranger compared to sitting in solitude, but other participants who were asked to imagine being in the experiment predicted precisely the opposite.

If our tendency to avoid conversation primarily stems from an affective forecasting error, then being prompted to talk to a stranger on one occasion should help people recognize the emotional benefits of sociability, making them more likely to initiate interactions with strangers in the future. To explore this possibility, Schroeder, Lyons, and Epley (2021) randomly assigned commuters in London to talk to a stranger, sit alone with their thoughts, or follow their normal routine during their train ride. Participants who were assigned to talk to a stranger reported that they would be more likely to talk to someone on their commute the following day, compared to those in the other two conditions. Of course, intentions do not always turn into actual behavior (for a review, see Sheeran & Webb, 2016), so it is unclear whether a single pleasant interaction with a stranger leads to longer-term changes in sociability. It is also worth noting that research assistants approached participants individually and asked them to change their commuting behavior, a strategy that does not lend itself to a larger scale intervention.

To address this scalability challenge, Sandstrom, Boothby, and Cooney (2021) designed a scavenger hunt app that was widely advertised in the media. Over the course of one week, the app prompted users to complete a series of missions. Half the users completed missions that involved interacting with strangers, while the other half completed missions that did not involve any interactions (e.g., observing a stranger from a distance). After completing missions that involved interacting with strangers, participants exhibited more positive expectations about these casual social interactions. A week after the game ended, these participants also reported initiating more interactions with strangers than they did before the study, although the effect was only marginal.

Underestimating One's Own Social Competence

As well as underestimating the emotional benefits of interacting with strangers, people may avoid interacting with strangers because they underestimate their own social competence. In one test of this idea, Schroeder et al. (2021) asked London commuters how they would feel about engaging in a conversation with a stranger in two different ways. Half of the commuters were asked to imagine how they would feel if they *tried* to start a conversation with a fellow commuter during their train ride. The other half were asked to imagine how they would feel after *successfully* starting a conversation. Participants expected to have a more positive commute when they imagined succeeding at having a conversation with a stranger than simply trying to do so, suggesting that people worry that they may struggle to initiate a conversation. More broadly, across five studies conducted in the field and lab, Sandstrom and Boothby (2020) found that people consistently underestimated how successfully they would be able to start and maintain a conversation.

If people avoid talking to strangers in part because they underestimate their own social skills, then providing them with tips on starting a conversation should make them more willing to approach strangers. In one set of studies, before interacting with a stranger, participants received tips on starting a conversation (e.g., talk about topics that the participant might have in common with the stranger, ask a question; Sandstrom & Boothby, 2021). Compared to participants who did not receive any tips, participants who received conversation tips expected to have a more pleasant experience talking to a stranger. However, participants who received tips were no more likely to start a conversation when given the chance to do so, suggesting that this strategy may have limited efficacy in promoting conversations between strangers.

Misreading Social Signals

Of course, people may consider not only whether they want to interact with a stranger, but also whether a stranger wants to interact with them. Making the latter judgment requires reading other people's social signals, and recent research suggests that people tend to systematically misread these signals (Epley & Schroeder,

2014). For example, commuters on Chicago public transit reported that they would be more interested in talking to their fellow passengers than their fellow passengers were in talking to them (Epley & Schroeder, 2014). This finding suggests that people may keep to themselves because they assume that others are not interested in chatting, leading everyone to exhibit greater reticence than they would prefer (a form of "pluralistic ignorance"; Miller & McFarland, 1991).

If an interpersonal misunderstanding plays a driving role in preventing social interactions between strangers, then enabling people to explicitly and accurately signal whether they are open to interacting should facilitate social interactions. Thus, we attempted to correct this interpersonal error by enabling people to use explicit social signals to convey their true level of interest in interacting. We recruited participants (N = 531) who were waiting at a major bus stop located on the UBC campus during daylight commuting times (Lok & Dunn, 2021a). We approached people who were waiting in line for the same bus, randomly assigning them all to the same condition. All participants were asked to wear a lanyard that included a card with a link to an online survey, which participants were instructed to complete after getting off the bus. In the control condition, all participants were asked to wear a black lanyard. In the social signaling condition, we explained that participants would be asked to wear a green lanyard if they were willing to have a brief conversation with someone new on the bus, and to wear a red lanyard if they preferred to keep to themselves. Most participants (68%) chose the red lanyard, indicating a preference to keep to themselves, while approximately one-third of the sample chose the green lanyard, indicating a willingness to form a new connection. Perhaps because most people used the explicit social signals to indicate a desire *not* to interact, participants in the signaling condition were no more likely to interact with a stranger than those in the control condition. On a typical bus, nine passengers wore lanyards, but given the size of the bus, participants reported being able to see one to three fellow passengers wearing lanyards. Thus, the efficacy of this intervention may have been limited because passengers who selected a green lanyard could not necessarily connect with their fellow "greenies."

This study provides some clues about the conditions that must be met for signaling interventions to have a chance of working. In particular, being able to signal a preference to make a new connection can only increase social interactions if other people express the same preference and are in close enough proximity with each other. There is also the possibility that a signaling intervention might backfire—reducing the likelihood of social interactions—if a majority of people indicate they do not want to interact with anyone. This suggests that social signaling inventions might work best in situations where many people are willing to interact and are surrounded by others who share the same preference.

We examined a situation that met these conditions by inviting almost 300 students to eat lunch at a popular food court on campus (Lok & Dunn, 2021a).

Four to six students were scheduled for each session, and all participants in each session were assigned to the same condition (social signaling versus control). At the outset, participants in the social signaling condition selected a green or red wristband to indicate whether or not they were interested in interacting; participants in the control condition received white wristbands. Then, participants were invited to spend 20 minutes eating their lunch at a designated long table within the café along with the other people in their session. Consistent with our assumption that most people would be willing to interact with other students in the food court, a slight majority (55%) indicated that they would like to have a conversation with a new person and make a new connection (versus keep to themselves). Students in the social signaling condition reported spending more time talking to other participants compared to those in the control condition, suggesting that our intervention was effective in altering behavior. As expected, additional analyses revealed that our effects were driven by those who selected a green wristband.

By asking people to select a wristband, we not only enabled them to convey their preferences about social interactions to others—but we also effectively reminded them that having a social interaction was an option. Thus, it is possible that simply being reminded that interacting with others is an option could underlie the benefits of our social signaling manipulation. To test this possibility, we replicated the lunch study with another 788 participants. We used the same instructions as before in our social signaling condition, but in the control condition, we explicitly reminded people that they should feel free to have a conversation with a new person and make a new connection or just keep to themselves. As in our original study, a slight majority of participants in our signaling conditions (52%) chose a wristband indicating that they wanted to interact with others. Yet, we found no significant differences between conditions in how much time people spent interacting over lunch. Comparing our results across studies, we observed that participants in the control group were more inclined to interact in our replication study (versus our original study), suggesting that simply providing a reminder that they *could* interact was sufficient.

Although we found limited evidence for the value of social signaling in our studies, a signaling intervention might still be useful in situations where (1) there is a large gap between perceived and actual interest in connecting and (2) those who are interested in connecting can easily do so. When we brought an additional 158 participants to the UBC food court and asked them to predict what percentage of people would choose the red versus green wristbands, their predictions were largely accurate. Thus, rather than assuming that people consistently underestimate how interested others are in interacting within a specific social context, it would be worthwhile to test whether people prefer to connect (versus keep to themselves), and what they think strangers would prefer within that social environment.

Integrative Summary

For the most part, the barriers that prevent us from connecting with strangers seem to live inside our heads. Exhibiting an intrapersonal error, we may underestimate the emotional benefits of talking to strangers and our own social skills. Exhibiting an interpersonal error, we may misread others' social signals, underestimating their interest in making new connections. Surprisingly, very few studies have directly tested whether these biases play a causal role in shaping our social behavior, and the studies that have begun to address this question have not produced very promising results. Indeed, trying to knock out these biases may be akin to a game of Whack-A-Mole; for example, eliminating intrapersonal biases may not help very much if interpersonal biases pop up and prevent people from engaging with strangers. More broadly, we would propose that changing people's minds about strangers may not be enough to change social behavior. In fact, avoiding strangers might be an evolutionary adaptation, given that interacting with strangers (versus ingroup members) would likely have posed a potential risk in our ancestral environment, perhaps making this suboptimal tendency more difficult to overcome even in our modern environment (see von Hippel & Smith, this volume).

Beyond Misperceptions: Other Ways to Promote Sociability

Social psychologists (including us) have devoted a great deal of effort to understanding individuals' cognitive biases and attempting to correct these biases in order to change social behavior for the better. But a more powerful approach may lie in accepting people's mindsets (however mistaken they might be) and altering situational features instead. For example, in an ultimately fruitless line of research, our lab attempted to convince wealthy people that their success might be due in large part to good luck and help from others (Whillans, Dunn, & Caruso, 2017). We hoped that this approach would push wealthy people away from their typical view of success as stemming purely from their own talent and hard work (an agentic mindset), toward a more communal view of success—which should theoretically promote a greater willingness to donate money to charity. But with every manipulation we designed, wealthy people clung more forcefully to their original agentic mindset, seeing themselves as the sole proprietors of their own success. So, we gave up on changing their minds, and instead changed the messages they saw about charitable giving. Indeed, when we re-vamped a charity's website by presenting charitable giving as an opportunity for individual achievement, wealthy people were substantially more likely to click "Donate Today" (Whillans, Caruso, & Dunn, 2017).

Drawing on this lesson, we propose that changing people's minds may be less effective than modern cognitively oriented psychologists often assume.

Instead, a more productive approach may lie in changing features of the situation. This situationist approach has deep roots in social psychology, of course, and classic work in the field emphasized how subtle contextual features could powerfully influence behavior in real-world settings (e.g., Asch, 1955; Latané & Darley, 1968; Milgram, 1963). In particular, Latané and Darley (1970) rejected the popular assumption that tragic cases of bystander apathy could be explained by psychological shortcomings of the bystanders themselves. Their Bystander Intervention Model outlines the series of decisions bystanders must make before helping a stranger, providing a framework for understanding the situational features that can shape such prosocial behavior. According to this model, bystanders must (1) notice a stranger in need, (2) interpret the situation as an emergency, (3) assume personal responsibility for helping, (4) decide how to help, and (5) initiate action. We believe that this classic model of helping strangers in emergencies can be productively extended to the more mundane prosocial action of engaging with strangers through friendly greetings and conversations in everyday life. We propose that in order to engage with a stranger, an individual must (1) notice the stranger, (2) interpret the situation as a social opportunity, (3) assume personal responsibility for engaging, (4) decide how to engage, and (5) initiate the interaction.

A strength of the Bystander Intervention Model is that it helps to explain why people may often ignore strangers (even strangers in desperate need of help). The model posits that people have to make it through *all five* steps of the decision tree in order to provide help. As Latané and Darley (1970) put it, "If an individual is to intervene in an emergency, he must make not just one, but a series of decisions. Only one particular set of choices will lead him to take action in the situation" (p. 31). This insight may help to explain why recent interventions intended to promote conversations between strangers have produced disappointing results: intervening at any one branch of the decision tree may make little difference if we fail to consider how engagement with strangers might be impeded at the other branches of the decision tree. Thus, in the remaining sections of this chapter, we extend the Bystander Intervention Model to develop a broad framework for understanding when and why strangers choose to engage with one another in everyday life (see Figure 6.1). This Stranger Engagement Model can potentially be applied to identify the conjunctive set of conditions that may be required in order to promote social interactions between strangers in specific settings.

Notice the Stranger

According to the Bystander Intervention Model, noticing a stranger in need of help represents the critical first step to engaging with the stranger. As Latané and Darley (1970) put it, "First, the bystander has to notice that something is happening. . . . He must stop concentrating on his private thoughts or the legs

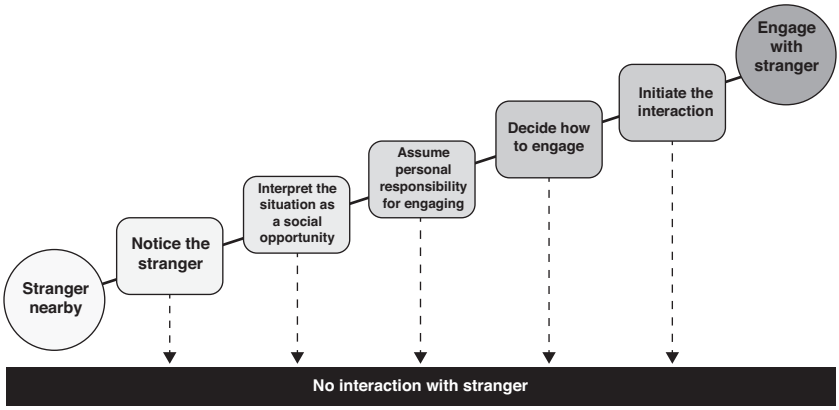

FIGURE 6.1 Stranger engagement model. A framework for promoting social interactions between strangers.

Source: Adapted from Latané and Darley's (1970) Bystander Intervention Model.

of the pretty girl walking down the street ahead of him and pay attention" (p. 31). Adopting this model, we would argue that one must first notice the opportunity to engage with a stranger in order to have any chance of chatting. The layout of physical spaces may play an important role in shaping whether people notice one another (for a review, see Hall, 1990). For example, residents who live close to a stairwell have more social interactions with their neighbors (Festinger, Schachter, & Back, 1963). Undergraduates living in dorms with communal bathrooms report exchanging more frequent greetings with their fellow residents compared to students living in dorms with private en suite bathrooms (Lok & Dunn, 2021b), and students living in buildings that include shared communal areas (e.g., lounges) are more likely to bump into other residents (Easterbrook and Vignoles, 2015). People are also more likely to know their neighbors in walkable (rather than car-dependent) neighborhoods (Leyden, 2003). Together, these findings suggest that physical spaces that lead people to cross each other's paths more often may promote casual social interactions by making people more likely to notice one another.

We would speculate that people may also be more likely to notice strangers who are accompanied by pets. McNicholas and Collis (2000) instructed a research assistant to go about her daily routine, with or without a dog. They found that almost all of her interactions with strangers (96%) occurred when she was accompanied by the dog. This finding was particularly striking because the dog had been explicitly trained to be as inconspicuous as possible and not to solicit attention from others. More broadly, people may be more inclined to chat with strangers who are accompanied by any positive, attention-grabbing stimulus, from a child to a unicycle.

The likelihood of our attention landing on the strangers in our midst may have diminished as a result of recent technological advances, particularly the development of smartphones, which enable a kind of "absent presence" (Gergen, 2002). Although smartphones can be a fruitful source of information and entertainment, phone use has been shown to undermine the likelihood that we will engage with others in our immediate social environment (Kushlev, Dwyer, & Dunn, 2019). For example, in one experiment, over 160 undergraduates were asked to sit in a waiting room along with a stranger while waiting for an experiment to begin (Kushlev, Hunter, Proulx, Pressman, & Dunn, 2019). In half the sessions, both participants were allowed to take their phones into the waiting room, while in the remaining sessions, phones were not permitted in the waiting room. Without access to their phones, only six students reported ignoring the other person in the waiting room. But this number more than quintupled when people had their smartphone: fully 32 people never engaged at all with the fellow student sitting in the same small room. This finding suggests that simply asking people to put away their phones in classrooms, conferences, and other settings might dramatically increase the likelihood that people will pay attention to the strangers around them.

Interpret the Situation as a Social Opportunity

According to the Bystander Intervention Model, noticing a stranger in need is not enough; one must also interpret the situation as an emergency. Likewise, even if we notice that there is a stranger nearby with whom we *could* engage, we must interpret the situation as an opportunity for social interaction. This decision should depend in part on our evaluation of whether the stranger is interested in talking to us. For example, consider again the students sitting together in a laboratory waiting room either with or without their phones (Kushlev et al., 2019). When both students had their eyes fixed on their own phones, they may have been less likely to take notice of each other at all, but they may also have been unlikely to assume that the other person was interested in talking to them. Indeed, Epley and Schroeder's studies on Chicago public transit point to the conclusion that people often underestimate how willing nearby strangers would be to chat.

If this is the case, then simply reminding people that it is acceptable to talk to strangers in a particular context may be sufficient for increasing sociability. For example, a coffee shop in Vancouver prominently displayed a poster saying, "Studies show that even 'trivial' interactions between strangers will have a positive impact on mental health. Try it out. Grab a seat next to someone you don't know—yet." Presenting this sort of injunctive norm—that talking to strangers is acceptable in a specific context—might be a simple, scalable, and effective strategy for promoting interactions between strangers by making it more likely that people will perceive the situation as offering an opportunity

for social interaction. This type of intervention might fit best in contexts such as a small neighborhood coffee shop, where people have some sense of shared identity. In addition, based on past research showing that injunctive norms are more powerful when supported by consistent descriptive norms (e.g., Cialdini, 2003), we would assume that injunctive messages about talking to strangers should be more effective in situations where talking to strangers is reasonably common.

What contextual factors may give rise to this type of descriptive norm? We would theorize that people may be more inclined to engage with "familiar strangers," individuals who they do not know, but have encountered previously. In a classic study, Moreland and Beach (1992) hired female confederates to stroll into a crowded lecture hall between 0 and 15 times over the course of the semester. By the end of the term, most students did not consciously recognize any of the women, but they were far more interested in connecting with the women who had entered their lecture hall more often. Along with other research on the mere exposure effect (e.g., Zajonc, 1968), this study points to the conclusion that strangers may be more inclined to interact with one another when their paths have crossed previously.

Assuming Responsibility

According to the Bystander Intervention Model, a person who notices an emergency must also assume personal responsibility for intervening. As Latané and Darley (1970) put it, "Even in a true emergency, it is not clear that everybody present should immediately intrude himself into the situation" (p. 32). In a similar vein, a person must decide to take the initiative in starting a conversation—a course of action that may be less normative than intervening to provide help in an emergency. Social psychologists have largely bypassed this problem by telling their participants to make the first move. In their experiments on Chicago public transit, for example, Epley and Schroeder (2014) instructed participants to start a conversation with someone new during their commute. When participants instigated a conversation, they were never rebuffed, suggesting that many people may be open to engaging in conversation—but a willingness to take the lead might be much rarer.

If strangers forego opportunities to talk because no one assumes responsibility for reaching out, then an intuitive solution is to appoint someone to be in charge of fostering connections. Although this strategy has not been tested empirically, many universities hire residence advisors to organize social events with the goal of helping students form new social connections. Building developers have also begun to apply this strategy outside the university context. In Vancouver, for example, residents of some buildings are appointed to serve as "social animators," hosting events—from Easter egg hunts to board game days to annual barbeques—to bring their neighbors together (Happy Cities, n.d.).

Drawing from these examples, we would speculate that people will engage more frequently with strangers when a particular individual or group is explicitly charged with initiating these interactions.

Knowing What to Do

In emergency situations, an important barrier to providing help is that people may feel they do not know exactly *how* to intervene (Latané & Darley, 1970). By comparison, chatting with a stranger might seem relatively easy, but recent research suggests that not knowing how to connect with a stranger poses a substantial psychological barrier. People doubt their own ability to start and maintain conversations, pointing to the potential value of increasing people's perceptions of their own social competence (e.g., Sandstrom & Boothby, 2021). Rather than attempting to alter people's self-views, however, a more effective strategy might lie in considering the situational context. We propose that the likelihood that people will connect with strangers in a particular context may depend heavily on whether the situation offers a clear social script. By providing a shared cultural understanding of what people are expected to do and say in a particular type of situation (e.g., Schlenker & Weigold, 1992), social scripts can greatly simplify the challenge of knowing how to engage with strangers. For example, people may be inclined to chat with strangers in dog parks because this situation affords an easy and familiar script for initiating conversations (e.g., "Is that a goldendoodle? What's her name?"). In contrast, there is no clear, shared cultural script for striking up a conversation with a stranger on public transit, perhaps explaining why research participants fear that their attempts to start such a conversation will fail (Epley & Schroeder, 2014; Schroeder et al., 2021). Indeed, when Epley and Schroeder (2014) extended their research to airport taxis, they found that people did not predict that chatting with a driver would be less pleasant than sitting in solitude, perhaps in part because these taxi rides offer a familiar script that includes at least some brief engagement between driver and passenger (e.g., "Where you headed today? Where are you coming in from?").

Given that shared cultural scripts make it easier to interact with strangers, we would propose that it may be possible for people to increase their day-to-day sociability by spending more time in settings that come preloaded with such scripts. For example, an older adult with few close social connections might benefit more from spending an hour per day at a dog park than from trying to screw up her courage to engage with strangers in more unscripted settings.

Initiating Action

Having noticed a problem, identified it as an emergency, decided to take charge, and figured out what to do, all that remains is to take action—but the initiation of action may be easily de-railed by practical considerations, such as safety

concerns. Correlational research has found that people who perceive very high (versus moderate) levels of crime in their neighborhood are much less likely to interact with their neighbors (Baba & Austin, 1991). On the basis of this finding, we would argue that an important part of increasing sociability lies in providing safe spaces for strangers to engage with each other. Leaving aside physical safety, people may also be concerned about the risk of being caught in a seemingly endless conversation with a dull stranger, pointing to the potential value of creating clear opportunities to cut off the conversation.

Of course, the decision to take action also rests on whether there is any benefit to be gained from talking to a stranger. In a new and unfamiliar environment, strangers can often be a valuable source of information. In one experiment, for example, Kushlev, Proulx, and Dunn (2017) found that students were extremely likely to talk to strangers on campus when they had to find an unfamiliar building—and were not allowed to rely on their smartphones to do so. Thus, creating situations in which strangers need to rely on each other for information may substantially increase the likelihood of engagement.

Integrative Summary

Following in the tradition of the classic Bystander Intervention Model, our Stranger Engagement Model is intended to provide an organizing framework for understanding the complex, multi-faceted judgments that underlie the seemingly simple decision to interact with a new person. We do not assume that each step in the model is fully independent of the others; for example, someone who is in charge of initiating interactions (e.g., a social animator) might be more likely to notice strangers and construe their presence as a social opportunity. We also do not assume that people consciously proceed through this sequential chain of decisions in a conscious or deliberative fashion. Instead of documenting the specific mental processes that any one individual exhibits, this model attempts to provide a comprehensive framework for understanding how to promote sociability within specific settings.

For example, consider a Bay area tech company that wants to rebuild its campus to promote social connections among its 5,000 employees. The company could increase the likelihood that co-workers will notice one another by, for example, replacing two large office towers with smaller buildings connected by winding outdoor paths that employees traverse in order to attend meetings or get coffee. To help employees construe encounters with unfamiliar co-workers as opportunities for social interaction, the company could build a small cafe or courtyard devoted to making new connections. To help people assume responsibility for engaging with strangers, each small building could appoint a social animator, charged with helping workers in each building form connections within and beyond their own building. To make it easier for people to know how to start conversations, employee badges could display a conversation starter,

such as "Ask me about . . . surfing." Finally, employees should be more likely to take advantage of these opportunities for social interactions if the company emphasizes the value of such interactions by, for example, sharing research showing that more socially connected workers are more productive (Olguín et al., 2008). Importantly, any one of these interventions alone might not change employees' sociability much, if at all. But by taking a comprehensive approach that incorporates each aspect of the Stranger Engagement Model, the company could potentially build a much more connected workforce.

Conclusion

Human beings are a highly sociable species, and yet we frequently forego opportunities to socialize with new people in daily life. Psychologists have uncovered a constellation of cognitive biases that might help to explain the tendency to avoid casual interactions with strangers. Individuals are prone to underestimate the emotional benefits of talking to strangers, as well as their own conversational competence (intrapersonal biases). Perhaps more importantly, individuals may also underestimate others' interest in talking to them (an interpersonal bias). On the basis of the assumption that changing people's minds will lead to changes in their behavior, researchers have tried to correct these cognitive biases, but these attempts have produced, at best, small effects on sociability.

The limited success of such empirically grounded interventions points to at least two possible conclusions. First, the cognitive biases that researchers initially identified might be less ubiquitous than initially assumed; for example, although Chicago commuters appeared to underestimate other people's willingness to socialize (Epley & Schroeder, 2014), we did not observe the same interpersonal bias among Canadian students at a university food court (Lok & Dunn, 2021a). Second, biased beliefs about interacting with strangers may exist *alongside* a behavioral tendency to avoid strangers, but we cannot assume that these beliefs *cause* this reticent behavior. In fact, it is entirely possible that people alter their beliefs in order to be consistent with their own behavior (e.g., Festinger & Carlsmith, 1959). To establish that affective forecasting errors and other biases play a causal role in shaping behavior, it is essential to alter these biases and then observe whether people change their social behavior accordingly.

We believe that such attempts have so far produced lackluster results, due in part to the complexity of everyday social decision-making. Harnessing classic research on the Bystander Intervention Model, we have argued that people must make a series of rapid, interrelated decisions in order to engage with a stranger. Thus, engaging with a stranger depends on a series of conjunctive events, and classic research shows that such conjunctive events are less likely to occur than people typically assume (Tversky & Kahneman, 1974). As our Stranger Engagement Model highlights, an intervention that perfectly overcomes a barrier at

one stage of the decision-making process (such as noticing a stranger) may have no detectable effect at all on downstream social behavior if a substantial barrier exists at another stage (such as deciding how to engage).

Compared to studying specific cognitive biases, adopting a broader, more holistic framework such as our Stranger Engagement Model is inherently "messier." For example, the same variable (presence of smartphones) can potentially influence decision-making at multiple stages of the model, and it may be difficult or impossible to isolate the effects at any one stage. By acknowledging the genuine complexity of human social behavior, however, we can potentially develop what Berkman and Wilson (2021) termed *practical theories*. According to Berkman and Wilson (2021), a practical theory must go beyond identifying a single mechanism, applying a broader lens in order to suggest "actionable steps toward solving a problem that currently exists in a particular context in the real world." We believe that our Stranger Engagement Model represents a practical theory that can be used to increase sociability in diverse real-world settings—a pressing goal at a time when loneliness has been identified as a pressing threat to public health (e.g., Jeste et al., 2020).

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7

HOW DOES THE QUEST FOR SIGNIFICANCE SHAPE HUMAN SOCIABILITY?

Arie W. Kruglanski and Molly Ellenberg

Abstract: The existential human need for recognition and significance has long been a topic of focus in social philosophy and psychology, and is one of the main motivations driving different patterns of human sociability, although the research programs and conceptual paradigms through which it has been addressed have rarely intersected. In this chapter, we connect the dots to outline our integrative significance quest theory (SQT) as a key component for understanding human sociability. We demonstrate that the SQT is consistent with the past literature in a number of different academic realms and is also supported by novel studies carried out in multiple settings. Finally, we discuss how the SQT contributions enhance and expand upon prior knowledge and propose directions for further study.

"Oh Speak My Name to the Memory of Men!"

Dante: Inferno, Canto VI

The quest for significance is the desire to matter, to be respected, and to have social worth. Human sociability would be unrecognizable without it. Over the course of human evolution, significance "questors" likely contributed to the welfare of the group gaining status and dignity in return (Willer, 2009). The latter conferred a selective advantage: high-status individuals were sought after; hence they were more likely to survive and procreate (see also von Hippel & Smith, this volume). Natural selection on the need for significance explains its prevalence in humans. Indeed, the *striving for recognition*, has been regarded as a "vital human need" by philosophers (Taylor, 1992, p. 26) and social scientists over the centuries.

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The *struggle for recognition* has been addressed by the likes of Aristotle, Mead, Fichte, Hegel, Marx, and Sartre. Hegel (1807/2018), in particular, famously distinguished between three forms of recognition: 1. The love and care that the infant receives from its adult caretakers, which promotes the "development and maintenance of the 'basic self-confidence' or 'trust in oneself'" (see also Mikulincer & Shaver, this volume); 2. "the universal dignity of persons" as accorded by the *legal system,* which grants *rights* to citizens in acknowledgment of their membership in the community; and 3. one's unique worth as an individual recognized for her or his *special* talents, attributes, and achievements (Anderson, 1995, pp. xiii–xvi).

According to Honneth (e.g., 1995, p. 160), the struggle for recognition and respect has driven major social movements that shaped the course of history, including the French, American, and Russian revolutions. It is not difficult to draw parallels to more recent movements (e.g., the Black Lives mater movement) that have focused on rectifying injustices and inequalities that curtailed individuals' and communities' sense of significance. However, we might want to draw a distinction between the individual's quest for personal/individual significance, and identitarian group movements like BLM that are fundamentally collectivist and seek recognition for defined identity.

Though philosophers have long viewed the quest for significance as a key facet of the human nature, they have not delved deeply into its workings, antecedents, and consequences. Psychology is perhaps better suited to examine these factors. Indeed, concern with people's self-worth has permeated numerous topics in social and personality psychology, and is a key issue in trying to understand varying patterns of human sociability. It appears in numerous research programs under different labels such as self-enhancement, self-protection, self-affirmation (Steele, 1998; Sherman & Cohen, 2006), self-esteem (Rosenberg, 1965), achievement motivation (McClelland, Atkinson, Clark, & Lowell, 1953; McClelland, 1985), the need to belong (Baumeister, & Leary, 1995; see also Baumeister & Maranger, this volume), social identity (Hogg, 1992, 1993; Tajfel & Turner, 1979; see also Gaffney & Hogg, this volume), normative influence (Deutsch & Gerard, 1955; see also Fiske, this volume), exclusion and inclusion (Williams, 2007), relative deprivation (Gurr, 1970; Pettigrew, 2001; Stouffer et al., 1949), and minority influence (Moscovici, 1985; see also Prislin & Crano, this volume). Though these areas of study all touch on people's quest for significance, they have been largely developed independently of one another. Each addressed a somewhat different aspect of this phenomenon and their deep commonality has escaped largely unnoticed. In what follows, we aim to address this gap in knowledge. Building on past insights and findings we outline now our significance quest theory (SQT) that examines the sources of this motivation, its activation conditions and its personal and social consequences for understanding this key feature of human sociability.

The Psychology of Significance Quest

Defining the Need for Significance

As already noted, we view the quest for significance as the motivation to have social worth, that is, the desire to be valued and esteemed by a community to which one belongs. Typically, though not invariably one's social worth is tantamount to one's self-esteem. As writers like Hegel (1807), Cooley (1902), or Mead (1934) noted, one's self-views are typically derived from the values one shares with members of one's ingroup, that are part and parcel of one's shared reality (Higgins, 2019). Self-esteem and social worth diverge, however, where one's membership group differs from one's reference group (Merton & Kitt, 1950). For instance, an immigrant may have low social worth in the host country (membership group), yet high self-esteem based on the values of their culture of origin (reference group) that they may uphold and affirm.

Antecedents

Evolutionary Origins

Altruism and prosocial behavior are known to carry evolutionary benefits. For example, there is indirect reciprocity for prosocial behavior not only by the behavior's recipient but also by other members of the community who grant the prosocial person high status and good reputation. Reputation in turn often elicits support and protection (see von Hippel & Smith, this volume). It also requires continuation of one's pro-social conduct in order to maintain one's status and reputation (Alexander, 1987). Furthermore, the reproductive benefit of altruism toward one's family and friends has promoted an "ingroup" morality that favors close community members over strangers (Bloom, 2013; see Gaffney & Hogg, this volume). Status, as defined for males in some societies by wealth (Borgerhoff Mulder, 1987), hunting skill (Kaplan & Hill, 1985; Gurven & von Rueden, 2006; Smith, 2004), or warriorship (Chagnon, 1988), has been found to be positively related to fertility (e.g., Betzig, 1986). Congruent with the evolutionary benefits of status and social worth are findings that the quest for status or significance, as defined by *ambition*, is a highly inheritable trait (Wooton, Davis, & Haworth, 2016).

Socialization

Evolutionary psychology suggests why QFS is universally a part of human nature and a key driver of different patterns of sociability. As with any trait, however, there are individual differences in this motivation. In part, these are explained by different patterns of socialization (see also Mikulincer & Shaver, this volume).

Essentially, the magnitude of QFS depends on the *value* one attaches to significance and one's *expectancy* of attaining it. Both value and expectancy are typically instilled in children by their parents and other adult care takers (Eccles et al., 1993). With regard to value, Higgins' work on regulatory focus (e.g., Higgins & Silverman, 1998) suggests that parenting that rewards desirable activities and withdrawing love, rather than punishing in reaction to undesirable activities, contributes to the child's development of a *promotion focus*, which emphasizes ambition and hence, the quest for significance. The child thereby places a higher value on setting lofty goals and "reaching for the stars" (Grant-Halvorson, 2010; see also Mikulincer & Shaver, this volume). With regard to expectancy, Dweck's (2006) work on fixed versus growth mindsets has found that for children with a fixed mindset, the expectancy and subsequent fear of failure curtails goal pursuit. In contrast, children with a growth mindset are relatively undaunted by failure, allowing them to set and pursue higher goals. They are, therefore, likely to be more ambitious and more committed to the significance quest.

Culture

Culture, too, may determine members' significance seeking tendencies, by highlighting the value of significance or expectancy of attaining it. Tight (more so than loose) cultures emphasize promoting and maintaining their normative value system (Gelfand, 2019). Because social worth is governed by norms, members of tight cultures should exhibit stronger significance seeking concerns than members of loose cultures (see also Fiske, this volume). Similarly in authoritarian and group-centric cultures, to be respected by one's group is of the utmost importance, thereby intensifying the individual's QFS (Kruglanski, Pierro, Mannetti, & De Grada, 2006; Kruglanski et al., in press). In Nazi Germany, for example, the Fuhrer's view of a person was crucial. The same was the case in Stalinist Russia, and in contemporary dictatorships like China and North Korea. Army generals, governmental ministers, officials, and others therefore strove to anticipate the dictator's wishes and demands in order to be granted significance by their leader (Rees, 2012).

With regard to expectancy, cultures that emphasize social mobility can inspire individuals' ambition for glory. As former President Barack Obama wrote in "The Audacity of Hope,"

> [The Constitution] despite being marred by the original sin of slavery—has at its very core the ideas of equal citizenship under the laws; and an economic system that, more than any other, has offered opportunity to all comers, regardless of status or title or rank.

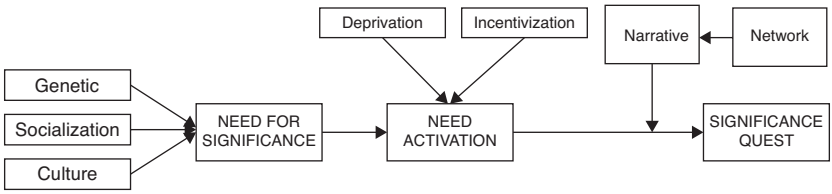
Indeed, this is the essence of the American Dream based on the ideology of Enlightenment—the idea that anyone can achieve significance regardless of the

circumstances of their birth. Western liberal societies based on this ideology produced far freer, more tolerant, and more flexible patterns of sociability than any other civilization in history (Pinker, 2014; see also Fiske, this volume).

Genetics, socialization, and culture thus contribute to individual differences in QFS. While some persons exhibit a seemingly insatiable need for significance, others are content to aim lower. Not everyone wants to be "a bride at every wedding, a corpse at every funeral, and a baby at every christening" as Alice Longworth Roosevelt characterized her father, Theodore. Yet no one is immune to disrespect, humiliation, and discrimination. In this sense, the quest for significance represents a universal motivational dimension.

Beyond individual differences, there are situational conditions that contribute to one's *state*-level QFS. The need is activated by significance loss, threatened significance loss or *deprivation*, or opportunity for significance gain (e.g., by becoming a hero, a martyr, or a billionaire as the case may be), that is *incentivization*. In striving to meet this need, the individual adopts a significance bestowing action as identified in a narrative shared within one's social network (Kruglanski, Bélanger, & Gunaratna, 2019). It is such a narrative that guides some individuals to seek significance by violence against the group's alleged enemies, and others by engaging in humanitarian, pro-social behavior. In all instances, significance is bestowed by a manifest commitment to a societal value (e.g., courage, honesty, wealth, or beauty) esteemed in the social life of one's reference group.

These ideas are graphically represented in Figure 7.1 and articulated and expanded upon as premises and hypotheses in the following section.



Premises and Hypotheses

Premise 1 (the Need for Significance): Humans have a basic need to have a sense of significance and mattering, to be recognized and respected.

Corollary 1 (to Premise 1): The Need for significance may be *acute*, that is situationally induced (i.e., a matter of "state"), or *chronic*, representing an enduring personality characteristic (i.e., a "trait").

Corollary 2 (to Premise 2): A person can attain or fail to attain significance (a) by dint of individual actions, and (b) by dint of their group membership (social identity).

Hypothesis 1a (from Premise 1, Corollaries 1, 2): The need for significance may be activated by a *loss of significance* suffered by the individual or their group.

Hypothesis 1b (from Premise 1, Corollary 1): The need for significance may be activated by an opportunity for *significance gain* afforded to the individual or their group.

Hypothesis 1c (from Premise 1, Corollary 1): The need for significance constitutes a dimension of stable individual differences.

Premise 2: The sense of significance (that gratifies the corresponding need) is conferred by matching/expressing a *cherished value* through one's actions (e.g., reflecting bravery, generosity, solidarity) or attributes (e.g., membership in a prestigious group, beauty, wealth)

Corollary 3 (to Premise 2): Different groups may cherish different *values* (not necessarily of the moral kind) contained in the group's shared reality (Higgins, 2019).

Premise 3: Where the need for significance is activated at a sufficient magnitude, it will drive behavior aimed at matching/expressing a group's cherished value.

Hypothesis 2 (from Premises 1,2,3): The need for significance will enhance the individuals' affirmation of their group's values.

Hypothesis 3 (from Premises 1,2,3): Significance-driven actions will be guided by *narratives* endorsed by the individual's *networks* that portray those actions as ways of realizing the group's values.

Premise 4: The main twin functions of networks are: (a) to validate the group's narrative, (b) to reward group members who follow the narrative.

Corollary 4 (to Premise 4): The more ample/salient are arguments *incompatible* with the group's narrative, the more pronounced will be attempts to *shore up* support for the narrative.

Hypothesis 4a (from Premise 4, Corollary 4): The more discrepant the group's position from that of the majority (i.e., mainstream society), the stronger the pressures in the group toward consensus and cohesion.

Hypothesis 4b (from Premise 4, Corollary 4): At a given level of discrepancy from the majority, the smaller the size of the group the stronger the pressures in the group toward consensus and cohesion.

Hypothesis 4c (from Premise 4, Corollary 4): The greater the discrepancy of the group position from that of the majority, the stronger the tendency of members committed to the group to *minimize* the group's exposure to the majority position.

(Hypotheses 4a, 4b, and 4c assume that consensus, cohesion, and opinion uniformity enhance the perceived validity of the narrative, whereas

discrepancy from the mainstream opinion, saliency of the mainstream opinion and paucity of support (small group size) weaken the perceived validity of the narrative).

Premise 5: The need for significance may be gratified via equifinal behaviors matching/expressing different cherished values, and/or by different equifinal behaviors aiming to match/express the same cherished value.

Hypothesis 5 (from Premise 5): In case a given behavior failed to result in gratifying the need for significance, other equifinal behaviors will be undertaken for that purpose.

Hypothesis 6a (from Premises 1, 2; Corollary 1): Matching/expressing a value cherished in a given social context will result in feelings of significance, and pride.

Hypothesis 6b (from Premises, 1, 2; Corollary 1): Failing to match/express a value cherished in a given social context will result in feelings of insignificance and humiliation.

Hypothesis 7 (from Corollary 3): Attributes or behaviors that earn a given degree of significance in a social context committed to given values, may earn a different degree of significance (more or less) in a social context committed to different values.

The Empirical Base: How QFS Shapes Patterns of Sociability

Activating the Quest

The need for significance, while universal, is not the only need to which individuals, even the most ambitious among them, must attend to promote sociability. No one can be constantly questing for significance. The need for significance must be activated, in order for QFS to come into play. Generally, QFS, like any other motivation, is activated through *deprivation* or *incentivization*. Deprivation amounts to significance loss, which can occur through rejection, failure, humiliation, or injustice. Significance loss can also be threatened unless one acted in a certain way (e.g., volunteered for a dangerous mission, made a costly sacrifice) (cf. Kruglanski, Chen, Dechesne, Fishman, & Orehek, 2009; Kruglanski et al., 2013, in press; Kruglanski, Chernikova, Rosenzweig, & Kopetz, 2014). We may term this conditional deprivation. In contrast, incentivization refers to a situational opportunity through which one can gain significance. Notably, there are individual differences in sensitivity to events that challenge one's sense of significance (e.g., Downey, Feldman, Khuri, & Friedman, 1994; Kernis, Lakey, & Heppner, 2008), as well as to opportunities for significance gain (Higgins, 1997).

Hypothesis 1a: Significance Loss

According to Hypothesis 1a, the need for significance may be activated by a *loss of significance* suffered by the individual or their group. There is considerable evidence consistent with this hypothesis, primarily gleaned from studies, which measured or manipulated individuals' sense of lost significance and examined how this related to their actions related to QFS.

Correlational Evidence

In field research carried out with detained members of the Sri Lankan terrorist organization the Liberation Tigers of Tamil Eelam (LTTE), a sense of lowered significance, assessed as feelings of humiliation and shame, was significantly correlated with endorsement of a violent struggle against the Sinhalese majority, their perceived oppressors. The latter, assessed via items, such as "Armed fight(ing) is a personal obligation of all Tamils today" and "Suicide bombers will be rewarded for their deed in the afterlife" (Webber et al., 2018), represents a significance enhancing narrative based on defending one's group's honor.

Jasko, LaFree, and Kruglanski (2017) also tested the significance quest theory in the context of individuals who have committed ideologically motivated (including the far-right, the far-left, and Islamic extremism) crimes (ranging from illegal protests to bombings) in the United States. Despite the variation in the individuals' ideological goals and their actions in pursuit of those goals, the analysis shows that the use of violence can be predicted in part by a sense of personal insignificance. Among the nearly 1,500 ideological extremists in the data set, the authors found that when individuals experienced *failure at work*, when they were rejected in *social relationships*, or when they were victims of *abuse*, they were more likely to resort to violence to pursue their ideological goals. Significance loss led to a more extreme need to gain significance, which led to a more extreme commitment to their ideology and a readiness to commit violence on its behalf.

Experimental Studies

The correlational studies provide important insight in their use of real-world data, but they cannot indicate causality. Experimental work concerning the impact of significance loss on the quest for significance expressed by enhanced commitment to individuals' values provides that needed evidence.

In one study (Orehek & Kruglanski, 2018, Study 4), participants were made to experience success or failure on an experimental task (made to appear important), and their sense of significance was measured via the Meaning in Life Questionnaire (Steger, Frazier, Oishi, & Kaler, 2006). Participants who

experienced failure scored lower on this measure than those who experienced success. Additionally, participants whose sense of meaningful existence was lowered by failure, self-identified as more interdependent (versus independent) than participants in the success condition. These findings suggest that a loss of significance induced by failure evokes individuals' QFS fulfilled by their sense of social connectedness with members of their network.

In another series of experiments, Leander, Agostini, and colleagues (2020) explored whether individual differences in higher education were related to people's reactions to significance loss, specifically in their endorsement of violence. College education negatively predicts attitudes toward, for instance, expansive gun use rights (Quinnipiac, 2013), the death penalty (Pew, 2018), military intervention in the Middle East (Pew, 2003), and authoritarian aggression (Carvacho et al., 2013). Indeed, Leander, Agostini, and colleagues (2020) found an interaction between experimentally thwarted goals (operationalizing failure or significance loss) and education level in predicting heightened endorsement of guns and war by non-college educated participants but not for those with college education.

Hypothesis 1b: Significance Gain

The vast literature on extremism, terrorism, and rebellion (see Kruglanski et al., 2019, for a review) makes it clear that significance loss (defined as grievance, injustice, or humiliation) is closely linked with violent action on behalf of one's cause. Therefore, evidence for our Hypothesis 1a is plentiful and relatively unsurprising. There is less literature considering whether the opportunity for significance gain can evoke similar actions.

Relevant to this possibility, there is experimental evidence that exposure to successful role models, and/or generally admired figures, inspires people to pursue higher goals. To explore this evidence with specific respect to QFS, Hasbrouck (2020) found that participants who were asked to write about how a personal hero had inspired and influenced them scored significantly higher on the 12-item Quest for Significance Scale (QSS, Molinario et al., in preparation) than participants who wrote about the last television show or movie they watched.

Delving more deeply into people's willingness to support a cause as it relates to significance gain, Jasko, Szastok, Grzymala-Moszczynska, Maj, and Kruglanski (2019) studied samples including members of a radical left-wing political party (Study 1), members of a social movement in defense of democratic rights (Study 2), feminist activists (Studies 3–4), environmental activists (Study 5), and participants and supporters of a doctors' hunger strike (Study 6). Across all of the studies, Jasko et al. (2019) found that people who perceived the cause as more personally important were also more motivated to engage in peaceful actions on behalf of the cause. Next, the researchers found that feelings of significance

(pride, meaning, and happiness) related to their involvement in the cause of their social reference group predicted their willingness to maintain such involvement. Even further, people who were engaged in radical and demanding (versus more moderate) forms of political behaviors were more committed to the cause, more willing to engage in all types of action for the cause, more willing to self-sacrifice for the cause, and most importantly, experienced stronger significance gain as a result of their involvement in the cause.

Hypothesis 1c: Individual Differences in the Quest for Significance

As previously mentioned, no one quests for significance incessantly, and the QFS is more pronounced for some people than for others. Our six-item Quest for Significance Inventory (QSI) (Molinario et al., in preparation) measures individual differences in QFS. This single-factor scale was found to be negatively correlated with life-satisfaction, self-esteem, presence of meaning in life, and presence of purpose in life. Moreover, it was positively correlated with measures of depression, fear of humiliation, cumulative humiliation, and loss of significance. Importantly, too, the QSI was positively correlated with measures of ambition and individual and collective narcissism attesting to the positive attraction of significance as inherent in significance quest.

The "How" of Significance Quest: Needs, Narratives, and Networks Working Together

When the *need* for significance is activated, an individual must determine the best way in which to meet that need in shaping their social strategies. As noted earlier, this is guided by a narrative that identifies a means to significance appropriate to given circumstances. Significance follows from exhibiting commitment to values cherished in one's social *network* or reference group (Merton & Kitt, 1950). Thus, significance is attained by following the dictates of one's network's *narrative* regarding cherished, sacred, and respected values (Atran, 2010a).

As such, the network serves a dual purpose in that it *validates* the narrative (by consensually embracing it), and it *rewards* (with significance) those who exemplify it or live up to its values, exerting both informational and normative influence, respectively, on its members (Deutsch & Gerard, 1955). Maintaining these types of influence require that networks regulating patterns of sociability take action to guard their sacred values, by rejecting those who deviate from the consensus (Kruglanski & Webster, 1991; Kruglanski et al., 2006; Schachter, 1951), by "rallying around the flag" when their narrative is challenged (Allen, 1975; Moscovici, 1985), and even by working to isolate themselves from a potential challenge to the narrative (Bittner, 1963; Coser, 1954, 1967; Kanter, 1968, Wilson, 1959). Evidence relevant to these possibilities is reviewed later.

Hypothesis 2: Activation of the Need for Significance Leads to Enhanced Support for a Group's Norms and Values

Hasbrouck (2020) obtained evidence that the quest for significance, measured as an individual difference variable, was positively correlated with the individual's highest ranked moral foundation (Graham, Haidt, & Nosek, 2009). In his liberal sample, the quest for significance was positively correlated with the two values of greatest importance to liberals, that is, Harm and Fairness (ibid.). In another study by Molinario, Jasko, Kruglanski, Sensales, and Di Cicco (under review), individual difference measures of the quest for significance were positively related to populist values for both Italian and American samples.

Hypothesis 3: Individuals Will Engage in Significance-Motivation Actions That Are Guided by Narratives Endorsed by the Individual's Networks

As previously expressed, the narrative and network elements described previously are tightly interwoven. Religious narrative is one example. In July of 2019, Islamic State fighters in Tunisia released a video pledging their allegiance to IS supreme leader Abu Bakr al-Baghdadi. In their pledge, they stated, "The pleasure of this world is small compared to the pleasures of the hereafter. . . . Jihad is one of the doors of Paradise whereby Allah takes away gloom and distress. Join your brothers the mujahideen" (Weimann, 2021, p. 15). These statements hint at the honor and recognition (significance) one would receive as a holy warrior, member of the IS network.

In the United States, Christian nationalists believe that the country is fundamentally Christian and should be governed as such. Christian nationalism promotes prejudicial narratives against various outgroups who do not conform to Christian nationalists' image of the United States. A recent study (Leander et al., 2020) found evidence for a serial mediation model in which significance loss (defined as a sense of disempowerment) predicted increased allegiance to Christian nationalism (the network), which predicted antisemitism (the narrative) and reduced recognition of hate crimes.

The quest for significance, extremism, and the power of one's network has been validated cross-culturally. In Sri Lanka, Morocco, and Indonesia, Jasko et al. (2019) found significant interactions between social network and QFS in relation to violent extremism: In more radicalized networks, the QFS, specifically related to the sense that one's group had not received the respect it deserved, was more predictive of violent extremism than it was in less radicalized networks.

As emphasized earlier, the QFS does not necessarily lead to antisocial behavior. Depending on one's network's narrative it can prompt pro-social behaviors as well. For example, when 387 Dutch adults were either instructed (or not) to think about

a recent failure in a domain in which they are typically competent, those who believed their network valued volunteer work and were primed to think about their failure were most willing to volunteer to support Middle Eastern refugees, exemplifying a humane, pro-social manner of asserting one's significance.

Hypothesis 4a: Differences Between the Opinion of the Majority and the Opinion of a Discrepant Group Drive Cohesion Within That Discrepant Group

The majority's different position from the position of the discrepant group can be construed as evidence against that group's position and consequently shape patterns of sociability. To counter such evidence, the group must come together in order to consensually reinforce their position. This hypothesis is supported indirectly by research on the social support for nonconformity—ergo, group cohesion that lends support for holding a position that is discrepant from that of the majority (Asch, 1955; Allen, 1975; Allen & Levine, 1968). For example, Asch (1955) found (and Allen (1975) replicated) that if even one group member agreed with a participant's discrepant position, the group significantly reduced their conformity to the majority position (see also Prislin & Crano, this volume). This research demonstrates that social support is integral to maintaining a position that is different from that of the majority.

Hypothesis 4b: The Smaller the Size of the Group Whose Narrative Is Discrepant from the Majority, the Stronger the Group Pressures Toward Consensus and Cohesion

A meta-analysis by Mullen, Brown, and Smith (1992) found that ingroup bias was negatively correlated with group size, demonstrating that, as Levine and Kaarbo (2001, p. 239) noted, "smaller minorities may be more highly identified with their faction and more biased toward it than are larger minorities."

Hypothesis 4c: Group Members Will Minimize the Group's Exposure to the Majority Position or Culture from Which It Deviates

Cells of suicide attackers, like cults, cut their members off from family and friends from their former lives (Kruglanski et al., 2013; Merari, 2010). Sociologists refer to this phenomenon by the term *renunciation*. It concerns the strengthening of individuals' relationship and commitment to their group by breaking their bonds with, or geographical isolation from, anyone who might disrupt the group's cohesion is an extreme example of manipulating sociability for an ideological purpose (see Bittner, 1963; Coser, 1954, 1967; Kanter, 1968; Wilson, 1959).

Consequences

Hypothesis 5: Should a Behavior or an Attribute Fail to Gratify the Need for Significance, Other Equifinal Behaviors Will Be Undertaken for That Purpose

As evidence for this hypothesis, Pedahzur (2005) described cases of Palestinian suicide bombers who felt unable to achieve significance in the traditional Palestinian society (e.g., because of infertility, divorce, or positive HIV diagnosis) and in response volunteered for their deadly mission in order to regain their lost significance. Volunteering for suicide missions serves as a substitute for traditional means of significance that for some reason failed to deliver.

Substitutability of means of significance is illustrated by extensive research on self-affirmation. Self-affirmation serves as a sort of psychological defense, against a self-concept challenge. And it restores the sense of self-worth upon affirmation of an important value (cf. Steele, 1998; Sherman & Cohen, 2006).

The sense of significance is dependent on one's demonstrable commitment to important cultural values; hence a loss of significance by failure to live up to a given value can be compensated for by demonstrating one's dedication to another value. Research by Atran and his colleagues (Atran, 2010b) demonstrates that such dedication is often proven by sacrifices one has made or is ready to make on behalf of a value deemed to be "sacred" in one's society. The more important (or "sacred") the value, the greater the significance bestowed for demonstrating a devotion to it.

Self-affirmation also occurs in response to self-uncertainty, which refers to a threatened sense of self-worth, or significance. Consistent with our hypotheses, McGregor and Marigold (2003, e.g., Study 3) found that inducing self-uncertainty resulted in enhanced conviction on socially important issues, and Hogg (2012) describes research where self-uncertainty increased the appeal of extreme, highly cohesive groups with clear cultural norms, through which one may restore their lost significance (see also Gaffney & Hogg, this volume).

Terror Management Theory [TMT] emphasizes the link between self-esteem (typically related to one's sense of significance or social worth) and one's cultural narrative or worldview (e.g., Solomon, Greenberg, & Pyszczynski, 1991). Awareness of one's mortality evokes the sense of fragility and insignificance. Therefore, the affirmation of one's value through support of one's cultural world view demonstrated in numerous terror management studies in response to mortality salience is consistent with the present hypothesis. Notably, TMT ties self-esteem to the fear of death exclusively, finding in hundreds of studies that manipulating mortality salience leads people to affirm their cultural narrative, which in turn improves their self-esteem (see Greenberg, Vail, & Pyszczynski, 2014, for a review). The SQT, however, holds that the loss of significance has the same effect, even with regard to situations other than death specifically, such as being

disrespected or treated unjustly, and that a similar effect occurs in situations where there is an opportunity for significance gain.

Hypothesis 6a: Matching/Expressing a Value Cherished in One's Social Group/Culture Will Result in Feelings of Significance and Pride

Constituting the fulfillment of an important need, engagement in a significance-earning behavior brings considerable satisfaction. In 2005, *Time* magazine correspondent Bobby Gosh interviewed Marwan Abu Abeida, a 20-year-old Iraqi of Abu Mussab al Zarqawi's insurgent group (Attawhid), the precursor of Islamic State. Marwan had served as a fighter for months but then took the next step in proving his commitment to the group's sacred values by volunteering to become a suicide bomber. When his name came up on the list, it

> was the happiest in his life . . . he claimed to be ready to die to be looking forward to it and praying for a large number of American casualties. His thoughts were occupied by the "final stage," after a mission would be assigned to him. He hoped for an important one.
>
> *Kruglanski et al. (2019, pp. 122–123)*

Research guided by the self-affirmation theory (Steele, 1988) showed that engagement in activities that enable people to affirm their values leads to increased self-esteem (e.g., Cohen, Aronson, & Steele, 2000; Fein & Spencer, 1997). Additionally, recent research on political activism (Skitka, Hanson, & Wisneski, 2017) showed people with higher moral convictions related to same-sex marriage and permission to carry guns on college campuses were related to stronger anticipated feelings of pride, which in turn predicted greater intentions to engage in activism regarding those moral convictions.

Hypothesis 6b: Failure to Match/Express a Value Cherished in a Given Social Context Results in Feelings of Insignificance and Humiliation

If significance brings joy, it stands to reason that insignificance should bring despair. In Emile Durkheim's (1897) classic work "Le Suicide," this burden of an unfulfilled need for meaning and significance can result in *anomic suicide*, which occurs in the absence of societal norms and definitions of legitimate aspirations (hence, the absence of a narrative as to the means to significance), and *egoistic suicide*, which occurs when one feels untethered from society and from respected others who can grant one significance (hence, the absence of a social network that lends one significance).

Evidence for this hypothesis comes from ample research including that on stigma (e.g., Link & Phelan, 2001; Major & O'Brien, 2005; Pachankis et al., 2018), exclusion, rejection, and ostracism (e.g., Baumeister, Twenge, & Nuss, 2002; DeWall & Bushman, 2011; Leary, 1990; Williams, 2007). Stigmatization is defined by Crocker, Major, and Steele (1998, p. 505) as exclusion or rejection of a person when she or he possesses (or is thought to possess) "some attribute or characteristic that conveys a social identity that is devalued in a particular social context." Such state is one of felt insignificance or lack of social worthy. This universally results in social pain (Baumeister, Twenge, & Nuss, 2002).

Hypothesis 7: Attributes or Behaviors That Earn a Given Degree of Significance in a Social Context That Highlights Given Values/Standards Should Earn a Different Degree of Significance in a Social Context Committed to Different Values/Standards

Whereas research has revealed the existence of universally cherished values (e.g., Schwartz, 1992), there are group and cultural differences in the weight assigned to various values in the way sociability is practiced. As such, social context can determine which actions or attributes earn significance. A shift in social context wherein one's previous source of significance is less valued or not valued at all. For instance, an immigrant or refugee who was a highly respected professional in their native country may feel a painful loss of significance upon finding that their license to practice that profession is not valid in their host country, requiring that they work in a job that affords them less social status. There is no shortage of stories such as these (Amir, 2010; Nguyen, 2018). As a particularly potent example, Viktor Frankl (2006/1946) recounts the venerated intellectuals, wealthy businessmen, and renowned artists he met in Nazi concentration camps, who were stripped of their former honor and turned into shivering inmates teetering on the verge of inhumanity.

A plethora of evidence across different psychological literatures is consistent with this proposition. For instance, Rosenberg (1965) reported that high school students from minority ethnic groups who lived in ethnically mixed neighborhoods had lower self-esteem than minority students in ethnically homogeneous neighborhoods. This suggests that attitudes and behaviors consistent with a particular minority group's norms and values would not lend a student as much significance in a neighborhood where not everyone shares those norms and values, versus in a more homogenous neighborhood. Similarly, Clark and Clark (1939) found that black children attending integrated, northern schools exhibited greater self-hatred than those in segregated southern classrooms.

General Discussion

In reflecting on his role in the advancement of attribution theory, Harold Kelley (2008, p. 214) noted: "*The theoretical fruit were hanging high in the tree, ripe and ready for picking, and I happened to be in the orchard at the top of the ladder. . . .*" The concepts and data on which the SQT is based were present in some form in the diverse research literatures that we have addressed. Specifically, the SQT suggests that a plethora of motivational constructs previously studied independently from one another, such as achievement motivation, competence motivation, status cravings, rejection sensitivity, self-enhancement motivation, the need to belong, and the fear of death, among others, share a fundamental commonality and have a profound influence on shaping patterns of human sociability. All contribute to the universal human need to feel that they matter, have worth, and are respected by those whom they respect, needs that can only be satisfied through sociability and intense engagement with others.

Without explicitly saying so, different research programs in social psychology have addressed the ways in which QFS may be activated, for instance, via exclusion, rejection, dissonance induction, failure, discrimination, self-uncertainty, and mortality salience among others. What is common among these differing processes and procedures is that they all induce a feeling of lost significance. However, whereas this prior research stressed defensive processes against threats to one's self-integrity, the SQT elucidates that as with any motivation, QFS can be activated by *incentivization* as well as *deprivation*. In other words, QFS may be sparked by an opportunity to gain significance, in addition to being sparked only by significance loss or threat of loss.

Moreover, terror management theorists, self-affirmation theorists, and others have all alluded to the idea that significance restoration can be achieved by demonstrating one's commitments to their personal and cultural values (e.g., Greenberg et al., 2014; Sherman & Cohen, 2006; Steele, 1998). McGregor's Identity Consolidation [IC] theory similarly suggests that an important way in which people cope with the threats due to personal uncertainty is "compensatory conviction about value relevant topics" (McGregor & Marigold, 2003, p. 839).

The SQT also provides an important clarification to these prior formulations that stress the restorative function of value affirmation, worldview defense, or identity consolidation: From the outset, an individual's self-worth is anchored in their group's values, and is affirmed with reference to group's norms of sociability. Embodying the values of one's group earns one worth and significance, and represent them results in rejection, and lowered sense of significance. Finally, the SQT explains that one's sense of significance may be subject to within context and between contexts fluctuations, such as is experienced by refugees or by ethnic minority students living in ethnically mixed neighborhoods, for instance.

Connecting the Dots

The SQT integrates prior insights in two ways roughly amounting to *intra-* and *inter-category* integrations. The former type of integration refers to identifying the functional common denominator of currently unconnected concepts. As noted earlier, different motivational constructs addressed in unrelated literatures all point to the same fundamental need for significance. The same applies to the disparate constructs functionally representing significance attainment/restoration based on affirmation or demonstrating one's commitment to an important societal value.

The inter-category integration refers to the concatenation of the *need*, the *narrative* and the *network* elements through which one satisfies their need for significance. In case of biological needs like hunger or thirst, the connection between the end and the means is innate. No one needs to be told specifically that in order to satisfy one's hunger one needs to eat, nor to drink when thirsty. The quest for significance, a quintessentially social need, is satisfied through means which are distinctly prescribed by, and unique to, the society/culture whose shared reality one has internalized (cf. Cooley, 1922; Mead, 1934). In other words, what constitutes significance is dependent on the cultural narrative embraced by the specific network of which one is member. As such, the motivational (need), the ideological (narrative) and the social (network) elements are inextricably interwoven.

The SQT relates these different concepts to real-world outcomes, demonstrated by the evidence for the major role that QFS may play in the radicalization and deradicalization of violent extremists, attitudes toward mass shooters, commission of ideological crimes, participation in political activism and engagement in suicidal terrorism. Thus, the SQT has both conceptual and empirical implications. Ultimately it suggests that the "failure" or "success" of a social action and meaningful patterns of human sociability cannot be fully understood without the consideration of the need which motivated it, the narrative which guided it, and the network which supported and rewarded it.

Directions for Future Research

Future work may fruitfully address quantitative dimensions of significance quest phenomenon and enhance the precision of the SQT's hypotheses. Future studies might find that the degree to which a given outcome will produce a significance loss or gain should be a function of the *extent* to which (1) it matches or deviates from a given value, (2) the place of the value in the hierarchy of one's group's narrative (i.e., its relative importance), (3) the degree of the group's cohesion and consensus around the group's narrative wherein that value hierarchy is embedded, and (4) the expected duration of the gain or loss. Moreover, the emotional impact of the significance gain or loss should depend on the

magnitude of the individual's trait-level significance quest and the degree of the loss or gain involved (see also Mikulincer & Shaver, this volume).

As we have noted throughout this chapter, research has neglected to address the potential for QFS to be activated as a result of the opportunity for significance gain, focusing instead on situations of actual or threatened significance loss. One possible way to address this gap would be to investigate *promotion* versus *prevention* mindsets (Higgins, 2012). It seems plausible that in activating the quest for significance, individuals with a prevention focus will be more responsive to significance losses and those with a promotion focus will be more responsive to significance gains versus gains. Specifically, the promotion orientation may instill a "unidirectional drive upward" (Festinger, 1954) that does not have a specific upper bound, and thus should be capable of producing a more extreme ambition for significance (Kruglanski et al., 2021) than the prevention orientation that merely strives to restore the status quo. Also, socialization practices that strengthen beliefs in growth possibilities, and hence the expectancy of positive outcomes, might contribute to significance strivings more so than practices that highlight fixed attributes and personal endowments (Dweck, 2006).

Future research could also investigate whether there are personality variables that predispose individuals to be more responsive to the network versus the narrative elements in activating QFS. For instance, characteristics such as conformism, self-monitoring, or dependence would privilege the influence of networks, whereas introversion would privilege that of narratives. In a similar vein, the role of culture and group type in promoting QFS could be profitably studied, especially as they relate to the power of networks and narratives (see also Fiske, this volume). As suggested earlier, for example, individuals in tight cultures (Gelfand, 2019) that attach greater importance to norms and values may be more devoted to adhering to those norms and values, resulting in a more pronounced QFS than individuals in loose cultures.

Coda

Like the mythological Proteus, the topic of significance quest has appeared in diverse forms and under diverse names across multiple domains of psychological (and philosophical) study. In this chapter, we have integrated this widespread and disconnected literature into a theory that ties them together, offering an integrated account of one of the main drivers of human sociability. Significance Quest Theory accounts for large bodies of extant empirical findings as well as with the results of novel inquiries in laboratory and field settings. The human need for significance and mattering is one of the preeminent social needs, responsible for much human suffering as well as joy, and for macro-level phenomena that chart the course of history.

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8

A SOCIAL IDENTITY ANALYSIS OF SOCIABILITY

Making, Breaking, and Shaping Groups and Societies

Amber M. Gaffney and Michael A. Hogg

Abstract: Warm feelings toward group members bind people to groups. Or do they? Sociability may be a product of group identification rather than a cause for it. In this chapter, we lay out a social identity analysis of sociability and suggest that because people feel favorably toward their important social identities, it creates sociability within groups. Processes of self- and social-categorization lay the foundation for feelings of warmth between certain ingroup members and a desire to affiliate with those ingroup members. We present the positive aspects of group identification and sociability (e.g., the promotion of mental and physical health, knowledge, esteem). We also suggest that the same motives that produce group identification and positively shape the human experience, can also be responsible for the darker side of collective behavior. The desire for affiliation with those who are similar reinforces worldviews and shapes a collective understanding of the world can drive intergroup polarization, stifle progress, feed populism, and pave a path for authoritarianism.

Key words: Social identity, self-categorization, cohesion, well-being, extremism

"Oh my name it is nothin'
My age it means less
The country I come from
Is called the Midwest
I's taught and brought up there
The laws to abide
And that the land that I live in
Has God on its side"

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Bob Dylan's "With God on Our Side" sets up a particular kind of warmth, amicability, and even prosociality that is only extended to those people who share a particular categorization: White, European American, Christian. Throughout the song, the "other" changes from Native Americans, to Germans, to Russians, to an unknown other against whom the ingroup must always fight. This fight against the other, even if it lands the planet in "chemical dust," is legitimatized because it is "us, the good people, who are with God" against "them, the other." This stark contrast shows that the sociable nature of humans is extended only to some people. For specific people, we are willing to fight and die. However, the nature of intergroup relations (and the point of this song) paints a picture in which a fight on the ingroup's behalf is often waged directly against the people we assume are different than us.

Desire for affiliation and preference for spending time with other humans rather than alone is not simply an individual characteristic. A simple definition of sociability as a "preference for being with others rather than being alone" or merely enjoying the company of others (see Cheek & Buss, 1981) ignores some fundamental issues surrounding the nature of sociability. Who are these others? How do we relate to them? What functions do they serve the individual and the larger collective? It is not uncommon for many of us to claim low levels of occasional misanthropy while simultaneously enjoying the company of our friends and colleagues who share our worldviews—a comforting form of self-validation.

Human desire for affiliation, and the sociability provoked by the desire, may very well be the mechanism that binds individuals into societies. Warmth, the feature that sits at the heart of sociability, is a fundamental trait of person perception. Perceived warmth is inherently tied to status relations and competition between groups (see Cuddy, Fiske, & Glick, 2008). Rather than examining sociability as merely an individual feature, in this chapter, we explore the unique function that the enjoyment of specific others—those who form identity groups—plays in shaping both the individual and collective selves. Here we suggest that sociability is both shaped and configured by group membership.

When and Why Humans Need Others

At the most basic level, social animals rely on their kind for survival. Humans need other humans to meet basic survival needs (safety, food and water, reproduction). Beyond survival (or perhaps because of the need for survival), Baumeister and Leary (1995) posit that humans also have a basic need to belong, which includes both "frequent interaction and intense caring" (p. 497; see also Baumeister & Maranges, this volume). Similarly, Brewer and her colleagues (e.g., Brewer, 1997; Brewer & Caporael, 2006; Leonardelli, Pickett, & Brewer, 2011) suggest that evolutionary concerns over survival create the need for humans to live communally and collaboratively. This manifests in a desire for inclusion in

social groups and interpersonal relationships. From this perspective, human sociability is an adaptive characteristic, which helps to ensure the survival of the species (see also von Hippel & Smith, this volume).

Beyond the tangibles of food, water, safety, and reproduction, humans use other humans to fulfill epistemic and existential needs (see also Kruglanski & Ellenberg, this volume). By affiliating with others, people learn the tools that they need for survival (e.g., what foods are appropriate to eat, geographical locations, social norms). Moreover, similar others can validate emotional responses and alleviate anxiety (e.g., Schachter, 1959). On an interpersonal level, people use others as a comparison source to garner knowledge about their own abilities and opinions (Festinger, 1954), to uplift the self, and to improve their weaknesses (e.g., Thornton & Arrowood, 1966; Wills & Suls, 1991; see also Fiedler & Hütter, this volume). Thus, the drive to socialize and surround one's self with other humans is necessary not only for basic survival but also because of the knowledge others provide about the world in general and the self in particular.

The interpersonal dynamics of sociability are clearly laid out in the literature (see also Prislin & Crano, this volume). However, the importance of sociability as both an antecedent for group behavior and an outcome of intra- and intergroup relations is also evident in the social psychological literature. The self-concept not only comprises idiosyncratic aspects of the individual personality (e.g., Glinda is high in sociability whereas Evanora is unsociable) or the knowledge the individual holds about the self from their close interpersonal relationships (e.g., desire for closeness to others based on early attachment bonds; see Ainsworth & Bowlby, 1992; see also Mikulincer & Shaver; and Reis et al., this volume), but a large portion of the self-concept contains the knowledge and evaluation of the self that people derive from specific social identities. Simply put, the groups to which people belong shape how people understand, interpret, and enact their identities (see Tajfel & Turner, 1986) and social identities both drive affiliative behaviors and set the normative patterns that dictate with whom people behave sociably and with whom they do not.

Social Identity as a Producer of Sociability

Groups: A Fundamental Part of the Human Experience

Important group memberships are the foundation of the human experience and encompass most interactions—including sociability and lack thereof. The social identity perspective provides an explanatory framework for empirically understanding the part of the self-concept people derive from their important group memberships. The groups that people use to define their self-concepts are often symbolic rather than real: Sports teams, political parties, followers of a band, movie genres, careers, physical location, gender, race, nation, families, and so

on. A group is thus a group to the extent that an assortment of individuals identify themselves and others using both their commonalities and the things that set them apart from other groups (see Hogg & Gaffney, 2018; see also Fiske, this volume). The social self-concept is both evaluative and epistemic. Important social identities help to dictate self-worth. When an individual's group does well, they do well; thus, people are motivated to defend, support, and uplift their important social groups (see Hogg, Abram, & Brewer, 2017; Tajfel & Turner, 1986). Moreover, group membership provides a contextual understanding of the world and the self within it.

When a social identity is psychologically salient, the normative features of that group (e.g., attitudes, values, behaviors, fashion, slang—the attributes which best define the group) are the lens through which the individual views both the self and others, which configures patterns of sociability. More specifically, when membership in a social group is psychologically salient, people define themselves and others with reference to the ingroup prototype. Take, for example, the current political climate in the United States. The instant a partisan's political party becomes salient, the defining features of the self become those that they share with other Democrats and the attributes that make Democrats both different from and better than Republicans (see Hogg & Turner, 1987; Gaffney, Rast, Hacket, & Hogg, 2014).

Social comparison is a key determining factor of a group prototype. People cognitively represent their groups by comparing the ingroup and a salient outgroup and also through comparing normative group members with peripheral group members (Gaffney & Hogg, 2017; Turner, Hogg, Oakes, Reicher, & Wetherell, 1987). Hence, group prototypes are dependent on social context. In an intragroup context, group prototypes are often moderate and reflect "average" group attributes. However, prototypes often become extremized as group members seek maximum distinction when comparing the ingroup to a relevant outgroup (Gaffney et al., 2014; Hogg, Turner, & Davidson, 1990).

The social identity perspective provides a broad framework for making predictions about both inter- and intragroup sociability. Although early scholars (e.g., Festinger, Schachter, & Back, 1950) argue that the sociability that exists within interpersonal relationships is what produces group cohesion, a self-categorization perspective suggests that sociability is a product of social identity and group belonging. Group-based member attraction occurs *because* members reflect extensions of the group prototype (Hogg, 1993). Rather than member-member attraction, it is the group identity to which members are attached to as an internalized extension of the self, and this in turn drives intragroup sociability (see also Brewer & Gardner, 1996).

In this chapter we propose that the desire for warmth and affiliation with fellow ingroupers and similar others is heightened in an intergroup context and is very rarely extended to the outgroup. All things being equal, people will choose to socialize with the people with whom they share a categorization.

This goes beyond a simple desire for the familiar—people seek out and affiliate with those people whom they find most informative for their self-concept. Ingroup members reinforce world views, validate collective experience, reduce conceptual self-uncertainty, and provide a roadmap for navigating the social world. The very motives that drive positive affiliative behaviors and ingroup sociability can also produce a dark side of sociability. These motives can often make people go to extremes on behalf of their groups. Ingroup cohesion and warmth can forge ethnocentrism, sectarianism, and populism, wedge partisan intergroup hostility, and stifle intergroup communication through echo chambers (Forgas, Crano, & Fiedler, 2021). The benefits that intragroup sociability create can help to explain some of the processes that seek to tear down society through division.

Motives for Group Identification

Self-esteem and Self-evaluation

An important premise of social identity theory is that groups allow people to feel good about themselves. As formally framed by the self-esteem hypothesis (Abrams & Hogg, 1988), groups can satisfy people's basic need to feel good about the self. This manifests in people actively attempting to protect their ingroup in the face of competing outgroups and in the desire to see their ingroup as not only different from but also better than relevant outgroups. This has clear ramifications for unsociable behavior directed toward outgroups. Yet, deep attachment to the ingroup and strong ingroup identification do not inevitably generate animosity toward the outgroup. Most frequently, intergroup discrimination is characterized by ingroup favoritism (Brewer, 1999), which in the case of dominant groups can lead to upholding systemic hierarchies between groups. This can happen both explicitly (e.g., direct and conscious acts of violence against marginalized communities) and more covertly (e.g., legislation that bans voting on Sundays).

The status relations and societal conditions that exist between groups determine social competition (Ellemers, 1993) and the desire for members of subordinated groups to seek to change unfair systems. Group identification supersedes relative status of a group—people can and do identify strongly with groups whether they are marginalized at a societal level or not (e.g., Reicher, 2004). Moreover, when people recognize (or come to believe) that their group is marginalized in an unfair system, they are willing to work on behalf of their group in an attempt to change the status relations. This willingness to work on behalf of the collective (even at expense to the self—see Klandermans, 2014) provides one of the clearest illustrations of sociability within groups.

Research on commons and social goods dilemmas also clearly illustrates sociability at the collective level. Paradigms used to study social dilemmas force individuals to decide between their own immediate needs and common resources.

De Cremer and Van Vugt (1999) found that among people concerned for their own individual welfare, increasing identification with the collective increased their contribution to the collective good. In an answer to the question: How do we motivate people with weak ingroup identification to contribute to the collective? Van Vugt and De Cremer (1999) provide evidence that weakly identified group members are more likely to increase their contribution in response to a punitive rather than a relational leader. People who identify strongly with their group, unsurprisingly, contribute to the collective regardless of leader type. Collective action research and commons and social dilemmas research both illustrate a clear pattern—caring for an overarching identity motivates people to behave sociably on behalf of the collective.

Distinctiveness and Assimilation

The optimal distinctiveness model (Brewer, 1991, 1993) posits that group identification can fulfill two basic human needs, which stem from human dependence on groups for survival (Leonardelli et al., 2011). Assimilation refers to a human desire to be included within a group. Distinctiveness is achieved when people identify with groups that they can easily differentiate from relevant outgroups (see Brewer, 1991). These two needs can compete with one another—overly inclusive groups are not satisfying because members' distinctiveness needs are not met. However, this is resolved through identification with groups that are optimally different from other groups—thus high assimilation does not necessitate low differentiation or even dissatisfaction. The desire to affiliate with others is maximized in groups that fulfill optimal distinctiveness needs. A group in which an individual finds themselves surrounded by similar others who are also different from other groups is one in which people express sociability toward the ingroup.

Uncertainty Reduction: The Epistemic Motive

Groups help individuals to feel favorably about themselves, to be surrounded by others who make them feel both assimilated and unique, and they also provide a significant amount of information about the social world. Uncertainty-identity theory posits conceptual self-uncertainty as yet another group identification motive (Hogg, 2007, 2012, 2021a, 2021b); see also Kruglanski & Ellenberg, this volume). Hogg and his colleagues argue that aversive uncertainties about the self and/or an important identity create a negative drive state. Not all uncertainties are created equal, and uncertainties that relate to the self-concept vary from person to person. Some people experience uncertainties regarding their interpersonal relationships (see Mikulincer & Shaver; and Reis et al., this volume), and others experience uncertainties regarding their individual personality (who they are and where they might end up in life). Others experience

uncertainty because of their position within a group or their group's position within society (Hogg & Mahajan, 2018). People may feel uncertain about their group's ability to survive and thrive. They may also feel uncertain about their own place in a group. Hohman, Gaffney, and Hogg (2017) showed that when strongly identified group members feel peripheral within their groups, this produces self-uncertainty. This in turn motivates strong group identification and sociable behaviors meant to establish the individual's prototypicality (e.g., prosocial behaviors on behalf of the ingroup, antisocial behaviors toward outgroups; see also Goldman & Hogg, 2016).

Such uncertainties are difficult to bear alone, and research in uncertainty-identity theory shows that group identification is one relatively simple method for reducing uncertainty. Self-categorization and social categorization depersonalize how individuals view the self and others, transforming people from individuals to representations of their respective group prototypes (Hogg & Turner, 1987). This process endows group members with the thoughts, feelings, and behaviors of the group—an important tool for reducing uncertainty. Because ingroup prototypes are both descriptive (they tell members what the group does) and prescriptive (they tell members what they *should* do), viewing oneself through the lens of the group prototype is inherently uncertainty reducing. Another reward for group membership is that knowing the self through the group reduces idiosyncratic uncertainties. As a result, when people experience uncertainty, they demonstrate a desire to join and identify strongly with self-inclusive groups (see Choi & Hogg, 2020 for a meta-analysis of 35 studies demonstrating the link between self-uncertainty and group identification). In the context of this chapter, conceptual self-uncertainty can motivate sociability, because group membership and affiliation with similar others provide valuable information about the self in the social context.

Entitativity

Some groups are more attractive to people experiencing uncertainty than others. In fact, we argue that such groups are almost by definition high in a form of sociability. When people experience uncertainty, they demonstrate a preference for groups with clear-cut norms, strongly delineated divisions with relevant outgroups, and high similarity among members—that is, when people are uncertain about who they are, they seek out groups high in entitativity (Hogg, Sherman, Dierselhuis, Maitner, & Moffitt, 2007). Entitativity is the structural quality of a group that allows a collection of individuals to be perceived as a distinct unit or entity (Campbell, 1958; Hamilton & Sherman, 1996).

Lickel and colleagues' (2000) work establishes four group types, which vary from highest to lowest in perceived entitativity. Intimacy groups (such as families and friendship groups) are characterized by clear intergroup boundaries, high member investment, similarity, low permeability, interdependence, and common

fate among members; thus, they are perceived as high in entitativity. Task groups comprise members who come together to reach a common goal (e.g., working groups), rather than being bound by interpersonal relationships. Members share similarity, a common goal, high member investment, and high interaction, but they are permeable and often dismantle after goal completion. Social categories, such as political parties, racial and ethnic groups, and shared nationality tend to have low interaction, but provide members with a sense of similarity based on the category, a common fate often based on shared history, low permeability, and often clear intergroup boundaries. Finally, loose associations, such as people who shop at the same grocery store, tend to have high permeability and less frequent interactions among members. An examination of these group types based on the criteria for entitativity critically suggests that the groups that are highest in entitativity should also have the highest degree of sociability among members. Whereas individuals within each group type may vary to the extent of their personal sociability, intimacy groups and task groups afford members with the most opportunities to be social with one another. Moreover, sharing a social category should also produce greater sociability toward other members of that category than toward members of other social categories.

People tend to believe that highly entitative groups are efficacious at achieving their goals (Clark & Wegener, 2009), which increases their attractiveness when people experience uncertainty. Another appealing feature of highly entitative groups is that they tend to be insular and exclusive (Hogg, 2021a, 2021b). Although this can help to satisfy motives for group identification by providing a fully inclusive group identity, it is also a feature of extremist and terrorist groups (Hogg, Kruglanski, & Van den Bos, 2013, Kruglanski, Pierro, Mannetti, & De Grada, 2006). Such groups are characterized by autocratic leadership, ideological orthodoxy, and strict norms, which often promote xenophobia, and violence directed toward both outgroups and ingroup members who stray from the group's rigid standards. In fact, people tend to more strongly endorse violence committed by groups high versus low in entitativity and find violence and prejudice directed toward an outgroup as more socially acceptable in highly entitative groups (Effron & Knowles, 2015).

Although entitativity characterizes extremist groups, entitative groups are not inherently extremist. Entitativity provides clear norms and often the normative content promoted is positive. Cruwys, Gaffney, and Skipper (2017) demonstrate that first-year university students primed with high uncertainty seek to join and identify with a group high (rather than low) in entitativity, whether or not the group promotes norms focused on education or norms of partying. The content of the norms is not important under uncertainty; it is the structure of the group and its entitativity that matters.

From a social identity perspective, entitativity helps to clarify a group's prototype, demarcating the ingroup from relevant outgroups. The more entitative

the group, the more people conform to the ingroup prototype and see the self as overlapping with other ingroup members and the ingroup prototype (see Gaffney & Hogg, 2017). Whereas the properties of highly entitative groups (e.g., shared fate, interdependence, similarity, cohesive norms) may promote sociable behaviors toward ingroup members, they also promote antisocial behaviors and attitudes toward outgroups.

Leaders: More Than Just Very Sociable Group Members

The sense of attachment to an important referent group's prototype extends to other group members who most closely embody the prototypical features of the group (Hogg & Gaffney, 2014). As extensions of the group prototype, they are the most liked (often loved) group members (see Hogg, 2001), and other group members seek them out for both affiliation and as sources of group normative information. Similarly, those group members who stray from the prototype in the direction of an outgroup (antinormative deviants) are disliked, cast as black sheep, censored, and derogated (Abrams, Marques, Brown, & Henson, 2000). Even within valued ingroups, there is thus a limit to sociability—group members who embrace important ingroup norms are valued and preferred for leadership positions, whereas those who are peripheral are often disliked and do not receive affable comradery within the group.

Ingroup members look to others who most closely approximate the prototypical features of the group. This makes prototypical ingroup members an important source of normative information, and thus they are sources of group-based influence (Hogg & Reid, 2006). People tend to either formally or informally elevate prototypical group members to leadership roles (Hogg & Van Knippenberg, 2003). For the last three decades, research on social identity and leadership has provided strong empirical evidence that people elect, support, trust, and find effective prototypical members and leaders over their peripheral counterparts (see Barreto & Hogg, 2017; Steffens, Munt, Van Knippenberg, Platow, & Haslam, 2021). Simply put, those who identify strongly with their groups view their ingroup prototype favorably, and favor members of the group who align closely with group prototypical features (Hogg, 2001; Hogg, Van Knippenberg, & Rast, 2012).

Ingroup members attribute a variety of other positive traits to their leaders, such as charisma (Hogg, 2001). Haslam and colleagues' (2001) work shows that group members perceived a male leader as more charismatic in an intergroup context if he affirmed the group identity (versus did not). From a social identity perspective, leaders are not necessarily born with traits that make them great leaders; rather, because they represent the group identity, group members attribute favorable traits to the leaders. This work indirectly implies that group members endow their prototypical leaders with favorable traits and likely assume that leaders are naturally sociable.

People pay attention to and engage in elaborative processing of persuasive messages that contain cues about the group prototype (Van Knippenberg, 2000). This special focus of attention confers prototypical leaders with the ability to shape group norms and the group prototype by deviating from it. Reicher, Haslam, and Hopkins (2005) argue that because of their prototypical status and because followers look to them for cues about group prototypicality, those leaders deemed as representative of the group identity have the ability to shape and change the group prototype in their own image, a common strategy for populist leaders who need to shape their propaganda to accommodate new realities (Forgas et al., 2021). Moreover, Abrams, Randsley de Moura, and Travaglino (2013; Abrams, Travaglino, Marques, Pinto, & Levine, 2018) show that whereas ingroup members sanction and censor other deviant group members, they provide deviant leaders with a transgression credit which essentially gives the leader a pass on their transgressions. Taken together, this strongly suggests that leaders who position themselves as representative of their groups can deviate from group norms, actually shift what it means to be prototypical, and not be sanctioned for deviant acts. The liking, warmth, and sociability that ingroup members extend to one another create different rules for their leaders—often because leaders help to make the rules.

This may leave the impression that only prototypical group members may attain leadership positions and create real and lasting change. However, recent history suggests that this is not the case. Occasionally the world (along with pundits and pollsters) is surprised when someone as seemingly non-representative of the Republican Party as Donald Trump is elected President of the United States. History provides several key examples of leaders who arose from positions, which seemingly lacked prototypicality and formal group identity and who went on to fundamentally transform the vision and shape of their groups, society, and history (for good or bad). Martin Luther King Jr., Marie Curie, Ruth Bader Ginsberg, Hitler, Lenin, Pol Pot, Emmanuel Macron all came from positions, which might be deemed (at one point in their early history) as peripheral in their respected nations, because of race, class, gender, or viewpoints. During times of heightened uncertainty, the prototypical leader advantage is weakened or disappears. Uncertainty provides a context in which ingroup members extend sociability to those who seem unlikely to receive it (Rast, Gaffney, Hogg, & Crisp, 2012). The uncertainty reduction motive that promotes the desire for a cohesive and self-inclusive group may also cause group members to overlook certain normative features in favor of focusing on specific leadership that will reduce the uncertainty.

A once deviant group member who attains a leadership position enjoys the benefits of the sociability extended toward prototypical group members. Gaffney, Sherburne, Hackett, Rast, and Hohman (2019) found that Republican voters perceived Donald Trump as more prototypical of the Republican Party after he was elected President than before he won the 2016 election. Because group

prototypes are consensually shared (i.e., group members must be in agreement regarding the core features of the group identity), democratically electing a leader legitimizes a once deviant candidate into a prototypical leader (Syfers, Gaffney, Rast, & Estrada, 2021). As a result, because Republicans reached consensus that Trump was their leader, they thus agreed that he was representative of the "new" Republican identity. Once elected, it may be difficult to view an ingroup leader as peripheral, because they are the embodiment of the group prototype. As a result, a democratically elected leader, who possesses deviant features, shapes the nature of how members view the prototype, positioning the leader's deviant features as normative attributes of the group and allowing them to reap the benefits of sociability.

Making and Shaping Groups and Societies

As already highlighted in the motives for group identification, social identities can shape patterns of sociability by being a positive source of self-esteem, can help to reduce conceptual self-uncertainty, provide significant information about the social world, and motivate people to work on behalf of a collective. Moreover, leadership, defined here as group-based influence, has the ability to motivate collectives to advance, improve, and innovate both their own group positions and society. Group memberships also have significant positive ramifications for mental and physical health and well-being (see Jetten, Haslam, & Haslam, 2012).

Being socially isolated from important others puts individuals at significant risk for clinical depression (see Cacioppo, Hawkley, & Thisted, 2010). On an interpersonal level, the loss of certain relationships (e.g., divorce, break-ups, death of an important other) can trigger depression (Tennant, 2002; see also Mikulincer & Shaver, this volume). On the therapeutic side, social support positively predicts willingness to seek treatment and to stick with it. Moreover, several therapeutic techniques rely on group-based strategies (e.g., group cognitive behavioral therapies) to treat depression (see Cruwys, Haslam, & Dingle, 2014).

Group memberships are a positive source of self-esteem for many individuals. Cruwys and colleagues (2014) reason that membership in important social groups should negatively predict depression. Moreover, because important social identities reflect a meaningful part of the self-concept, they drive sociability through group-based influence. This means that to the extent that group prototypes promote social connectivity and depression reduction, increasing group identification will increase the effectiveness of group-based depression therapies (Cruwys et al., 2014). Although one model of mental health posits that mental ill-health isolates individuals from social connectivity, Saeri, Cruwys, Barlow, Stronge, and Sibley (2018) show that social connectivity is a stronger predictor of mental health outcomes than mental health predicting social connectivity. Social connectivity through group membership may thus act as a "social cure" (see Jetten et al., 2012) for mental health issues such as anxiety and depression.

Presumably entitative groups, which promote connectivity and healthy behaviors among members, should strongly promote favorable mental health outcomes and treatments.

Recovery groups such as Alcoholics Anonymous and Narcotics Anonymous rely on promoting ingroup sociability, and the transition from an addict identity (e.g., "smoker") to that of a recovery-based identity (e.g., "former smoker") (Buckingham, 2017). Subjective norms present in such groups can help to guide positive behavior and the cognitive representation of the social self as a part of a recovery group providing the benefits of social identification: Increased efficacy, self-esteem, and collective agency. Indeed, membership in at least one recovery group positively predicts favorable recovery outcomes and negatively predicts relapse (Best, Irving, Collinson, Andersson, & Edwards, 2017). Taken together, this work suggests that recovery groups created through self-categorization and social categorization promote intense sociability and a form of a social cure for addiction.

The social connectivity that comes from group identification may also promote health and exercise behaviors. For example, increasing group identification with an exercise referent group promotes increased efficacy with physical activity and favorable expectations regarding physical activity (see Grant, Hogg, & Crano, 2015). Possessing an exercise-specific identity (e.g., *I am a runner, I am a cyclist, I am a crossfitter, I am an exerciser*) is associated with greater self-efficacy in physical activity as well as frequent exercise behaviors (see Brouwer, 2017).

Forming a positive social identity also helps to buffer against the negative physical and psychological consequences of stigma. Although it is often difficult to derive a positive sense of identification with a group identity that is devalued by society, it can be beneficial for health and well-being outcomes (Jetten, Haslam, Cruwys, & Branscombe, 2018). For example, identifying with a body positivity movement and categorizing oneself as "fat" can help to revise the stigma surrounding body type and become a positive source of esteem for people who categorize themselves into this identity.

In a similar way, Cooper, Smith, and Russell (2017) provide evidence that autism identification negatively predicts anxiety and depression and positively predicts self-esteem. Stereotypes surrounding autism cast people with autism as inherently unsociable; however, according to this work, positively identifying oneself as a member of the autism community (a sociable act) is associated with favorable physical, psychological, and social outcomes.

Leadership plays a key role in fostering healthy behaviors and promoting positive group identification. Physical activity leaders (e.g., coaches, team leaders, exercise group leaders), who actively advance their group's interests, lead for the good of the group, and are clearly representative of the group's identity (i.e., those who practice "identity leadership") help to foster participation in exercise groups over time by increasing members' identification with their groups (Stevens et al., 2020).

Leading from the position of "us" rather than "me" has the ability to mobilize the collective. In the context of a raging global pandemic, leaders who foster collective behavior may suppress concerns about restrictions of individual rights by actively promoting the greater good of the collective (Haslam, Steffens, Reicher, & Bentley, 2021). Since 2020, the world has seen several instances of leadership failures that have fueled partisan divisions in managing Covid-19 (Abrams, Lalot, & Hogg, 2021). However, the leaders who have worked on behalf of their groups and nations provide models of how group identity can be harnessed to promote collective behaviors for the good of their communities.

Group memberships are the building blocks of society. Because humans are social animals, our need for social connectivity and sociability is fundamental to every aspect of the human experience. Group identification is thus crucial for survival, health, and happiness. However, the same self-categorization and social categorization processes, which can elevate the human experience through connection, are also the mechanisms that can forge divisiveness—there can be a dark side to group-based sociability.

Breaking and Shaping Groups and Societies

Group identification can promote sociable behaviors that help to positively shape the human experience. However, these same motives also produce a form of sociability that is only extended to the ingroup. Collective sociability is not only used to promote group interest, but it also may be used to punish the perceived wrong-doer. Take for example the January 6, 2021 attack on the U.S. Capitol. A collection of people felt that an "illegitimate" election prevented President Trump from "rightfully" stepping into his second term as U.S. President. This "usurpation" not only prevented Trump's second term, but also "denied" insurrectionists their voting rights. As a result, they felt justified in attacking the U.S. Capitol in an attempt to overthrow the "unequal" status relations—a perception that is clearly explained by Tajfel and Turner's original argument regarding status relations and social competition. Although there is absolutely no evidence of a rigged election, the majority of the people who showed up at the U.S. Capitol on January 6 believed that their collective identity had been the victim of a major injustice.

Many of the same motives that drive group identification and inclusive sociability within groups can also polarize groups and societies, when each side exposes themselves to drastically different versions of reality. Increasing reliance on social media for connecting social identities also helps in the creation of echo chambers. Because group identification is an effective means of uncertainty reduction, leaders can both exploit and fuel collective uncertainty to better their own positions by deepening divisions between groups and strengthening support for authoritarian regimes. For example, although people brought up in democracies tend to prefer democratic leaders (both in organizations and in politics),

uncertainty promotes the acceptance of autocratic leadership that eliminates the uncertainty (Rast, Hogg, & Giessner, 2013).

Keeping in mind that people attribute a variety of positive features including prototypicality, to their ingroup leaders, it is likely that most people view their ingroup leader's autocratic tendencies as strengths. Whereas 50% of the United States might label former President Trump an autocrat, it is very likely that his supporters simply do not view his personality as such. Thus, one group's autocrat may be another group's pillar of democracy.

Populism, as an ideology, often precedes autocratic leadership and authoritarian regimes (Inglehart & Norris, 2017; see also Forgas et al., 2021). Although populism is a vaguely defined construct at best (see Crano & Gaffney, 2021), most definitions include a moral majority (the people) juxtaposed with an elitist class comprising the wealthy and/or politicians (Mudde, 2007). The desire to bring down the elite cements the very nature of populist thought to the conceptualization of who is "with us" and who is "against us"—that is, populism is an exercise in intergroup relations. People who feel that their anger at being deprived is representative of their groups and nations as a whole, express stronger populist attitudes and more willingness to work in the service of advancing a populist agenda than do those who do not feel that their anger is shared by their nation as a whole (Gaffney, Hackett, Rast, Hohman, & Jaurique, 2018; Hogg & Gøtzsche-Astrup, 2021). Because populist movements are often (but not always) led by a charismatic figurehead (e.g., David Duke, the former Ku Klux Klan leader; Marine Le Pen of the French National Front, Geert Wilders of the Netherlands' Party for Freedom), the role of leadership in such movements is key to their success and their demise (Crano & Gaffney, 2021). Economic uncertainty and uncertainty over a changing society that might leave behind a once majority class (see Inglehart & Norris, 2017) can engender support for strong-style leaders, which trend toward authoritarianism. Their aggressive methods are attractive because they are conducted seemingly on behalf of the movement and offer a clear-cut definition of the outgroup.

The role of uncertainty in promoting ingroup sociability and outgroup hate is well documented in the social psychological literature (for a review, see Hogg, 2021a, 2021b). The desire for structure (e.g., Kruglanski & Webster, 1996; see also Kruglanski & Ellenberg, this volume) and to reduce uncertainty (Hogg, 2007; Van den Bos, 2009) makes cohesive, insular groups highly attractive. When such groups promote radical behaviors as a legitimate means for goal attainment, members are willing to go to extremes on behalf of the group (Kruglanski, Burson, & Soll, 2013). Again, this can be viewed as an extreme form of intense sociability on behalf of the collective. However, its destructive potential against outgroups shows how the combination of radicalized norms and uncertainty can produce a destructive ideology.

Extremist movements are to some degree populist movements (e.g., Hogg, 2021b; Hogg & Gøtzsche-Astrup, 2021), and are almost always characterized

by conspiratorial thinking (Douglas & Sutton, 2018). When people feel that they lack economic and social control, they adopt a mindset, which provides explanatory power in a confusing world that they often believe is out to get them (van Prooijen & van Vugt, 2018). Conspiratorial thinking, when reinforced by ingroup sociability, provides a clear worldview, which casts the ingroup as victimized by the government, elites, and other nefarious entities (or simply "them"). This can motivate violent action on behalf of seemingly outlandish beliefs. The belief that the U.S. Democratic Party and Hollywood elites traffic children for sex can lead followers to violent action in an attempt to remedy the problem. The belief that some people contain the DNA of lizards can lead a father to slay his own children. And the belief that a rigged election cost Donald Trump a second term can lead a violent mob to engage in collective action to attack police officers, government officials, and politicians. The last example shows the power of echo chambers that restrict incoming information to only those who are "one of us" in promoting the false perceptions of a large ingroup majority (Kim, 2019). This form of extreme ingroup sociability is marked by a consensual understanding of the world. If we all believe it, it must be true. Those who do not believe must be against us.

In politically polarized times, partisans exist in two realities. Each reality is fully supported by information from those with whom they have developed sociable relationships: Ingroup members. Any critique is "fake news" and is cast as coming from the outside and having ill intentions toward "us" (Hornsey & Imani, 2004). Echo chambers fueled by a restrictive confirmation bias sustain lack of intergroup communication, fuel polarization and conspiratorial thinking (see Barberá, Jost, Nagler, Tucker, & Bonneau, 2015). Although leaders have the ability to positively shape norms and even build relational identities across groups (see Rast, Hogg, & Van Knippenberg, 2018), current events suggest a trend toward divisiveness and lack of willingness to extend and sociability across the aisle to the outgroup.

Concluding Comments

Sociability is typically considered a fundamental and essential part of being human. It is often conceptualized in the psychological literature as a personality attribute or behavioral property that varies between people (some of us being more sociable than others) and is thus a focus of research on interpersonal processes and close relationships. We do not disagree with much of this; but we do feel that there is more to sociability and it can be viewed slightly differently through the lens of the study of group processes, intergroup relations, and social identity dynamics.

Who one is sociable with, and why, may be contextually configured by the contours of group membership, the relations within and between groups, and the motivations associated with self-definition as a group member. Alongside

the benefits of sociability, we suggest that there is also a dark side to it. People can simultaneously be authentically sociable regarding their ingroup but only inauthentically sociable (or entirely anti-sociable) vis-à-vis outgroups. Maybe these two sides of sociability are inextricably linked. Ingroup sociability builds cohesion, solidarity, mutual trust, and a sense of oneness; outgroup lack of sociability can build distrust, hostility, and even hatred.

In this chapter, we defined sociability through the lens of social identity theory. Our analysis suggests that motives for group identification play a significant role in how groups express sociability. Sociability and ingroup identification can provide a safe haven for stigmatized identities and promote favorable self-esteem, and even health and well-being. Moreover, particularly when a representative leader is at the helm of a well-functioning group, collectives can positively change the world. However, the same processes, which reduce uncertainty and provide clearly defined and valued identities, can promote a dark and extreme form of sociability—one that fosters intergroup animosity. The groups in which people reside and the cherished social identities and patterns of sociability contain keys for understanding human advancement and destruction (see also Baumeister & Maranges, this volume).

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PART 3

Sociability and Sharing Knowledge

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9

TRAFFICKING IN INFORMATION

The Sociable Agent, Shared Reality, and the Limited Role of Truth

Roy F. Baumeister and Heather M. Maranges

Abstract: The human self has multiple functions that are generally absent in other species, including acting as a moral agent and an economic one. This chapter and talk will focus on the self as information agent. Human societies create a shared body of information that is the basis for much activity and interaction, and considerable time and effort are put into maintaining this shared information, which is a key component of culture and often a driving force behind both cultural stability and change. Eight functions of the information agent were outlined in a recent article, only the first of which (curiosity, or the drive to acquire new information) is found among other species. The other seven comprise communicating one's knowledge and mind contents to others; passing along information obtained from others; criticizing and refining information; maintaining shared reality, which sometimes entails prioritizing consensus over accuracy and truth; and the more advanced ones, including disseminating false information, withholding information strategically, and working with others to shape the collectively shared understanding. The latter includes censorship of scientific findings and theories.

That humans are social animals has become a truism. Thinkers from Aristotle to Aronson have made this point, and modern biological research has backed it up.

Yet human sociality is different from that of most social animals. The differences led one of us to characterize human beings as essential *cultural* animals rather than (merely) social animals (Baumeister, 2005). Classifying humans (correctly) as *social* animals emphasizes how they are similar to a great many other social animals. Those similarities are genuine and important. But saying humans are cultural animals highlights what is unique and special about humankind. To

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be sure, several dozen other social species have some degree of culture, in the minimal sense of learned behavior shared through the group and passed on to further generations. De Waal (2002) has made this point effectively. But as de Waal readily acknowledged, these bits and beginnings of culture are a far cry from the human version. These are mostly minor and limited patterns of behavior, and the members of the species who do not participate suffer no penalties and can dispense with them entirely. In contrast, culture is part of daily life for all humans, permeating and transforming nearly everything they do—even altering such basic animal processes as eating, sex, and health. Humans are the only *essentially* cultural animals.

This chapter focuses on one aspect of the human self that is not only centrally important but also almost unprecedented in nature and evolution. The human self is what can be called an *information agent* (Baumeister, Maranges, & Vohs, 2018). Like other social animals, it engages in frequent interactions with other members of the species. Unlike them, many of those interactions consist of *trafficking in information*. The human animal not only collects information but also frequently exchanges information with others. It works together with other humans to build a stock of common knowledge.

We adopt the term "doxa" from the sociologist Bourdieu (1977) to denote the collectively shared beliefs and knowledge. The original definition invoked "that which goes without saying," which is hugely important in communication. Something goes without saying because both people know it and know that each other knows it (and perhaps further recursions). Shared implicit understanding makes communication much more efficient and effective.

Information agency adds a crucial but neglected aspect to theory of self. That the human self is a moral agent has long been recognized: It makes choices based on moral principles and knowingly cultivates a morally virtuous reputation. Economists have likewise elaborated the understanding of the self as economic agent. It calculates gain and loss and alters behavior so as to maximize resources, except when being irrational (which happens far more often than economists like to think). There may be other aspects of human agency also. But being an information agent is important and central to what the human self is and does.

As scientists, we naturally assume that the topmost goal of collecting information is to get at the truth. But it is essential to question whether this is in fact the topmost goal of creating a shared worldview across a group or society (see also Krueger et al., this volume). In laboratory studies of group problem-solving tasks, there is a reassuring pattern that has been labeled "truth wins" (e.g., Kirchler & Davis, 1986; Laughlin & Ellis, 1986). That is, when group members initially disagree and discuss, the member who has found the objectively correct answer tends to prevail, and others gladly switch from their wrong answer to the correct one. Outside the laboratory, however, truth may prevail rather less reliably (see also Fiedler & Hütter; and Kruglanski & Ellenberg, this volume).

Theory and Background

Most creatures live solitary lives, but social animals capitalize on the advantages of group action so as to survive and reproduce better. Sociality is thus a highly adaptive step forward in evolution (see von Hippel & Smith, this volume). Culture is another. Culture can be considered a better way of being social. Culture includes advanced forms of cooperation, such as economic trade and group planning. Central to culture, however, is the sharing of information (the doxa). Cultural societies build a collective understanding of both the physical and the social world.

Most animals do not build shared understandings because, lacking theory of mind, they assume that their experience of the world is fully objective and therefore already shared by all actors in it. Having lost the solipsistic illusion that everyone has the same awareness, humans seek to rebuild consensus by sharing information.

The human self thus became an agent for whom exchanging and discussing information is a major component of social life (Baumeister et al., 2018). As far as we know, in most human cultures, social interaction begins typically with an exchange of subjective self-reports (e.g., "How are you?"). Even if such greeting-inquiries are often dismissed with minimal disclosure (e.g., "Fine, thanks"), the ritual greeting does affirm the central importance of informational exchange among humankind. It acknowledges that exchanging reports on one's thoughts and feelings is central to human sociality.

Building a doxa facilitates communication. Each new fact of information does not have to be explained from the ground up. Instead, it merely needs to be integrated into what is already known collectively. There are other advantages, including facilitating progress. Each new generation can build on what it learned from previous generations, rather than starting over from nothing. Joint action can occur rapidly and efficiently, as opposed to needing to work everything out starting from no common understanding.

Indeed, one can make the case that communication is the original and definitive human trait. Some animals have limited communication, but none has language, whereas all known human societies do. Humans have long defined themselves in terms of their high intelligence, as reflected in their self-conferred name *homo sapiens*. It was not difficult to develop a speculative evolutionary account, in which big brains evolved because of solving problems, and then the upright posture was adopted to protect the brain, thereby freeing the hands for tool use, and so forth. However, the discovery of Australopithecus indicated that upright posture preceded the expansion of brain size (see also von Hippel & Smith, this volume). On the basis of the work by Suddendorf and Corballis (1997) and others, Baumeister and Masicampo (2010) proposed that the upright posture was at least partly selected because it freed the hands for communicating. (Gestural communication long preceded speech, and other great apes have

been seen attempting to communicate with each other by gesture.) Crucially, communication created a social environment in which high intelligence would be advantageous. After all, brains consume many calories, and for many animals evolving toward greater intelligence would likely increase the risk of starvation, because increased intelligence would not bring in enough calories to offset the increased demand for them by the larger brain. Communication vastly increases the amount of information available to the brain. It can therefore do more, including improving the acquisition of more resources—thus being able to sustain the higher caloric load.

As scientists, we are prone to the assumption that discovering the truth is the purpose of thinking and intelligence. But truth is not the only important aspect of a doxa. There are two other important dimensions (Baumeister et al., 2018). One is consensus, that is, the degree to which the knowledge is held in common by members of the group. The other is conceptual coherence, which takes the form of completeness and consistency. The various bits of information in the doxa should ideally not contradict each other (consistency), and they should explain all that needs explaining (completeness). Truth may be sacrificed for either of those other goals, as we shall see.

The Information Agent in Action

This section elucidates what the information agent does. These are intended to flesh out a key aspect of human sociality. The human self traffics in information. Trafficking is here unpacked so the nine various aspects and activities are explained.

Obtaining Information

The first of the nine activities of an information agent involves seeking and acquiring information. This is not sufficient to explain what the human self does—though it is probably the full extent, or almost, to which other animals are involved in information. Consider the now taken for granted fact that infants gaze longer at novel versus known stimuli—humans are curious from the beginning. Curiosity is a basic drive to gain information about the environment. It probably goes back far in evolution. The usefulness of brains lies in their capacity to process information, so obtaining information is an important first step. A drive to obtain information would clearly be adaptive.

Loewenstein (1994) proposed that curiosity is stimulated by a subjective information gap. It thus reflects some degree of metacognition: The person knows that there is something specific that the person does not know. Whether human curiosity surpasses that of other animals is unclear, though the greater metacognitive powers of human beings would suggest that human curiosity is in some fashion more advanced. Still, animals do seem to exhibit selectively specific curiosity.

Curiosity may exist and operate in solitary minds. But in human social life, it takes on additional dimensions. In particular, one seeks to fill out gaps not just in one's personal knowledge but also in the group's shared knowledge (the doxa). Gaining information that is valuable to the group can bring prestige and other social rewards. Even just knowing some "juicy gossip" can bring positive emotions because one anticipates the reward of telling it to others. The juicy part means that it is relevant to shared understandings that are highly interesting to many.

Communicating Information

With this second aspect of the information agent, the analysis moves beyond individuals and nonhuman animals. The human information agent not only wants to get information: He or she wants to share it with others, specifically with other members of the group (see also Prislin & Crano, this volume). Suddendorf's (2013) analysis of the evolutionary gap between humans and other primates emphasized the desire to communicate the contents of one's mind to others as one of the two most profound psychological differences (the other being recursion).

The emphasis on sharing the contents of one's mind points to a preference for sharing information about oneself. Tamir and Mitchell (2012) showed that people were willing to sacrifice small amounts of money in order to answer questions about themselves, as opposed to questions about Barack Obama (who was then president) or trivia questions. They also showed that disclosing information about oneself increased brain activity in regions associated with neural reward. Apparently, people like to talk about themselves.

People also desire to share information that is socially relevant and helpful to other group members. Feinberg, Willer, Stellar, and Keltner (2012) showed that people felt better after giving relevant information to someone (a fellow student) who was about to play economic games with a new partner, and they would even give up some of their own reward money to send this information (see also Regan & Lyubomirsky, this volume). The information was that they had seen the impending partner play these games in a selfish, uncooperative manner on previous trials with other people, so the information would warn the new person not to be taken advantage of. The researchers explained their findings in the context of gossip theory. Dunbar (1996) explained that human societies are founded on cooperation, but not everyone is cooperative, and in a large group, it is impossible for an individual to keep track of everyone in terms of who is versus is not cooperative. Gossip thus enables members of the group to share information about this. In any case, it is clear that people are often motivated to share information about other people.

Telling others what one has learned is a valuable step toward building a doxa (i.e., a common stock of shared information). Many animals seek to learn about

the world, as was noted in the previous section, but for humans that is not enough. They feel the urge to share what they have learned with each other.

Relaying and Circulating Information

The first two steps provide the beginnings of a socially constructed stock of information. Members of the group are motivated first to seek out new information and second to tell others what they have learned. So far, so good. A problem arises with large groups, however. Each time you learn something, you want to tell it to others—but if there are 150 people in your group, you might have to repeat the same news 150 times, and that may require you to approach people you are not friends with or of whom you are possibly afraid.

The systemic solution to the problem of disseminating information through the group is that people are motivated to pass along information garnered from others. Someone who discovers something therefore only has to tell a few people. Assuming they pass it on to a few others, who in turn pass it on further, information can spread effectively through the group and become part of the group's shared knowledge. One does not have to say the same things 150 times.

Social psychology has a long history of research on how people pass along information. The seminal studies by Allport and Postman (1947) on rumor transmission observed how information typically changes as it is relayed from one person to another. The most common changes involved simplification, such as omitting details and complications. They found that up to 70% of details were lost in the first telling of the rumor. Other changes assimilated the new information to what was already known, including expectancies, stereotypes, and other preferences. These can bias and distort the information—but they help it fit in well with what is already collectively known and believed. Thus they help build and sustain the doxa. Crucially, this process sacrifices accuracy in favor of consensus. The full complexity is lost in favor of a message that is easily repeated and easily grasped (see Krueger et al., this volume).

Probably the most systematic and institutionalized form of relaying information involves teaching. Teachers typically operate as links in the chain of passing along information: The teachers learn their information from some people and pass it on to other people, especially children and young adults who have not yet mastered the common stock of knowledge.

Teaching appears to be universal, or close to it, among human societies. As far as we know, all countries in the modern world have schools for disseminating common knowledge to children. Even simple human societies that do not have schools practice various forms of teaching, by which adults communicate knowledge to younger and new members of the group. Although nonhuman animals sometimes communicate information to others, such as by emitting a loud cry upon noticing the approach of a possible predator, it is far from clear that they do this with a deliberate intention to inform their fellows. Reviews

of animal communication have concluded that there is no solid evidence of intentional teaching of generalizable knowledge (again, unlike all known human societies) (Csibra & Gergely, 2009; Hoppitt et al., 2008). Intentional teaching is thus a uniquely human trait, or very nearly. It is certainly a central part of the informational agency of the human mind.

Part of the discrepancy between humans and other animals with respect to teaching has to do with the much-advanced human mastery of theory of mind. Humans can mentally represent the mental states of others and hence recognize explicitly that they themselves know something that another person does not. Teaching thus provides a social extension of the metacognitive aspect of curiosity. The curious person discerns a gap in his or her own knowledge—and seeks to fill it. The teacher discerns a gap in someone else's knowledge—and seeks to fill it. Note that that advance, which no doubt required sophisticated new brain hardware, is well suited to building a doxa. People can spot that someone else lacks crucial pieces of the shared knowledge and quickly fill them in.

Gossip has much lower prestige than teaching, but it likewise helps circulate information through the group. Gossip is often regarded as a form of aggression that seeks to discredit rivals and trash their reputations, and no doubt sometimes that is a fair assessment (e.g., Baumeister, Reynolds, Winegard, & Vohs, 2017). It may however serve other, more positive functions than aggressive ones. Often, gossip focuses on norm violations and their consequences, and so gossip helps circulate information about what the group's norms are and how dangerous it is to violate them. Many people report learning useful lessons from gossip (Baumeister, Zhang, & Vohs, 2004). Gossip builds and strengthens the doxa, though it sometimes sacrifices truth for motivated consensus.

Refining and Improving the Information

Obtaining information, telling others, and passing it along are important ways to build the shared social knowledge, and may be sufficient to establish and build (?) a doxa. But the human information agent does more than that. Not all information is equally valid, of course. Some is simply wrong. Other information may be inconsistent with what is already known, which suggests that either the existing knowledge or the new knowledge needs to be adjusted.

In short, information agents refine and improve information. When new information arrives, they subject it to critical scrutiny. They integrate it with what is already known (whereby inconsistencies may emerge). They raise alternative interpretations or point to incompatible evidence.

The broader point is that the human mind is not so much designed to find the truth on its own as to build knowledge collectively, with ample social interaction. Many people make errors in observation and interpretation. Yet by telling their thoughts to others, they expose their ideas to social correction (see also Fiedler; and also Krueger et al.'s discussion of Bernoulli's "law of large

numbers," this volume). Indeed, an influential article by Mercier and Sperber (2011) contended that reasoning is not so much for finding truth as for arguing. Various thought patterns that have been regarded as flaws in the knowledge-acquisition system, such as confirmation bias, do not look like flaws if the purpose is to argue for one's side. When people argue, they, or at least the onlookers, can evaluate both positions and decide which is more likely correct. To be sure, "correct" often means consistent with collectively shared beliefs, so again truth may take a back seat to consensus. As classic historical examples, many discoveries such as the earth's rotation around the sun and circulation of the blood in the body were initially rejected, with considerable hostility, because they clashed with widely held religious views. Still, at the time, there was strong consensus that the religious doctrines were true, and when new positions were inconsistent with them, the default assumption was that the new views must be wrong. It reportedly took the Church 300 years to admit that Galileo had been right in upholding the Copernican theory about earth's rotation around the sun. Such examples are powerful instances of consensus trumping truth.

The Galileo example suggests that within a society there may be alternate versions of consensus, indeed alternate doxae, that coexist and compete. During those 300 years, the scientific community was building a consensus that the earth revolved around the sun (even using such frameworks as basis for correctly predicting eclipses), while the religious community maintained the view of earth as immobile center of the universe. No doubt in modern society, there are multiple doxae in place. This development may be accelerating now that online news sources have embraced various political extremes so that people can live in their own bubble or so-called echo chamber, nicely shielded from contrary views.

The human information agent is therefore not simply a creature who uncritically accepts and passes along information from others. Indeed, people become quite astute at evaluating the quality of information obtained from others and evaluating which other people are the most credible sources of information. These skeptical evaluations of others are in evidence already among 4-year-old children (Koenig, Clément, & Harris, 2004; Koenig & Harris, 2005). DiFonzo and Bordia (2007) concluded that as rumors about an organization spread among members of the organization, such as possible changes in top management or overall prospects, they tend to become more rather than less accurate. The improvement in accuracy occurs because the high importance attached to such issues motivates people to question the rumors and look for converging evidence.

Shared Reality Effects

In theory, a group could develop common knowledge without knowing that is what they are doing. Presumably many social animals do just that, insofar as they all come to understand certain key facts about their physical environment

(e.g., the boundaries of their territory) and their social environment (e.g., the status hierarchy). Bees famously spread information about food by dancing, so a group of bees may end up all with the same information—but again it is highly unlikely that any of them, even the dancer, has any inkling that information is being shared and a common stock of beliefs is being created. The whole process is hard-wired, so to speak.

In contrast, humans are often quite aware that that is what they are doing. They are motivated to build up a set of beliefs that is shared with others. Both conscious and unconscious processes contribute to building it. People are highly attuned to what others know, and to conflicts and disagreements between their knowledge and others' knowledge.

Shared reality begins with shared attention. Shteynberg's (2015) review of shared attention findings notes that shared attention processes can be observed even in human babies and continue to be influential throughout life. People are highly sensitive to what others are noticing. When multiple people attend to the same thing, they remember it better. Jointly attended goals are pursued more zealously and with greater success than solitary goals. Joint attention intensifies both emotional reactions and cognitive processing. Moreover, according to the Theory of Collective Learning (Shteynberg, Hirsh, Bentley, & Garthoff, 2020), the fact that humans experience a shared reality means they not only learn from others, but they also learn *with* others to collaborate in uniquely human ways. Collective attention (joint attention across group members) creates common knowledge that facilitates cognitive alignment, representation of common knowledge, which facilitates social cohesion, and application of common knowledge, which facilitates cognitive collaboration.

Shared reality has been the focus of a program of research by Higgins (2019). In the first studies, participants altered their descriptions of a stimulus person to an interaction partner based on the preferences and biases of the partner. They subsequently remembered the stimulus person as they had described him, thus in biased fashion, rather than in line with their original impression (Higgins, Rholes, & Jones, 1977). The authors described this as "saying is believing." The implication is that consensus trumps accuracy: People cling to the shared belief rather than the more accurate original one. Subsequent work has confirmed that interpretation. People show these effects more strongly with ingroup than with outgroup partners, and if the partner fails to get the message (indicating that reality was not shared, that the consensus has failed), the participant reverts to the original, more accurate impression (for review, see Echterhoff, Higgins, & Levine, 2009).

Evidence for building consensus is evident in research on anticipatory change. Many studies showed that attitudes can change after group discussion—but then researchers discovered that attitudes begin changing before the discussion. Typically, people shift their attitudes to be closer to those of their interaction partners, or sometimes toward more moderate and neutral attitudes (e.g., Cialdini, Levy,

Herman, & Evenbeck, 1973). Both shifts suggest that people modify their inner opinions so as to be more compatible with those of their group. This occurs prior to getting any new information. It's all about consensus, truth is less relevant.

The appeal of shared knowledge is evident in the remarkable finding that people feel more like they know things that they in fact do not know, as long as other members of their group know those things. Sloman and Rabb (2016) showed that people rate their understanding higher insofar as other group members understand it. This exemplifies the sociality of information: Apparently, having the information in the group is almost as good as knowing it oneself, even when one does not.

The classic studies using the so-called hidden profile paradigm likewise exemplify the power of shared reality (e.g., Stasser & Titus, 1985). Groups were tasked with a hypothetical hiring decision. All members received three bits of information favoring one candidate and a single bit favoring the other. Crucially, the three were the same for all group members, whereas the single bit was different for each of the eight or so. The total profile of information thus favored the latter candidate, but it was hidden in the sense that the preponderance of information favoring the latter would only be found if the members brought up their separate knowledge. Typically, the group failed to discover the superior candidate, because they mainly talked about their common knowledge of the three bits favoring the lesser candidate. The ostensible purpose of committees is to bring in different perspectives and knowledge, but that failed repeatedly. Committee members apparently were more focused on building consensus than on bringing up information they alone knew.

It is also revealing that the hidden profile failure seems to increase with age. Normally, one expects children to get better at finding the truth as they get older. Gummerum, Leman, and Hollins (2014) used the group decision task with children. The 7-year-olds tended to bring up their idiosyncratic information and therefore reached the correct decision. In contrast, the 9-year-olds mainly talked about what they knew in common and hence made the poorer decision. The 9-year-old thus showed the adult pattern more than the 7-year-olds. But the adult pattern favored consensus over accuracy. Apparently, as children grow up, they learn to prioritize having the same thoughts and beliefs as other group members, rather than asserting their individual perspective and reaching an accurate decision.

This shared reality is essential to valuation and maintenance of social, economic, and educational institutions that constitute culture. Take, for example, money. Although some currencies are backed up by gold, this has become increasingly rare. Instead, the shared understanding of the purpose of money imbues it with value: money is the reward for work, a common commodity through which work is made capital, and therefore should be accepted by anyone in the cultural group. This facilitates division of labor and broad economic

cooperation. Indeed, Shteynberg, Radua, Kempton, and Mehta (2021) demonstrate that respect for and valuation of a group's currency is driven by knowledge that many other group members know about the currency rather than by liking of the currency. Across three experiments, people (students) were introduced to the new currency of their ingroup (university), rated how much they liked the currency, and read that they should tear up a dollar bill of this new university currency at the end of the experiment. Participants also reported their negative feelings about tearing it up. The more ingroup members who knew about the money, the slower and more hesitant people were to destroy it. The worse they felt about having to destroy it, the slower they were to destroy it. Importantly, hesitancy did not depend on how much they liked the money, but it did depend on the extent to which they identified with the ingroup. That is, people who had high, versus low, ingroup self-esteem were slowest to destroy the money.

Withholding Information

The urge to tell one's information to others may be basic and widespread, but nevertheless there are occasions when people strategically withhold information. People seem to recognize that knowledge confers power and that sometimes it serves their interests to keep knowledge to themselves. Sometimes this takes the form of protecting ingroup knowledge from outgroups. There is a long history of protecting group secrets from others, such as when soldiers or spies are captured by enemies and wish to keep vital information from falling into enemy hands. Likewise, organized crime groups enforce a strong code of silence to prevent important knowledge from being revealed to law enforcement groups when members of the group are captured and imprisoned.

Although torture may have originally been used for entertainment, it has long been used to extract information from reluctant captives. How effective it is for that purpose is debated, though certainly it sometimes works. Moreover, it has also been used to obtain false but useful information, such as by coercing false confessions of wrongdoing that can be used to justify government policies.

Even before they can actively deceive another agent by providing false information (which occurs around 5 years old), children as young as 3 years block a competitor from seeing information that would benefit the competitor (Peskin, 1992; Sodian, 1991). For example, when interacting with a character who expressed preference for one particular sticker that the participating child also preferred, only 29% of 3-year-olds knew to lie or be silent about their sticker preference, whereas 87% knew to stand in front of the stickers (Peskin, 1992).

People also sometimes withhold information as a way of maintaining power in the group. In large groups where there are rivals and full cooperation is not the norm, it is advantageous to have more information than others. In that context of knowledge being power, people would lose their advantage by sharing

their information (De Dreu, Nijstad, & van Knippenberg, 2008). An experiment by Bălău and Utz (2016) showed that information sharing is often strategic. Participants imagined being department heads at an event-planning company and made decisions about what information to share with subordinates. They withheld more information after being primed to think of power as opportunity to reach goals than when thinking of power as ethical responsibility.

A review by Webster et al. (2008) focused on withholding information in (mainly business) organizations. This is recognized as a common problem, with at least 11% of knowledge requests not fulfilled, and it can impair the organization's effectiveness, especially when some people do not have the requisite information to perform their roles effectively. They indicated that there are multiple strategies such as "playing dumb" (feigning ignorance when asked for information), evasive hiding (offering other information than is requested, or stalling), and rationalized hiding (asserting they are prohibited from disclosing confidential information). The most common reason for withholding information is to increase one's power, such as by using information for one's own strategic advantage over organizational rivals. Other motives include territoriality, reciprocation (if others have rebuffed one's requests for information), and organizational culture.

Withholding information can serve other, more basic goals, too. Like other animals, some of humans' most basic motives are to find and maintain relations with suitable mates, which necessitates intrasexual competition. Unlike other animals, humans engage in information agency to pursue those mating goals. For example, women are not only more likely to spread reputation-damaging information (e.g., sleeps around, has an STD) about an attractive romantic rival, but they are also more likely to withhold reputation-enhancing information (e.g., donates to charity, has a high IQ) about her (Reynolds, Baumeister, & Maner, 2018).

Disseminating False Information

People talk (and write) a great deal, thus sending out much information. Some of that information is false. Some of the false information may fall in the category of honest mistakes. In that case, the information agent's motivations and actions resemble those already covered, only that the person is unwittingly spreading false rather than true information. Of greater interest is the deliberate dissemination of false information. The everyday term for it is lying. But it can be subtler than that.

Lying is quite common. Diary studies by DePaulo, Kashy, Kirkendol, Wyer, and Epstein (1996) found most people lie at least once per day, and a student sample averaged two per day. One in every three to five social interactions contained a lie, and lies were told to 30–40% of interaction partners. At a grander level, false information is often disseminated by governments and news

organizations, and some think this has been increasing. Former President Trump was widely lambasted for making countless false statements, presumably many of which involved deliberate lies. For example, his early assurances that masks were not needed to prevent COVID infections were presumably never grounded in reality. Meanwhile, the news media have also lost credibility despite their ostensible commitment to reporting facts. The *New York Times* had built a reputation for truthful reporting over decades but may have switched to a different business model. Gurri (2021) estimated from Google searches that the *Times* published over 3,000 stories developing the false story that the president was colluding with the Russian government to undermine American democracy. Ultimately, formal investigations into these accusations yielded no evidence.

Insofar as the information agent participates in the process of building a socially shared stock of knowledge, lying is counterproductive for the group, because it results in falsehoods being incorporated into the doxa. However, lying may be advantageous to the individual.

A thoughtful analysis of deception by von Hippel and Trivers (2011) noted that the efficacy of deception will create a selection advantage for those who can spot liars and hence not be swayed to do what they want. This is analogous to a military arms race, as both liars and targets (marks) have strong incentives to keep up and gain an edge. The liar needs to maintain credibility to increase chances that the target will believe the lie. Meanwhile, the target benefits by being able to spot lies and therefore not be swayed to act on the basis of false information. This is especially true in an influence situation, because the target of lying-based influence may be swayed to do something to his or her disadvantage. (After all, if influencers could get others to comply with their wishes by telling the truth, they would presumably do so. Targets are best served by doing what is best for themselves, which is likely based on the objective truth rather than the lies that benefit the liar, which is why they lie.)

Recent studies on dishonesty by Ariely and colleagues (Barkan, Ayal, & Ariely, 2015; Mann, Garcia-Rada, Houser, & Ariely, 2014; Mazar, Amir, & Ariely, 2008) have concluded that people will indeed lie, but only to a limited degree. Their conclusion is that people lie only up to where they can continue to regard themselves as basically honest people. We note, however, that self-assessments have no direct payoff in terms of survival and reproduction. Combining this view with the von Hippel and Trivers (2011) position makes this plausible, however. People restrain their lying, not ultimately for the sake of self-regard, but for the sake of social esteem. Others need to respect them. Becoming known as a liar will reduce one's ability to influence others in the future and may compromise one's welfare in other ways. Hence they need to sustain respect by being known as honest and truthful, which is facilitated by being mostly honest and truthful and, when that strategy is not compatible with one's goals, being defensible. We think what matters is not self-appraisal but reputation (and potential reputation).

The reality of lying again indicates that contributing to collective truthful understanding is not the first goal of the information agent. Lying is an abuse of the system of information sharing, but an eminently understandable one. In a sense, it is primitive version of what the next two sections cover.

Suppressing Information

If lying is a self-serving but destructive counterpart to sharing truthful information, then suppressing information is the self-serving but destructive counterpart to relaying information obtained from others. A doxa is built up by many people sharing information (along with critical refinement). However, some groups may benefit by keeping certain information from entering into the common stock of shared knowledge.

Undoubtedly the most common form of information suppression occurs when influential groups gain resources by means of widely shared beliefs that affirm their right to claim resources produced by others. In previous centuries, religious elites often stifled dissent, sometimes with brutal methods such as torture and burning heretics alive (as punishment for having dissented and as warning to other potential dissenters). Particular religious views helped maintain compliance with the existing power structure (both religious and political), and contrary views held the danger of weakening the elites' hold on power and resources. In the 20th century, suppression of information and dissent came to be exercised by political authorities without the religious aspect. Even today, in some countries, criticism of the government carries the risk of imprisonment and perhaps worse.

Scientists pride themselves on pursuing the truth, but suppression of scientific findings has often appealed to religious or political groups. Here in the 21st century in the United States, the censorship and suppression of scientific findings continue to occur and may indeed be increasing.

In modern life, suppression of information is highly specific, targeting groups and ideas that clash with the approved or "politically correct" viewpoint. But suppression of information in general occurs too. Most histories of Western civilization celebrate Guttenberg's invention of the printing press, which enabled information to be reproduced and disseminated far more rapidly than any previous system. (The main previous systems were lectures, literally readings, in which the professor would read from a book while the students would write down what he said for themselves; and scribes, who would copy handwritten books with their own hands (hence, *manuscripts*), thus gradually producing additional copies. The printing press made both obsolete, though university lectures continue in various formats.) Historians are right to remark on the importance of this innovation. What is less often reported, however, is that many countries banned printing presses. The ruling elites saw the potential for information and knowledge to spread through their societies—and immediately took steps to stop that from happening. The underlying sentiment was presumably that if the

masses gained knowledge, this could destabilize the society and eventually compromise the comfy position of the elites. We think this was a quite reasonable and correct fear. Nevertheless, banning the printing press has been an impressive rear-guard action to stop technology from changing society by spreading information. Ultimately, trying to stop the spread of either false or true information seems futile, though as censorship gains favor in modern society, abetted by technology, such efforts may be quite successful for periods.

Controlling the Doxa

The final function of an information agent involves exerting control over the group's stock of shared knowledge and belief. To be sure, not all information agents do this, but some do it aggressively—especially in conjunction with other people. Individuals may occasionally contribute some important insight, fact, or finding to the collective stock of knowledge, but exerting long-term control over it is typically the work of social groups. It may be a form of group lying—or of groups working together to uncover collectively shared false beliefs.

In a large modern society, power often depends on widely shared beliefs. Swaying the group's beliefs can thus greatly benefit a subgroup. For example, European history saw many instances of mobilizing public opinion against the Jews, including the pre-Crusade pogroms, the perennial suggestion that the Jews were causing the bubonic plague by poisoning wells, blaming a Jewish conspiracy for causing Germany to lose the First World War, and various hostile purges during the Soviet Union's reign. The ruling parties often benefited, at least in the short run, from these manipulations of public opinion. Benefits including confiscating the Jews' wealth and deflecting blame for domestic problems onto scapegoats.

Conclusion

Many activities of the human self reflect the social uses of information. People not only seek information, but they also share it, critique, refine it, and sometimes use illicit informational strategies such as spreading false information, withholding or suppressing information, and seeking to manipulate the group's shared beliefs for their own advantage. Among humans, information is not merely stored in individual minds but in the group, as such being passed along to new members. The doxa (shared beliefs) is not solely focused on the truth, though truth is often favored. People also seek consensus (most of the group agrees) and completeness, and sometimes truth may be sacrificed for these other social goals.

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10

SOCIABILITY AND ADVICE TAKING

Chances (and Pitfalls) of the Wisdom of Crowds

Klaus Fiedler and Mandy Hütter

Abstract: The wisdom of crowds (Surowiecki, 2004) highlights the superiority of collective over individual knowledge. The very demonstration, and the conviction, that an average estimate of many people is more accurate than an individual estimate (Galton, 1907) and that collective decisions are more likely correct than individual decisions is not only consistent with Bernoulli's (1713) law of large numbers but also lies at the heart of democratic systems. It is also central to sociability (getting along with others) conceived as a precondition of all collective problem solutions. In this chapter, we review novel research and theorizing on advice taking—an intriguing paradigm of social rationality research. The pertinent literature and the reported findings corroborate the wisdom of crowds, the surplus value of social advice over mere anchoring, and the reality created through social conventions. Yet, findings also testify to dangers of taking misleading and biased advice for granted and point to metacognitive deficits in separating the wheat (valid advice) from the chaff (invalid advice). Finally, however, it is essential to understand the tradeoff between purely epistemic and sustainable social functions of adaptive social cognition, as evident from a discussion of the relationship between advisor and advisee.

Key words: critical assessment; debunking; wisdom of crowds; law of large numbers; advice taking; metacognitive myopia; advisor expectation

Introduction

Crucial to the basic idea of democracy is not only the commitment to legal equality, political freedom, and human rights but also the belief in the wisdom of crowds (Surowiecki, 2004). In democratic systems, each individual has

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exactly one vote, regardless of his or her origin, status, income, or appearance. This tallying (Dawes, 1979) or equal-weighting scheme of democratic decisions should not be solely understood as a gesture of fairness or ethical generosity, which has come to dominate a latent oligarchic conviction that letting the poshest or the smartest or the most handsome people decide would lead to better outcomes and more satisfaction. If that were really the case, then evolution would have long erased the luxury of democracy. It makes more sense to presume that democratic societies actually believe that it is very hard for even the best individuals to outperform the simple beauty of tallying, as prominently expressed in a seminal article by Galton (1907), "the middlemost estimate expresses the vox populi." Galton was amazed by the accuracy with which a crowd of farmers' average guess reflected the actual weight of an ox at a fair. That Galton considered the median rather than the mean to be the best estimate is fully compatible with a primitive tallying interpretation of "vox populi."

Thus, the notion of sociability—the topic of this volume—must not be confined to the positive tone of cheerful and gregarious social behavior or the rewarding feeling of getting along with others, although this is certainly a major function of socializing (Snyder, 1984). A major motive of our interest in other people is need for approval, social support, self-disclosure, and intimacy, finding an audience for self-presentation and reciprocation of prosocial and antisocial behavior (see also Dunn & Lok; and Gaffney & Hogg, this volume). Yet, an equally important motive is epistemological in nature. We need other people as information sources, advisors, epistemic authorities (Kruglanski et al., 2005), and social comparison partners that allow us to validate our own beliefs and opinions (see Baumeister & Maranges; Kruglanski & Ellenberg; and Prislin & Crano, this volume). Collective intelligence not only provides an opportunity to participate in knowledge acquired by others so that the wheel must not always be invented anew by every individual person. But it also constitutes a chance for several people (and groups) to jointly outperform individuals. After all, many people jointly see, hear, notice, understand, recall, and create more than what individual people can accomplish.

The Social Bernoulli Law of the Large Number

This belief in the superiority of collective over individual wisdom and problem-solving is an essential aspect of the sociability motive. It is reminiscent of Bernoulli's (1713) law of large numbers, which states that with increasing size of a sample, the average of a sample of estimates approximates the true value of the underlying population mean. Thus, even though single intelligence test items afford only a lousy measure of intelligence, with a single-item reliability of, say, 0.15, the summed reliability of 20 items of this kind reaches a reliability of over .80. This is a blessing feature of the probabilistic world, in which singular

diagnostic observations are of little value but aggregating over samples of several observations nevertheless allow for a reasonable level of diagnosticity.

As explained in Rosenthal's (1987) monograph on judgment studies, the same logic that holds for the reliability gained from increasing test length—measurable in terms of the common Spearman-Brown formula (Novick & Lewis, 1967)—also holds for an increasing number of judges (Einhorn, Hogarth, & Klempner, 1977). Indeed, a group of *N* judges or advisors can be considered as a test with *N* items, each of which contributes a highly restrictive portion of reliability. Yet, together, *N* persons' estimates or opinions can jointly reach a clearly higher accuracy level than individual persons' estimates. Because the unsystematic error components in individual judge's estimates are uncorrelated, they cancel each other out, whereas the valid components all reflect the same latent attribute, the visibility of which increases with increasing *N*. It is no surprise that research on advice taking lends support to this beneficial feature of the social version of the Bernoulli law, which is at the heart of the wisdom of crowds, and one of the drivers for human sociability.

Averaging Gains in Advice Taking

A long tradition of research on collective forecasting and advice taking corroborates the intuition that averaging the estimates provided by a growing number of *N* individual judges serves to increase accuracy (Yaniv, 2004). The accuracy increment gained from every added judge is however subject to asymptotic deceleration (Johnson, Budescu, & Wallsten, 2001). Aggregating a small number of two to six individual estimates is typically enough to maximize accuracy. The failure to further exploit the wisdom of more judges could reflect the fact that the disadvantage of increasing coordination loss soon exceeds the advantage of a shrinking standard error $\sigma/\sqrt{N}$ that can be expected from increasing *N*. Nevertheless, an averaging advantage was "demonstrated in diverse domains, ranging from perception (line lengths) and general knowledge tasks (historical dates) to business and economics (sales or inflation forecasts)" (Yaniv, 2004, p. 75; Zarnowitz, 1984). The lesson to be taken from research on advice taking is that averaging is the key to accuracy, whereas the egocentric tendency to give more weight to one's own opinion than to other available opinions is considered a chief obstacle for collective judgments (Yaniv & Kleinberger, 2000).

Thus, a long tradition of research on group decisions and opinion aggregations corroborates the superiority of collective wisdom—at least under reasonable boundary conditions specified later. Although the basic averaging advantage reflects a statistical truism (Bernoulli, 1713; Rosenthal, 1987), it is worth noting the practical value of advice taking in the context of many everyday tasks (Bonaccio & Dalal, 2006), such as estimating historical dates (Herzog & Hertwig, 2009), temperature (Harries, Yaniv, & Harvey, 2004), or calories of food items (Hütter & Ache, 2016).

Even within participants, accuracy can be enhanced by averaging the same persons' repeated judgments (Ariely et al., 2000; Vul & Pashler, 2008). This is particularly the case when the task instructions emphasize the need to make a second judgment from a different perspective, in a paradigm that Herzog and Hertwig (2009) labeled "dialectic bootstrapping." Thus, even a single person can exploit the "crowd within" (Vul & Pashler, 2008) by emulating the judgment of someone who sees the world from a completely different angle (but see Steegen, Dewitte, Tuerlinckx, & Vanpaemel, 2014). To illustrate this dialectic method, suppose you first make an estimate of the outcome of your favorite soccer team's next match from your own perspective and then from the perspective of an opponent team's fan. Dialectic bootstrapping means to improve your predictive accuracy by averaging over both antithetic judgments.

There are at least two good reasons why averaging over heterogeneous or antithetic opinions can be expected to be particularly informative. First, the averaging advantage (i.e., the cancellation of error variance through aggregation) only functions if individual opinions are stochastically independent. Combining conditionally dependent opinions of people who rely on each other or on the same information sources is like including the same items on a test twice or several times, foregoing all aggregation benefit. A decisively dialectic instruction to provide opponent or divergent estimates may function as a remedy against such interdependency. Second, dialectic task instructions may enhance the likelihood of bracketing (Larrick & Soll, 2006), making two estimates discrepant enough that the true value lies in between. Thus, if my weather forecast for a summer weekend is 30°C and a worst-case scenario leads to an antithetic second forecast of 15°C, most reasonable temperatures lie in between these two brackets. As Larrick and Soll (2006) have shown, bracketing by a second estimation implies that the average of both estimates will be closer to the correct value than the first estimate alone, unless the distance of the second from the first estimate is at least three times the inaccuracy of the first estimate. Thus, by taking both boundary conditions to heart, stochastic dependency of judgments and bracketing, it should be easy and straightforward for social individuals to benefit from the wisdom of crowds, exploiting one of the intrinsic benefits of human sociability.

How Sociability Can Undermine the Aggregation Benefit

Yet, the same harmony-driven, homophile motivation that facilitates social advice taking can prevent people from radically dialectic thinking and reasoning. Committed to their social bonds, groups, and identity constraints, people are not primarily oriented toward democracy (i.e., equal weighting of all opinions) or rationality (i.e., informative value of opinions). They rather tend to use others' opinions in a biased and egocentric way (Bonaccio & Dalal, 2006). For instance, they give more weight to opinions of people who share their own knowledge background, that is, of similar people influenced by the same media and

information sources. To use a term coined by Kahneman and Tversky (1973), they fall prey to an illusion of validity, increasing rather than reducing their reliance on redundant knowledge (Yaniv, Choshen-Hillel, & Milyavsky, 2009), because convergent opinions raise a sense of consistency and validity. They fail to recognize the informative advantage of independent and potentially conflicting opinions and overrate the spurious consistency of interdependent opinions. Instead, they tend to discount dissenting opinions that might contribute independent information (Harries et al., 2004). In a similar way, research on group polarization showed that opinions become more extreme and distorted as a direct result of more extensive group discussion—a clear case of increased sociability producing greater judgmental distortions (Forgas, 1977).

Pitfalls of Advice Taking

Not only the surplus meaning of genuinely social opinions but also the impact of superficial judgment heuristics on advice taking is evident from a series of four experiments conducted by Hütter and Fiedler (2019). Participants of a web-based experiment first provided their own best estimates of 18 historical dates. Then they were provided with another estimate (40, 42, or 52 years deviating from the correct year) that was either framed as a social advice ("Stefanie thinks that the year of the three emperors was 1848") or as a randomly chosen numerical anchor. Finally, participants could revise their estimate in light of the social advice or advice anchor, respectively. Dependent variables were the proportional accuracy gained from the advice and the weight of advice, defined as WOA = (initial estimate—final estimate)/(initial estimate—advice).

While the final estimates were generally much closer to the correct value than the first estimates (due to the anchors' relative accuracy), the surplus value of genuine social advice was compelling (see Figure 10.1). In the first experiment,

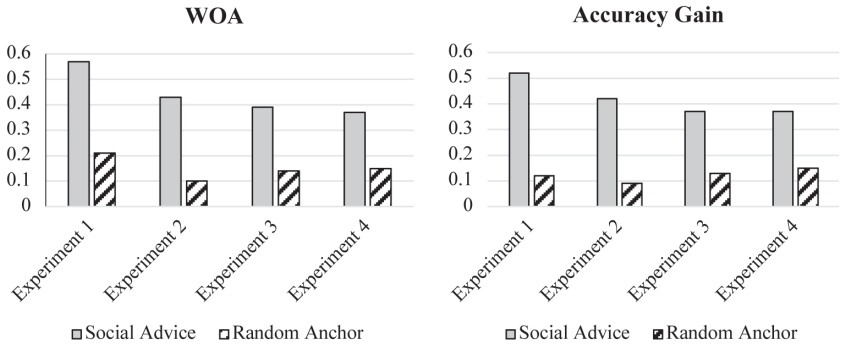

FIGURE 10.1 Mean weight of advice (WOA) and mean accuracy gain obtained in four experiments by Hütter and Fiedler (2019).

the average weight given to advice was WOA = .57 as opposed to WOA = .21 for the anchor. The corresponding accuracy gain was .52 (reflecting an inaccuracy decrease from 81.84 to 36.14 years on average) for advice compared to .12 (from 83.95 to 67.54 years on average) for anchors. However, note that both the stronger impact of the same numerical stimulus framed as social advice than as a mere anchor and the above-zero impact of either advice type are of theoretical importance. Participants not only revised their estimates in the direction of an unknown other person, who was arbitrarily selected, but also followed to a sizeable degree a randomly drawn numerical anchor. From this research, it appears as if advice takers are uncritically suggestible to all kinds of quantitative stimuli, quite independent of any quality control of the validity of the advice or the anchor. We shall elaborate on this memorable finding in our discussion of metacognitive myopia.

Hütter and Fiedler (2019) replicated the same findings in several controlled lab experiments (see Figure 10.1), along with additional evidence for the extra impact of social advice on subjective confidence. In one experiment (Experiment 2), the inclusion of an additional condition with computer-generated advice showed that the weight of advice depended on aboutness rather than the human origin of the advice. Because computer-generated estimates were also about the historical event to be estimated, like the human advice but unlike the random anchor, they exerted a similar impact on the participants' revised judgments as the advice ascribed to human beings.

From Naïve Advice Taking to Metacognitive Myopia

Hütter and Fiedler's comparison with the computer-generated advice suggests that their paradigm did not involve much reasoning about the underlying motives or effort that went into the production of advice. The fact that we told participants that the computer-generated advice was based on a knowledge base comparable to that of an average participant was enough for it to exert the same impact on final judgments. One may conclude from this finding that the traditional advice taking paradigm, the judge-advisor-system (JAS; Sniezek & Buckley, 1995) is not well suited for the investigation of social aspects of advice taking, because advisees receive no information about the advisor or the advice they gave.

Prior evidence on accuracy gains from averaging relies on a paradigm, in which estimates or opinions are only subject to imperfect reliability but not to systematic bias or invalidity. For instance, calorie estimates (of a boiled egg, a glass of wine, or a coffee with two lumps of sugar) may vary in accuracy due to unequal experience just as the ox-weight estimates provided by Galton's farmers on the fare varied around the true value, but they are hardly subject to systematic over- or underestimation. Because all these estimates are only contaminated with unsystematic noise, there are good reasons to expect massive

accuracy gains from averaging. If, however, estimates are contaminated with systematic bias, such as overpriced bargains by people who have vested interests or deceptively low duration estimates by agents of a construction firm, then it would be naïve to believe that averaging produces accuracy. When invalid information (deception; collective mistakes; fake news) is at stake, adaptive advice taking does not call for uncritical averaging but for stringent monitoring and the critical ability to separate the wheat (valid advice) from the chaff (invalid advice).

While sociability certainly facilitates averaging and trust in the wisdom of crowds, it is more likely to foster harmony and equality than critical assessment. The same sociability motive that encourages people to listen to others and trust in their advice, to follow the majority and embrace a "compromise and conformity works" rule cannot be expected to energize competition and adversarial debates about the potential invalidity of opinions and counter-opinions (see also Baumeister & Maranger; Gaffney & Hogg; Krueger et al., this volume). Thus, the same sociable broad-mindedness that helps people exploit the wisdom of crowds renders them naïve and lenient in monitoring and controlling the validity of information at the metacognitive level. In other words, sociability intensifies a problem termed metacognitive myopia (Fiedler, 2000, 2012; Fiedler, Hütter, Schott, & Kutzner, 2019) by potentially leading people to use advice even when it is explicitly said to be invalid (Schultze, Mojzisch, & Schulz-Hardt, 2017).

Metacognitive Myopia

In the advice-taking paradigm, when there is a conflict between a valid and an invalid source of advice, meta-cognitive quality control is sorely needed (Ackerman, 2014). Adaptive intelligence calls for the ability to separate the wheat from the chaff. Rather than falling prey to an equality bias (Mahmoodi, Bang et al., 2015) and giving equal weight to any advice, it is essential to engage in strict and contrastive discrimination between valid and invalid advice. Such sharp reasoning is often at variance with sociability and harmony motives. It calls for the kind of backbone that Hannah Arendt (1963) had in mind when she wrote about the obligation to be disobedient and non-conformist. The need to discriminate between, and to assign clearly unequal weights to, critically assessed opinions runs counter the sociability motive to please other people.

Equality Bias in Collective Decisions

For an experimental illustration of this drawback of sociability, or equality bias, let us consider the following investigation by Fiedler, Krüger, Koch, and Kutzner (2020). Pairs of participants were asked to jointly make a choice between two lotteries, A and B, based on both partners' experience with a sample of 50 observations about the lotteries. One partner, the valid sampler, could on every

trial solicit a random draw from one of the two lotteries, followed by feedback about that lottery's winning or non-winning outcome. Because the aim of the task was to figure out the lottery with the higher winning rate, it was appropriate to sample outcomes conditional on lotteries. The other participant, who was assigned the invalid sampler's role, could select on every trial a winning or non-winning outcome, followed by feedback about the lottery (A or B) to which the chosen outcome belonged. Note that this reverse sampling of lotteries conditional on outcomes is logically inadequate, because $p(\text{A} \mid \text{win})$ can widely diverge from $p(\text{win} \mid \text{A})$. Like the probability $p(\text{right-handed} \mid \text{deaf})$ that a deaf person is right-handed is much higher than the reverse probability $p(\text{deaf} \mid \text{right-handed})$ that a right-handed person is deaf, due to unequal base-rates $p(\text{deaf}) < p(\text{right-handed})$, a high rate of $p(\text{A} \mid \text{win})$ may simply reflect a high $p(\text{A})$ base-rate. Rather than being the better lottery, A may only be frequently played.

In two experimental conditions, A was the better lottery, $p(\text{win}|\text{A})=.60 > p(\text{win}|\text{B})=.40$, which was clearly visible to valid samplers. However, the base-rates $p(\text{A})$ and $p(\text{B})$ were manipulated such that the expected A rate of winning outcomes was either $p(\text{A}|\text{win}) = .88$ or $p(\text{A}|\text{win}) = .27$. Indeed, individual judgments confirmed that valid samplers correctly recognized the higher winning rate of A than B, whereas invalid samplers either corroborated the A advantage or greatly underestimated A relative to B, depending on the base-rate condition. Finally, valid and invalid samplers were asked to discuss their experience and then to jointly decide which one of the two lotteries they wanted to play. One might expect these discussions to reveal that only valid samplers had gathered appropriate evidence for the differential value of lottery A and B and that dyads should therefore base their decision on valid samplers' and discard invalid samplers' estimates. This should be particularly the case in dyads high in epistemic vigilance, according to an assumption propagated by Sperber et al. (2010) that epistemic vigilance—the alerting function of the presence of and potential conflicts with other people—constitutes an asset of sociability, a catalyst of critical reasoning.

However, the results of three experiments (asking for the superiority of lotteries or educational programs in an organization) consistently showed that dyads fell prey to a conspicuous equality bias. Metacognitive myopia prevented them from a rational strategy to follow the valid and discard the invalid sampler's advice. While all participants' individual estimates were highly sensitive to whatever contingency they had sampled from, only the valid samplers recognized the superior winning rate of lottery A, whereas invalid samplers inferred a link between lottery A and winning from the reverse conditional $p(\text{A}|\text{win})$. Depending on the base-rate manipulation, they estimated lottery A's winning rate clearly higher when $p(\text{A})$ and $p(\text{A}|\text{win})$ was high rather than low, although the lotteries' winning rates were invariant. An equality bias—a failure to discriminate between valid and invalid advice—was evident in final dyadic estimates and decisions

that were strongly biased toward invalid advisors' opinions. The higher/lower the elusive $p(A|win)$ observed by invalid samplers (reflecting a base-rate fallacy), the higher/lower were the dyads' preferences for lottery A. Indeed, because the base rate manipulation affected invalid but not valid samplers' individual estimates, the former contributed more to predicting the final dyadic judgments than the latter. In one experiment, the regression weights of invalid and valid advisors were $\beta_{invalid} = .72$ and $\beta_{valid} = .36$, respectively. In another experiment, the corresponding regression weights were $\beta_{invalid} = .64$ and $\beta_{valid} = .17$, respectively. Something conceptually similar was demonstrated in the group-induced attitude polarization paradigm, where more extreme and more committed group members tend to bias group decisions toward extremity, and their distorting effect increases with extended discussion (Forgas, 1977).

Would epistemic vigilance moderate this sobering result? Would at least participants in those dyads that exhibited enhanced epistemic vigilance be more sensitive to advice validity? For an empirical check, Fiedler et al. (2020) included in one experiment a seven-item measure of the degree to which discussions in 57 dyads were epistemically vigilant. Dyads were considered vigilant if they provided (on average, over both discussants) high ratings prompted by such questions as how incompatible both partners samples were, how strong the discrepancies were that had to be reconciled, whether the information value of the samples rather than the eagerness of discussant determined the joint evaluations. The internal consistency was satisfactory, $\alpha = .76$. As it turned out, epistemic vigilance did not ameliorate or eliminate metacognitive myopia and the dyads' susceptibility to invalid advice.

Naïve Susceptibility to Invalid Advice Concerning Health Risks

In another series of experiments exploring how patterns of sociability might influence decisions, Fiedler et al. (2019) compared the relative impact of two different advisors on individual advice takers' estimates of health risks, conceived as conditional probabilities. For instance, they estimated the likelihood of a suicide attempt given depression, of young women developing anorexia given a gender role conflict, or that a person has contracted HIV given a positive test result. The manner in which an advisor had determined the risk estimate from a relevant study sample (in which a rare health outcome was either represented at its true rate or dramatically overrepresented) indicated whether the advice was valid or invalid. Participants in different experiments provided either only one estimate per risk problem, after receiving one or two advisors' estimates, or they could revise an initial estimate (based on their own opinion or an initial advice) in a second estimate informed by another advice. Across several experiments and a variety of risk estimates, risk estimates were generally sensitive to advisor estimates and to the statistical information of the advisors' alleged study samples.

However, they were remarkably insensitive to advice validity. Inflated estimates by advisors whose study samples dramatically overrepresented the risk event had a similar effect as, or sometimes even stronger effect on participants' own subsequent estimates than correct advisor estimates derived from unbiased samples. Even participants who had already formed a prior risk estimate based on valid advice distorted their final judgments when later exposed to invalid advice, derived from a biased sample that only deviated from the unbiased sample in terms of a dramatic oversampling of the critical event. Participants were misled by invalid advice regardless of whether advisors were introduced as experts of laypeople, and even when participants themselves rated invalid advice to be low in validity. Consistent with the notion of an equality bias, it was as if sociability came along with a pseudo-fairness norm to pay the same attention to every individual's opinion, independent of a critical assessment of his or her validity.

In one experiment, a perseverance effect highlighted the metacognitive naivety with which invalid opinions affected advice takers' risk assessments. In this experiment, explicit instructions blatantly revealed the invalidity of an advice, the invalidity of which participants did not have to recognize themselves. Specifically, participants were told,

> *Sometimes experts have used a sample in which the proportion of a critical attribute is the same as in the population. If, for instance, the prevalence of anorexia in the population is 5%, then such a sample also includes 5% of cases with anorexia and 95% of cases without anorexia. These samples support correct estimations. . . . If, however, experts rely on a sample in which anorexia and no anorexia occur at the same 50–50% rate, then this sample does not allow for an accurate estimate. For the normal 5% rate of anorexia is severely overrepresented at 50%. If one then determines the anorexia proportion within the subset of cases that have a gender-role conflict, then anorexia is also misrepresented in this subset of cases. Such 50–50% samples are therefore misleading and not suitable for an accurate estimate . . . With regard to the final task, this means that given a sample size of 1400, the number of included anorexia cases should be only 70.*
>
> *[p. 326]*

An extra pilot study made sure that participants accepted the debriefing as intelligible and comprehensive. Nevertheless, the debriefing did not prevent participants from letting an unambiguously wrong advice influence their final risk estimate. Such a perseverance effect (Ross, Lepper, & Hubbard, 1975) highlights the irrational nature of advice takers' metacognitive myopia.

Reality Created Through Social Conventions

Thus, although sociability clearly facilitates the benefitting from the wisdom of crowds, the same openness to knowledge sharing and opinion averaging that

allows people to profit from advice taking also seems to inhibit adversarial functions that call for social discrimination and competition. Although openness for advice taking is essential to overcome egocentrism and to exploit an inherent aggregation advantage, naïve advice taking can be dysfunctional, irrational, and simply dangerous. Human history is replete with people following false advice and the emergence of often long-lasting consensual illusions. This is especially the case in populist political movements where fake news and conspiracy theories produce dangerous tribal beliefs and allegiances (Forgas, Crano, & Fiedler, 2021). After all, the mistake committed by people who became victims of con artists, who suffered serious financial losses, or who subscribed a deceptive contract, is rarely suboptimal averaging of advice. These costly mistakes are most likely due to naïve trust in deceptive advice they should have disregarded or discarded. Adopting an invalid opinion about the health risks in the Fiedler et al. (2019) research or acquiescence to a shady compromise in dyadic decision-making as in Fiedler et al. (2020) can be expensive and dangerous.

A growing body of recent research on misinformation and fake news (Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012) corroborates people's limited ability to ignore or to get rid of invalid information, consistent with the notion of metacognitive myopia. A recent meta-analysis of debunking research (Chan, Jones, Hall Jamieson, & Albarracín, 2017) shows that explicit debriefings and corrections of invalid beliefs are only partially successful; erroneous beliefs persist and persevere to a threatening degree.

Given all these unwanted side effects of sociability and openness to other people's opinions, the question that suggests itself is why ontogenetic and phylogenetic learning does not help people overcome such irrational tendencies. Why do people exhibit perseverance effects, continue to follow unequivocally invalid advice? Why does fake news persist, and why is debunking not met with sustainable success? Or more generally, why did evolution not equip us with better remedies to meta-cognitive myopia (see also von Hippel, and Baumeister & Maranger, this volume)? Only modest cognitive competencies would be required to avoid such blatant weaknesses; they hardly exceed our cognitive capacities.

There are several answers to these captivating questions. First, precise metacognitive monitoring and control of one's own and one's social companions' judgments may be an unrealistic ideal, simply because normative benchmarks for accuracy and validity are rarely ever available. Most of the time, there is no optimal or rational way to estimate risks, to invest money, or to win a soccer match. The task settings used in pertinent experiments (such as the conditional probability tasks used by Fiedler et al., 2019) may be rare exceptions, quite unrepresentative of those everyday tasks that have shaped the evolution of our metacognitive competencies.

The second answer is related to the lack of ultimate normative solutions mentioned in the last answer. "Reality" and "correct solutions" are often a

matter of social convention or social reification rather than logical or empirical proof. To illustrate this point, consider the strong and persistent finding from the lie detection literature that is the focus of a meta-analysis by Hartwig and Bond (2011). The unwarranted weight lie detectors give to virtually all lie detection cues—such as gaze, speech hesitations, adaptors, or verbal cues—when they judge the veracity of communications is systematically higher than the cues' actual diagnosticity. Although no single cue bears a strong and stable correlation with actual truth, there is wide agreement among lie detectors and consistency within lie detectors in utilizing cues for subjective judgments of truth. However, crucially, this collective agreement on veracity clichés creates a conventional reality that is elusive but nevertheless consequential. Because of the collective belief that gaze aversion reveals lying, or that emotional language is diagnostic for truth, it is prudent and highly adaptive to follow the conventions, although they are unwarranted. Still, for a job applicant or a politician to appear credible, it is advisable to look into others' eyes and to interlard emotional cues in one's speech. Thus, to the extent that reality is created by convention (or conventional error), it may be wiser to trust in the convention than to engage in critical monitoring and control.

Third, it is possible that the costs of more critical and less naïve assessment with others' opinions exceed their benefits. In the long run—or maybe in a long-term historical perspective—the benefits of getting along with others, embracing a truth bias and taking trust for granted, and the relief from logical stringency and sharp competition may be worth more than the occasional detection of a deception or mistake (see von Hippel & Smith, this volume). In other words, maintaining group agreement, cohesion, and harmonious sociability may sometimes have higher evolutionary survival value than seeing reality as it is. Attempts to overcome metacognitive myopia may come along with unwanted side effects that may be even more costly and unpleasant.

Advisor Expectations

These side effects can be due in particular to impaired social relationships between advisees and advisors, who belong to their social network and with whom they interact repeatedly. Advisors often have clear expectations as to the attention and weight that advisees should give to their advice. Violations of these expectations may create severe interpersonal costs, even when these violations are motivated by rational concerns and normative standpoints. Surprisingly, research into advice taking has paid little attention to the sources of advice and their affective experience. Only recently, researchers have started looking at advisors' affect, cognition, and socially motivated behavior. For instance, Blunden, Logg, Brooks, John, and Gino (2019) found that advisors whose advice was disregarded report feeling offended and less close to the advisee compared to advisors whose advice was followed. As a consequence, advisors whose advice

was discounted were less willing to give advice again in the future. Belkin and Kong's (2018) research documents that advisors whose advice was not followed were even prone to deceive the advisee in a subsequent interaction. If advisees risk losing access to their information sources by giving too little weight to advice, underweighting may incur an important interpersonal cost that does not matter only from the perspective of sociability, but also impacts the quality of judgments and decisions.

However, the costs of underweighting constitute only half the story. Several studies conducted by Blunden et al. (2019) concern domains in which the advisor expected full adoption. Thus, while these studies suggest that advice should be regarded and adopted full-heartedly in order to maintain a good relationship with the advisor, this need not always be the case. That is, if advisors are uncertain about the true value, maintaining a good relationship may require much lower weighting. Ache, Rader, and Hütter (2020) went away from scenario studies and instead developed a paradigm that allowed information exchange between advisee and advisor pairs. They also assessed advisees' and advisors' certainty concerning their judgments. They found that the level of weighting expected by the advisor was contingent on their certainty. For difficult items, advisors wanted their advice weighted less than for easy items. Advisees, however, gave more—not less—weight to advice for difficult items than for easy items. Ache and colleagues (2020) showed that both underweighting and overweighting incurred interpersonal costs: Advisors evaluated the advisee more negatively and were less willing to give advice again the more advisees deviated from the desired weighting.

Similar interpersonal turbulences as a consequence of irregular patterns of sociability can arise as a consequence of consulting multiple advisors. Blunden et al. (2019) found that to the degree that consulting additional advisors implies discounting of an advisor's judgment (counter to the advisor's expectations), the social relationship could be impaired. In the end, violating advisor expectations may even have the potential to limit and undermine the validity function of advice sampling (Hütter & Ache, 2016).

Concluding Remarks

Thus, advisees face an incredibly difficult task, having to balance at least two motives that often call for opposite behaviors: validity and sociability. Pursuing purely epistemic motives or weighting opinions in an egocentric manner can cause damage to the relationship with the advisor, unless one knows how to buffer against such interpersonal costs (Belkin & Kong, 2018). In the long run, however, epistemic concerns and sociability motives may not serve mutually exclusive goals. By maintaining constructive relationships, one secures continuous access to socially shared information, which enhances the quality of one's judgments and decisions (compared to having no access to advice sources) in a sustainable way.

Thus, to round up the parabolic message conveyed in this present article, we started from wisdom of crowd as a blessing feature of the probabilistic world. Just by aggregating over multiple opinions, it is possible to cancel out error variance and improve the accuracy of judgments and decisions. However, blind opinion averaging and naïve reliance on the wisdom of crowds can result in irrational inferences and wrong decisions when individual opinions are subject not only to unsystematic error but also to systematic bias. Whenever advice can be invalid, misleading, or deceptive, and different individuals succumb to the same invalid sources, the metacognitive ability to separate the wheat from the chaff becomes a crucial competence of adaptive cognition. However, finally, we have seen that adaptive and rational behavior is not solely an epistemic question. Getting along with each other and creating sustainable networks is at least as important in the long run as metacognitive quality control. Thus, future research on advice taking has to master the trade-off between two fundamental aspects of adaptive cognition, exploiting the epistemic advantage of collective knowledge and fostering long-term social bonds.

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11

POWER AND SOCIABILITY

Joachim I. Krueger, David J. Grüning, and Tanushri Sundar

Abstract: Social life unfolds with tension between power and sociability. As both, the need of power and the need of sociability, are crucial to interpersonal transactions, the person faces a dilemma. The satisfaction of one need limits the satisfaction of the other. We approach this dilemma by first reviewing two seemingly contrary biases in social perception: self-enhancement and social projection, which are respectively associated with power and sociability needs. We then discuss the ultimatum game, which puts the mandates of power and sociability into direct conflict. Data suggest that observers are sensitive to both material and social interests when judging players' character. We then introduce a general power game, describe it with the *theory of moves*, and review observer-judgment data. We conclude with an exploration of how individuals might incorporate reputational concerns into their social decision-making and speculate about how the power-sociability dilemma can be transcended if it cannot be solved.

Note. We thank Forgas Joseph and Hume David for the inspiration they have given in spoken and written words, respectively.

> A number of porcupines huddled together for warmth on a cold day in winter; but, as they began to prick one another with their quills, they were obliged to disperse. However the cold drove them together again, when just the same thing happened. . . . In the same way the need of society drives the human porcupines together, only to be mutually repelled by the many prickly and disagreeable qualities of their nature.
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> —Arthur Schopenhauer (1851/1964, p. 226)

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The concept of *sociability* pervades the mission, the methods, and the self-construal of social psychology (Allport, 1924). Humans are a social species. For sociologists, this is axiomatic (Gumplowicz, 1885/2017; Tönnies, 1887/2012). Many biologists agree (Wilson, 2004), classing humans with bees or ants, which are social animals in that individual organisms cooperate with one another to achieve collective aims. But bees or ants are not sociable.[1] For humans, the concept of sociability suggests a need to be in the presence of and engaging with other humans, and it entails the idea that this need better be satisfied (Simmel, 1910/1949). Sociability is associated with affiliation, inclusion, love, conformity, and cooperation (see von Hippel & Smith, this volume). Desirable as it is, sociability can be hard to attain. The characters in the works of Sartre, Strindberg, and Bergman fail at this task. They are social, but they are not sociable.

When social psychologists refer to the human being as a social animal, they often look to Aristotle. The Macedonian sage regarded humans as "political animals," creatures placed in hierarchically organized societies. If humans are free and live in a *polis* (or 'republic'), they can participate in the processes of debate, persuasion, influence, deception, and expulsion (Rosler, 2013). In this context, the psychology of sociability taps into a wide range of capacities associated with social intelligence, social perception, and social behavior. The pleasure of being in one another's company is part of a larger social ecology.

Social psychology has not done well owning and developing the concept of sociability. The work has been largely left to the trait school of personality psychology (Cheek & Buss, 1981). The trait model casts sociability as a facet of extraversion (Lucas et al., 2000), and perhaps the most important one (McCrae & Costa, 1987). One might wonder what would be left of extraversion if sociability were excised. Zorba the Greek (Kazantzakis, 1946/2016) might be dancing alone on a Cretan beach, hollering jubilantly. But what kind of extraversion would that be? Dancing the *Hasapiko* with the "Boss," however, and teaching him the joys of dancing, Zorba displays extraversion in its prototypically sociable form.

The trait approach, valuable as it is, has limitations. Being a paradigm of measurement and prediction, trait psychology has little to say about various kinds of mentalizing that characterize and guide social behavior. This is where a social psychology of sociability can be useful. In this chapter, we take up Aristotle's concern with the political dimension of social life. We assume that the social world is not benign, no matter how fervently we want it to be. Status and power hierarchies are ubiquitous (Anderson et al., 2015), and they are often unstable; they are constantly challenged, defended, and re-negotiated (Sapolsky, 2017). Some people try to cope with this uncertainty by following Machiavelli. Prioritizing power, they might fake sociability instead of enjoying it (Christie & Geis, 1970). Others find enlightenment in the writings of the young Frederick II of Prussia, who, before becoming an autocratic king, sought to refute

Machiavelli in his "Anti-Machiavel," declaring the Florentine's doctrine "monstrous" (Lifschitz, 2021).

Irrespective of personality and politics, most people experience some conflict between self-assertion and affiliation, that is, between power and sociability (see Baumeister and Maranges; and Kruglanski & Ellenberg, this volume). We take this to be the fundamental social dilemma: how can an individual get ahead in a group without sacrificing supportive, respectful, or loving relations with other group members? If there were no conflict, one might even have it both ways and be loved by those whom one has subdued. Such harmony between power and sociability is elusive. In Book IX of the Iliad, Achilles tells Odysseus what his mother, the goddess Thetis, had told him: he could either join the fight against the Trojans, die a hero, and his "glory will live forever," or he could return to his ancestral Thessaly, enjoy a full and peaceful life, but his "glory will die" (Mitchell, 2011, p. 440). Moved by reputational greed and a thirst for vengeance, Achilles sacrificed sociability—and life. The rest is history—or legend.

To explore the power-sociability dilemma, we take a social-cognitive and game-theoretical approach. In the first section of this chapter, we review two elements of social perception, namely, self-enhancement and social projection, which are associated with power and sociability, respectively. In the second section, we review the social perception of players' competence and morality in the ultimatum game with a view to power differentials. In the third section, we introduce a general power game, analyze it in light of the *theory of moves* (Brams, 1994, 2011), and present social-perception data, again with a view to competence and morality. In the final section, we ask how individuals might allow perceived social perception, that is, reflected appraisals, to enter into their decision-making within dilemmas. We close with a reflection on personal freedom, a concept that transcends the basic dilemma.

Social Perception and the Basic Dilemma

The perception of others and the perception of the self are entangled with one another (Alicke, Dunning, & Krueger, 2005). People do not have impartial bird's-eye views of themselves or others, much as they might try. Comparisons spontaneously arise, creating a push and pull of perceived similarity and difference (Suls & Wheeler, 2000). Social-cognitive theory has long framed any evidence of comparative effects as evidence of bias. This implies that noncomparative judgment is both desirable and possible (Krueger, Freestone, & MacInnis, 2013). The assumption that social perceivers are less than rational has led to a flood of claims about contradictory biases such as false consensus and false uniqueness or over- and underconfidence (Krueger & Funder, 2004). Research on self-enhancement (or overconfidence) and social projection (a.k.a., "false consensus") is of interest here because these biases respectively tap into motives of power (e.g., self-assertion, competition, dominance, and individuation) and

motives of sociability (e.g., affiliation, cooperation, and collectivization) (see also Fiedler & Hütter, this volume).

Self-enhancement is typically taken to serve the motive of feeling good about oneself (Alicke & Govorun, 2005; Brown, 2012). The better-than-average variant of this bias captures a positive differentiation of the self from others (Krueger et al., 2017). Self-enhancement thereby becomes conflated with other-diminishment (Heck & Krueger, 2015). To say that one person is better than another person is to assert oneself, and others tend to object ("You think you're better than me?"). Observers may judge self-enhancers as more competent than others, but only if their claims to superiority are backed up by evidence. Regardless of whether these claims are correct or not, perceivers rate self-enhancers as less moral than others (Heck & Krueger, 2016). Broadcasting self-enhancing claims carries the risk of alienating potential social partners and is thus a threat to affiliation and sociability.

Social projection is seldom studied through a motivational lens. Most research focuses on the informational value of self-referent information for predictions about others under conditions of uncertainty (Krueger, 2007; Krueger & Grüning, 2021; Thielmann et al., 2018). This approach construes social projection as an accuracy-enhancing prediction heuristic. Yet, motivational accounts have always been considered. In their seminal work on the false consensus effect, Ross, Greene, and House (1977) noted that people may wish to assume consensus (i.e., similarity) with others because similarity implies social connectedness and the normalcy of the self. Heck and Krueger (2020) provided direct evidence for this idea, showing that individuals who, to their own surprise, failed at a reasoning task were particularly likely to project this outcome to others.

In short, self-enhancement as a claim of superiority is related to the concept of power and overconfidence (Larrick, Burson, & Soll, 2007; Moore, 2020). Highly confident individuals are most likely to enter competitions for material gain and vie for leadership positions (Kamas & Preston, 2012). The many who fail turn out to be overconfident after the fact, a judgment that amounts to an outcome bias (Baron & Hershey, 1988; Krueger, Heck, Evans, & DiDonato, 2020). In contrast, social projection is associated with sociability and affiliation as it is well established that perceived similarity predicts liking (Byrne, 1961; Clement & Krueger, 1998).

In its simplest form, the power-sociability dilemma suggests a hydraulic system in which self-enhancement and social projection are perfectly negatively correlated. As one bias increases, the other must decrease. Indeed, early theorists equated self-enhancement with the bias of false uniqueness, that is, the logical opposite of false consensus. But it soon became clear that this cannot be so (Krueger, 2000). If the distribution of scores is negatively skewed, that is, if there are few spectacular failures, most individuals are better than average, and seeing themselves as such is neither an error nor does it reveal a false sense of uniqueness. Even symmetrical distributions do not entail a necessary mapping

of self-enhancement on false uniqueness, as self-effacers would also be considered victims of the false-uniqueness fallacy.

Although there are constraints on the joint emergence of self-enhancement and social projection, the two biases can—and often do—co-exist. One statistical consideration is that self-enhancement is typically conceptualized and measured as a difference in mean level, whereas social projection takes the form of a correlation between self- and other judgments over items (traits, behaviors, preferences, etc.). Barring ceiling or floor effects, individuals can construe themselves both as better than others (with more favorable judgments on most items) and as similar to them (with a positive covariation between self and other judgments; Krueger, 2002). We estimated the statistical relationship between self-enhancement and social projection in two data sets. Each data set comprised judgments from about 200 respondents, which we separated into 10 equally large subsamples, computed self-enhancement as the difference between average self-ratings and average ratings of the average other, and social projection as the correlation between self-ratings and other ratings over respondents. In one data set, raised in a lecture course in the spring of 2021, respondents rated their own and others' happiness. In the other data set, respondents had to rate own and others' contentedness (Heck & Krueger, 2015). The two correlations were positive though small (r = .29 and .14). With a more rigorous bootstrapping procedure involving 1,000 re-samples from each data set, the mean correlations between the two biases were lower, that is, −.02 and −.24, but still not wildly negative.

In social dilemmas as well, self-enhancement and social projection can co-occur. In the volunteer's dilemma, for example, most respondents claim they are more likely than others to make a prosocial sacrifice, while their estimates of the likelihood of others' volunteering are positively correlated (over scenarios and over respondents) with their own decisions (Krueger, Ullrich, & Chen, 2016). Indeed, social projection can facilitate cooperation (Krueger, 2013). In a prisoner's dilemma, players who project at the limit, that is, players who believe others choose the same strategy as they themselves choose, will cooperate rather than defect because mutual cooperation is payoff-dominant, that is, better for everyone (Krueger, DiDonato, & Freestone, 2012). Such findings are reminiscent of Codol's (1975) classic demonstration that most people think they are superior to others in that they more faithfully follow group norms. Taken together, these findings suggest that the dual tendencies to enhance the self and to project it not only fuel the fundamental social dilemma but also create room for a positive balance. We now turn to two experimental games to explore strategies people use to balance the needs of power and sociability.

Social Perception in the Ultimatum Game

The most powerful weapon in chess is to have the next move.

—David Bronstein

Once introduced by Güth, Schmittberger, and Schwarze (1982), the ultimatum game has become a Petri dish for hypotheses, experimental data, and inferences about human nature and strategic behavior. The game itself is simple. One player proposes a proportional split of a small sum of money to a responder. If this responder accepts the offer, both players receive the proposed share of the money. If the responder rejects the offer, presumably because it is insultingly low, neither player receives anything (see also Forgas, this volume).

Most research has focused on responders' decision-making. Once offers fall below 30% of the purse, many prefer to receive nothing (Camerer, 2003). Why? Theories and speculations abound. Classic game theory, as Güth himself anticipated, is at a loss because a rational player prefers any amount of money to no money at all. When the responder rejects an offer, other values, emotions, or motives, such as righteous anger and the will to punish, must have come into play (Gabay et al., 2014; Xiao & Houser, 2005). What we see is the wrath of the responder. Social game theory (Krueger et al., 2020) adds a formerly neglected element of social perception to the puzzle. The theory states that players know their behavior is meaningful in the eyes of others (and thus their own eyes). Transactional behavior signals real and claimed power, and it provokes moral evaluation. The theory assumes that humans care about how their behavior reflects on them (Wu, Balliet, & van Lange, 2016). This is not an entirely new idea. David Hume (1740/1978) based his theory of motivation on it.

Here we report findings from an informal study conducted as part of a class project to highlight how observers evaluate decisions in the ultimatum game along the dimensions of competence and morality (see also Tham, Hashimoto, & Karasawa, 2021). Participants (N = 198) rated a proposer, an accepting responder, or a rejecting responder. All participants read about offers that were fair ($5 out of $10), a little selfish ($4), or greedy ($1). On 7-point scales, they rated both players on the adjectives competent, intelligent, and rational, as well as on the adjectives moral, benevolent, and principled. In each condition, ratings were reliable ($\alpha > .81$ for competence and $\alpha > .78$ for morality). Adjective ratings were therefore averaged to form composite scores of competence and morality. Figure 11.1 shows the mean ratings for the three experimental conditions. In what follows, we report descriptive statistics and effect sizes. Unless otherwise noted, these effects are statistically significant at the conventional .05 level.

Judgments of competence and morality were positively correlated with each other in each condition (mean r = .52; after Fisher's r-Z-r transformation). Participants evaluated the proposer (morality: η^2 = .60; competence: η^2 = .27) and the accepting responder (morality: η^2 = .12; competence: η^2 = .26) positively to the extent that offers were generous. They evaluated a rejector positively to the extent that offers were low (morality: η^2 = .20; competence: η^2 = .39). This latter finding may seem surprising because it indicates rejectors received more credit for those judgments that are *easier* to make.

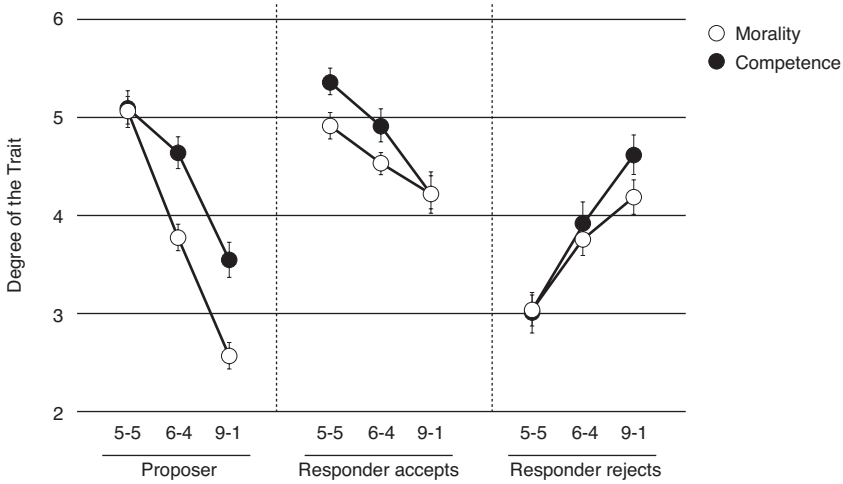

FIGURE 11.1 Means (and error bars) for competence and morality judgments in the 3 (Proposer versus Responder accepts versus Responder rejects) × 3 (5–5 versus 6–4 versus 9–1) × 2 (Morality and Competence) design.

Knowing that observers prefer to see a fair exchange, ultimatum gamers can anticipate how they will be viewed given their choices. Proposers concerned about their moral reputation may bias their offers toward generosity, and responders might reject material offers they could otherwise obtain. Such reputational concerns might help explain the standard finding that the outcomes of ultimatum games depart from strict game-theoretic rationality.

Note that the ultimatum game contains a sequential power play. Proposers have the right of the first move and thus the power to frame the deal. They have an opportunity to do better—if only by a little—than responders, who will likely accept a 6:4 split. Yet, the proposers' power is fleeting. Once exercised, it is gone. Responders have the power of the last word, being able to claim reputational compensation by rejecting a bad deal. A mildly unattractive deal may be a bitter pill because it leaves a material disadvantage without much reputational compensation. The opportunity to offer such a deal could leave the proposer with a small power advantage.

Our participants attributed a small (d = .19) and nonsignificant power advantage to the proposer (M = 5.20 versus M = 4.75 for the responder). When asked which role they themselves would prefer to play, a slight (and marginally significant) majority of 56% opted for the proposer role. Individual role preferences predicted the differences in the power judgments for the two roles (r = .44). As power is distributed in the ultimatum game, choosing a role is hard. Choosers must determine which type of power they prefer.

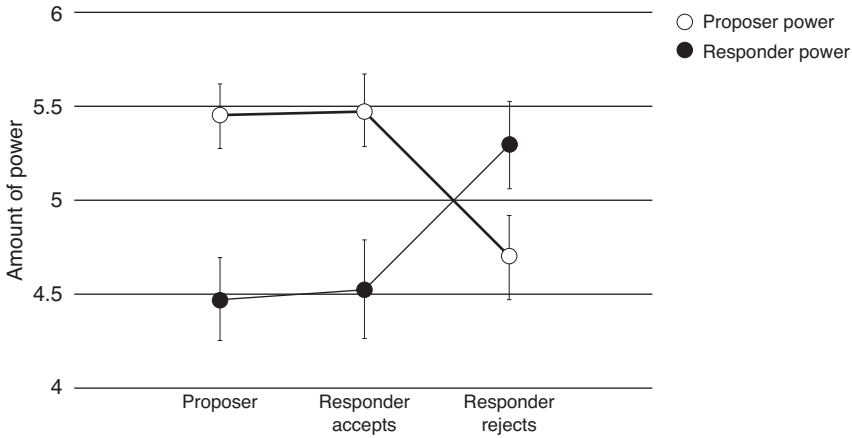

FIGURE 11.2 Means (and standard errors) of power judgments for a proposer (white) and a responder (black) in the three between-subjects conditions (Proposer versus Responder accepts versus Responder rejects).

The differences in the results obtained in the three conditions provide further insights. When participants had observed the decisions made by a proposer (d = .39) or had seen an accepting responder (d = .31), they considered the proposer to be more powerful. In contrast, they considered the responder more powerful when they had seen offers rejected (d = .20, marginally significant; see Figure 11.2 for all means).

Participants also rated themselves on the same trait adjectives, allowing us to see self-enhancement and social projection. In each condition, they rated themselves as more moral (M_{self} = 5.26 versus M_{player} = 3.91, d = 1.10) and as more competent (M_{self} = 5.42 versus M_{player} = 4.26, d = .82) than other players. Social projection was captured by the correlations—across respondents—between self-ratings and ratings of others (mean r = .26). The degree of social projection showed little variation over conditions or between the dimensions of competence and morality. Again, self-enhancement and social projection (perceived similarity) can, to a degree, go together.

The ultimatum game shows off the complexity of the power-sociability dilemma. Whereas the proposer is a gambler who tries to find a balance between a comparative material advantage and social acceptance, the receiver is left to using power as a costly signal of own strength or virtue. The two are in conflict. It takes more spite to reject a mildly unfair offer, but the moral credit is greater for rejecting highly unfair offers. The wrath of Achilles at Agamemnon's overbearing claim on the captured Princess Briseis is a Homeric example of determination in defeat without the benefit of moral credit. Achilles, it seems, won the admiration of the Greeks *because* he let them down, *not*

in spite of letting them down. Heroes are rarely remembered for their high morality.

A General Power Game

> *The fundamental concept in social science is Power, in the same sense in which Energy is the fundamental concept in physics.*
>
> —Bertrand Russell (1938/2015, p. 4)

The ultimatum game teaches us that power is rarely a one-way street. Often, all interactants have access to particular power moves. The data suggest that moral and affiliative concerns can be relevant input to a player's choice. Many games described by game theory contain a conflict between doing well and getting along well, that is, the conflict between power and sociability. We now describe a simple and generic power game, where conflict arises from players' misaligned preferences. Figure 11.3 shows the game in matrix form.

Once preferences have been identified, the *theory of moves* identifies each player's primary and secondary interest (Brams, 1994, 2011; Krueger, 2020). The primary interest is given by a strategy, either one's own or the opponent's, that yields consistently better results than its alternative. The secondary interest is a player's contingency plan. How will they respond once the opponent has shown their hand? In the power game, row is primarily interested in having an accommodating partner (if column chooses strategy A, row realizes either 4 or 3 points). Row's contingency plan is to play tit for tat. If column accommodates, row will too (4 > 3); if column challenges, so will row (2 > 1). Column's primary interest is also to have an accommodating partner, which yields 3 or 4 points. Column's contingency plan is to anti-reciprocate; if row accommodates, column's best response is to challenge; if row challenges, column will accommodate.

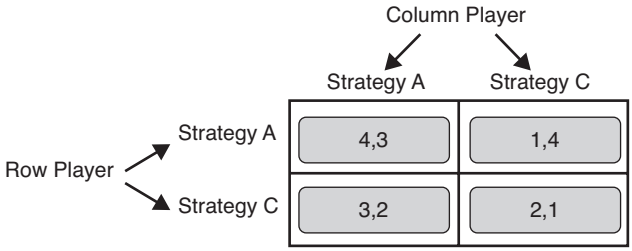

FIGURE 11.3 The preference matrix for the general power game. Higher numbers are preferred. The numbers to the left of the comma designate the preference of the Row player. Strategy A may be termed "accommodation"; strategy C may be termed "challenge."

Both players have cause to be sensitive to their expectations regarding the other's move (Pruitt & Kimmel, 1977)—and they are (Boone, Declerck, & Suetens, 2008; Pletzer et al., 2018). Game-theoretically rational players (Luce & Raiffa, 1957) are indifferent between strategies if the probability of the other player being accommodating is $\frac{P-S}{R-S-T+P}$, where R is the payoff for mutual accommodation, T denotes unilateral challenge, P mutual challenge, and S unilateral accommodation (Rapoport, 1967). With the presented payoff matrix, both players are indifferent when p(Strategy A) = .5. However, if the other player's p > .5, row favors accommodation, whereas column prefers to challenge. This asymmetry reflects the misalignment of the players' secondary interests. Both players may also ask how probable it is that the other player chooses the same strategy they themselves choose (Krueger, 2013). This probability, which reflects social projection, leaves players indifferent if it is $\frac{T-S}{R-S+T-P}$. Here, the two players' outlooks are aligned. The more they project to the other, the more they gravitate to the mutually rewarding outcome of shared accommodation. In this way, the power game is similar to the prisoner's dilemma.

When presenting students (N = 373) with the power game, we found that assuming the role of the row player, 47.7% preferred accommodation, whereas 75.5% wished to accommodate when in the role of the column player. This difference was significant (by McNemar's test), amounting to an odds ratio of 2.89. Figure 11.4 shows the distribution of joint frequencies for random pairing. The numbers in brackets show the coincidences when players are paired with their own preferences. The similarities between the random and the *et ipsum* pairings show that participants would not do better when playing against themselves.

With a slight plurality finding themselves in the situation of a challenging row and an accommodating column, respondents failed to take full advantage of the efficient outcome of mutual accommodation. Interestingly, the probabilities of the pairings were more highly correlated with row's (r = .87) than with column's payoffs (r = .05) as though respondents implicitly identified with row.

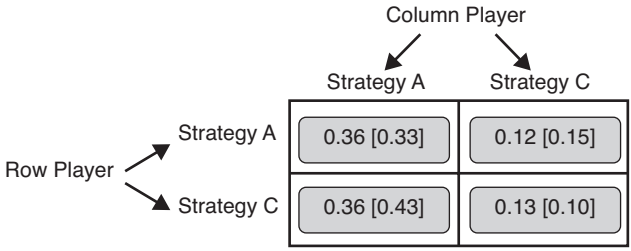

FIGURE 11.4 Coincidence matrix as estimated from student preferences in either role.

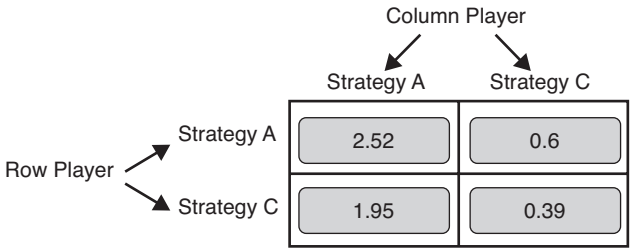

FIGURE 11.5 Estimated expected values for the general power game.

Indeed, row's expected value is greater (2.99) than column's (2.47) because column accommodates above the Nash equilibrium. Figure 11.5 shows the expected values for the four possible outcomes as the product of the sum of the two payoffs times the [unbracketed] probability estimates shown in Figure 11.4. The sum of these four values, namely 5.46, represents the expected value of the entire game. This result is slightly better than the value obtained with Nash equilibrium play, that is, with each player selecting a strategy at random.

Why did respondents fail to play efficiently? Why did not more row players choose accommodation? Loss aversion might play a role (Tversky & Kahneman, 1991). An accommodating row runs the risk of being suckered by a challenging column, whereas column's greatest fear is a counterchallenge from row. Despite the apparent advantages for row, respondents did not perceive a strong power differential between row (M = 4.16) and column (M = 4.00), d = .09, on 7-point scales. They did, however, consider row (M = 4.13) to be happier than column (M = 3.74), d = .24. Judgments of happiness, it seems, were more sensitive to the likely outcomes, which favored row, whereas judgments of power reflected the ability to bring these outcomes about. Here, row had no advantage as there was no power of coercion. The dissociation between power and happiness also emerged in the low correlations between the two sets of ratings, which were .29 for row and .28 for column.

As in the ultimatum game, we assessed perceptions of morality and competence with ratings on six trait-descriptive adjectives. We limited this investigation to perceptions given row's accommodation. Again, the two short scales had adequate internal consistency (all $\alpha > .77$). Correlations between morality and competence ranged from .26 to .57. That is, the two dimensions of judgments were discriminable but seemed to share an evaluative core. When the players achieved mutual accommodation, row was seen as less moral (M = 4.48) than column (M = 5.10), d = .49. Respondents seemed to recognize column's sacrifice of foregoing a challenge, which would have returned a better value. There was no difference in perceived competence (M = 4.93 and 4.82, respectively, for row and column), d = .07. When column challenged and row accommodated, perceptions markedly diverged. Now, column was seen not only as far

less moral (M = 3.25) than row (M = 4.34), d = .58 but also as more competent (M = 4.77 versus 3.77), d = .55. Under these circumstances, perceptions of morality and competence diverge in a way that highlights the column player's social dilemma. Challenging an opponent is a tough stance, analogous to defection in the prisoner's dilemma, and observers view defection as smart but mean (Krueger et al., 2020). Whether this logic extends to the row player remains to be seen in future studies including perceptions of row in the role of the challenger.

Thinking, Shallow and Deep

The power game, more so than the ultimatum game, requires players to ask how far ahead they are willing and able to think. Whereas in the ultimatum game, only the proposer needs to anticipate the responder's move, in the power game, both players stand to gain from correctly predicting the other's move. This prediction, in turn, requires a prediction of the other's prediction of one's own move. Once triggered, recursive predictions descend into a bottomless rabbit hole (Grüning & Krueger, 2021; Stahl & Wilson, 1995). Game theory solves this problem by finding the probabilistic Nash equilibrium (Nash, 1950). The rabbit hole's infinite regress leaves a random choice as rationality's last resort. As we have seen, however, this equilibrium is inefficient, and on average, our participants did a little better due to the column player's willingness to accommodate.

It seems unlikely that the difference between row and column player's willingness to accommodate is due to the same participants thinking more deeply and further ahead when taking the role of the column player than when taking the role of the row player. Recursive mentalizing is hard, and people tend to give up fast (Burchardi & Penczynski, 2014). Yet, most think they have looked more deeply into the hole than others (Grüning & Krueger, 2021). When they do, they express self-enhancement. As it stands, the data are most compatible with the view that participants used a loss-aversion heuristic in both roles.

Can Social Value Orientations Resolve the Social Dilemma?

It is reasonable to think that social preferences affect behavior in the power-sociability dilemma. We may ask how much individuals care about the payoffs accruing to others or about the differences between their own and others' payoffs (Messick & McClintock, 1968). That is, we can ask if the values of benevolence or fairness (i.e., inequality aversion) affect social decision-making to tilt the balance away from power and toward sociability. We now revisit the ultimatum and the power game in light of van Lange's (1999) social value model. The model assigns a weight of 1 to a person's own payoffs and scales the weights

apportioned to benevolence (w_1) and fairness (w_2) relative to the person's self-interest, with 0 and 1 bounding the theoretically and empirically likely range. Applying these values, a person transforms the objective payoffs into subjective payoffs. The subjective payoff for a unilateral competitive move, for example, is $T' = T + w_1S—(w_2|T—S|)$. This value transformation, rooted in Thibaut and Kelley's (1959/2017) early work on interdependence, is linear. It is not assumed, in contrast to social preference models prevalent in economics, that payoffs are psychologically represented as concave utility functions (Fehr & Schmidt, 1999). Also, unlike behavioral economic models, the social value model does not differentiate between favorable (guilt) and unfavorable inequality (envy).

In the ultimatum game, a benevolent player, who places a positive weight on the other's payoff ($w_1 > 0$) but no weight on the inequality of outcomes ($w_2 = 0$), is at best indifferent to making or receiving different offers. For any $w_1 < 1$, proposers prefer to offer less, while responders prefer to be offered more. Proposers who care about fairness will prefer an even split if $w_2 > .5$, otherwise make a minimal offer (see Krueger, Vohs, & Baumeister, 2008, for a mathematical proof that the social value model cannot account for intermediate levels of cooperation; the model predicts all-or-nothing decisions). Responders who care about fairness are indifferent between accepting and rejecting an offer if $w_2 > \frac{R}{10-2R}$, where 10 is the total available and R is the offer.

For the power game, a benevolence weight of 1 turns accommodation into a weakly dominating strategy for each player. Anything less preserves the structure of the game as given by the objective payoffs. As to the value of fairness, the row player would not perceive accommodation as the weakly dominating strategy unless $w_2 < -.5$, that is, unless the row player values unfairness. In contrast, accommodation weakly dominates for the column player if $w_2 > .5$, which lies within reach. This is an interesting result as it shows that in the asymmetric power game, social values can affect choices in different ways. Following convention, we have assumed that a social dilemma is mitigated inasmuch as cooperation increases and outcomes become more efficient. From this perspective, mitigation comes at the expense of the need for power. We now return to the question of how the dilemma may play out if both needs, power and sociability, are accepted as part of the natural human endowment.

Reputational Concerns Amplify the Social Dilemma

The ultimatum game and the power game pit people (players) against others whose choices can only be guessed; and they invite a concern about their own reputation. The social animal cares about what others think, be they real observers or a set of internalized standards. Yet reputations staked on power and sociability pull in opposite directions. Consider again a responder who rejects a 9:1 ultimatum and a column player in the power game who responds to row's accommodation

with a challenge. The ultimatum rejector faces a choice between a comparatively small material gain and a large reputational gain along both moral and competent lines. Many individuals in this situation opt for the reputational gain, but then turn to their material interests when the stakes increase (Andersen, Ertaç, Gneezy, Hoffman, & List, 2011). The challenger in the power game, however, sacrifices the morality bonus for the material gain and the reputation of competence.

The power-sociability dilemma is a fascinating, and perhaps tragic, fact of life. Human beings have much at stake: material interests and security, competitive advantage, interpersonal connectedness, cooperation, respect, and love. Some of this is just in a person's head in the form of a pleasing self-image or a favorable self-assessment—which can be inaccurate. As we have seen, social decisions are affected by these concerns in complex and, at times, surprising ways. The basic dilemma can be nudged and navigated, but it cannot be nullified. If sociability were the only concern, equality might be attained, as in the ultimatum game, or not, as in the power game. But any society of sociable saints is vulnerable to invasion by self-seeking exploiters, those who relish power for their own sake, or those who degrade the common good out of spite (Fowler, 2005), as did Achilles. Moreover, what seems to pass as a moral stance can be destructive. In both our sample games, the conventionally admired value of equality can precipitate states in which everyone is worse off.

Caring about what others think is often lauded as the socially responsible attitude (Mayeroff, 1965), but it hardly passes the bar to become a moral imperative. Brazilian author Joaquím Machado de Assis (1881/2020) satirized obeisance to public opinion when having his hero Brás Cubas declare that one "merely needs to consider the salutary effects of public opinion in order to conclude that it is the very finest achievement of the flower of mankind, that is, of the greater number" (p. 186). In psychology, clinical mastermind Albert Ellis (1962/2010) was among the first to claim that an inflated concern with reflected appraisals is a hallmark of neurosis (see also Brinkmann, 2019).

The Incredibly Enduring Social Dilemma

If reputational concerns fuel the social dilemma, so do the messages pervading contemporary culture. On the one hand, there are the moral mandates of cooperation, social mindfulness, and empathic concern that saturate familial, ecclesiastical, educational, and political narratives. The social institutions of family, church, school, and government have an interest in making us fit for society by making us fit in. A rocked boat is a danger to all, and so we are told not to rock it (Cialdini & Goldstein, 2004). On the other hand, there are the siren songs of individual achievement, self-actualization, and power, which, we are told, characterize the excellent person. Many institutions contribute to the excellence-and-power narrative, but the personal coaching industry—taking its inspiration from Maslow (1954)—casts the message in its purest form. Two

examples shall make the point. One example comes from the irrepressible Tony Robbins, who helps his clients to "Ignite. Achieve. Excel" (www.tonyrobbins.com). This is hard to do on one's own, so Robbins announces that he will make you "Skyrocket to success with a coach." Does it work? The testimonials say it does. Writes one "I feel on top of the word. I feel incredible, I feel motivated, I feel empowered. I am the master of my destiny." The power motive is a greedy master. Another example comes from the start-up company "BetterUp" (www.betterup.com), known for its employment of the Duke of Sussex as its Chief Impact Officer. Better.up describes its mission as to "unlock greater potential, purpose, and passion" (www.betterup.com/en-us/about-us). This narrative is seductive because it suggests that each client harbors inner greatness. It just needs to be "unlocked." This message tickles the power motive.

From Conflict to Freedom

A solution to the social dilemma, however fragile, requires a balance of the dual instincts to be sociable and to be powerful. A strong need of power can threaten group cohesion. The Ju/'Hoansi Bushmen of Southern Africa, for example, demand that successful hunters share their bounty, and they mock them even when they do (Suzman, 2021). In more hierarchal societies, some leaders succeed by cultivating a generous attitude toward others on their own accord (Zenger & Folkman, 2020). Meanwhile, less dominant individuals can also gain favor and influence by projecting their own likability and sociability, a strategy found among chimpanzees as well (Newton-Fisher, 2004; Sandel, Langergraber, & Mitani, 2020).

The inverse challenge arises when the need of sociability becomes too strong. Individuals who habitually seek acceptance, inclusion, and connection readily appease, cooperate, and conform, often to their own psychological and material detriment. These individuals need to find the courage to rebel (Gino, 2019) and assert themselves by cultivating a sense of what might be called "controlled eccentricity" (Krueger, 2019). The maintenance of a balance of power and sociability requires costly self-regulation. The person who achieves it may be said to be free. Freedom is a delicate achievement. It requires wisdom and work, and perhaps a passing familiarity with the relevant research literature.

Note

1. "Human beings find it profitable to live in communities, but their desires, unlike those of bees in a hive, remain largely individual; hence arises the difficulty of social life and the need of government" (Russell, 1938/2015, p. 165).

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PART 4

Sociability and Relationships

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12

AN ATTACHMENT PERSPECTIVE ON HUMAN SOCIABILITY

Interpersonal Goals, Mental Representations, and Information Processing

Mario Mikulincer and Phillip R. Shaver

Abstract: Attachment theory has become one of the leading approaches to conceptualizing and studying human sociability in general, and social relationships in particular. In this chapter, we explore the relevance of attachment theory for understanding individual differences in interpersonal behavior. We begin with a brief summary of attachment theory and an account of the two major dimensions of attachment styles in adulthood, attachment anxiety, and avoidance. We then review evidence concerning the associations between these dimensions and interpersonal behavior, proposing that the associations are mediated by both attachment-related cognitive-motivational predispositions (goal structures, mental representations of self and others, mental scripts concerning social transactions) and patterns of information-processing biases during social interactions.

Attachment theory (Bowlby, 1973, 1980, 1982) has become one of the leading approaches to understanding how human beings manage the demands of social life and relate satisfyingly or not to other people (see Cassidy & Shaver, 2016, and Mikulincer & Shaver, 2016, for overviews of attachment). In thinking about the motivational bases of human sociability, Bowlby (1982) postulated a species-universal, dynamic information-processing neural program—the *attachment behavioral system*—that organizes social behavior in ways that increase the likelihood of survival and reproductive success. According to Bowlby (1973), this psychobiological program or system contains experientially modifiable parameters, which (in addition to modest genetic influences) account for relatively stable individual differences in what has come to be called *attachment style* or *attachment orientation* (see Mikulincer & Shaver, 2016, for a history of this construct).

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We begin this review with a brief summary of attachment theory and an account of the two major dimensions of attachment style in adulthood, attachment-related anxiety, and avoidance. We then review evidence concerning the associations between these dimensions and various aspects of interpersonal behavior, proposing that the associations are mediated by both attachment-related biases in the cognitive-motivational predispositions (interpersonal goals, mental representations of self and others, mental scripts) that people bring to a given interpersonal exchange and patterns of information processing during the exchange (how people attend to and interpret information, appraise their own and others' reactions, and decide among response alternatives).

Basic Concepts in Attachment Theory and Research

According to attachment theory (Bowlby, 1982), human beings are born with a biologically evolved psychobiological system (*attachment behavioral system*) that motivates them to seek proximity to supportive others (*attachment figures*) in times of need as a way of reducing anxiety and obtaining protection from threats (see also von Hippel & Smith, this volume). According to Bowlby (1988), attachment figures function as a "safe haven" in times of need—that is, they provide protection, comfort, and relief—and a "secure base," encouraging autonomous pursuit of non-attachment goals while remaining available if needed. In this way, attachment figures provide a sense of attachment security (confidence that one is a worthy and lovable person and that others will be supportive when needed). Provision of this sense of security normally terminates proximity-seeking bids and allows a person to function effectively in a wide array of non-attachment activities, such as exploration, learning, interpersonal exchanges, and sexual mating.

Bowlby (1988) assumed that although age and development increase a person's ability to gain comfort from internal, symbolic representations of attachment figures, no one at any age is completely free from reliance on actual others, a basic feature of human sociability. The attachment system therefore remains active over the entire life span, as indicated by adults' tendency to seek proximity to and support from attachment figures when threatened or distressed (Zeifman & Hazan, 2016). Moreover, people of all ages are capable of becoming emotionally attached to a variety of close relationship partners (e.g., siblings, friends, romantic partners, coaches, and leaders), using such people as "stronger and wiser" (Bowlby, 1982) attachment figures—that is, as safe havens in times of need and secure bases from which to explore and develop skills—and suffering distress upon prolonged or permanent separation from these key people (Zeifman & Hazan, 2016).

Although all human beings are born with the capacity to seek proximity and comfort from attachment figures in times of need, important individual differences arise in the context of relationships from birth on. According to Bowlby (1973), these individual differences are shaped by reactions of attachment figures

to one's bids for support in times of need and from storing representations of these reactions in *attachment working models* of self and others. When attachment figures are reliably available when needed, sensitive to one's attachment needs, and willing and able to respond warmly to one's bids for proximity and support, a person of any age feels more secure and self-efficacious and is better able to explore the physical and social environment curiously and to enjoy life's many challenges and opportunities (see also Kruglanski & Ellenberg, this volume). However, if one's key attachment figures have not been reliably available and supportive, this sense of security is not attained, doubts about one's lovability and worries about others' motives and intentions are raised, and affect-regulation strategies other than healthy proximity seeking are adopted (*secondary attachment strategies* characterized by *avoidance* and *anxiety*).

In social-psychological studies of adolescents and adults, tests of attachment theory have focused on a person's *attachment orientation* or *attachment style*—a systematic pattern of relational expectations, emotions, and behaviors conceptualized as the psychological residue of a person's unique attachment history (Fraley & Shaver, 2000). Beginning with Ainsworth, Blehar, Waters, and Wall (1978) studies of infant attachment, and followed up in hundreds of adult attachment studies, researchers have found that attachment orientations can be measured along two orthogonal dimensions: attachment-related *avoidance* and *anxiety* (Brennan, Clark, & Shaver, 1998). Attachment-related *avoidance* is rooted in a person's distrust of relationship partners' goodwill, which causes the person to maintain behavioral and emotional distance and independence from others. Attachment-related *anxiety* is based on self-doubt and worries that relationship partners will not be available in times of need. In contrast, people who score low on these two dimensions are said to be secure with respect to attachment or to have a secure attachment style. The two dimensions can be measured with reliable and valid self-report scales (e.g., Brennan et al., 1998) and are associated in theoretically predictable ways with relationship quality and affect-regulation strategies (see Mikulincer & Shaver, 2016, for a comprehensive review of thousands of studies; see also Forgas, this volume).

We have proposed that the two-dimensional space defined by attachment anxiety and avoidance is important for understanding the nature of human sociability, and specifically, the different strategies people use to deal with threats and stressors (Mikulincer & Shaver, 2016). Those who score low on these dimensions hold more positive beliefs about self and others, use more effective affect-regulation strategies, and enjoy higher levels of psychological well-being than people who score high on either avoidance or anxiety (Mikulincer & Shaver, 2016). People who score high on measures of either attachment anxiety or avoidance differ from their more secure peers in using less effective coping strategies, and they differ from each other in adopting different affect-regulation strategies which we, following Cassidy and Kobak (1988), call "hyperactivating" or "deactivating" (of the attachment behavioral system).

Those who score high on attachment anxiety typically adopt *hyperactivating attachment strategies*—energetic, insistent attempts to induce a relationship partner, viewed as insufficiently available or responsive, to pay more attention and provide more effective and satisfying care and support. Hyperactivating strategies include clinging, controlling, and coercive responses, cognitive and behavioral efforts to establish proximity, and overdependence on relationship partners as a source of protection (Shaver & Mikulincer, 2002). In contrast, people who score relatively high on attachment-related avoidance tend to adopt *attachment-system deactivating strategies*, manifested in suppression or inhibition of proximity-seeking inclinations and actions, and a tendency to try to handle distress alone. These strategies involve maintaining physical and emotional distance from others, being uncomfortable with intimacy and interdependence, and suppressing threat- and attachment-related thoughts and feelings (Cassidy & Kobak, 1988).

Attachment-Related Differences in Interpersonal Behavior

Individual differences in attachment anxiety and avoidance are important for understanding characteristic differences in a wide variety of interpersonal reactions and behaviors. For example, attachment insecurities encourage negative, dysfunctional construals of social interactions. In studies assessing the quality of daily interactions over the course of 1 to 3 weeks, for example, attachment-related avoidance was associated with lower levels of relationship satisfaction, intimacy, self-disclosure, supportive behaviors, and positive emotions, and attachment anxiety was associated with more intense negative emotional reactions and more frequent feelings of rejection (e.g., Sadikaj, Moskowitz, & Zuroff, 2011; Tidwell, Reis, & Shaver, 1996). There is also evidence that more attachment-insecure people (anxious, avoidant, or both) have more problems in disclosing and expressing their needs, thoughts, and feelings to relationship partners and in relying on constructive strategies to resolve relational tensions and conflicts (e.g., Garrison, Kahn, Sauer, & Florczak, 2012; Tran & Simpson, 2009).

Attachment insecurities are also related to more negative reactions to interpersonal transgressions and offenses (see also Reis et al., this volume). Research indicates that less secure people tend to react to actual or imagined partner criticism, rejection, or betrayal with stronger negative emotions (e.g., anger, sadness) and heightened release of the stress hormone cortisol (e.g., Carnelley, Israel, & Brennan, 2007; Dewitte, De Houwer, Goubert, & Buysse, 2010). Moreover, several studies have found that attachment insecurities are associated with more "exit" responses (attempts to harm a partner or leave a relationship) and "neglect" responses (ignoring the partner and refusing to discuss a problem)—the two most relationship-destructive responses to a partner's negative behavior (e.g., Pizzano, Sherblom, & Umphrey, 2013). In addition, more insecure people tend to be less inclined to forgive a hurtful partner and more likely to withdraw or seek revenge

(e.g., Kachadourian, Fincham, & Davila, 2004). There is also evidence that more avoidant people felt less grateful than less avoidant people during interactions in which someone behaved positively toward them (e.g., Beck & Clark, 2010).

Attachment insecurities also interfere with prosocial attitudes and behavior during interactions with people who are distressed or in need. Numerous studies have found that attachment anxiety and avoidance are associated with lower scores on self-report scales measuring sensitivity and responsiveness to a relationship partner's needs (e.g., Millings, Walsh, Hepper, & O'Brien, 2013) and with actually behaving less supportively toward a distressed partner (e.g., Monin, Feeney, & Schulz, 2012). There is also evidence that both dispositional and experimentally augmented attachment security (*security priming*) are associated with heightened empathy and compassion for a stranger or romantic partner who is seeking care and support (e.g., Mikulincer, Shaver, Gillath, & Nitzberg, 2005; Mikulincer, Shaver, Sahdra, & Bar-On, 2013).

In this chapter, we focus on the fact that attachment anxiety and avoidance affect sociability by biasing interpersonal predispositions (interpersonal goals, beliefs about self and others, mental scripts) and biasing the ways in which people attend to, interpret, and respond to information that arises during a social interaction. In the following sections, we review evidence concerning these attachment-related cognitive and motivational biases.

Attachment-Related Cognitive-Motivational Predispositions

People enter social interactions with knowledge and attitudes acquired during past interactions with the same partner(s), or they transfer and apply to new partners their knowledge and attitudes based on previous relationships (Brumbaugh & Fraley, 2006). These personal predispositions are manifested in the various components of sociability tendencies: (a) the goals people typically seek during social interactions; (b) the beliefs they hold about their own social value, lovability, and social skills; other people's likely motives and intentions; and (c) procedural knowledge (*mental scripts*) concerning the ways in which interpersonal exchanges typically unfold. These predispositions can bias the acquisition and use of social information during a particular interpersonal exchange via top-down, schematic processes that favor attention to and encoding of information that fits existing knowledge and disfavor or dismiss information that disconfirms expectations. These predispositions are the main vehicles by which attachment insecurities affect a person' sociability within interpersonal interactions and relationships.

Interpersonal Goals

During security-enhancing interactions with supportive attachment figures, a person learns that closeness and dependency are rewarding and can contribute

to growth and autonomy. As a result, secure people do not view dependency and autonomy as antagonistic goals. Getting close to others and relying on them does not threaten a secure person's sense of autonomy, and maintaining an autonomous stance does not induce worries about separation, rejection, or abandonment.

In contrast, attachment-insecure people are less able to balance or coordinate closeness and autonomy. They are compulsively driven to pursue goals that serve unmet attachment needs, reduce their fears, and allow them to maintain their habitual defenses. On the one hand, avoidant people tend to organize their goals around desires for optimal distance and self-reliance, and they are averse to closeness and interdependence. As a result, avoidance interferes with a person's own intimacy-promoting behavior, responsiveness to a partner's bids for intimacy, and commitment to a lasting relationship. On the other hand, people who are anxious with respect to attachment tend to adopt interpersonal goals compatible with their intense desire to be loved and cared for, and they are frequently afraid of rejection and abandonment. As a result, attachment anxiety can be expected to result in intense and even intrusive bids for closeness or intimacy, strong negative reactions to minimal signs of a partner's distancing, and premature commitment to close relationships as a means of maintaining closeness and dependency. However, due to their fear of rejection, attachment-anxious people are often ambivalent about initiating bids for intimacy or committing to a relationship because they fear that a partner may reject or abandon them if they reveal their unsatisfied hunger for love and care.

Numerous studies have provided empirical support for these ideas. For example, more avoidant people are more likely to emphasize distance-related goals during interpersonal exchanges, whereas more anxious people are more likely to emphasize closeness-related goals—mainly avoiding distance (e.g., Collins, Guichard, Ford, & Feeney, 2004; Emery, Muise, Dix, & Le, 2014). These tendencies tend to be most strongly manifested in interpersonal contexts that activate attachment-related fears and defenses. For example, Overall and Sibley (2009) found that when more avoidant people felt they had insufficient independence or control in their relationships on a given day, they tended to inhibit closeness-related goals during that day.

There is also evidence that attachment-related avoidance is associated with intimacy aversion and less willingness to ask for proximity and support in times of need (e.g., Lynch, 2013; Rowe & Carnelley, 2005). And studies have revealed that attachment anxiety is associated with rejection aversion, fear of being alone, and excessive reassurance seeking as a means of mitigating fears of separation and abandonment (e.g., Shaver, Schachner, & Mikulincer, 2005; Spielmann et al., 2013). Attachment-anxious people's difficulties in regulating their intense need for love and care have also been documented in studies showing that they are more likely to exhibit intrusive behaviors, such as stalking and unwanted pursuit (e.g., Lavy, Mikulincer, & Shaver, 2013).

These difficulties tend to be exacerbated by anxious people's relational ambivalence. In a series of studies, Mikulincer, Shaver, Bar-On, and Ein-Dor (2010) found that attachment anxiety was positively associated with both explicit and implicit measures of relational ambivalence. Moreover, in experimental contexts that highlighted separation from a romantic partner and the negative aspects of relational closeness, attachment-anxious people exhibited a paradoxical preconscious tendency to approach relational closeness, along with heightened signs of implicit motivational ambivalence toward closeness. In addition, attachment-anxious people reacted to contexts that highlighted relationship initiation and the positive aspects of relational closeness with a paradoxical preconscious tendency to avoid relational closeness, along with heightened signs of implicit motivational ambivalence toward closeness. This relational ambivalence seems to increase confusion and indecision about beginning a promising relationship or ending a painful one.

A large body of evidence also shows that attachment insecurities are cross-sectionally and prospectively associated with lower scores on scales measuring relational commitment (e.g., DeWall et al., 2011). Attachment insecurities are also associated with lower relational investment, lower willingness to make sacrifices for a relationship, and higher attraction to alternative partners (e.g., Impett & Gordon, 2010; Overall & Sibley, 2008). Moreover, there is evidence that security priming (e.g., recalling a warm and loving relationship) increases relationship investment (e.g., Gillath & Shaver, 2007).

Mental Representations of the Self

Bowlby (1973) argued that episodes of attachment-figure availability can promote perceptions of the self as valuable, lovable, and special, because one is actually valued, loved, and regarded as special by the caring attachment figure. In contrast, frustrating interactions with unsupportive an attachment figure can challenge and tarnish these positive self-representations. Indeed, adult attachment research has consistently shown that attachment insecurities are associated with negative self-representations (see Mikulincer & Shaver, 2016, for a review). For example, more attachment-anxious individuals report lower self-esteem (e.g., Schmitt & Allik, 2005), view themselves as less competent and efficacious (e.g., Cooper, Shaver, & Collins, 1998), and hold less optimistic expectations concerning their ability to cope with stress (e.g., Berant, Mikulincer, & Florian, 2001; see also Regan & Lyubomirsky, this volume). Moreover, both attachment anxiety and avoidance are associated with having a less coherent model of self (Mikulincer, 1995).

Mental Representations of Others

According to attachment theory, people with different attachment styles should also differ in their perceptions of other people (Bowlby, 1973). Whereas

security-enhancing interactions with available and responsive attachment figures promote a positive view of others, emotionally painful, frustrating interactions with unavailable or rejecting attachment figures contribute to negative views of others (Mikulincer & Shaver, 2016). Indeed, numerous studies have found that attachment anxiety and avoidance scales are associated with lower esteem for others (e.g., Luke, Maio, & Carnelley, 2004), more doubts about others' trustworthiness (e.g., Hofstra, van Oudenhoven, & Buunk, 2005), and less respect for relationship partners (e.g., Frei & Shaver, 2002). Moreover, attachment insecurities are associated with lower ratings of relationship partners' responsiveness, dependability, authenticity, and faithfulness (e.g., Beck, Pietromonaco, DeVito, Powers, & Boyle, 2014) and more negative expectations concerning a partner's behavior (e.g., Rowe & Carnelley, 2003; see also Reis et al., this volume). For example, using a lexical decision task, Baldwin, Fehr, Keedian, Seidel, and Thomson (1993) found that, as compared with secure people, insecure people reacted with shorter decision times (reflecting greater cognitive availability) to words naming negative partner behaviors (e.g., rejection, criticism) than to words naming positive partner behaviors (e.g., acceptance, love).

Mental Scripts

As we explained in our comprehensive review of adult attachment research (Mikulincer & Shaver, 2016), mental representations of interactions with warm, loving, and supportive attachment figures are embodied in a relational if-then script, which Waters, Rodrigues, and Ridgeway (1998) called a *secure-base script*. This script is thought to include something like the following if-then propositions:

> If I encounter an obstacle and/or become distressed, I can approach a significant other for help; he or she is likely to be available and supportive; I will experience relief and comfort as a result of proximity to this person; I can then return to other activities.

Secure people might hold a strong, well-developed secure-base script for interacting with others, which can, by itself, mitigate distress, promote optimism and hope, and help in dealing with life's exigencies. However, insecure people, who have a history of painful and frustrating attachment relationships, lack the coherent and encouraging knowledge base for constructing a coherent secure-base script and automatically applying it during social interactions.

There is evidence for the hypothesized attachment-related differences in the psychological reality of the secure-base script in adults. For example, Mikulincer, Shaver, and Avihou-Kanza (2011) found that people who score higher on self-report scales measuring attachment anxiety or avoidance were less likely to include elements of the secure-base script (support seeking, support provision,

and distress relief) when writing about projective-test pictures of a troubled person or when narrating their own dreams about distressing events. People who scored relatively high on the attachment anxiety scale tended to omit or deemphasize the final step in the script (relief and return to other activities), whereas those who scored relatively high on the avoidance scale tended to omit the part about seeking and benefitting from others' support. That is, anxious study participants more often wrote about an injured person who was seeking support without achieving relief. Avoidant participants more often wrote about a person achieving relief without seeking or receiving support. Similar findings were obtained in studies involving content analyzed dreams about romantic partners (Selterman, Apetroaia, & Waters, 2012) or narratives about current relationships (McLean, Bailey, & Lumley, 2014).

Summary

Overall, it does seem, as theorized, that attachment anxiety and avoidance bias a person's sociability tendencies, including their interpersonal goals, beliefs about self and others, and mental scripts representing interpersonal exchanges. However, more research is needed to determine how the cognitive-motivational predispositions a person brings to a given social interaction influence his or her behavior during the interaction, and to determine whether and how these predispositions mediate the connection between attachment insecurities and sociability.

Attachment-Related Differences in the Processing of Social Information

In this section we review evidence concerning the influence of attachment styles on the processing of information about interpersonal exchanges as they unfold in time. In analyzing attachment-related biases in the processing of information during interpersonal exchanges, we consider the cognitive steps that lead to behavior according to sequential models of social information processing (e.g., Dodge, 2011). In these models, the initial steps of processing involve attention, perception, and interpretation of incoming informational cues, and the later steps involve appraisals of the current situation, the other person's reactions, and the self in relation to this person, as well as decisions among behavioral alternatives in the given situation (response generation, response evaluation, and enactment). Attachment insecurities seem to biases all of these cognitive processes.

Early Stages of Information Processing

There is extensive evidence linking attachment insecurities to biases in the perception of others' nonverbal messages. For example, Noller and Feeney (1994)

asked one member of a newlywed couple to send a set of nonverbal messages expressing particular emotions (e.g., sadness, anger) to the other member of the couple and found that more anxious or avoidant spouses were less accurate in decoding their partner's nonverbal messages. Similarly, using Ekman and Friesen's (1975) facial action coding system, Magai, Distel, and Liker (1995) found that less secure people were less accurate in decoding facial expressions of emotions.

In another study of facial affect decoding, Fraley, Niedenthal, Marks, Brumbaugh, and Vicary (2006) used a "morph" movie paradigm in which participants were shown computerized movies of faces in which a neutral facial expression gradually changed to an expression of a particular emotion. Participants were instructed to stop the display when they perceived that the emotional expression was coming into view, and to judge which emotion the face was expressing (anger, happiness, or sadness; see also Forgas, this volume). More attachment-anxious individuals tended to perceive the onset of facial expressions of emotion earlier and made more errors in judging which emotion the face was expressing. These results suggest that anxious people's difficulties in decoding emotions might be a result of heightened vigilance to emotional cues and a tendency to make premature judgments.

Studies examining avoidant people's defensiveness also provide evidence concerning the early stages of social information processing. For example, Fraley, Garner, and Shaver (2000) examined the hypothesis that avoidant people would direct attention away from, or encode in a shallow way, distressing attachment-related information. Participants listened to an interview about the loss of a relationship partner and were asked later to recall details of the interview, either soon after hearing them (Study 1) or at various delays ranging from half an hour to 21 days (Study 2). An analysis of forgetting curves revealed that (a) avoidant people initially encoded less information about the interview, and (b) people differing in attachment styles forgot encoded information at the same rate. Thus, avoidant defenses sometimes act preemptively, by blocking threatening material from being encoded. In a subsequent study, Fraley and Brumbaugh (2007) found that more avoidant individuals performed worse on tasks assessing both explicit and implicit memories of information about the death of a sister. Interestingly, this memory deficiency was found even when participants' motivation was increased by offering a monetary reward for recalling the loss-related information.

Appraisals of Others During Interpersonal Exchanges

Adult attachment studies have shown that people who differ in attachment style also differ in the way they interpret and explain others' negative behaviors (e.g., Collins, Ford, Guichard, & Allard, 2006). For example, Collins (1996) presented study participants with six hypothetical vignettes about a romantic partner's

troubling behavior (e.g., "Your partner didn't comfort you when you were feeling down") and asked them to write an open-ended explanation of each event and to complete an attribution questionnaire assessing the extent to which the behavior was attributed to internal, stable, controllable, and intentional causes. Insecure individuals' explanations emphasized the partner's poor intentions and negative traits. Specifically, more insecure people were more likely to believe that their partner's negative behaviors were caused by the partner's lack of love, to attribute these behaviors to stable and global causes, and to view them as unkindly motivated. A similar attributional bias was noted by Mikulincer (1998), who found that insecure people tended to attribute hostility to another person even when this person did not show signs of having hostile intentions.

Biases have also been recorded in insecure individuals' appraisals of other people's supportiveness. For example, Collins and Feeney (2004, Study 1) experimentally manipulated a partner's supportiveness and assessed attachment-related differences in perceptions of this support. Dating couples were brought into the lab; one member of the couple was informed that he or she would perform a stressful task (deliver a speech while being videotaped); and the couple was then observed for 5 minutes. Next, couple members were separated, and support was manipulated by having the non-speech-making partner copy either two clearly supportive notes or two ambiguously supportive notes to send to the partner who would be giving a speech. The speech-giver then read the notes and rated both their supportiveness and the partner's behaviors during the prior interaction. Insecure participants rated the ambiguous notes as less supportive and more upsetting and inferred more negative intent than did secure participants. More important, after receiving the ambiguous notes, insecure participants rated their prior interaction as less supportive than would be expected on the basis of judges' ratings of the interactions. These findings were replicated in a second study (Collins & Feeney, 2004, Study 2) in which partners were allowed to write their own authentic notes.

These attachment-related biases in interpretation of others' behavior are also evident in Zhang and Hazan's (2002) study of person perception. Participants received a list of positive and negative traits and were asked to estimate the number of behavioral instances they would need to confirm or disconfirm the possession of each trait by a hypothetical romantic partner. Anxious participants required less evidence to confirm the presence of both positive and negative traits and disconfirm the possession of negative traits. That is, anxious individuals tended to make positive and negative judgments about others relatively quickly and without much behavioral evidence. More avoidant participants requested more behavioral evidence before concluding that others possessed positive traits or that their possession of negative traits was disconfirmed. This finding suggests that avoidance fosters hard-to-refute, negative appraisals of others' traits.

Moreover, it implies that avoidant people's negative views of others may be more stable and pervasive than those of anxiously attached people, who tend to be more receptive to others' positive behaviors.

Although attachment-insecure people tend to engage in biased perception of their partner's behavior, there are also times, of course, when the partner of an insecure individual actually fails to be supportive (Mikulincer & Shaver, 2016). That is, partners of insecure individuals may actually be less supportive than partners of secure people. Rholes, Simpson, Campbell, & Grich, 2001) proposed that, beyond choosing less supportive partners, insecure individuals may actively interfere with their partner's supportive behavior, thus confirming and strengthening their negative expectations. In particular, insecure people's disbelief in others' support may cause them to be suspicious concerning a partner's helpful actions, which in turn may induce the partner to withdraw support over time. In addition, insecure people may repeatedly claim that a partner is unsupportive, until the partner accepts this view and begins to behave non-supportively.

The involvement of both cognitive biases and a partner's actual behavior in determining an insecure person's perception of support was documented by Rholes et al. (2001) in their study of the transition to parenthood. Married women described both their attachment orientation and the support they perceived themselves receiving from their husband approximately 6 weeks prior to the birth of their first child and again 6 months after delivery. In addition, husbands provided information about how supportively they had behaved toward their wife. The more attachment-anxious wives perceived their spouses as less supportive at both times than the husbands claimed to be. This association could not be explained by husbands' reports of the support they offered their wives. However, findings also provided some evidence for the accuracy of anxious women's negative appraisals: Men married to more anxiously attached women reported being less supportive than men married to less anxious women. Taken together, these findings suggest that, although anxious women's hyperactivating strategies may negatively bias their memories of a partner's helpful behavior, there is at least some degree of accuracy in their complaints about their husband's lack of support.

Attachment studies of married individuals provide further information about the role of a partner's actual characteristics in influencing perceptions of support (e.g., Woodhouse, Dykas, & Cassidy, 2009; see also Reis et al., this volume). For example, Gallo and Smith (2001) found that secure couples (i.e., those in which both partners were securely attached) were more confident about the availability of their spouse's support than doubly insecure couples (those in which both spouses had insecure attachment styles). No significant difference was found between secure couples and mixed couples (in which only one partner was secure), implying that the negative perceptions of the insecure partner can sometimes be mitigated by the attachment security of the other partner.

Moreover, since secure spouses tend actually to be more sensitive, responsive, and supportive than insecure spouses (Mikulincer & Shaver, 2016), this finding suggests that insecure people's perceived support from a spouse can be positively affected by the spouse's truly supportive behaviors. Overall, it seems that insecure people's negative appraisals stem from both intrapsychic processes (negative biases in interpretation and memory of a partner's helpful behavior) and dyadic processes (actual partner lack of support, perhaps as responses to the insecure people's behavior).

Self-appraisals During Interpersonal Exchanges

Attachment insecurities also bias people's self-appraisals during social interactions. For example, attachment anxiety has been found to be associated with self-defeating, hopeless interpretations of negative experiences in relationships—an attribution pattern that includes taking responsibility for a disappointing interaction and attributing it to a temporally stable lack of ability, skill, or personal value (e.g., Gamble & Roberts, 2005). Studies assessing beliefs about personal control have also shown that attachment-anxious individuals often feel relatively hopeless about their relationship prospects (e.g., Fass & Tubman, 2002).

Several studies have found that less secure individuals rely more on external sources of validation for their sense of worth, whereas relatively secure individuals rely on an internal, relatively stable sense of worth (e.g., Park, Crocker, & Mickelson, 2004; see also Kruglanski & Ellenberg, this volume). For example, Srivastava and Beer (2005) asked people to take part in four weekly small-group discussions and, following each group session, to rate their own likeability and the extent to which they liked each other person in the group. Participants who were more liked by others following a group session had more positive self-evaluations at a later session. However, this dependence on others' liking was mainly found among participants scoring high on attachment anxiety. For more securely attached group members, self-evaluations were relatively high regardless of what other group members thought.

In two experimental studies, Carvallo and Gabriel (2006) provided initial evidence that the self-esteem of avoidant individuals may also rely on external sources of validation. In these studies, half of the participants were provided with positive information about others' approval, and their post-feedback self-esteem was then compared with that of participants who received no feedback. Findings indicated that more avoidant people reported higher state self-esteem after receiving positive feedback than no feedback. As usual, the self-esteem of more secure participants was not significantly affected by the experimental feedback. These findings were conceptually replicated by MacDonald and Borsook (2010), who asked participants to interact with either a highly positive or mildly negative research confederate.

Response Generation

Insecure individuals' egocentric focus on their attachment-related worries and defenses can interfere with the generation of effective responses that can heighten the quality of interpersonal exchanges. They are so preoccupied with avoiding rejection (in the case of anxiously attached people) or preventing unwanted interdependence (in the case of avoidant people) that they might lack the resources necessary to coordinate their responses with their partner's responses, thereby frustrating or alienating their partner during the interaction.

This lack of flexibility was documented by Mikulincer and Nachshon (1991) in a study of self-disclosure. More secure individuals reported disclosing more personal feelings to a close relationship partner than to a stranger. However, anxious people reported heightened self-disclosure even to a stranger, and avoidant people reported relatively low levels of self-disclosure even to their closest relationship partner (see also Forgas, this volume, on mood effects on self-disclosure). Lack of interpersonal coordination is also evident in Whiters and Vernon's (2006) findings that attachment-anxious people were more likely to experience embarrassing situations in which they failed to follow a coherent script in social interactions. Snir and Wiseman (2010) asked dating and married couples to engage in a joint drawing task and found that more anxious and avoidant participants were less likely to evaluate the joint drawing session as smooth and positive.

Studies have also shown that less secure people are less likely to use humor as a strategy for improving and smoothing social interactions. Specifically, insecure people are less likely to endorse a social style of humor (i.e., use humor to amuse others and heighten social connections) and more likely to endorse maladaptive styles of humor, such as hostile humor and self-defeating humor (e.g., Sar-El, Mikulincer, & Doron, 2013). Insecure individuals are also less likely to engage in behavioral mimicry—a form of non-verbal behavior that promotes liking and smoothness of social interactions (e.g., Chartrand & Bargh, 2002). Hall, Millings, and Bouças (2012) assessed the extent to which participants engaged in implicit behavioral mimicry of a confederate presented by video. Following a rejection-related prime, less secure people showed less mimicry of face-rubbing gestures. Attachment orientation did not predict baseline levels of face-rubbing, demonstrating that attachment orientation had an effect on mimicry, not on overall behavior expression.

Another aspect of interpersonal coordination includes attempts to modify internal states to make them congruent with a partner's behavior. Gabriel, Kawakami, Bartak, Kang, and Mann (2010) found that more secure participants who interacted with a low self-esteem partner were more likely to lower their self-esteem and thus feel more similar and connected to the interacting partner, even when paying the price of lowering their sense of self-worth. In contrast,

less secure participants were less likely to display this kind of synchronization, probably in an effort to protect their fragile self-esteem.

Insecure people's lack of interpersonal coordination can also be related to their lack of attention to a partner's signals and feelings of alienation from him or her. Rholes, Simpson, Tran, Martin, and Friedman (2007), for example, found that more avoidant people knew less about their romantic partners' emotional lives and were less interested in knowing their partner's thoughts and feelings. Furthermore, Rholes et al. (2007) found that anxiously attached people were less attentive to their partner's needs, focusing on meeting their own emotional needs rather than attending to the emotional needs of their partner.

Concluding Remarks

An enormous, and constantly growing, body of evidence shows that attachment orientations are profoundly related to strategies of sociability, including people's goals and wishes in interpersonal encounters; their mental representations of the self, others, and social interactions; the ways in which they process incoming information during these interactions; and their pattern of interpersonal experiences and behavior. The findings reviewed here, as well as many more reviewed in our comprehensive book about adult attachment (Mikulincer & Shaver, 2016), clearly show that sociability involves a long chain of mental states and observable behaviors—a chain in which the links often involve reciprocal causal arrows. A person's mental representations of self and others in a given interpersonal exchange are residues of actual experiences in relationships, sometimes amplified or shaped by imagination, rumination, worries, and wishful thinking. These mental representations color subsequent concerns, goals, and action plans, which influence actions, which not only shape joint self-partner situations, which are again interpreted, remembered, and ruminated upon in ways colored by pre-existing scripts or models, but which can also alter those scripts or models (usually only slightly, unless similar information is repeatedly incorporated). All stages of this complex process have begun to receive researchers' attention, but we need more sophisticated research aimed at examining these recursive processes over long periods of time in different kinds of relationships.

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13

SOCIABILITY MATTERS

Downstream Consequences of Perceived Partner Responsiveness in Social Life

Harry T. Reis, Guy Itzchakov, Karisa Y. Lee, and Yan Ruan

Abstract: Extensive research has documented people's desire for social partners who are responsive to their needs and preferences, and that when they perceive that others have been responsive, they and their relationships typically thrive. For these reasons, perceived partner responsiveness is well-positioned as a core organizing theme for the study of sociability in general, and close relationships in particular. Research has less often addressed the downstream consequences of perceived partner responsiveness for cognitive and affective processes. This gap in research is important, because relationships provide a central focus and theme for many, if not most, of the behaviors studied by social psychologists. This chapter begins with an overview of the construct of perceived partner responsiveness and its centrality for relationships. We then review programs of research demonstrating how perceived partner responsiveness influences three core social-psychological processes: self-enhancing social cognitions, attitude structure, and emotion regulation. The chapter concludes with a brief overview of how deeper incorporation of relationship processes can enhance the informativeness and completeness of social psychological theories.

By now there can be little doubt that social connections represent some of the most powerful influences on human well-being and health (see also Dunn & Lok; Fiske; Kruglanski & Ellenberg; Mikulincer & Shaver; and von Hippel & Smith, this volume). Many literature reviews have come to this conclusion (e.g., Lyubomirsky, King, & Diener, 2005), and several large-scale meta-analyses have empirically verified it (e.g., Holt-Lunstad, Smith, & Layton, 2010; Holt-Lunstad, Smith, Baker, Harris, & Stephenson, 2015). The weight of evidence has led the editors of the Public Library of Science to editorialize that "social

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relationships are key to health, and to health policy" (2010, p. e100334). Also, no less a leader in public medicine than Vivek Murthy, Surgeon General of the United States under Barack Obama, and recently reappointed to the same position by President Joe Biden, testified to this association, in his 2020 book "Together," subtitled "The healing power of human connection in a sometimes lonely world." It would seem, then, that the scientific case for the importance of relationships is clear, compelling, and well-established.

Impressive as these findings may be, research has not yet identified just what it is about social connections that fosters illness and unhappiness on the one hand, or that promotes health and well-being on the other hand. Under the heading of relationship science, a large number of principles and processes have been articulated and studied, all of them designated in empirical studies as characteristics of well-functioning human relationships. These include, for example, rapport, empathy, trust, compassion, love, intimacy, social support, chemistry, social-network participation, marriage, secure attachment, communal sharing, autonomy support, relatedness, shared identities, emotional acceptance, commitment, belonging, and inclusion of other in the self. This list is but a sampling of some of the more popular constructs.

All of these constructs have well-deserved positions in the literature; nevertheless social psychologists have not articulated overarching themes that can tie them together into a theoretically integrated framework—in other words, rather than emphasizing what makes each construct distinct and different, we might instead ask, what properties do they share in common? How do their everyday manifestations engender behavior that influence people in similar ways? Identifying and attending to their common elements might facilitate research and theory that will move relationship science to a deeper, more cohesive, and more useful understanding of precisely how relationships foster health and well-being.

Elsewhere, the first author has discussed these questions in terms of a search for core organizing principles—that is, a systematic examination of overlapping elements that can integrate related yet seemingly disparate phenomena and contribute to a cumulative science (Reis, 2012). The first author has further proposed that perceived partner responsiveness could serve in that role. Perceived partner responsiveness refers to the belief that an interaction partner understands, values, and is willing to care for and support oneself (Reis & Clark, 2013; Reis & Shaver, 1988). Perceived partner responsiveness emerges from repeated interactions in which one partner reveals important aspects of the self and the other partner responds knowingly, respectfully, encouragingly, and supportively. By emphasizing self-disclosure and response processes, perceived partner responsiveness is grounded both in enacted patterns of interaction and in the individual's motivated interpretations of those interactions (Reis, Clark, & Holmes, 2004).

Extensive research has demonstrated the relevance of perceived partner responsiveness to relationship well-being. For example, perceived partner responsiveness has been shown to underlie effective social support (Maisel & Gable, 2009) and

its role in lowering all-cause mortality (Selcuk & Ong, 2013), while promoting gratitude (Algoe & Zhaoyang, 2016), caregiving (Canevello & Crocker, 2010; Lemay & Clark, 2008), attraction to potential romantic partners (Birnbaum & Reis, 2012; Reis, 2020), constructive conflict resolution (Overall & McNulty, 2017), beneficial outcomes of sharing personal good news (Gable, Gosnell, Maisel, & Strachman, 2012; Reis et al., 2010), sexual satisfaction and desire (Birnbaum et al., 2016), forgiveness (Pansera & La Guardia, 2012), security-enhancing interactions (Simpson, Rholes, Oriña, & Grich, 2002), effective partner support of goal-directed activity (Rusbult, Kumashiro, & Reis, 2010), sleep quality (Selcuk, Stanton, Slatcher, & Ong, 2017), pain management (O'Neill, Mohr, Bodner, & Hammer, 2020), marital conversations about end-of-life care (Moorman, 2011), and coping with COVID-related stress (Slatcher, 2021). Partner responsiveness is likely to be a key variable not only in personal relationships, but also in all forms of human sociability in general.

These and many other studies document *inter*personal benefits of perceived partner responsiveness. In this chapter, we aim to discuss research that explores more *intra*personal processes, namely, how perceived partner responsiveness helps individuals engage the environment in a less defensive, more open-minded manner. The basic premise underlying these programs of research is that when people feel understood, appreciated, and supported by significant others, their sense of psychological safety and acceptance is enhanced, thereby attenuating the need to enact self-protective defenses and orientations (cf., Murray, Holmes, & Collins, 2006; see also Mikulincer & Shaver, this volume).

In a larger sense, the goal of this chapter is to argue that social-psychological research has often neglected the role that relationships play in many of its most popular constructs. By examining downstream consequences of perceived partner responsiveness in three central areas of social-psychological work—egoic self-enhancement, open-mindedness, attitude structure, and emotion regulation—we hope to illustrate how recognition of the central role that relationships play in everyday life can enhance the informativeness and completeness of social psychological theory, and in practice become positive influences in society.

Lessening Egoic Self-enhancement

Motives to protect and enhance the favorability of self-perception are well established in the social-psychological literature (see also Gaffney & Hogg; and Kruglanski & Ellenberg, this volume). Extensive research spanning several particulars shows that people generally construe themselves, their activities, and their circumstances in an overly positive light. As Sedikides recently put it, in an overview of this large literature, "self-protection and self-enhancement rush in where [objectively realistic] self-assessment fears to tread" (2020, p. 264). This propensity is revealed in such staples of every introductory social-psychology textbook as the tendency to evaluate the self more favorably than average others

(Alicke & Govorun, 2005; Brown, 2012), to perceive more control over events than is warranted (Langer, 1975; Thomson, Armstrong, & Thomas, 1998), to claim greater personal responsibility for success than for failure (Mezulis, Abramson, Hyde, & Hankin, 2004), to view one's own relationships as superior to others' relationships (Reis, Caprariello, & Velikovic, 2011; Van Lange & Rusbult, 1995), and to overperceive one's personal contributions to a group product (Ross & Sicoly, 1979). These and other related tendencies are typically interpreted as manifestations of a broad self-serving bias that protects and enhances self-worth and facilitates constructive engagement with the environment (Sedikides & Gregg, 2008; see also Prislin & Crano, this volume).

Some of the most persuasive evidence that these biases reflect self-serving motives comes from research in which the self anticipates threat—for example, potential failure on a diagnostically significant task or the anticipation of a stressful public performance (Campbell & Sedikides, 1999). Conversely, self-related concerns can be relaxed by making a person's positive attributes salient, thereby minimizing these biases (Sherman & Cohen, 2006; Steele, 1988). In most of the existing studies, the individuals intrapersonally affirm their own abilities, values, or expected future (see also Baumeister & Maranger; and Krueger et al., this volume). In the research to be described, we wanted to provide evidence for the impact of *interpersonal* threats and affirmations. That is, when significant others are perceived to be salient affirming sources of responsive support and encouragement, the need to self-enhance is lessened, presumably because the social environment provides positive feedback about one's worth and value to others (Leary & Baumeister, 2000). On the other hand, when people feel that significant others are misunderstanding, nonvalidating, and uncaring, self-enhancing biases become stronger, reflecting the need to compensate for the perceived nonresponsiveness of others, a principle that may have impact on all forms of sociability.

I next describe three studies from our lab that demonstrate these principles (Reis, Lee, O'Keefe, & Clark, 2018).

The "Better-Than-Average" Effect

Across diverse traits and skills, people tend to describe themselves more favorably than an average peer (Alicke & Govorun, 2005; Brown, 2012), in large part reflecting the operation of self-enhancement motives (Guenther & Alicke, 2010). We used Guenther and Alicke's (2010) paradigm, asking a sample of 182 undergraduates to compare themselves to the average student at their university on a series of 23 generally positive trait adjectives, such as cooperative, faithful, motivated, and intelligent. Prior to providing their ratings, participants were randomly assigned to one of four conditions. In the key *responsive* condition, they were asked to describe someone "who esteems you, shortcomings and all" and is "interested in what you are thinking and feeling." In the *unresponsive*

condition, they described someone who is "often critical and seems unwilling to be helpful and supportive." The study included two control conditions. In one, participants were asked about a neutral acquaintance: "someone who knows you, but not particularly well." In the *organizer* condition, participants described an organizational tool that helps them get through the day. In all conditions, participants completed a series of specific open-ended questions so that they would spend around 15 minutes thinking and writing about their assigned topic.

All trait ratings were provided on −9 (much less than the average student) to 0 (about average) to +9 (much more than the average student) scales. Of course, everyone is better or worse than average on various traits, but across the full set of ratings, if there were no self-serving bias, we would expect the group average to approximate 0. Across all 23 trait adjectives, participants in the *responsive* condition rated themselves significantly less favorably (M=2.09) than participants in the *unresponsive* condition (M=3.10). Mean ratings in the two control conditions fell in between these extremes: for the *acquaintance* condition, M=2.67; for the *organizer* condition, M=2.48. It is noteworthy that this pattern opposes what might have been expected from a mood-contagion perspective: participants who spent their writing time describing a very positive social relationship nevertheless rated themselves less favorably than participants whose writing was less positive. The fact that average ratings were still greater than the scale midpoint suggests that some self-enhancement bias was still present. However, the point to be made here is that perceived responsiveness lessened this tendency whereas perceived unresponsiveness increased it.

Overclaiming Credit for Housework

When cohabiting partners' report on their personal contributions to household tasks, they generally take more credit than is objectively warranted—in other words, when both partners' reports are added together, they typically sum to more than 100% (e.g., Marini & Shelton, 1993; Press & Townsley, 1998; Ross & Sicoly, 1979). Most people do not find household tasks particularly fulfilling or enjoyable, and for that reason, inflated claims of responsibility are usually thought to be examples of a self-serving bias, consistent with the idea that contributing to the common good is considered virtuous and tends to foster positive affect (Sun, Stevenson, Kabbani, Richardson, & Smillie, 2017).

In this study, 772 cohabiting participants reported on the percentage of personal responsibility they took for 26 common household activities. We included tasks traditionally done by women (e.g., cooking meals, washing, mending clothes) and by men (e.g., auto repair, yard work). Prior to providing these estimates, we adapted a responsiveness-priming procedure from Lemay, Clark, and Feeney (2007), which is less suggestive than the one used in the previous study. In the *responsiveness* condition, participants were asked to recall and list two kind or considerate things that their partner had done for them in

the past week. In the *unresponsive* condition, participants listed 10 such things. Because it should be easy for most people to recall two kindnesses, the former task was expected to activate feelings of perceived responsiveness. However, recalling 10 such acts is likely to be challenging, which therefore was expected to prime doubts about the partner's responsiveness (Schwarz, 1998). In a *control* condition, participants were asked to list five qualities of an acquaintance they "don't have strong feelings about, one way or the other."

Not surprisingly, and consistent with much prior work, women reported a higher percentage of contributions than men did (M_{Women} = 59.4%; M_{Men} = 54.7%). Nevertheless, participants in the responsive condition claimed less credit for their contributions, M = 52.4%, than participants in the unresponsive condition, M = 62.3%. The mean for the control condition fell in between these values, M = 54.4%. There was no interaction effect. Namely, the difference across conditions was significant and similar for both sexes. Once again, if the results had simply matched the valence of the priming manipulation, we should have seen that listing ten kind or considerate acts would result in lower estimates than listing only two kind or considerate acts. The fact that the results did not conform to this pattern supports our theorizing about how responsiveness lessens the need to self-enhance.

Hindsight Bias

People sometimes claim that they knew something all along, even though direct evidence indicates that they did not. This effect, commonly known as hindsight bias, is theorized to reflect the desire to see oneself as well-informed and perceptive, presumably because these qualities enhance self-worth and a sense of coherence and control about the world (Roese & Vohs, 2012). Consistent with the results of the prior two studies, we reasoned that when perceived partner responsiveness is activated, the need to claim knowledgeability would be lessened, given the salience of affirming feedback from significant others. On the other hand, we expected that unresponsiveness would heighten this tendency, given the threat to self-conceptions that misunderstanding and non-validation entail.

To assess hindsight bias, we used a paradigm developed by Werth and Strack (2003), in which participants were given a series of 13 engaging trivia questions (e.g., "How many paintings did Pablo Picasso create?") along with the correct answer (1,885). We then asked how they would have responded, had they not been told the correct answer. Because these questions are of a sort whose answers are typically unknown but credible, the closer the response to the provided answer, the greater the indication of hindsight bias. Thus, we calculated the absolute discrepancy between participants' responses and the correct answer, divided by the standard deviation, and then summed across all questions, resulting in an index for which higher scores (i.e., responses more divergent from

the provided answer) denote a greater willingness to acknowledge that one had not known the correct answer—that is, less hindsight bias.

Prior to receiving the 13 trivia questions, participants completed the same responsiveness-priming task as in the overclaiming-credit-for-housework study. A total of 579 persons were randomly assigned to one of the three conditions. As expected, hindsight was least in the *responsive* condition (M=15.0), intermediate in the *control* condition (M=14.6), and lowest in the *unresponsive* condition (M=13.2). In short, after being reminded of a responsive partner, the tendency to claim knowledgeability was minimized; in contrast, a reminder about an unresponsive partner amplified hindsight bias.

Interim Summary

Taken together, these studies indicate that self-serving biases in social cognition can be ameliorated by activation of thoughts about responsive social partners, whereas thoughts about unresponsive others amplify these tendencies. To the best of our knowledge, this is the most direct evidence yet available demonstrating that an *inter*personally based affirmation can have similar effects to *intra*personal self-affirmations, suggesting a close link between patterns of sociability and internal mental processes (see also Baumeister & Maranger; and Krueger et al., this volume). Although our protocol differs in scope from the standard self-affirmation procedure—the latter is based on abstract personal values, whereas responsiveness refers to specific relationships and interactions—the underlying mechanisms appear to include common elements. Critcher and Dunning (2015) found that self-affirmations alleviate distress and defensiveness by broadening people's perspectives so that their self-assessments are less firmly tethered to momentary experiences of threat. We repeated their experiment using the same responsiveness inductions as in the prior two studies, and obtained the same result: Perceived responsiveness led participants to adopt a broader perspective, but perceived unresponsiveness fostered a more narrow focus. Thus, reminders of others' valuation and care can lessen the ego-protective defenses that people frequently adopt to make the self seem worthy and the world appear more manageable.

Attitudinal Openness and Structure

Many of the entities about which we have attitudes include both positive and negative elements. For example, we can like Joe Biden's approach to the pandemic but dislike his immigration policies. This sort of inconsistency can create an aversive evaluative conflict, termed felt or subjective ambivalence, typically characterized by indecisiveness (Priester & Petty, 1996). As a result, people typically strive to resolve their inconsistent beliefs and restore a sense of cognitive consistency and a coherent view of the world (Itzchakov & Van Harreveld,

2018). These processes reflect the operation of the "psychological immune system," which helps us reinterpret attitudinal objects and situations in a way that seems less threatening and precludes the necessity of changing one's beliefs (Clark & Wegener, 2013). For example, people often focus their attention on attitude elements that support their pre-existing beliefs while excluding contradictory items, defensively dismissing conflicting information, or over-perceiving the correctness of their positions relative to other positions (Itzchakov, Amar, & Van Harreveld, 2020).

Because the formation, maintenance, and revision of attitudes have interpersonal roots in addition to their more commonly studied intrapersonal origins, Itzchakov and Reis (2021) explored the role of perceived partner responsiveness on open-mindedness and attitude structure—specifically, attitude ambivalence and correctness. As described earlier, when people feel understood and valued by others, they can more readily process information openly and non-defensively. This is because perceived responsiveness lessens concerns about self-worth and acceptance by others, diminishing the need to defend the coherence and correctness of one's point of view (Leary & Baumeister, 2000; Reis et al., 2004; Sedikides & Gregg, 2008; see also Baumeister & Maranger; and Kruglanski & Ellenberg, this volume). Hence, we hypothesized that people would be more open-minded in considering opinions that differ from their own, more willing to tolerate this internal consistency, and behave in an open-minded manner. In contrast, we expected that when people feel that others do not understand or value their beliefs, they will process information more defensively, such as by focusing on attitude-congruent elements (Frey, 1986), discounting the significance of attitude-inconsistent input (Ottati, Price, Wilson, & Sumaktoyo, 2015), or bolstering their initial attitudes (Festinger, Riecken, & Schachter, 1956). This should result in a more closed-minded approach to attitude-inconsistent elements, lesser tolerance of opposing elements within the attitude, and an enhanced sense of correctness of their existing beliefs.

Itzchakov and Reis (2021) conducted a series of four experiments and one diary study examining these hypotheses. Because the four experiments used very similar designs, we describe the general method here. Upon arrival in the lab, participants were asked their attitude about the focal issue—for example, genetically modified food—after which they were asked to write a short essay advocating for their position. They were then randomly assigned to *responsiveness*, *unresponsiveness*, or *control* condition, using either one of the manipulations described in the prior section of this chapter. Subsequently, they completed several dependent measures. For parsimony, for each measure, we will describe results from a single study. However, most of these measures were included in at least two of these experiments and yielded consistent results.

Open-Mindedness

Open-mindedness, defined as the willingness to consider multiple points of view, allows people to be aware of, without necessarily endorsing, divergent perspectives. Receptiveness to novel information has been linked to more effective information processing and decision-making, and higher creativity (see Petty, Briñol, Loersch, & McCaslin, 2009, for a review), as well as better ability to work collaboratively with others (see also Fiedler & Hütter, this volume). We operationalized open-mindedness with a standardized thought-listing protocol (Cacioppo & Petty, 1981). Participants were given 2.5 minutes to list as many thoughts as came to mind about the topic identified at the beginning of the study (e.g., genetically modified food). We then tabulated the ratio of the number of thoughts opposing their initial attitude divided by the total number of thoughts listed. The scores ranged from 0 (no attitude-incongruent thoughts) to 1 (only attitude-incongruent thoughts).

Results supported the hypotheses. Participants listed a significantly greater proportion of thoughts opposing their initial attitude in the *responsive* condition (M=0.31) compared to the *control* condition (M=0.23), which in turn had a larger proportion of opposing thoughts than in the *unresponsive* condition (M=0.13).

We found similar results in a diary study conducted with M.B.A. students. At four intervals, each separated by three days, participants were asked to describe a conflict experienced at work. Responsiveness was assessed with the perceived responsiveness questionnaire, which asks about how understanding and validating other people in general are (Reis, Crasta et al., 2018). In this analysis, open-mindedness was indicated by listing a relatively greater proportion of thoughts that opposed their initial position on the conflict. Consistent with the lab experiments, fluctuations in perceived responsiveness predicted fluctuations in open-mindedness such that greater responsiveness was accompanied by listing a larger proportion of opposing thoughts.

These results complement findings from a two-week daily diary study in which undergraduates were asked to report on behaviors indicative of openness to considering alternative points of view—for example, whether they had read an article or listened to a podcast that disagreed with their beliefs or whether they had conceded a point of difference to another person (Reis, Lee et al., 2018). In that study, the more responsive participants felt their friends had been on a given day, the more they reported having been open-minded in their behaviors.

Simultaneous Accessibility and Ambivalence

Another way in which to conceptualize open-mindedness refers to simultaneous accessibility, or, the degree to which attitude-consistent and attitude-inconsistent

thoughts come to mind quickly and equally quickly (Newby-Clark, McGregor, & Zanna, 2002). Simultaneous accessibility is a precondition for objective ambivalence—the recognition that an attitude object has both desirable and undesirable features, which, as mentioned earlier, creates an evaluative conflict and the possibility of internal confusion and mixed feelings. We reasoned that open-mindedness would increase simultaneously accessibility; however, because responsiveness lessens the salience of threats to self-worth, this sort of mental conflict should be less uncomfortable, allowing people to be aware of their mixed feelings without experiencing epistemic uneasiness.

Three experiments supported this conjecture. Following Newby-Clark et al.'s (2002) procedure, we recorded response latencies to a pair of split-semantic differential items asking separately about positive and negative feelings and thoughts about the attitude object. A standardized index based on these latencies confirmed that responsiveness increased levels of simultaneous accessibility—that is, quicker and equally quick latency to both positive and negative items. Moreover, we assessed objective ambivalence using a split-semantic differential scale (Kaplan, 1972) and found that perceived responsiveness increased it—that is, awareness of both the attitude object's positive and negative features was enhanced. However, and key to our theorizing, subjective ambivalence—the extent to which participants felt indecisive, mixed, and torn regarding their attitude—was unrelated to simultaneous accessibility and objective ambivalence. That is, in the *unresponsive* and *control* conditions, simultaneous accessibility was significantly correlated with evaluative discomfort (i.e., subjective ambivalence; rs = .50 and .35, respectively), consistent with the idea that conscious recognition of opposing elements of an attitude can be uncomfortable. But in the *responsive* condition, simultaneous accessibility and subjective discomfort were uncorrelated (r = .03). Simply stated, responsiveness allowed participants to be aware of contradictory elements of their attitudes without experiencing mental discomfort. Thus, responsiveness makes it easier for people to be aware of, yet tolerating, apparent contradictions in their attitudes.

Attitude Correctness and Behavioral Intentions

The flipside of ambivalence and openness is certainty: the subjective assurance in the correctness and validity of one's attitudes. When people are open-minded, they appreciate that their positions might not be correct, which promotes consideration of alternative points of view (Petrocelli, Tormala, & Rucker, 2007). Defensiveness, in contrast, closes them to examining opposing perspectives, an orientation that can be justified by bolstering conviction in the correctness of one's attitudes. Following the same theoretical thread as described earlier, we hypothesized that when feelings of unresponsiveness are activated, beliefs about the correctness of one's attitudes would be magnified, but when people feel understood and appreciated, this certainty would be diminished.

This experiment was similar to the earlier studies but did not include a control group. As expected, participants in the *responsive* condition reported being significantly less certain of the correctness of their attitude toward genetically modified food (M=4.19) than participants in the *unresponsive* condition (M=5.70). Moreover, attitude correctness mediated participants' intentions to enact behaviors consistent with an open-minded perspective (e.g., reading newspaper articles or blogs that expressed counter-attitudinal viewpoints). In the *responsive* condition, these behavioral intentions were significantly greater (M=5.68) than in the *unresponsive* (M=5.09) condition. A formal test of mediation confirmed this finding.

Additional results from the diary study with M.B.A. students corroborated this finding. Fluctuations in responsiveness were associated with fluctuations in perceived correctness and behavioral intentions: the more responsiveness participants experienced in their social networks, the less convinced they felt about the correctness of their own position regarding the conflict at work, and the more willing they were to enact behaviors associated with the opposing position.

Interim Summary

Attitudes are among the most venerable subjects in the pantheon of social-psychological research (see also Prislin & Crano, this volume). Traditionally studied as an intrapersonal construct, our research shows that the structure of attitudes also has interpersonal roots, linking patterns of sociability to crucial internal mental representations, something that, to be sure, early attitude researchers recognized (see Briñol & Petty, 2012, for a review). Our studies reveal that when people feel that others have been responsive, they become more open-minded, engendering a weaker sense of attitude correctness accompanied by greater awareness of, and tolerance for, simultaneously pro and con beliefs. One implication of these findings points to the contemporary upsurge of factionalism in political ideology (Finkel et al., 2020). Perhaps this trend reflects citizens' sense that the social environment has become less responsive to their needs and values (see also Itzchakov, Weinstein, Legate, & Amar, 2020), a theme that underlies much of the current resurgence of populism (Forgas, Crano, & Fiedler, 2021). If so, targeting responsiveness might provide a means of increasing openness and tolerance, with corresponding gains in cooperative resolutions of differences of opinion.

Emotion Regulation

Expressing emotions to others is a popular strategy for modulating and maintaining emotional states—for example, by soliciting support or advice, by seeking help to downregulate the emotional aftermath of an upsetting experience, or

by savoring a happy occurrence (Clark, Fitness, & Brissette, 2001; Gross, 2015; Rimé, 2007; see also Fiske; and Forgas, this volume). The usual targets of these emotional expressions are close relationship partners, but because close relationships inherently entail risk and vulnerability (Murray et al., 2006), the nature of these relationships influences the extent to which partners are willing to open up about their emotions. Clark et al. (2001) speculated that the critical variable here is responsiveness: does one believe that a partner understands and values oneself and can be expected to respond in a caring, supportive manner? Emotional openness, after all, signals others about something that one needs or desires. When responsiveness is high, emotion expressers can anticipate that partners are accepting enough to acknowledge the legitimacy of their emotions, knowledgeable enough to provide a response that matches their needs and concerns, and caring enough to want to help, even if that help is costly. In contrast, when partners are not perceived as responsive, the risks of emotional openness may often appear to outweigh the possible benefits. In this case, self-protection motives and emotional reserve predominate.

Although emotion scholars generally recognize that most emotion-regulation attempts occur in an interpersonal context (Ekman & Davidson, 1994), existing research has tended to emphasize intrapersonal strategies. To help fill this gap, we conducted a pair of studies (Ruan, Reis, Clark, Hirsch, & Bink, 2020). The first study was a laboratory experiment focused on the expression of anxiety in the context of a specific, anxiety-provoking task. The second study was a daily diary study examining the expression of various emotions in everyday life. In both studies, we expected that perceived responsiveness would encourage emotional openness.

In the lab experiment, to induce responsiveness, participants (all of whom were in an ongoing romantic relationship) completed the same "list 2 (10) kind or considerate things your romantic partner has done for you in the past week" described earlier. In the control condition, they were asked to list five things in their partner's dormitory room. Next, participants were told that they would video-record a five-minute speech about their personal strengths and weaknesses for obtaining their dream job, and that expert evaluators would judge these recordings. This task, known as the Trier Social Stress Test, elicits high levels of anxiety among college students, especially as they anticipate their speech. After receiving these instructions, while the experimenter prepared the camera for recording, we asked them to send an email to their romantic partner describing their feelings about the experimental task. There were no instructions or time limits as to what they could write.

We used the Linguistic Inquiry and Word Count (LIWC; Pennebaker, Boyd, Jordan, & Blackburn, 2015) to code the content of participants' emails. In the *responsive* condition, participants used a significantly percentage of negative emotion words (M=2.19%) and anxiety words in particular (M=0.67%) than in the *control* (Ms=1.85% and 0.41%) and *unresponsive* (Ms=1.70% and 0.41%)

conditions. The latter two conditions did not differ from each other. We also had coders blind to conditions code the emails for expressions of anxiety and social support-seeking. Once again, expressions of anxiety (M=2.68) and support-seeking (M=2.47) were significantly more common in the *responsive* condition than in the *control* (Ms=2.15 and 1.54) and *unresponsive* (Ms=2.20 and 1.64) conditions (which again did not differ significantly; although we cannot be certain, we suspect that this lack of difference occurred because a fair number of participants in this condition were able to list 10 kind or considerate things with little difficulty, thus obviating the intended effect of the manipulation). These results indicate that responsiveness encouraged participants to be emotionally open about this stressful task they were about to complete.

Our diary study took a different approach. Herein, at study onset, we asked participants to identify someone with whom they had a meaningful relationship and with whom they expected to communicate daily for the next two weeks. Forty-one percent named a romantic partner, 54% chose a close friend or roommate, and 8% picked a family member. Then, each night, on a series of 11 items, participants described the extent to which that target person had been responsive to them on that day (e.g., "Today [target person] really listened to me"). Separately, participants indicated whether they had experienced each of eight different emotions on that day—joy, pride, gratitude, contentment, excitement (in a positive way), anger, anxiety, and sadness—and, if so, whether they had expressed these emotions to their target person.

Analyses demonstrated that daily fluctuations in perceived partner responsiveness were associated with similar fluctuations in emotional openness, suggesting a close link between patterns of sociability and affective reactions (see also Fiske; and Forgas, this volume). That is, on days when participants felt that their partners had been somewhat more responsive than average, they were more likely to express emotion to them. Similarly, on days when participants felt that their partners had been less responsive than usual, they were less likely to express emotion to them. Interestingly, although these effects were strong for all emotions, they were appreciably stronger for the positive emotions than for the negative emotions, suggesting that people may be relatively more reluctant to share positive than negative experiences when a partner's understanding and encouragement seem unsure and risky. This finding dovetails with other research, in which we have shown that people's willingness to share personal good news with relationship partners depends on their expectation of an enthusiastic, supportive response (see Gable & Reis, 2010, for a review).

Discussion and Conclusion

A growing body of research has documented people's desire for social partners who are responsive to their needs and preferences, and when they perceive that others have been responsive, individuals and their relationships typically thrive

(see also Dunn & Lok, this volume). The research reviewed in this chapter contributes to this literature by showing that perceived partner responsiveness has beneficial effects in lessening defensiveness and promoting a more open-minded perspective toward alternative points of view and other persons. In principle, the motives underlying egoic self-enhancement and attitudinal fixedness are functional because they support agentic striving and self-esteem—that is, seeing the self as competent and worthy, and the world as a safe and coherent place (Sedikides & Gregg, 2008). However, and somewhat ironically, these tendencies often interfere with people's ability to develop their skills, advance their interests, and interact with others in a harmonious, enjoyable manner (see also Regan & Lyubomirsky, this volume). Thus, understanding the processes that intensify or attenuate self-enhancement and open-mindedness has the potential to provide researchers and lay people alike with insights useful for more flexibly engaging the social environment.

Calling attention to research that might provide insights for helping people lessen their self-limiting motives is not a novel endeavor, of course. Over the years, social psychologists have conducted many such studies. This work has mostly focused on intrapersonal processes. For example, when people explicitly reaffirm their personal values and beliefs, such as by listing personal strengths or the tenets they most value, self-serving biases tend to wane (Sherman & Cohen, 2006; Steele, 1988). The current work on perceived partner responsiveness contrasts with this approach in its more interpersonal focus, emphasizing not "what *I* value" but rather "how others value *me*." By shifting from self as subject to self as object, perceived partner responsiveness directs attention to the relationship with the other, a difference in perspective that has significant consequences. For example, other-focused orientations, as opposed to self-focus, are associated with greater communal motivation, higher levels of empathy and compassion, more and more effective support-giving, and stronger feelings of responsibility among leaders (Le, Impett, Lemay, Muise, & Tskhay, 2018; Rameson & Lieberman, 2009; Scholl, Sassenberg, Scheepers, Ellemers, & de Wit, 2016). We would expect, therefore, that perceived partner responsiveness interventions would have stronger implications for these phenomena than would self-affirmation interventions.

More broadly construed, this research is part of a budding movement to examine social-psychological processes in terms of their relationship context (Clark, Lemay, & Reis, 2020). Self-serving motives, attitudes, and emotion regulation may seem to be characteristics of individuals—and have traditionally been studied as such—but our work suggests the value of taking a broader view, emphasizing how these processes reflect people's response to their relational circumstances (see also Mikulincer & Shaver; and von Hippel & Smith, this volume). Increasingly, social psychologists recognize that the relationship context in which behavior takes place—who one is with, who one is thinking about, and who might be affected by one's actions—can profoundly influence many,

if not most, social-psychological processes. Consider such venerable social-psychological topics as conformity, moral judgments, helping, social comparison, social identity theory, or aggression. In each case, one might reasonably expect behavior to differ as a function of the relational context; indeed, for many core social-psychological processes, research already so demonstrates (see Clark et al., 2020, or Reis, Collins, & Berscheid, 2000, for reviews; see also Prislin & Crano, this volume). As just one example, the factors that influence helping in high communal-strength relationships differ from those that affect helping strangers, and people reason differently when making moral judgments about their loved ones than about anonymous others.

In this light, it may seem incongruous that the integration of relationship science into social psychology is at best described as nascent. Unquestionably, and to our benefit, more and more research into the nature and impact of relationships appears in social-psychological journals. However, this work tends to be balkanized, in the sense that it is rarely integrated with other topics studied by social psychologists (and even less into other psychological disciplines). This is readily apparent in textbooks, where one, or perhaps two, chapters on relationships are now standard. However, these chapters are typically stand-alone; that is, they present content that is not systematically integrated with the content of other chapters and topics. By so doing, authors fail to recognize how the reality of these processes in everyday life reflects the predominant focus of the social world—namely, our social and close relationships. In our view, deeper integration of this sort is likely to enhance the informativeness, completeness, and usefulness of social-psychology theory and research.

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14

MOOD EFFECTS ON SOCIABILITY

How Affect Regulates Relationship Behaviors

Joseph P. Forgas

Abstract: Affective states play a crucial role in managing human sociability and relationships. Indeed, as Zajonc (1980) suggested, affect is the primary dimension of interpersonal behavior, yet the evolutionary functions of affective states in regulating sociability remained poorly understood until quite recently. This chapter describes recent evolutionary approaches that suggest a functionalist explanation for the regulatory benefits of mild positive and negative mood states on social behavior, and the subtle relationship between feeling, thinking, and interpersonal behavior will be discussed. On the basis of recent experimental research, the chapter suggests that both positive and negative moods offer significant adaptive benefits for human sociability by fine-tuning interpersonal strategies. First, theories linking affect to cognition and behavior will be discussed. Next, a series of experiments will be described demonstrating that moods function as useful regulatory inputs to sociability, influencing social judgments and impressions, interpersonal trust, the detection of deception, as well as actual strategic behaviors. These results support theoretical models that predict that moods may spontaneously promote thinking and behaviors that are most appropriate in a given situation. The relevance of these findings for our understanding of sociability, as well as recent affect-cognition theories will be discussed, and the practical implications of the adaptive benefits of moods for promoting more effective social strategies in a number of applied fields will be considered.

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Introduction

The unique human capacity for sociability—to engage in complex, well-coordinated, and ultimately successful interactions with others, and to build rewarding relationships—is arguably the foundation of the evolutionary success of our species (see also Prislin & Crano; and von Hippel & Smith, this volume). We also know from a growing body of recent research that affect is one of the primary dimensions of interpersonal behavior, with affective reactions often preceding, influencing, and even dominating patterns of sociability (Zajonc, 1980, 2000; see also Fiske, this volume). However, the precise evolutionary functions of mood states in regulating sociability and interpersonal behavior and the mechanisms linking moods to cognition and behavior have remained relatively neglected until recently.

This chapter will discuss the adaptive regulatory benefits of positive and negative mood states in fine-tuning strategies of sociability, and in particular, the subtle links between feeling, thinking, and social behavior will be explored. On the basis of recent experimental research, the chapter suggests that in many cases moods are capable of conferring significant adaptive benefits in regulating how people perceive and interpret each other, how much they trust others, how they select interpersonal strategies, and how they use language and communication to achieve their social objectives.

The experiments reviewed in the second half of the chapter will demonstrate that mild, temporary moods function as useful and adaptive inputs to guiding both the *content* (valence) and the *process* of managing social interactions. Mood states were found to fine-tune both cognitive and social performance, influencing impressions, judgments, as well as interpersonal communication, language use, and trust in social partners. These results are consistent with evolutionary theoretical models that predict that positive and negative moods have an important adaptive function in regulating sociability.

Affect and Sociability

Fluctuating affective states are ever-present in our lives, and color and filter everything we think and do during our waking hours (Zajonc, 2000). This chapter deals with the adaptive, evolutionary role of mild affective states and moods in guiding our interpersonal strategies and reactions to the manifold challenges of everyday life. In this discussion we will emphasize the potential relevance of an evolutionary approach that assumes that affective states serve important adaptive functions, operating like evolved functional "mind modules" that spring into action in response to various social and environmental contingencies (Buss, 2019; Forgas, Haselton, & von Hippel, 2007; Frijda, 1986; Al-Shawaf & Lewis, 2017; Tooby & Cosmides, 1992). Both negative and positive moods can have beneficial, adaptive effects on sociability. The chapter will survey

experimental findings offering counterintuitive evidence especially for the often useful and adaptive consequences of mild negative moods for social cognition, judgments, and interpersonal behavior, complementing evidence for the well-demonstrated benefits of positive affect (Fredrickson, 2009).

It is somewhat surprising that in empirical psychology, affect has for long remained the least understood and most neglected component of the historical tripartite division of the human mind into cognition, affect, and conation (Hilgard, 1980). In Western thought, affect was traditionally juxtaposed with reason and rationality, and affective reactions were often considered as a more primitive, archaic, and dangerous force that has an invasive quality capable of compromising rational thinking and behavior (Koestler, 1967). In philosophy the view that affect is dangerous and primitive can be traced all the way to the works of classical thinkers including Plato. This approach found its most explicit manifestation in Freud's psychoanalytic speculations, which also saw affect as a dynamic invasive force that requires considerable psychological energy to be properly controlled.

It is only in the last few decades that this distorted view of affect has undergone a major revision, thanks largely to important advances in neurophysiology and evolutionary psychology. After decades of neglect, renewed psychological interest in the empirical study of affective phenomena emerged in the early 1980s. On the basis of growing research in neuropsychology and neuroanatomy, we now know that affect often serves as an essential and adaptive input that has a key influence on determining how we respond to social situations (Forgas, 1995, 2002).

Frijda (1986), Adolphs and Damasio (2001), and Zajonc (1980) were among the first to recognize that affective reactions often constitute the primary and dominant form of responding to social situations (Unkelbach, Forgas, & Denson, 2008), operating as adaptive signals that guide subsequent cognitive and behavioral reactions. This view is also supported by earlier evidence showing that affective features also constitute the primary dimension that determines how people cognitively represent and store their everyday social experiences (Forgas, 1979). It turns out that mental representations of the social world are primarily organized in terms of how people feel about them, as social "stimuli can cohere as a category even when they have nothing in common other than the emotional responses they elicit" (Niedenthal & Halberstadt, 2000, p. 381).

Beyond the Hedonic Principle

Since positive affective states are hedonistically desirable, their evolutionary functions could be relatively easily discerned. We know that positive affect often promotes creativity, flexibility, co-operation, and improves life satisfaction (Forgas, 1994; 1998a, 1998b, 2002; Forgas & George, 2001; Fredrickson, 2009). The enduring human search for happiness has been a recurring theme for

philosophers for centuries (Haidt, 2007), and the "hedonic principle" of seeking pleasure and avoiding pain was long considered the quintessential "simple and sovereign" theory to explain all human behavior (Allport, 1985). In elaborating this principle, utilitarian philosophers like Jeremy Bentham and David Hume developed a "hedonistic calculus" to estimate the likely positive and negative consequences of alternative courses of action (Haidt, 2007).

Our own epoch is characterized by perhaps the most emphatic individual and cultural pursuit of happiness. There appears a single-minded obsession in modern Western cultures with the benefits of positive affective states, and the rejection of negative affect as undesirable and dispensable. One consequence of this hedonistic bias is that the adaptive evolutionary functions of negative affective states remain poorly understood and often denied. Yet it is rather intriguing that four of the six primary emotions of *Homo sapiens* are also negative—fear, anger, disgust, and sadness (Ekman & Cordaro, 2011). Negative affect, and sadness in particular, has long represented a major challenge for evolutionary theorizing (Andrews & Thomson, 2009). Fredrickson (2009) and others have made a convincing case for the adaptive befits of positive affect, but a parallel case for the benefits of negative mood and its effects on promoting more accommodative processing received far less attention. This chapter seeks to restore balance by offering a more objective evolutionary perspective, focusing on the adaptive functions and benefits of both positive and negative affective states on human sociability.

This approach is consistent with evidence for the important adaptive functions of a number of negative emotions (Al-Shawaf, Conroy-Beam, Asao, & Buss, 2015; von Hippel, 2018). For example, fear is instrumental in avoiding dangerous situations, anger helps to focus attention and energy on overcoming obstacles, and disgust plays a key role in avoiding contamination and illness. But what might be the adaptive function of sadness, perhaps the most common, ubiquitous, and problematic of our negative affective states? Managing and controlling sadness is probably the major challenge facing most professional psychologists and counselors, and much of the psychology profession is employed in alleviating sadness.

We also need to consider the question: even if sadness could be controlled, would it be desirable to remove all sadness from our social lives? Without sadness, happiness would not be quite the same. More importantly, we cannot ignore the essential adaptive functions of sadness, still incompletely understood (Andrews & Thomson, 2009; Ciarrochi, Forgas, & Mayer, 2006). Rather than defining sadness as problematic, the research presented here suggests that negative moods should be considered as an important adaptive component of the human affective repertoire.

There are also interesting historical considerations suggesting the adaptive benefits of sadness and negative moods. In many previous epochs of human history, experiences of sadness and melancholia were considered legitimate and

even useful and desirable affective states that deserve to be cultivated (Sedikides, Wildschut, Arndt, & Routledge, 2006). The majority of great artistic works in Western civilization deal with experiences of sadness, loss, and tragedy as instructive and worthy of rehearsal. From classic Greek tragedies through dramas by Shakespeare, Chekhov and Ibsen, and the great novels of the 19th century by Tolstoy, Flaubert, Dostoyevsky, Dumas, Dickens, and Balzac, all were committed to exploring the landscape of sadness. It seems that rehearsing and vicariously experiencing artistic elicitations of sadness and tragedy have long been considered instructive and even ennobling in previous epochs. This cultural history reinforces the view advanced here that negative affect is often useful and adaptive, and we need to understand how it impacts on social behaviors and communication. Exploring just how affect may influence sociability will be considered in the next section.

The Adaptive Functions of Mood in Sociability

Affective states often appear to function as built-in evolutionary signals that promote adaptive changes in the behavior of the organism, and can often directly trigger functional affective displays including facial expressions that are universally recognizable (Darwin, 1872; Ekman & O'Sullivan, 1991; Frijda, 1986). In a highly social species such as humans, the experience and spontaneous display of affective states typically elicits automatic reactions from others that require no prior cognitive elaboration, as demonstrated by research on mirror neurons. For example, sadness may signal withdrawal from competitive and potentially costly activities (Andrews & Thomson, 2009; Buss, 2019), and was found in some studies to invite empathy, support, and care from the immediate social network (Buss, 2019; Watson & Andrews, 2002).

In a similar way, affective states and moods also have important internal signaling functions. For example, the experience of negative affect may function as an internal trigger either to increase effort and attention or to disengage from and abandon costly and potentially hopeless activities. Negative mood may also be an important corrective signal to reduce the habitual human tendency for unrealistic optimism and overconfidence (Nesse & Williams, 1994; Stevens & Price, 1996; von Hippel, 2018). The last few decades produced intriguing experimental research demonstrating the specific mechanisms that link moods to cognitive and behavioral responses involved in sociability, to be considered in the second half of the chapter.

It is the influence of moods rather than distinct emotions on sociability that will be of greatest interest here, as moods are more common, more enduring, and produce more uniform and reliable cognitive and behavioral consequences than do more intense, conscious, and context-specific emotions (Forgas, 2002, 2006). Moods are low-intensity, diffuse, and relatively enduring affective states without a salient antecedent cause and therefore little conscious cognitive

content. In contrast, emotions are more intense, short-lived, and usually have a definite cause and conscious cognitive content (Forgas, 1995, 2002). Unlike emotions that arise in response to specific environmental challenges, moods are ever-present and represent a subtle stream of affective reactions that accompany every aspect of human sociability.

Theories Linking Affect and Sociability

Two distinct kinds of cognitive and behavioral effects of mood have been identified. The experience of a mood reliably produces preferential access to mood-consistent contents from memory (Bower, 1981). This pattern of subconscious affect infusion into thinking and behavior can have an adaptive influence on the *valence and content* of cognition and ensuing social behaviors. The second adaptive function of moods is their impact on the *process* of cognition, automatically triggering a more or less attentive, accommodating, concrete, and detail-oriented information-processing style (Bless & Fiedler, 2006; Schwarz, 1990). This evolutionary effect is essentially like a *cognitive tuning* function—subtly shifting the information-processing strategies employed along a processing continuum (Schwarz, 1990). These two theoretical accounts will be briefly summarized next.

Valence and Content Effects

Moods may have an adaptive effect on the content of thinking and behavior by selectively triggering more mood-consistent thoughts, memories, judgments, and associations. Two complementary mechanisms may explain the process of affect infusion, a memory-based *affect priming* mechanism, and the inferential *affect-as-information* model.

The affect-priming account (Bower, 1981) suggests that affective experiences are embedded in an associative network of memory representations, and will selectively prime memory contents previously linked to that affect. Mood-consistent constructs are more likely to be retrieved and used to inform cognitive and behavioral responses. Several studies found that people induced to feel good or bad selectively remember more positive or negative details respectively from their childhood as well as their recent past (Bower, 1981) and are prone to interpret social behaviors in a mood-consistent manner (Forgas, Bower, & Krantz, 1984; Forgas & Bower, 1987). This effect occurs most reliably when a task requires open, constructive processing and greater reliance on retrieval processes, as is the case with many interpersonal judgments, inferences, associations, and interpersonal behaviors (e.g., Bower & Forgas, 2001; Forgas, 2002; Forgas & Eich, 2013).

A complementary *affect-as-information* model (Schwarz & Clore, 1988; Clore, Gasper, & Garvin, 2001) suggests that rather than computing a judgment on the basis of recalled features of a target, individuals may . . . ask themselves: "how do I feel about it? [and] in doing so, they may mistake feelings due to a

pre-existing state as a reaction to the target" (Schwarz, 1990, p. 529). Thus, mood congruence in sociability is caused by an inferential error or mistaken association, linking an underlying mood to an unrelated social stimulus (Clore & Byrne, 1974). Such misattribution is most likely when "the task is of little personal relevance, when little other information is available, when problems are too complex to be solved systematically, and when time or attentional resources are limited" (Fiedler, 2001, p. 175; Forgas & Moylan, 1987; Schwarz & Clore, 1988).

Affect infusion into interpersonal behavior is greater when a person needs to engage in more constructive, open thinking that requires the use of new information to construct a response, according to the Affect Infusion Model (AIM; Forgas, 1995, 2002). When there is a pre-existing response available in response to a familiar situation, or when reactions are dominated by a pre-existing motivational objective (e.g., trying to look good at a job interview), affect infusion is unlikely according to the AIM. The AIM also makes a counter-intuitive prediction, now confirmed in numerous experiments: paradoxically, affect infusion is increased when more lengthy, extensive and constructive thinking is required to deal with a more complex, unusual or unfamiliar social situations (Forgas & Eich, 2013).

Information-Processing Effects

Affect may also have an adaptive influence on the *process* of cognition and behavior, that is, *how* a task is performed (Clark & Isen, 1982; Forgas, 2002). It has long been suspected that negative mood may function as an evolutionary alarm signal, promoting a more effortful and vigilant processing style (Sinclair & Mark, 1992; Schwarz, 1990; Schwarz & Bless, 1991). Evolutionary explanations highlight *functional principles* suggesting that affective states exist for the purpose of modulating the degree of attention and effort required to deal with more or less demanding situations (Forgas et al., 2007). This principle has been developed into a comprehensive theory by Bless and Fiedler (2006), who explicitly argue that different moods have the evolutionary function of recruiting qualitatively different processing *styles*.

Following the processing dichotomy introduced by Piaget, Bless, and Fiedler (2006) argue that sadness and negative moods call for *accommodative, bottom-up* processing, directed at discovering new, relevant details in the external world in a challenging situation. In contrast, positive moods recruit *assimilative, top-down* processing, signaling that reliance on existing schematic knowledge and heuristics is adequate (Bless, 2000; Bless & Fiedler, 2006; Fiedler, 2001). In an adaptive pattern, *accommodation* in sad mood thus involves increased attention to new information, greater sensitivity to external norms and social expectations, and a more concrete, piecemeal and bottom-up processing style.

Numerous experiments now confirm that negative affect, by facilitating the processing of new external information, can improve memory performance,

reduce judgmental mistakes (Forgas, 1998a, 2011a, 2011b; Forgas, Goldenberg, & Unkelbach, 2009), and also improve the effectiveness of various social behaviors such as persuasive communication (Forgas, 2007). Experimental evidence for the adaptive benefits of negative affect will be summarized in the next section, demonstrating the adaptive benefits of moods for interpersonal trust, judgments, and impressions, as well as strategic interpersonal behaviors. The prediction is that different mood states, by spontaneously triggering different information-processing and behavioral strategies, recruit a thinking style that is most appropriate in a given situation (Al-Shawaf, Lewis, Wehbe & Buss, 2019; Bless & Fiedler, 2006; Forgas & Eich, 2013), consistent with the view that affective states "exist for the sake of signaling states of the world that have to be responded to" (Frijda, 1988, p. 354).

Evidence Linking Mood and Sociability

Moods may have an important beneficial effect on sociability by automatically promoting access to more mood-congruent information, and also by recruiting different information processing strategies. The experiments reported here typically employ a two-stage strategy. Participants are first induced to experience a sad, negative affective state (e.g., exposure to sad versus happy movies, music, autobiographic memories, or positive or negative feedback). Once affect is induced, in the second stage, described as a "separate study," mood effects on participants' responses to various experimental tasks are assessed, including memory, judgmental, or behavioral measures.

Mood Effects on Evaluating Others

Perceiving and interpreting others is one of the basic pre-requisites of effective sociability. Perhaps the most common influence of mood states on the way people evaluate others is the mood congruence effect. People experiencing a positive mood tend to form more positive, generous, and optimistic impressions of others, and negative mood has the opposite effect. Several early studies found that mood states can readily infuse the valence of unrelated social judgments and behaviors in a mood-consistent direction (Clore & Byrne, 1974; Feshbach & Singer, 1957; Razran, 1940). These effects were reliably linked to memory-based affect priming mechanisms in the early 1980s, confirming that the constructive interpretation of others is selectively influenced by the mood state of the judge at the time.

Mood congruence in sociability judgments. For example, Forgas et al. (1984) found that the interpretation of videotaped social behaviors by happy and sad judges showed a significant mood-congruent pattern. While watching exactly the same videotape, happy observers "saw" more skilled and positive behaviors, and observers induced into a negative mood saw more negative,

unskilled behaviors, suggesting that moods have a profound effect on how people come to interpret even the most basic observed social actions. The same smile may be interpreted as friendly and positive in a good mood, while judges in a negative mood see it as conceited and condescending. These effects occur because constructive social judgments necessarily rely on the greater availability of mood-primed information (Bower, 1981).

Similar effects were found in an experiment by Forgas and Bower (1987), where happy and sad judges were asked to learn about, and form judgments of target characters featuring both positive and negative characteristics. There was a significant mood-congruent effect on judgments. An analysis of reading and judgmental latencies confirmed that consistent with the affect-priming model, judges took longer to read and encode mood congruent details into a richer primed base of associations, but were significantly faster in producing mood-congruent judgments.

Mood effects on sociability judgments also tend to be significantly greater when more complex, ambiguous, or demanding targets requiring longer and more elaborate processing are evaluated. As the Affect Infusion Model predicts, the reason for this non-obvious effect is that more elaborate and constructive processing increases the likelihood that selectively primed mood-congruent contents will be accessed and used in the judgment. This prediction was confirmed in several studies, showing greater mood-congruent effects when evaluating atypical rather than typical people, and atypical rather than typical couples and relationships (Forgas, 1993, 1995).

Mood effects on social accuracy. In addition to influencing the *valence* of social evaluations, mood states may also influence how carefully information is processed, impacting on the accuracy of sociability judgments. Many common judgmental errors in everyday life occur because people are imperfect and often inattentive information processors. It turns out that mild negative mood, by triggering a more thorough and accommodating processing strategy, tends to reduce some of the common judgmental errors that occur as a result of simplified, heuristic processing.

For example, the *fundamental attribution error* (FAE) or *correspondence bias* refers to a pervasive tendency by people to infer intentionality and internal causation and underestimate the impact of situational forces in their social judgments (Jones & Davis, 1965). This error occurs because perceivers are captivated by salient information such as the actor and neglect information about situational constraints. Several experiments found that sad participants were less likely to succumb to the fundamental attribution error than those in a positive mood (Forgas, 1998b; Forgas, Bower, & Moylan, 1990). Consistent with the adaptive processing influence of negative affect, it appears that those in a negative mood processed the available information more carefully, took longer to compute a judgment, and also remembered the details better later on. A statistical mediation test was consistent with the assumption that

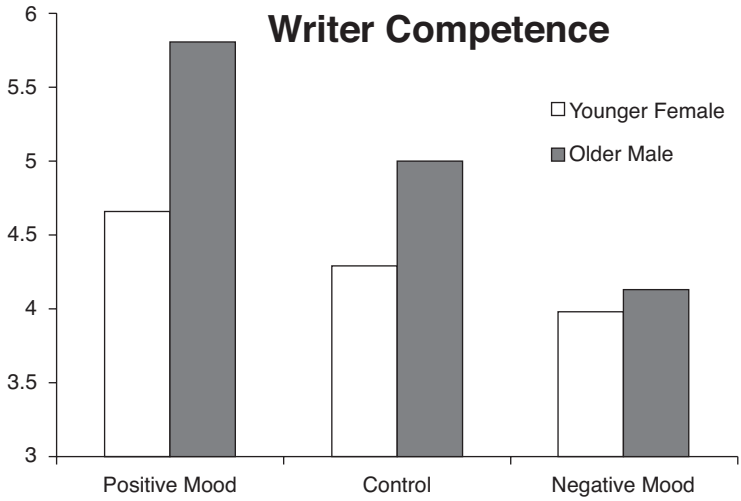

FIGURE 14.1 Sad mood reduces the halo effect: Judgments of a writer's competence were less influenced by their photo showing their age and sex when judges were in a negative rather than in a positive mood (after Forgas, 2011b).

mood-induced processing differences resulted in the reduction of judgmental errors and improved memory performance.

Other experiments found that negative affect reduced such common judgmental heuristics such as *halo effects* and *primacy effects* in impression formation. In one study we found that halo effects (evaluating a person based on their appearance) were eliminated by negative mood, while positive mood increased reliance on appearance (Forgas, 2011b; Figure 14.1.)

The adaptive benefits of sad mood on judgmental accuracy were also confirmed by studies exploring primacy effects—the tendency to rely on early information when forming impressions (Asch, 1946; Luchins, 1958). Primacy effects often disappear when judges process every detail equally carefully. Since negative mood can reliably trigger more accommodative processing (Bless & Fiedler, 2006; Forgas, 2002, 2007), primacy effects were also reduced by a sad mood as confirmed by several experiments (Forgas, 2011a), a finding conceptually also consistent with prior evidence for directed forgetting induced by positive mood (Bäuml & Kuhbandner, 2009).

Mood effects on interpersonal fairness. One of the key requirements of effective sociability is to act in a fair and reasonable way toward others. Evolutionary psychologists have long speculated about the possibility that negative affect may provide hidden social benefits by arousing and promoting interpersonal sympathy and fairness (Buss, 2019; Forgas et al., 2007; Tooby & Cosmides, 1992). Whereas positive affect may trigger more optimistic, confident, assertive,

and selfish behaviors, sadness and negativity may promote more cautious, pessimistic assessments and more attentive and accommodative processing. Several experiments evaluated mood effects on the level of selfishness versus fairness people display in strategic interactions such as the dictator game and the ultimatum game.

In the *dictator game* allocators distribute resources (e.g., money, raffle tickets) between themselves and a partner in any way they like. In the *ultimatum game*, their partner has a veto power: if a distribution is rejected, neither side gets anything. In principle, actors should maximize benefits to themselves, but in reality allocators are also influenced by norms of fairness in their decisions (Güth, Schmittberger, & Schwarze, 1982). Affect may influence fairness, as negative mood might recruit more accommodative and externally focused strategies that increase attention to external norms of fairness.

In several experiments (Tan & Forgas, 2010) we found that happy players were significantly more selfish and kept more resources (e.g., raffle tickets) to themselves, while sad allocators were much fairer. Similar mood effects on fairness were also found in the more complex decisional environment in the ultimatum game. Again, those in a sad mood allocated significantly more resources to others than did happy individuals, and this pattern was also confirmed in a serial allocation task, with eight different partners (Figure 14.2). Sad individuals also took longer to make allocation decisions than did happy individuals, consistent with their expected more accommodative and attentive processing style.

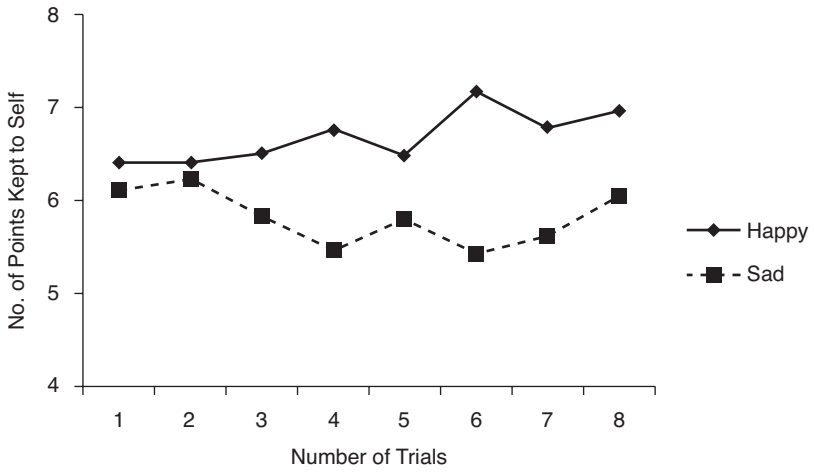

FIGURE 14.2 The effects of sad mood on selfishness versus fairness in the ultimatum game: sad participants kept fewer rewards to themselves and gave more to their partners, consistent with their greater attention to external fairness norms (after Forgas & Tan, 2013).

Fairness was also more important to sad *recipients* in the ultimatum game, who were more likely to reject unfair offers than did happy recipients (Forgas & Tan, 2013). These results further support the evolutionary idea that moods have important regulatory and adaptive consequences in decisions involving sociability.

Mood Effects on Trust and Gullibility

One of the most critical factors that affect sociability is the question of trust. Human beings rely on each other for information, support, and, indeed, survival, but at the same time, must also be on the lookout for attempts to exploit their potential gullibility (Buss, 2019; von Hippel, 2018). In every social episode, interactants must continuously assess the trustworthiness and reliability of their partners (Forgas, 1979). There is now a growing body of literature suggesting that subconscious fluctuations in mood states might play a critically important role in calibrating the level of trust—gullibility versus skepticism—that people experience when interacting with others. In particular, negative affect may function as an evolutionary "alarm signal" against excessive trust and gullibility, as also found for aversive smells that reduce gullibility (Schwarz & Lee, 2018), potentially producing more attentive information processing and actually improving the ability to detect deception (Lane & DePaulo, 1999). Positive mood in contrast has the opposite effect, promoting trust and in many cases, gullibility. We shall look at this literature next.

Mood effects on detecting deception. How does mood influence people's ability to detect deception? In one experiment, we asked happy or sad participants to detect deception in the videotaped statements of people who were accused of theft, and who were either lying or telling the truth (Forgas & East, 2008b). Results showed that those who experienced a negative mood were generally less trusting and more dubious. More surprisingly, negative mood also improved judges' actual ability to correctly identify deception (Figure 14.3).

Mood effects on trusting emotional communication. In a similar way, negative mood also appears to reduce trust and increase skepticism about the genuineness of interpersonal communication. When happy or sad participants judged the genuineness of positive, neutral, and negative facial expressions, those in a negative mood were significantly less likely to accept facial expressions as genuine.

In a follow-up study, happy or sad judges judged the genuineness of facial expressions of the six basic emotions (i.e., anger, fear, disgust, happiness, surprise, and sadness) as displayed by trained actors. Negative mood again reduced, and positive mood increased people's trust of the facial displays as genuine. It seems that moods have an adaptive regulatory influence on interpersonal trust. As a result of triggering a more accommodative and attentive processing style, negative mood seems to reduce, and positive mood seems to increase trust (Figure 14.4).

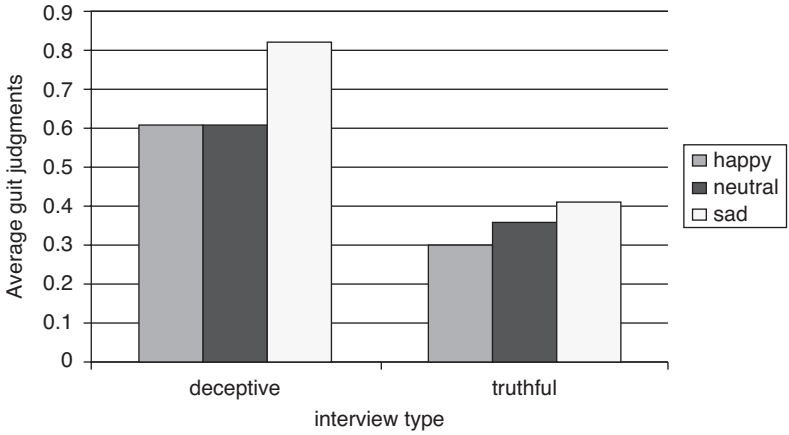

FIGURE 14.3 Sad mood improved judges' ability to discriminate between truthful and deceptive targets (after Forgas and East (2008b)).

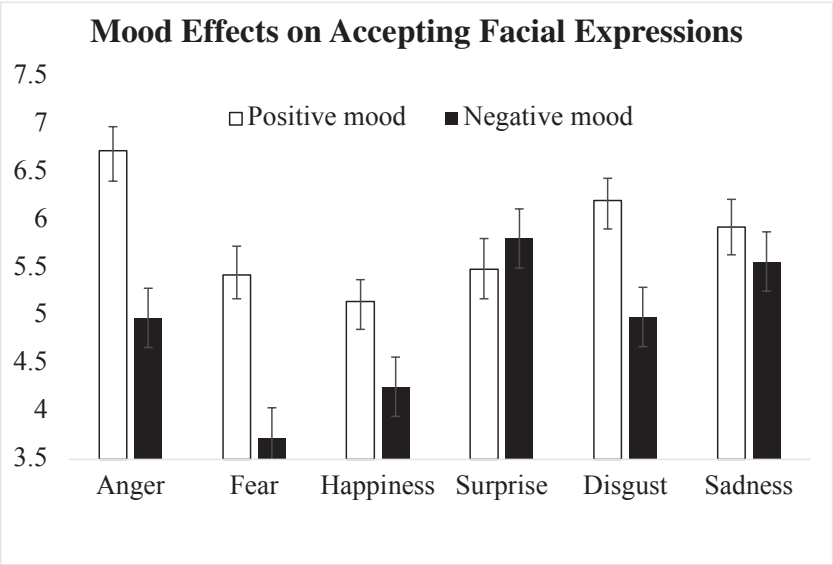

FIGURE 14.4 Sad mood judges are significantly more skeptical and are less likely to accept facial expressions of basic emotions as genuine compared to happy judged (except for the "surprise" expression where differences due to mood were nonsignificant) (after Forgas and East (2008b)).

Mood effects on gullibility. Verbal messages we receive from others lie at the heart of human sociability. The human tendency to trust such second-hand verbal information is an important source of both social integration, and the development and maintenance of a coherent view of the world (Buss, 2019; Harari, 2014). Does mood influence people's willingness to accept or reject dubious verbal information received from others?

In one experiment happy or sad judges estimated the likely truth of a number of ambiguous verbal statements (urban legends and fake news), such as "power lines cause leukaemia" or "the CIA murdered Kennedy" (Forgas & East, 2008a). Positive mood increased trust, but negative mood increased skepticism. These effects were most marked when considering new and unfamiliar claims that required on-line constructive processing, as predicted by the Affect Infusion Model (Forgas, 1995). In a related study, negative mood participants were better able to distinguish between the true and false claims they had seen previously, while those in a positive mood rated all previously seen claims as true, consistent with the "what is familiar is true" heuristic. In this case, sad mood again conferred a clear advantage by promoting a more accommodative processing style, reducing unwarranted trust and gullibility (Fiedler & Bless, 2001).

Mood effects on truth judgments. As the previous experiment suggests, decisions about trusting the truthfulness of received messages are often influenced by reliance on heuristics, such as the apparent *cognitive fluency* of a claim. In one experiment, happy or sad participants judged the truth of 30 ambiguous statements presented with high or low visual fluency (against a high or low contrast background). Overall, neutral and positive mood subjects were more trusting and rated fluent claims as more likely to be true than disfluent claims. However, negative affect eliminated this fluency effect (Figure 14.5). The role of moods

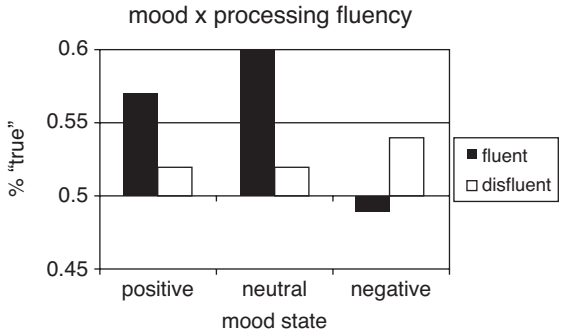

FIGURE 14.5 When in a sad mood judges were less likely to be influenced by irrelevant cues such as the cognitive fluency and visibility of different messages in evaluating their truthfulness (after Koch & Forgas, 2012).

in regulating interpersonal trust may be very important in many real-life situations, as many such judgments (such as believing or disbelieving your partner, your boss, or believing fake news) typically occur in affect-rich contexts.

Mood effects on bullshit receptivity. Effective sociability is fundamentally based on the human ability to observe, interpret, and find meaning and pattern in the behaviors of others. Generally, humans are inclined to search for, and often find meaning and pattern even in intentionally random or meaningless information. This tendency reflects the evolutionary imperative that missing an existing pattern is usually far more dangerous and costly than mistakenly identifying a non-existent one.

Remarkably, humans—including academics who surely should know better (!!)—are often inclined to trust and believe even intentionally meaningless, nonsense information as long as it confirms their pre-conceived ideas. This tendency has been strikingly demonstrated recently among academics committed to postmodernist theorizing, as several editors of leading "scholarly" journals in this field were happy to accept for publication articles that were intentionally written to be completely meaningless (Sokal & Bricmont, 1998).

Post-modern "scholars" are not alone in finding meaning where there is none. In another intriguing study, Pennycook, Cheyne, Barr, Koehler, and Fugelsang (2015) found that people perceive some meaning and depth even in completely vacuous, computer-generated verbiage. Can sad or happy mood influence people's receptivity to see meaning in nonsense information? In one study we asked happy or sad participants to rate the meaningfulness of two kinds of verbal "bullshit" (nonsense) communication, including vacuous computer-generated random New Age pronouncements (e.g., "Good health imparts reality

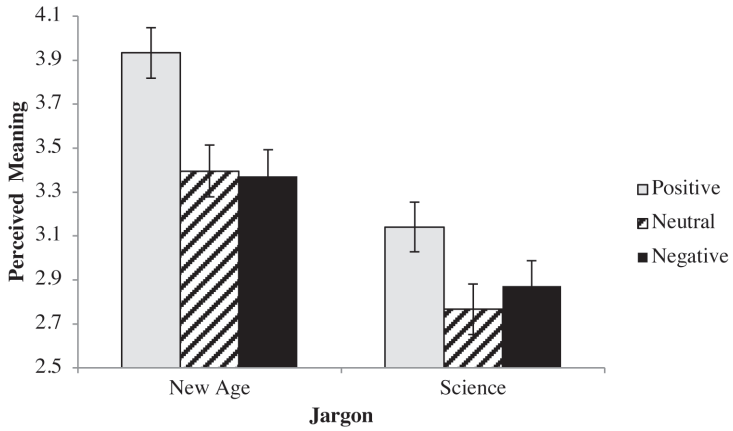

FIGURE 14.6 Mood effects on trust and accepting meaningless new age and scientific nonsense jargons as meaningful: participants in the positive mood condition were more trusting and gullible than those in the neutral and negative mood conditions.

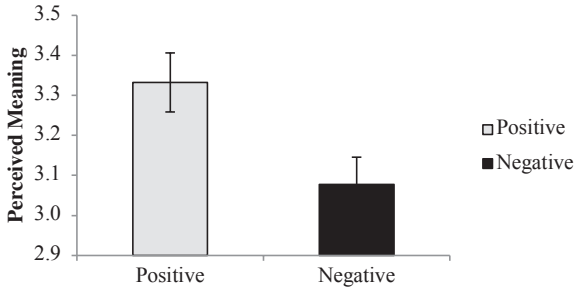

FIGURE 14.7 Mood effects on visual bullshit receptivity: Sad mood reduced the tendency to perceive meaning in abstract expressionist paintings compared to positive mood.

to subtle creativity"; see URL http://wisdomofchopra.com/), as well as meaningless psychological jargon phrases (e.g., "subjective instrumental sublimations"; Forgas, 2019; Forgas, Matovic, & Slater, 2018).

As expected, those in a positive mood perceived more "meaning" compared to those in a sad mood in these gibberish texts (Figure 14.6). Negative mood judges were not only less trusting and gullible, but also slower to make decisions, and had better memory for the statements than did those in a positive mood, confirming the predicted mood-induced processing differences. The same pattern was confirmed in another study using uninterpretable, abstract visual rather than verbal stimuli (abstract expressionist paintings; Forgas et al., 2018): happy participants were more likely than sad participants to find meanings in these uninterpretable images (Figure 14.7).

Overall, positive and negative moods do seem to play an important regulatory function influencing trust and gullibility. Positive moods signal familiar, safe and non-challenging situations, recruit more assimilative and simplified processing strategies, and increase trust and the tendency to accept communications from others at face value. Negative mood in contrast calls for more accommodative, systematic, and attentive processing, increases skepticism and the detection of deception, and reduces trust and gullibility, consistent with moods operating like evolutionary "mind modules," fine-tuning our reactions to the everyday challenges presented by the demands of human sociability.

Mood Effects on Interpersonal Strategies

Engaging in an interaction with another person—the very essence of sociability—is one of the most complex, subtle, and demanding tasks humans face in everyday life. Moods are likely to play a very important role in fine-tuning the interpersonal strategies people adopt in such social situations. As we have seen, positive

and negative moods have a dual function in influencing thinking and behavior, influencing both the valence and content of cognition, as well as the processing strategies people adopt in regulating sociability.

Improved access to mood-congruent information informing subsequent behaviors has been documented in a number of studies (Bower, 1981; Fiedler, 2001; Forgas, 1995). For example, people in a sad mood are more likely to think of negative words (Ruiz-Caballero & Gonzalez, 1994), and generate more negative associations (Bower, 1981). Overall, sad affect also triggers more negative and pessimistic judgments (Forgas & Bower, 1987; Forgas et al., 1984; Sedikides, 1995). These effects are accentuated when more open, constructive thinking is required to deal with more complex or unfamiliar social stimuli (Forgas, 1994, 1995). Next, a number of experiments will be briefly reviewed illustrating how mood states have a significant effect on influencing specific strategies essential for effective sociability.

Mood effects on requesting. One of the most difficult tasks in strategic social interactions is requesting, an inherently complex and risky communicative move that is characterized by uncertainty and the dual dangers of being refused or giving offence. Requests should be formulated with just the right degree of politeness versus assertiveness to maximize compliance without risking giving offence. Moods may have a significant influence on how requesters evaluate the felicity conditions of the request situation. By selectively priming access to more negative information, negative mood may promote a more cautious and pessimistic request strategy (Forgas, 1999a).

This pattern was confirmed in several experiments (Forgas, 1999a) showing that negative mood promoted more polite, elaborate, and considerate requests, whereas happy participants produced more assertive and direct requests. These mood effects on requesting were stronger in difficult rather than easy situations that required more elaborate, substantive processing, as predicted by the Affect Infusion Model (Forgas, 1995). These mood effects were also confirmed in real-life situations (Forgas, 1999b, Exp. 2), when happy or sad participants were asked to "get a file from a neighboring office." The words used in their requests were surreptitiously recorded, and results confirmed that negative mood participants used significantly more polite, elaborate, and hedging request formulations (Figure 14.8).

Mood Effects on Self-Disclosure. Self-disclosure is also a critically important strategy in managing sociability and increasing intimacy between interactants (see also Mikulincer & Shaver; and Reis et al., this volume). Deciding how much to disclose, when, how, to how many people, and how often can make or break a relationship, indicating that the management of self-disclosure is an essential skill in fostering sociability. Moods again may play an important adaptive function in fine-tuning self-disclosure strategies, by selectively priming positive or negative thoughts or associations

In several experiments, happy or sad participants were asked to exchange self-disclosing information with a "partner" in an on-line interaction. We found

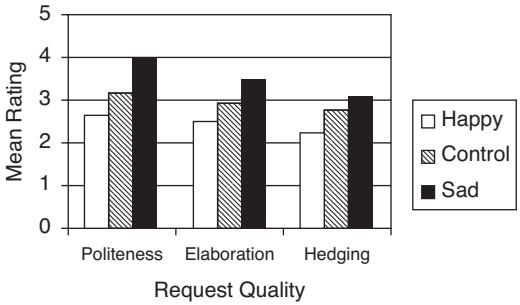

FIGURE 14.8 Mood effects on naturally produced requests: Sad mood increased the degree of politeness, elaboration, and hedging in the request formulations used (after Forgas, 1999b).

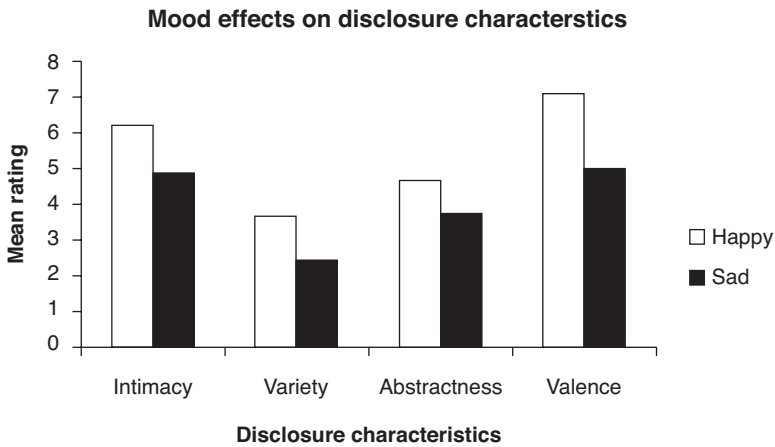

FIGURE 14.9 Affective influence on self-disclosure: Sad mood promotes a more pessimistic and cautious self-disclosure style, resulting in less intimacy, variety, abstraction, and greater negativity in what is disclosed.

that negative mood resulted in less intimate, less varied, and less abstract disclosures, consistent with affect infusion significantly impacting on disclosure strategies (Forgas, 2011c; Figure 14.9). It seems that negative affect produced a more pessimistic, cautious, and avoidant interpersonal style (Buss, 2019; Andrews & Thomson, 2009).

Mood Effects on Conversational Performance. Language is a key medium of sociability. By triggering more or less attentive and accommodative processing strategies, moods may influence people's ability to use language effectively. In a series of experiments (Koch, Forgas, & Matovic, 2013), we found that in

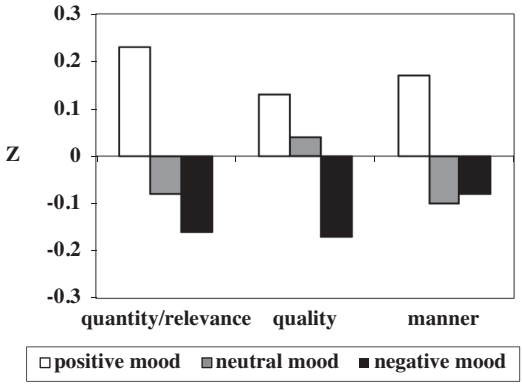

FIGURE 14.10 Mood effects on conversational competence and adherence to Grice's (1975) conversational maxims (Z-scores): sad mood produced fewer violations of the maxims of *quantity/relevance*, *quality*, and *manner* in spoken language compared to positive mood (after Koch et al., 2013, Exp. 1).

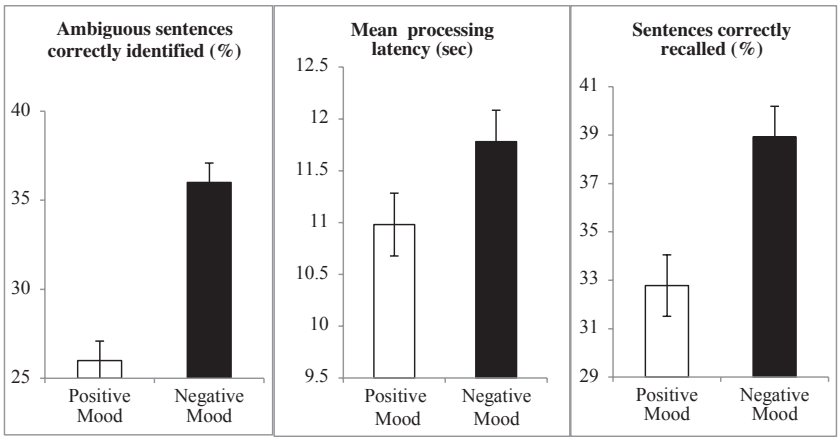

FIGURE 14.11 Sad mood improved the ability to detect ambiguous communication (left panel), and also resulted in longer and more elaborate processing (middle panel), and better memory for the target information compared to positive mood (after Matovic et al., 2014).

conversational situations, participants in a negative mood communicated significantly more effectively and complied significantly better with Grice's (1975) normative maxims than did participants in a positive mood (Figure 14.10).

Mood effects on Language Comprehension. Moods may not only influence language *production*, but can also impact on language comprehension. In

two experiments (Matovic, Koch, & Forgas, 2014) we found that negative mood significantly improved people's ability to correctly detect imperfect, "bad" sentences (ambiguous anaphora; Figure 14.11), linked to mood-induced differences in processing style. Consistent with the predicted signaling effects of moods, negative mood actually produced slower and more attentive processing, mediating mood effects on language comprehension.

Mood Effects on Persuasion. Mood effects on sociability may also influence complex interpersonal performances, such as persuasive communication. Despite much prior interest in responding to persuasive messages, affective influences on the *production* of persuasive messages received little past attention (but see Bohner & Schwarz, 1993). In these experiments, we predicted and found that accommodative processing recruited by negative mood promoted more concrete and factual thinking and produced higher quality, more concrete, and more persuasive arguments than happy participants (Figure 14.12). Of course, the real test of a persuasive argument is whether it actually persuades a recipient. We confirmed this in a follow-up experiment: naïve participants were more persuaded by the more effective arguments written by sad rather than happy persuaders, actually producing greater attitude change (Forgas, 2007).

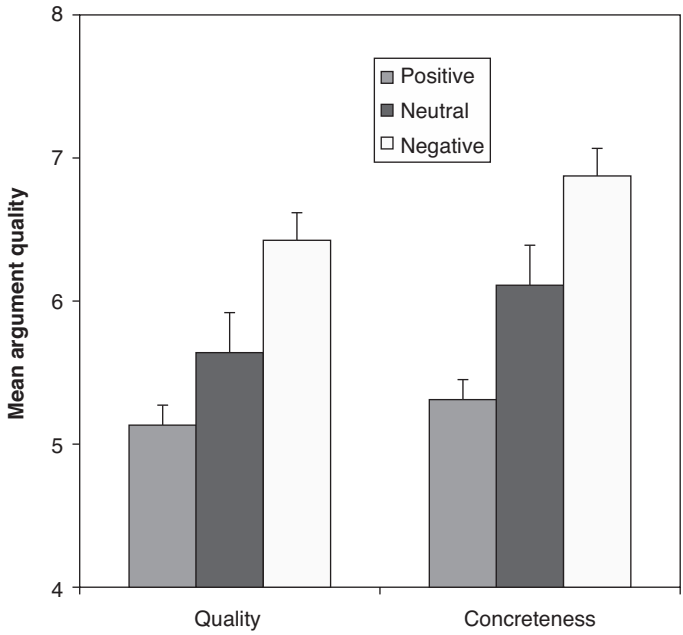

FIGURE 14.12 Sad mood improved the quality and effectiveness of persuasive messages, consistent with sad mood participants using more concrete and specific formulations (after Forgas, 2007, Experiment 2).

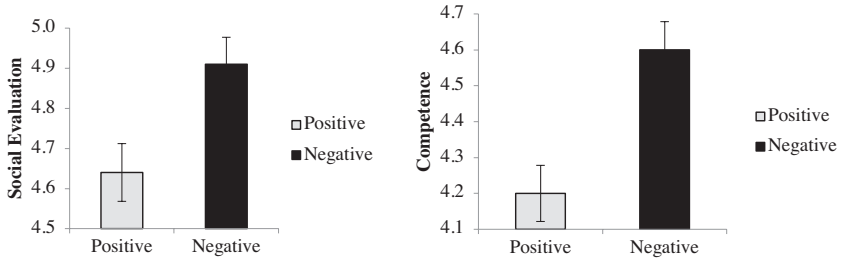

FIGURE 14.13 Ingratiating messages produced by communicators experiencing a sad mood were significantly more effective and resulted in more favorable ratings of the communicators by recipients on measures of *social evaluation* and *competence* compared to messages written by happy ingratiators (after Matovic & Forgas, 2018a).

Subsequently, we found that people in a negative mood were actually more effective in persuading a partner to volunteer for an experiment, and this effect was mediated by a slower and more accommodative thinking style adopted in negative mood.

Mood effects on Ingratiation. Just like persuasion, ingratiation is also a complex and challenging interpersonal task in managing sociability (Jones, 1964). In several experiments (Matovic & Forgas, 2018a), we found that negative mood improved communicators' ability to effectively use ingratiation tactics such as flattery, conformity, and self-promotion. Ingratiating messages written in a sad mood were actually more effective in creating a positive impression than messages written in a positive mood (Figure 14.13). An analysis of response latencies (processing strategies) and memory data also showed that these improvements were due to negative mood promoting a more extensive and attentive style. These results further confirm that moods play a significant adaptive influence on fine-tuning strategies of sociability (Forgas, 1998a, 1998b; Forgas et al., 2005).

Summary and Conclusion

This chapter argued that underlying affective states, such as mild and often subconscious moods, can play a significant adaptive role in regulating strategies of sociability. There is now growing evidence that even dysphoric moods, often considered undesirable and superfluous, do provide important benefits in many everyday situations (Buss, 2019; Andrews & Thomson, 2009). This approach is broadly consistent with evolutionary theories going back to Darwin that suggest that the human affective repertoire has been largely shaped by processes of natural selection. According to such a functionalist view, mood states may operate as adaptive evolutionary algorithms that promote situationally appropriate responses in particular situations (Buss, 2019; Frijda, 1986; von Hippel, 2018).

It now seems that all of our affective states—including the unpleasant ones—may operate as efficient "mind modules" that produce tangible benefits in responding to different social situations (Al-Shawaf & Lewis, 2017; Al-Shawaf et al., 2015; Tooby & Cosmides, 1992). Several studies confirm that people in a negative mood are less prone to judgmental errors (Forgas, 2011a, 2011b, 2011c), are fairer and more sensitive to social norms (Forgas, 1999a, 1999b), and are better at a variety of verbal communication tasks requiring attention to situational details (Forgas, 2007; Koch et al., 2013; Matovic et al., 2014). Positive mood in turn promotes more positive associations, greater self-confidence and more assertive strategies. In other words, both positive moods, and temporary states of dysphoria are a normal part of the human affective repertoire, and help to regulate sociability (Frijda, 1986).

Sociability lies at the core of the human evolutionary experience (Baumeister & Maranges; von Hippel & Smith, this volume). This chapter argued that over evolutionary time, affective states became adaptive, functional triggers that automatically promote information-processing and behavioral strategies that are appropriate in a given social situation. Mild sadness is typically an affective response that occurs in problematic, difficult of unfamiliar situations, and by triggering more cautious, pessimistic and attentive reactions, it promotes a more appropriate response. A key implication of this line of research is that accepting the entire range of our affective reactions, including the negative ones, may be more conducive to human well-being than single-mindedly focusing on the pursuit of positive affective states alone.

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